

THE HARBOR, THE PERSON, AND THE ECONOMY

Figure revision atlas

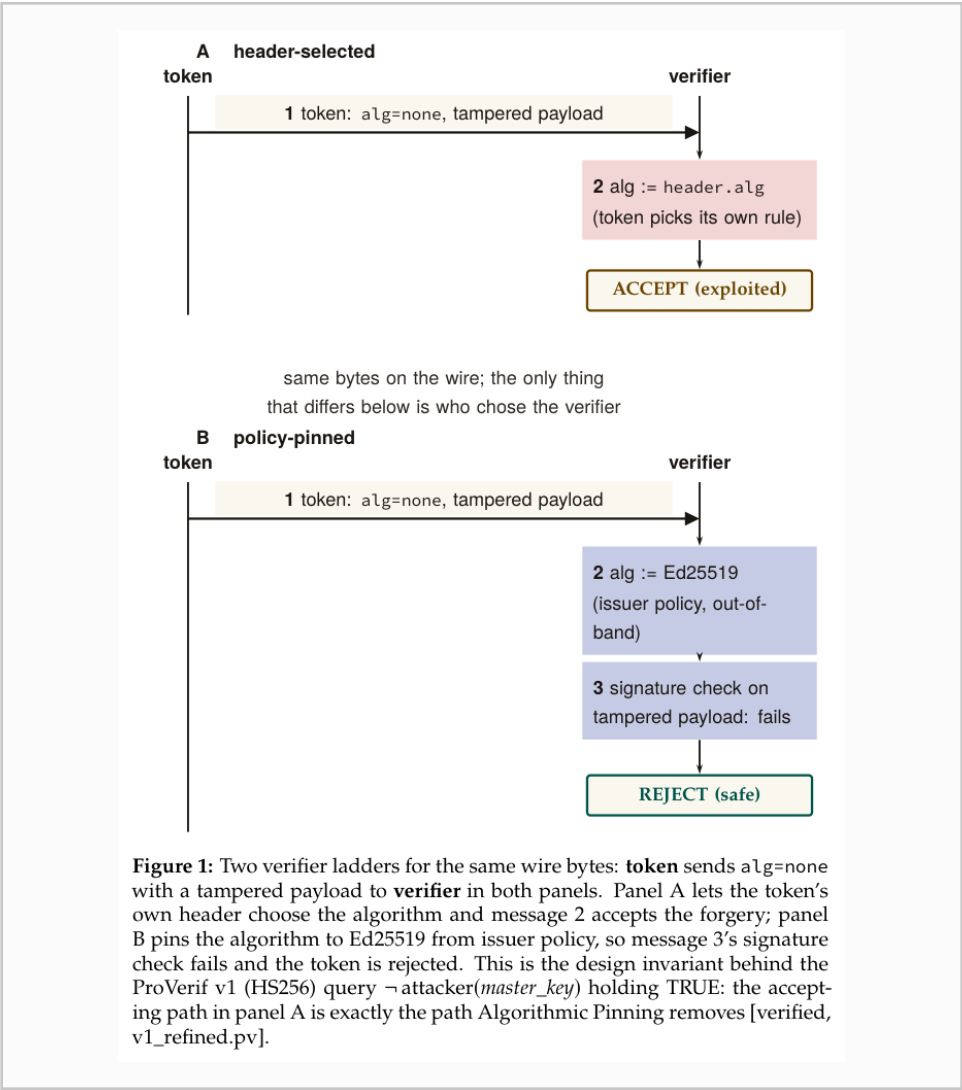
Old and new, one figure per spread

56 redrawn figures · 67 new figures/tables · 1 former figure replaced by a table

Rendered from the stacked figure PRs at 0dfe4a5a6.

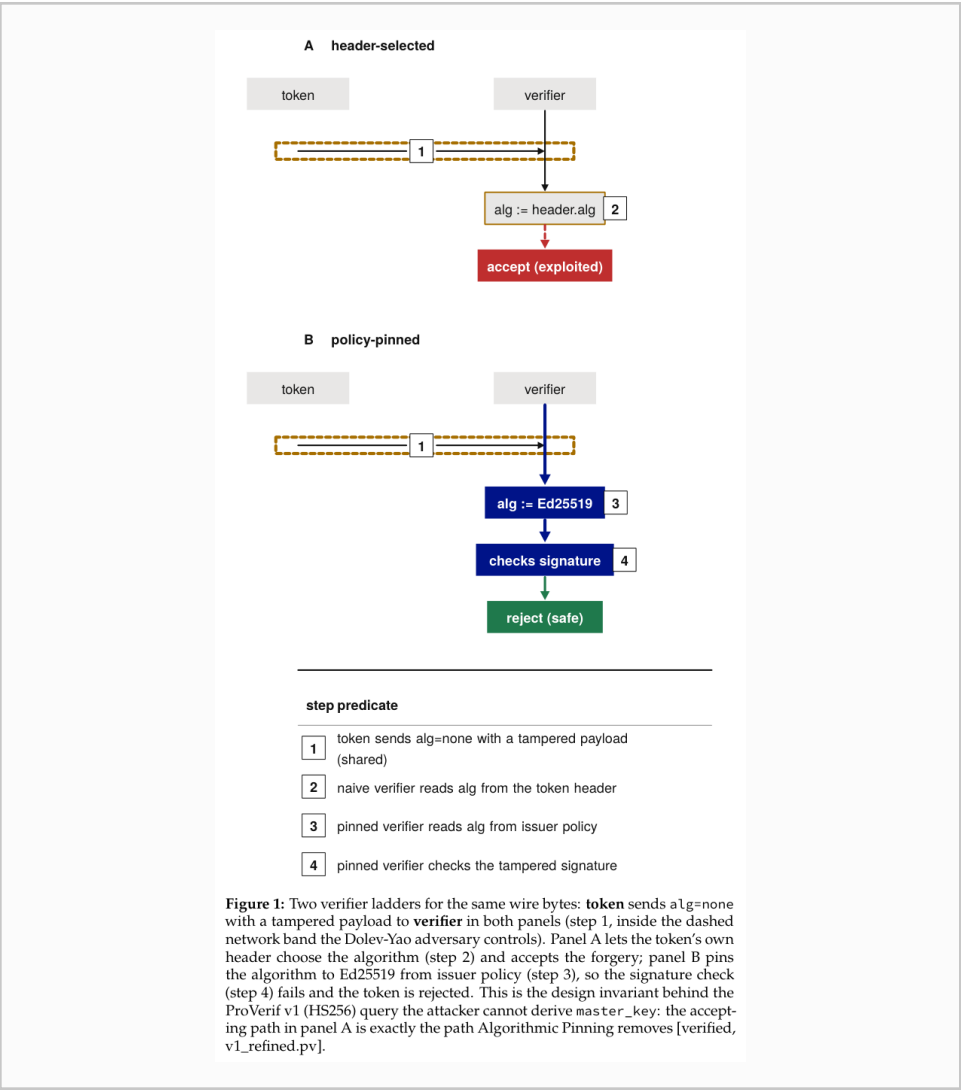
LEFT: BEFORE · RIGHT: AFTER

BEFORE



QA: FLAGGED: B9

AFTER

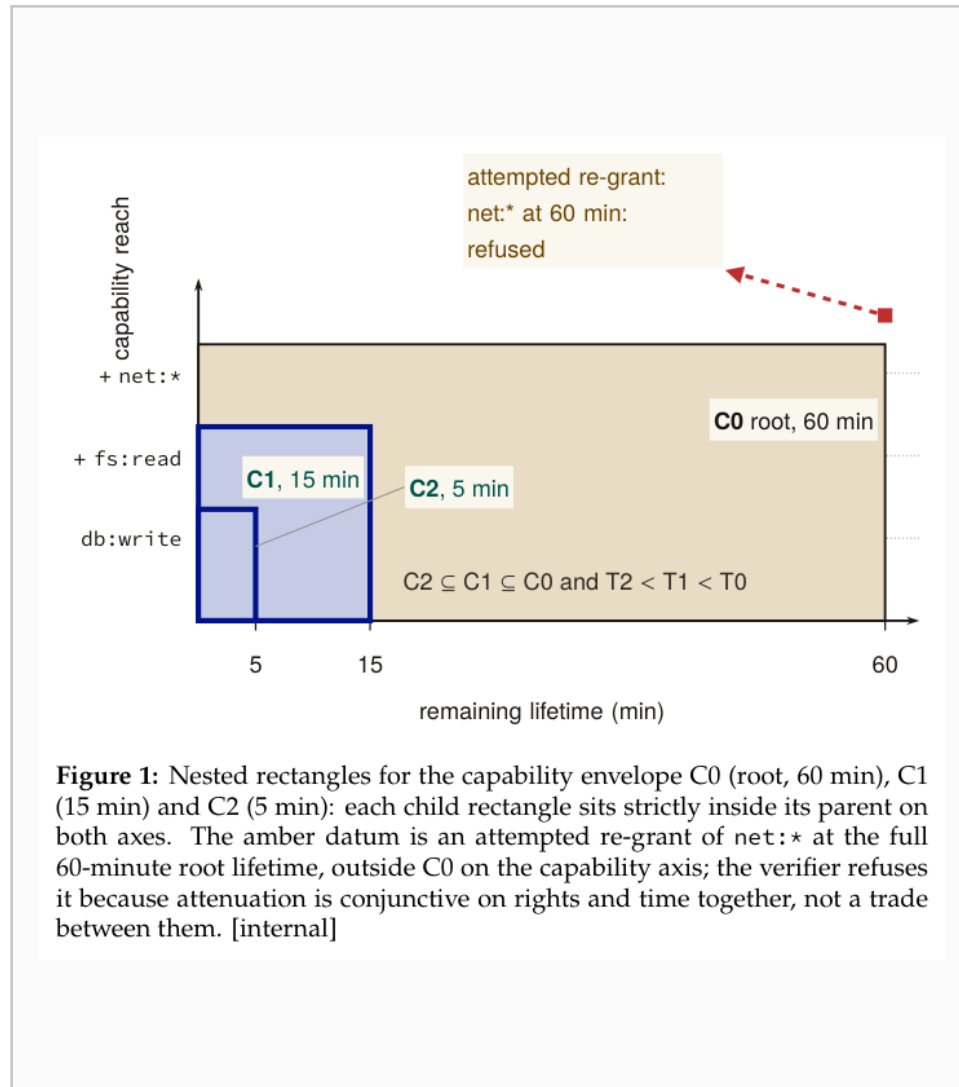


QA: PASS

Two verifier ladders for the same wire bytes: token sends {alg=none with a tampered payload to verifier in both panels (step 1, inside the dashed network band the Dolev-Yao adversary controls). Panel A lets the token's own header choose the algorithm (step 2) and accepts the forgery; panel B pins the algorithm to Ed25519 from issuer policy (step 3), so the signature check (step 4) fails and the token is rejected.

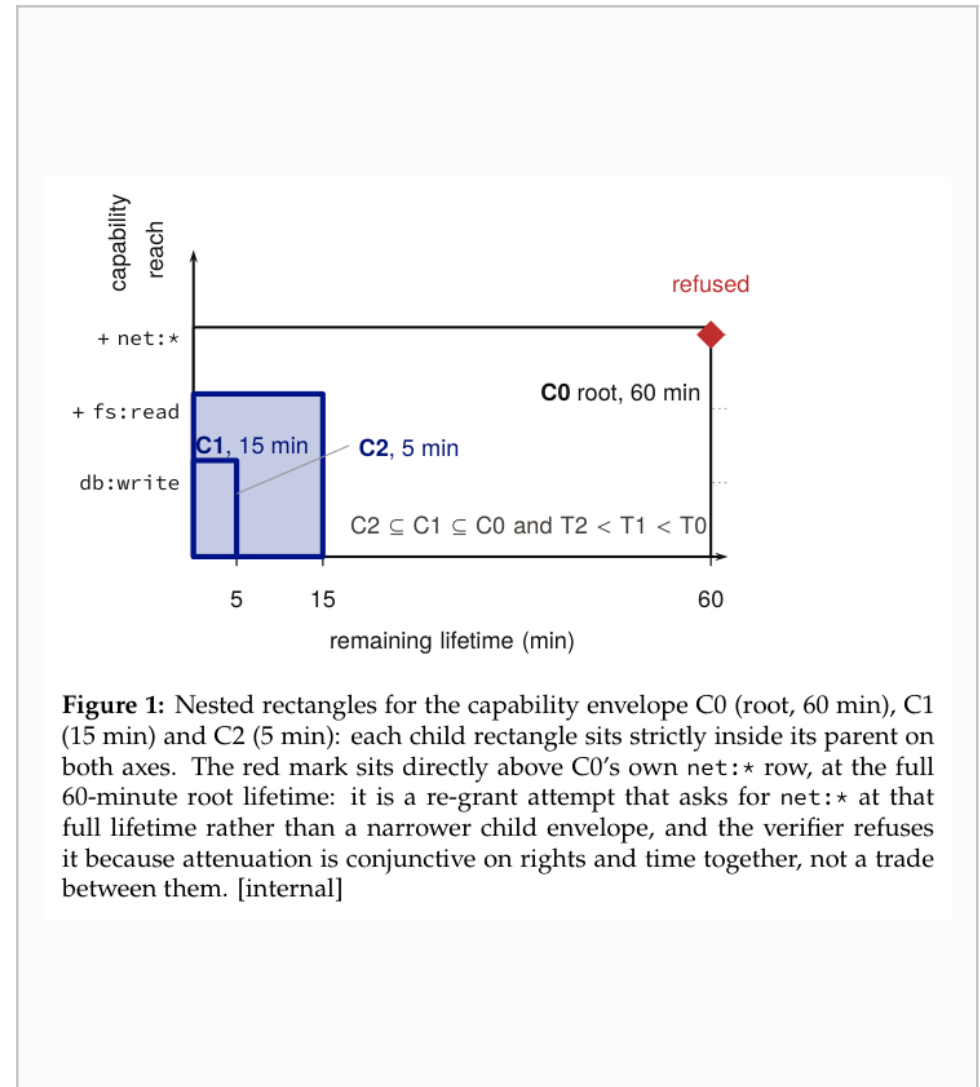
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BEFORE



QA: PASS

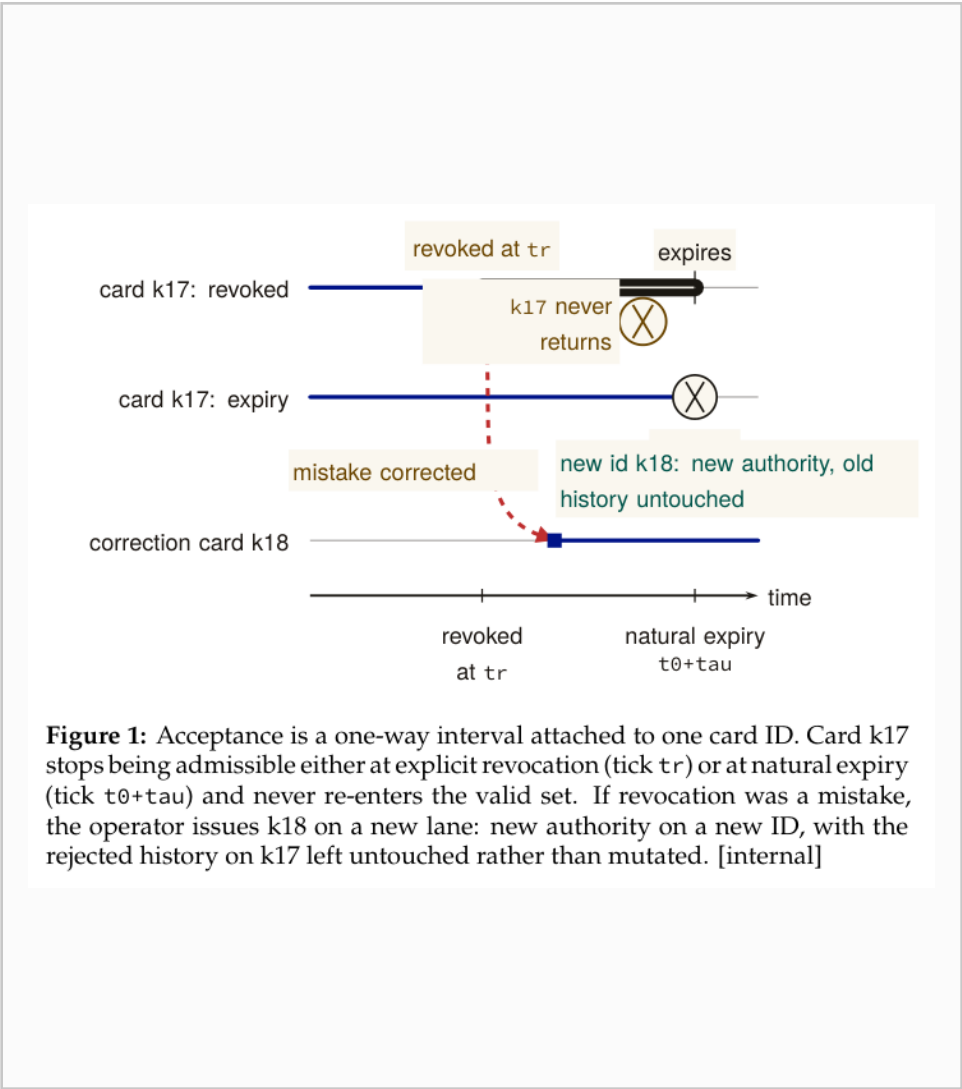
AFTER



QA: PASS

Nested rectangles for the capability envelope C0 (root, 60 min), C1 (15 min) and C2 (5 min): each child rectangle sits strictly inside its parent on both axes. The red mark sits directly above C0's own {net:* row, at the full 60-minute root lifetime: it is a re-grant attempt that asks for {net:* at that full lifetime rather than a narrower child envelope, and the verifier refuses it because attenuation is conjunctive...

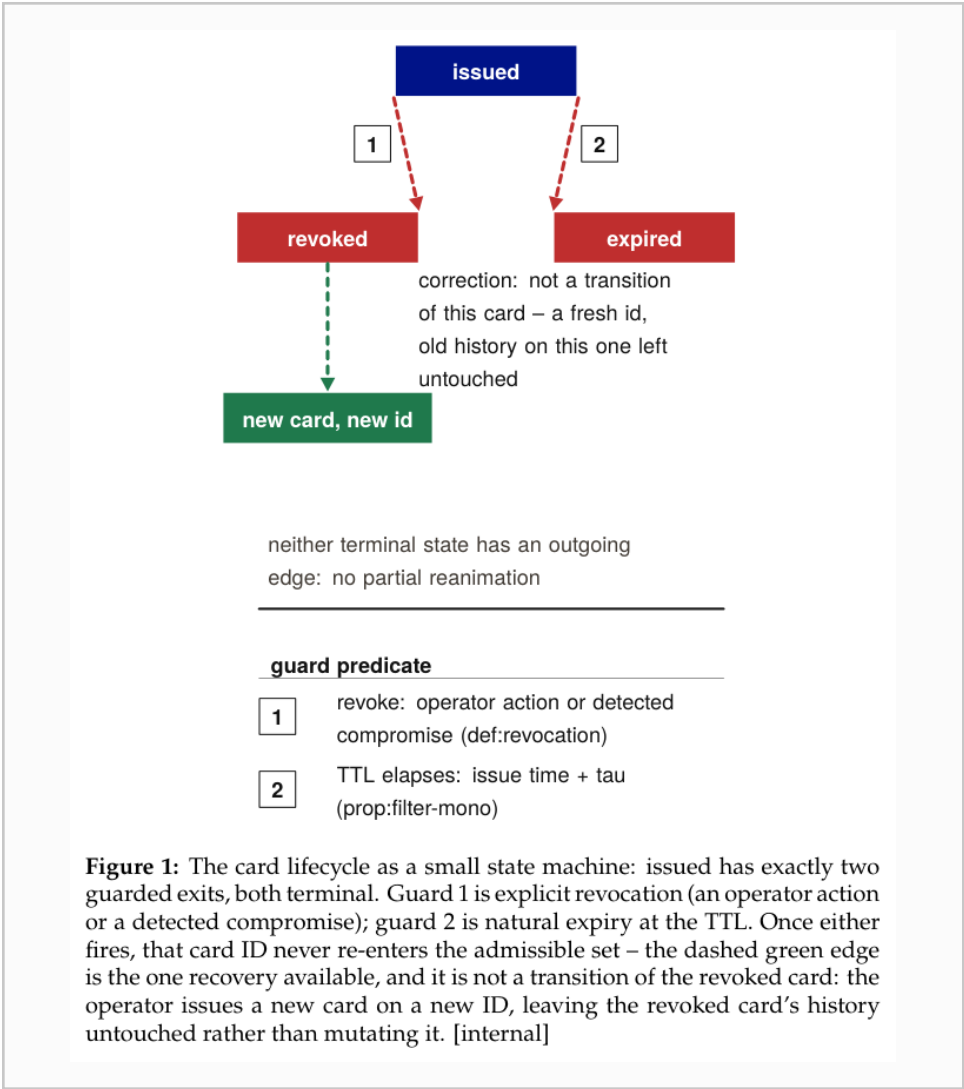
BEFORE



QA: FLAGGED: T2, T3, B3

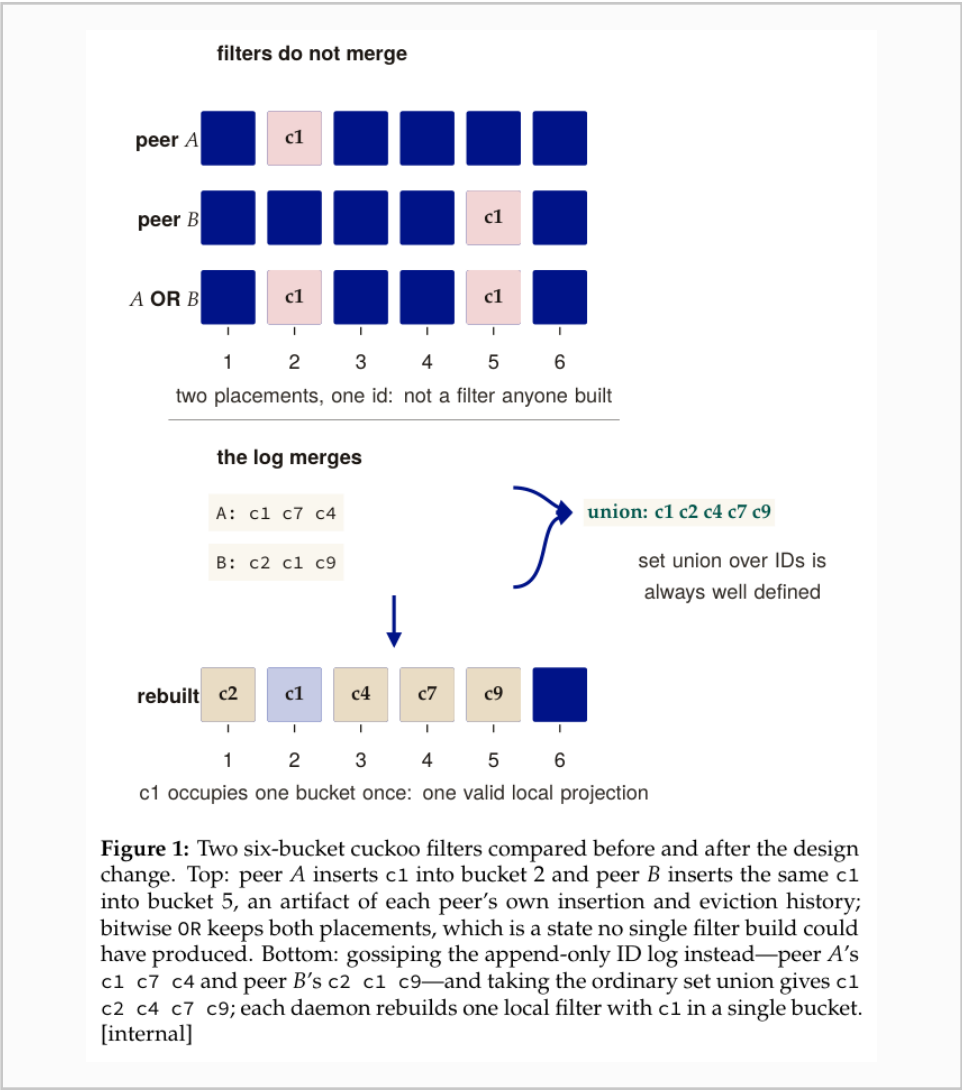
The card lifecycle as a small state machine: issued has exactly two guarded exits, both terminal. Guard 1 is explicit revocation (an operator action or a detected compromise); guard 2 is natural expiry at the TTL. Once either fires, that card ID never re-enters the admissible set -- the dashed green edge is the one recovery available, and it is not a transition of the revoked card: the operator issues a new card o...

AFTER



QA: PASS

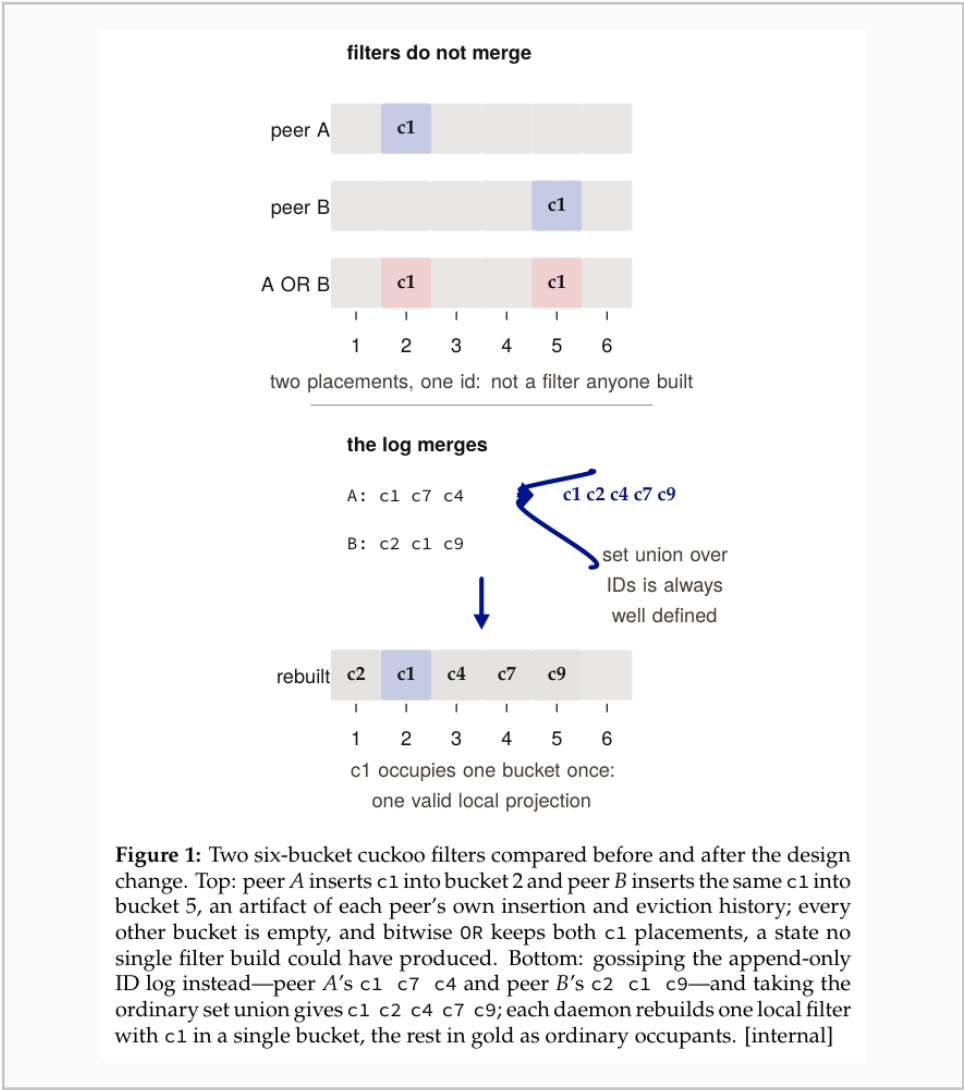
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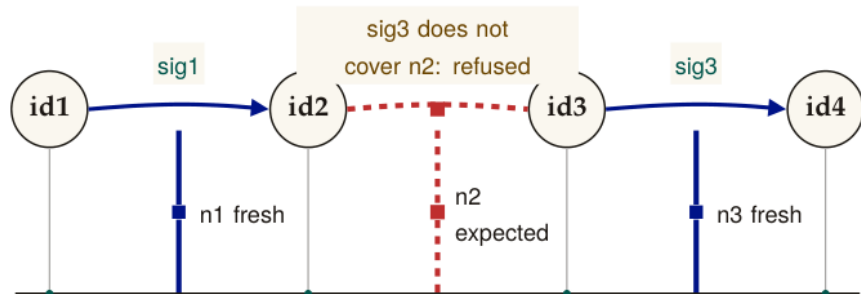
Two six-bucket cuckoo filters compared before and after the design change. Top: peer A inserts {c1 into bucket 2 and peer B inserts the same {c1 into bucket 5, an artifact of each peer's own insertion and eviction history; every other bucket is empty, and bitwise {OR keeps both {c1 placements, a state no single filter build could have produced. Bottom: gossiping the append-only ID log instead---peer A's {c1 ...

AFTER



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BEFORE



same h at every hop

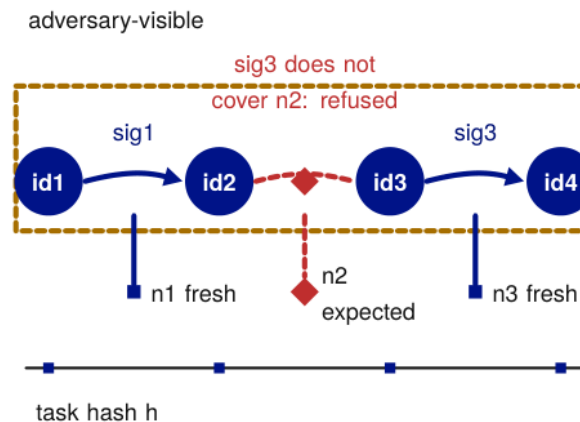
each ordinary crossing binds parent, child, a fresh nonce, and the same task hash h; the amber crossing reuses sig3 (bound to n3) where n2 was expected, so verification fails there

Figure 1: The delegation chain as a signed provenance braid across four identities id1–id4. Ordinary hops 1 and 3 bind parent, child, a fresh nonce (n1, n3), and the shared task hash h below. Hop 2 shows a splice attack: an attacker moves the legitimate signature sig3 (computed over id3, id4, and n3) into the id2–id3 position, where the verifier expects a signature over n2; the check fails at exactly that crossing, so the discontinuity is visible at one adjacent-identity boundary rather than anywhere else in the chain. [internal]

QA: PASS

The delegation chain as a signed provenance braid across four identities id1–id4, drawn inside the network the Dolev-Yao adversary observes. Ordinary hops 1 and 3 bind parent, child, a fresh nonce (n1, n3), and the shared task hash h below. Hop 2 shows a splice attack: an attacker moves the legitimate signature sig3 (computed over id3, id4, and n3) into the id2–id3 position, where the verifier expects a signature over n2; the check fails at exactly that crossing, so the discontinuity is visible at one adjacent-identity boundary rather than anywhere else in the chain. [internal]

AFTER



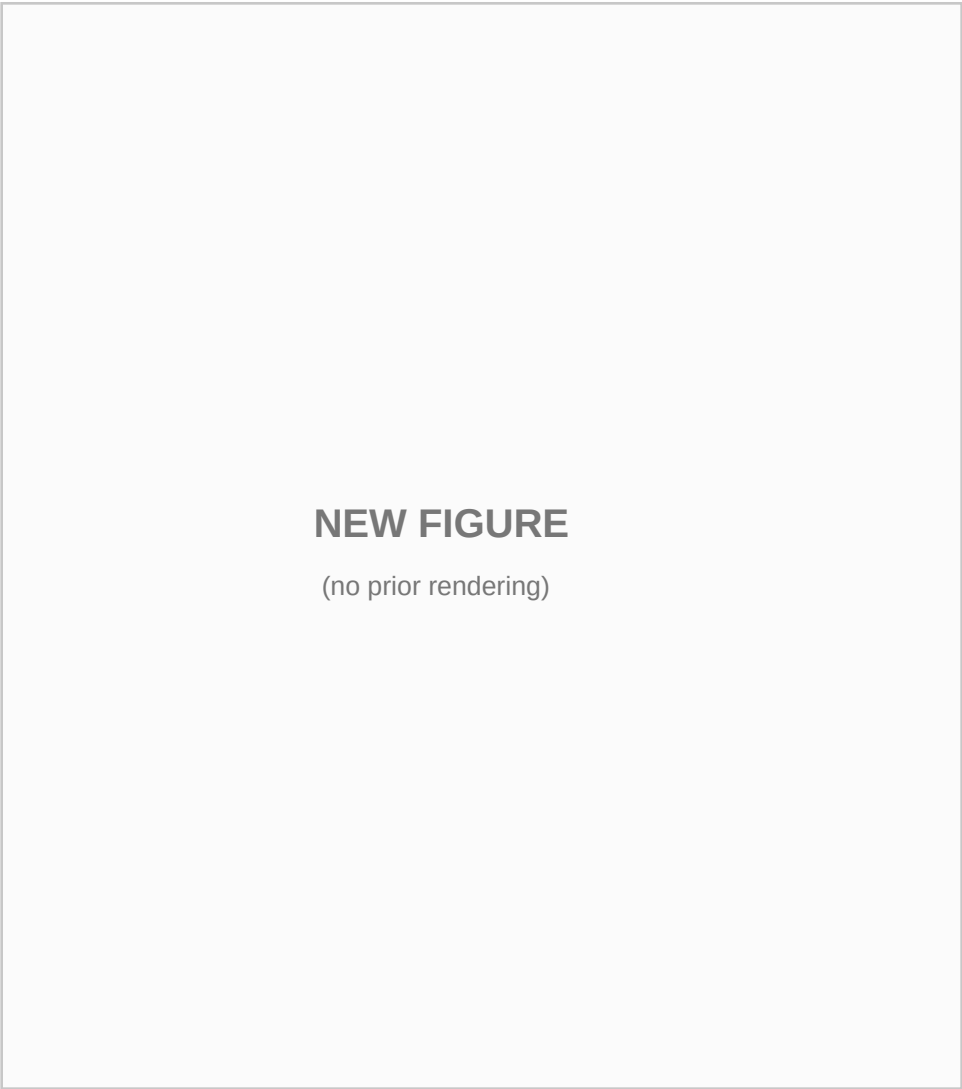
task hash h

each ordinary crossing binds parent, child, a fresh nonce, and the same task hash h; the breach crossing reuses sig3 (bound to n3) where n2 was expected, so verification fails at that crossing only

Figure 1: The delegation chain as a signed provenance braid across four identities id1–id4, drawn inside the network the Dolev-Yao adversary observes. Ordinary hops 1 and 3 bind parent, child, a fresh nonce (n1, n3), and the shared task hash h below. Hop 2 shows a splice attack: an attacker moves the legitimate signature sig3 (computed over id3, id4, and n3) into the id2–id3 position, where the verifier expects a signature over n2; the check fails at exactly that crossing, so the discontinuity is visible at one adjacent-identity boundary rather than anywhere else in the chain. [internal]

QA: PASS

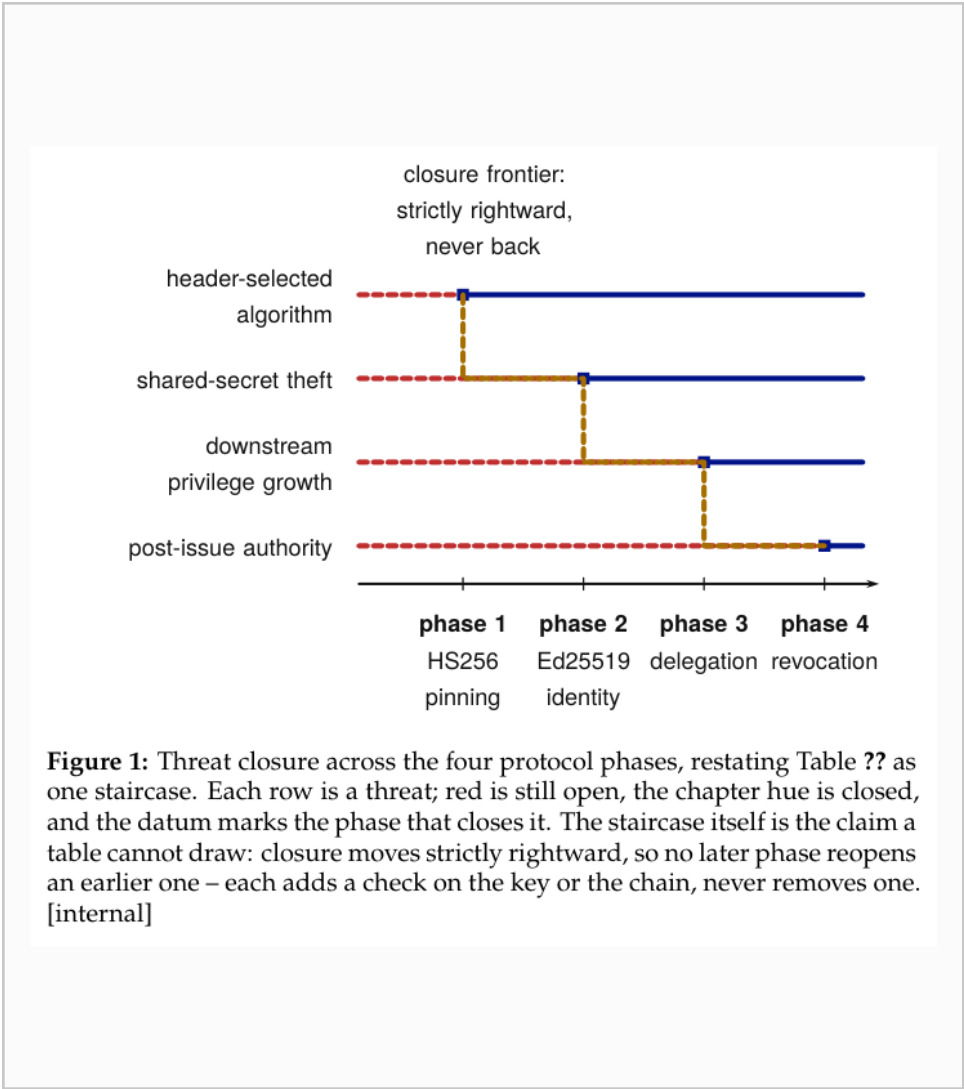
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QA: —

Threat closure across the four protocol phases, restating Table-{tab:anchor-four-phases as one staircase. Each row is a threat; red is still open, the chapter hue is closed, and the datum marks the phase that closes it. The staircase itself is the claim a table cannot draw: closure moves strictly rightward, so no later phase reopens an earlier one -- each adds a check on the key or the chain, never removes one. [i...

AFTER



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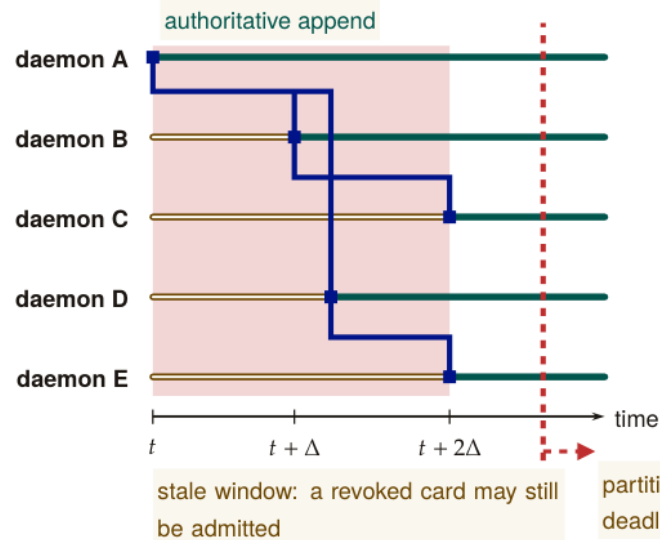


Figure 1: A revocation advances as an observation frontier, leaving a distinct stale-exposure interval on every verifier lane. This execution converges after two rounds; the protocol claim is only expected $\Theta(\log m)$ dissemination under connected reliable rounds. The dashed partition marker separates that expectation from a guarantee: an unbounded partition removes every finite delivery deadline.

QA: PASS

A revocation advances as an observation frontier, leaving a distinct stale-exposure interval on every verifier lane: daemon A's append is authoritative at t , and each lane's amber segment (still stale) turns to the chapter hue at its own wake datum. This execution converges after two rounds; the protocol claim is only expected $\Theta(\log m)$ dissemination under connected, reliable rounds. The dashed red line marks a sec...

AFTER

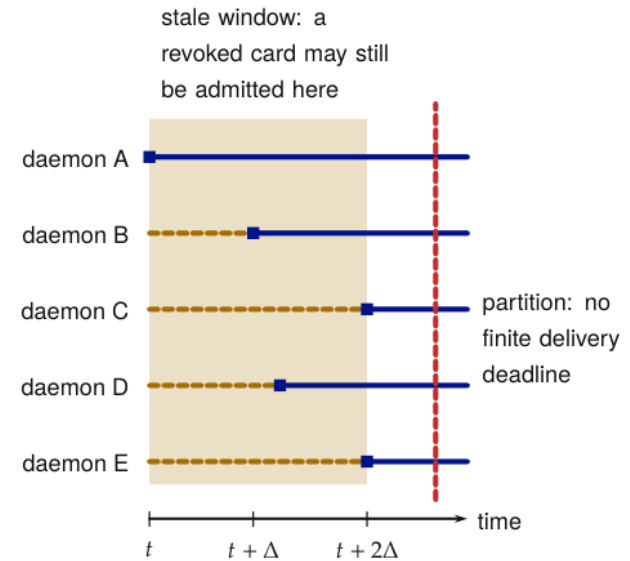
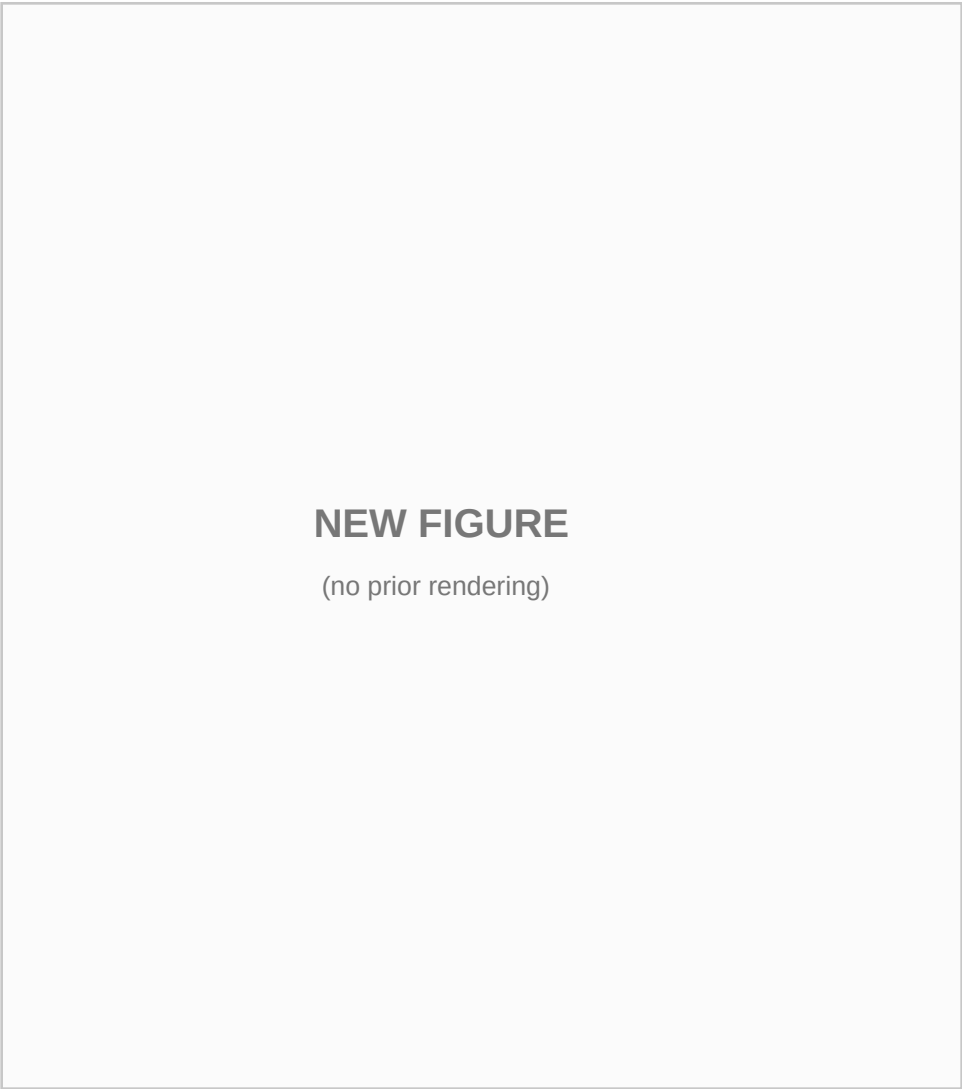


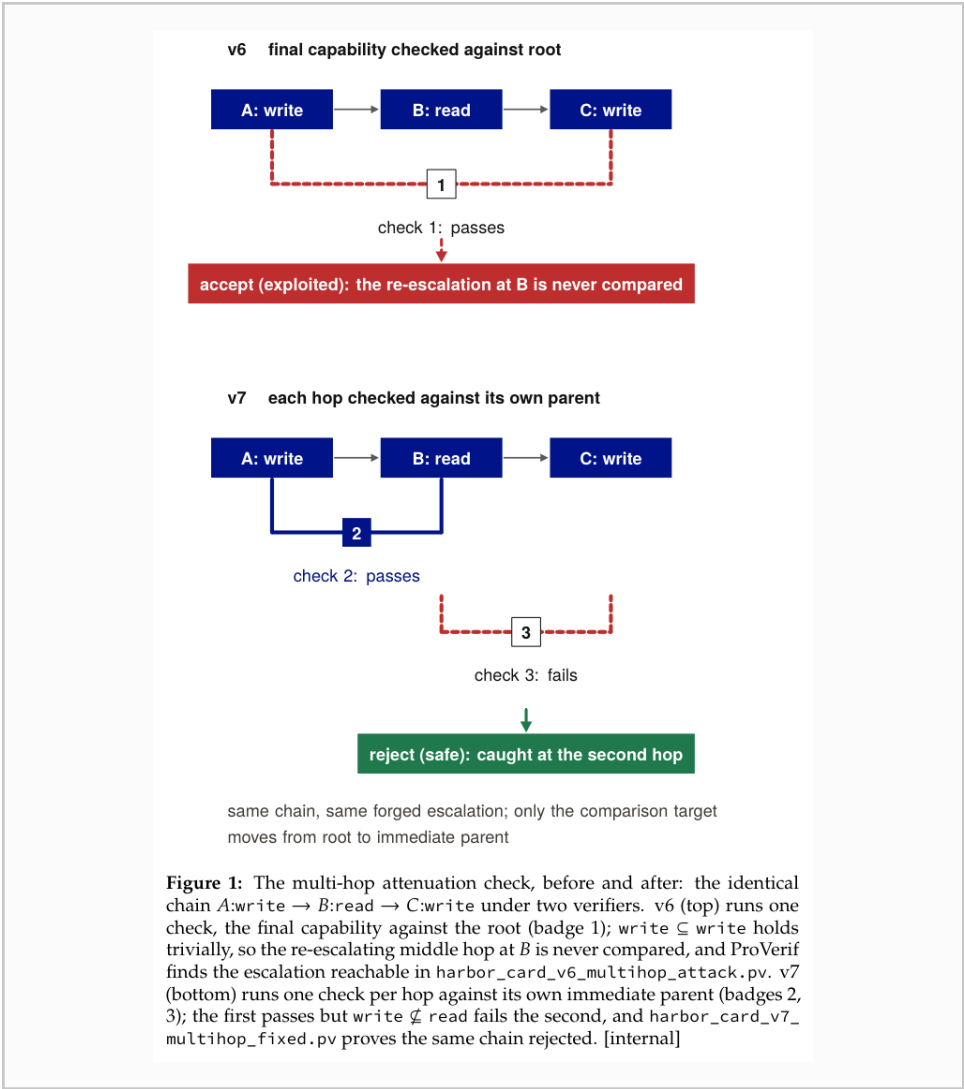
Figure 1: A revocation advances as an observation frontier, leaving a distinct stale-exposure interval on every verifier lane: daemon A's append is authoritative at t , and each lane's amber segment (still stale) turns to the chapter hue at its own wake datum. This execution converges after two rounds; the protocol claim is only expected $\Theta(\log m)$ dissemination under connected, reliable rounds. The dashed red line marks a second, separate claim: an unbounded partition removes every finite delivery deadline, regardless of how many rounds have passed. [internal]

QA: PASS

BEFORE



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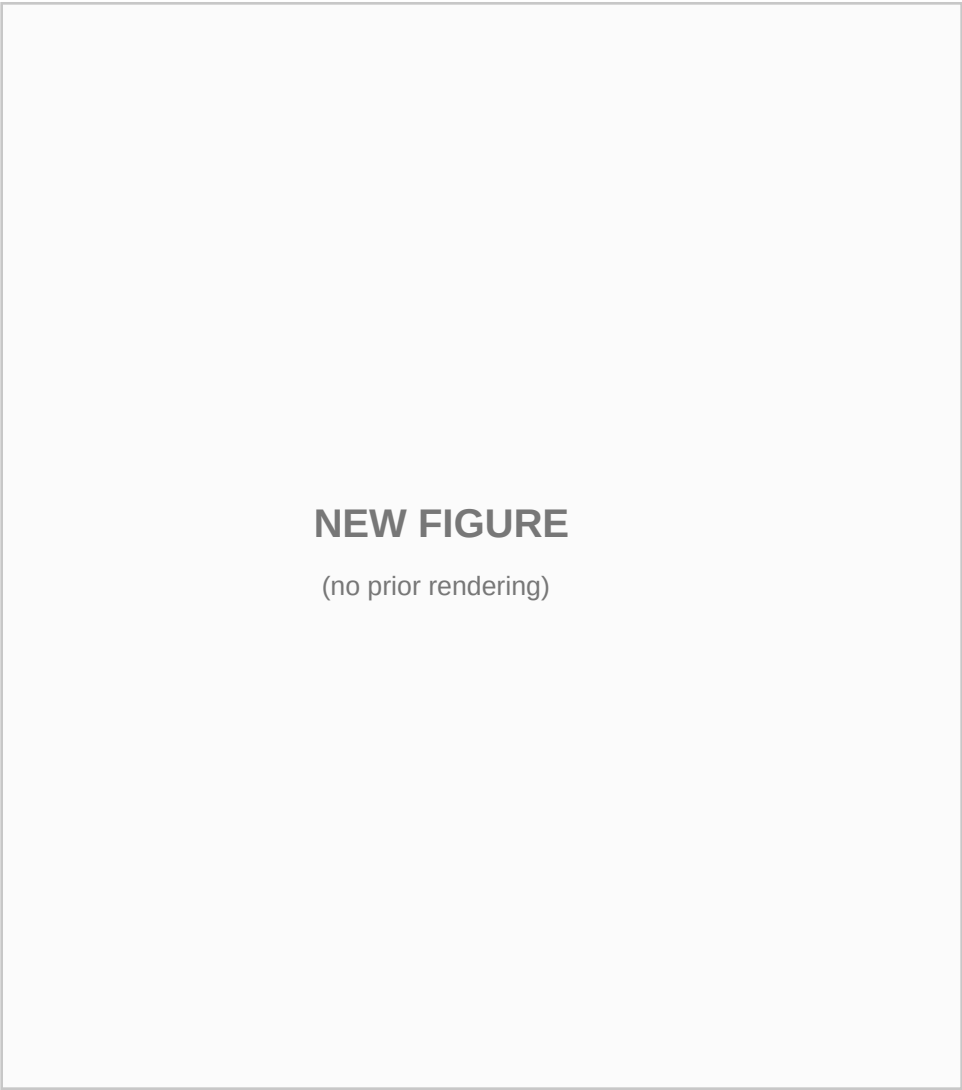


QA: —

QA: PASS

The multi-hop attenuation check, before and after: the identical chain $A:\text{write} \rightarrow B:\text{read} \rightarrow C:\text{write}$ under two verifiers. v6 (top) runs one check, the final capability against the root (badge 1); $\text{write} \sqsubseteq \text{write}$ holds trivially, so the re-escalating middle hop at B is never compared, and ProVerif finds the escalation reachable in `{harbor_card_v6_multihop_attack.pv}`. v7 (bottom) runs one check per hop...

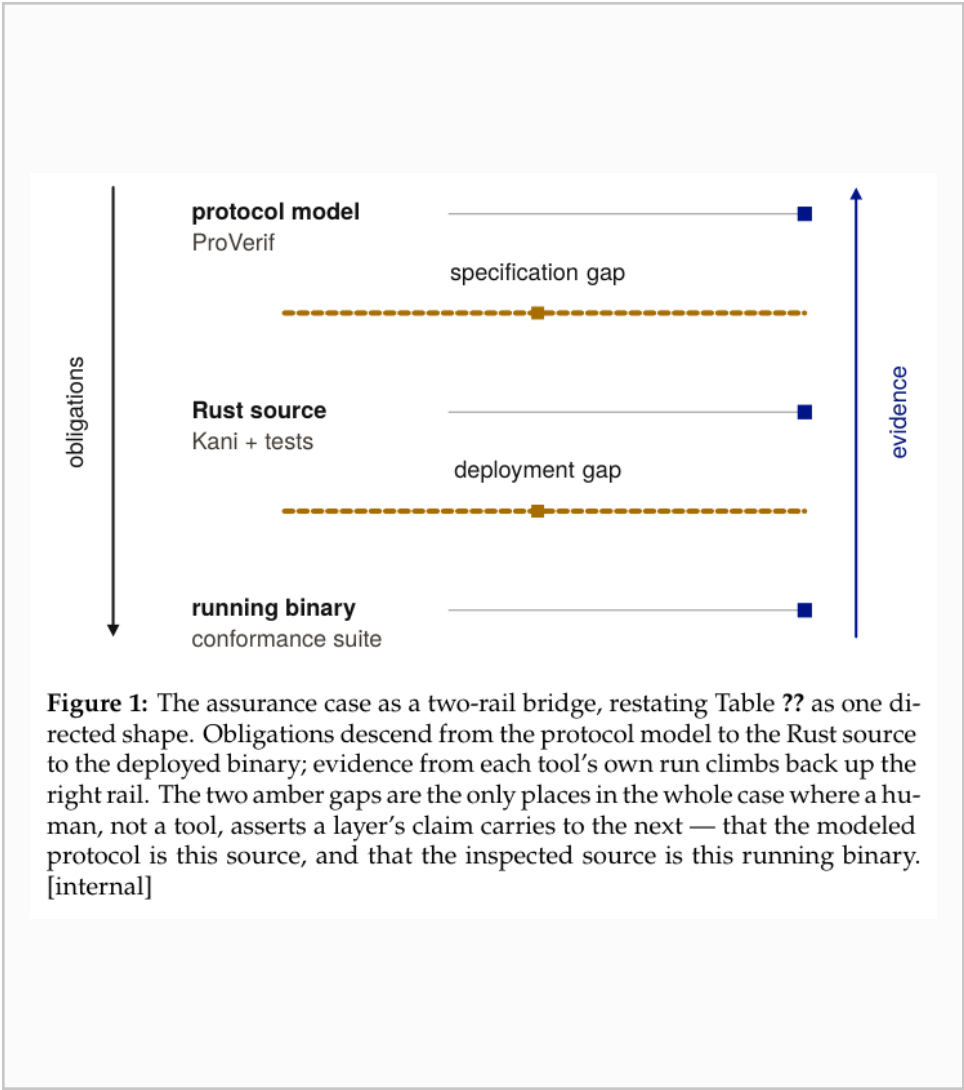
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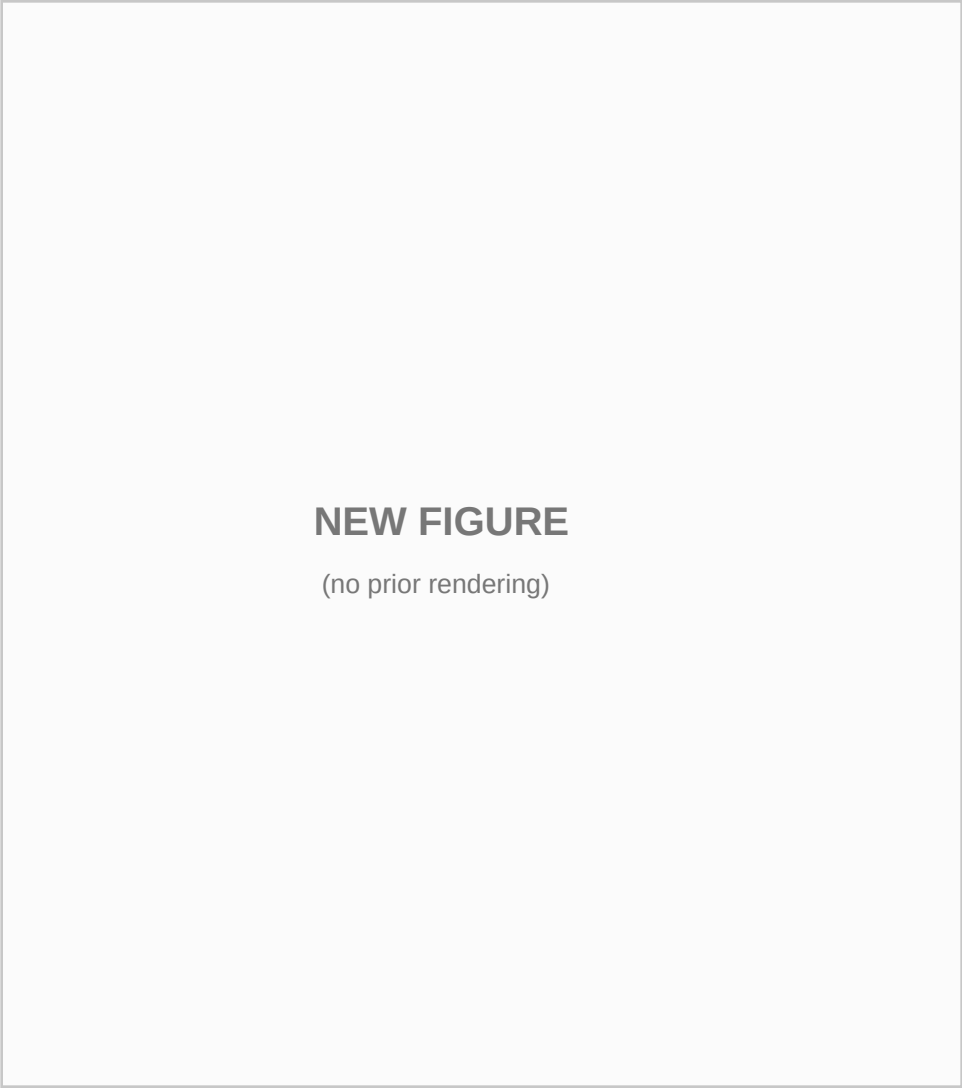
The assurance case as a two-rail bridge, restating Table~{tab:anchor-verification-stack as one directed shape. Obligations descend from the protocol model to the Rust source to the deployed binary; evidence from each tool's own run climbs back up the right rail. The two amber gaps are the only places in the whole case where a human, not a tool, asserts a layer's claim carries to the next --- that the modeled proto...

AFTER

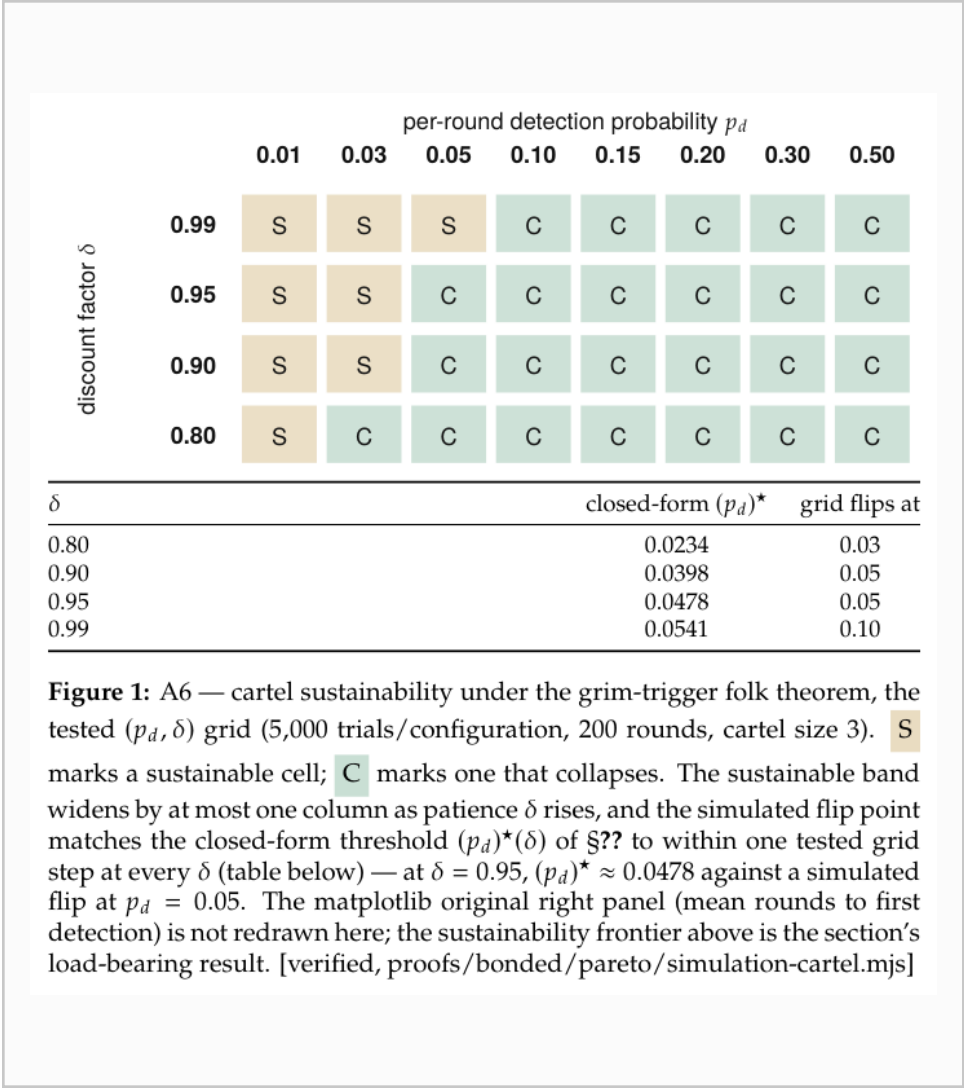


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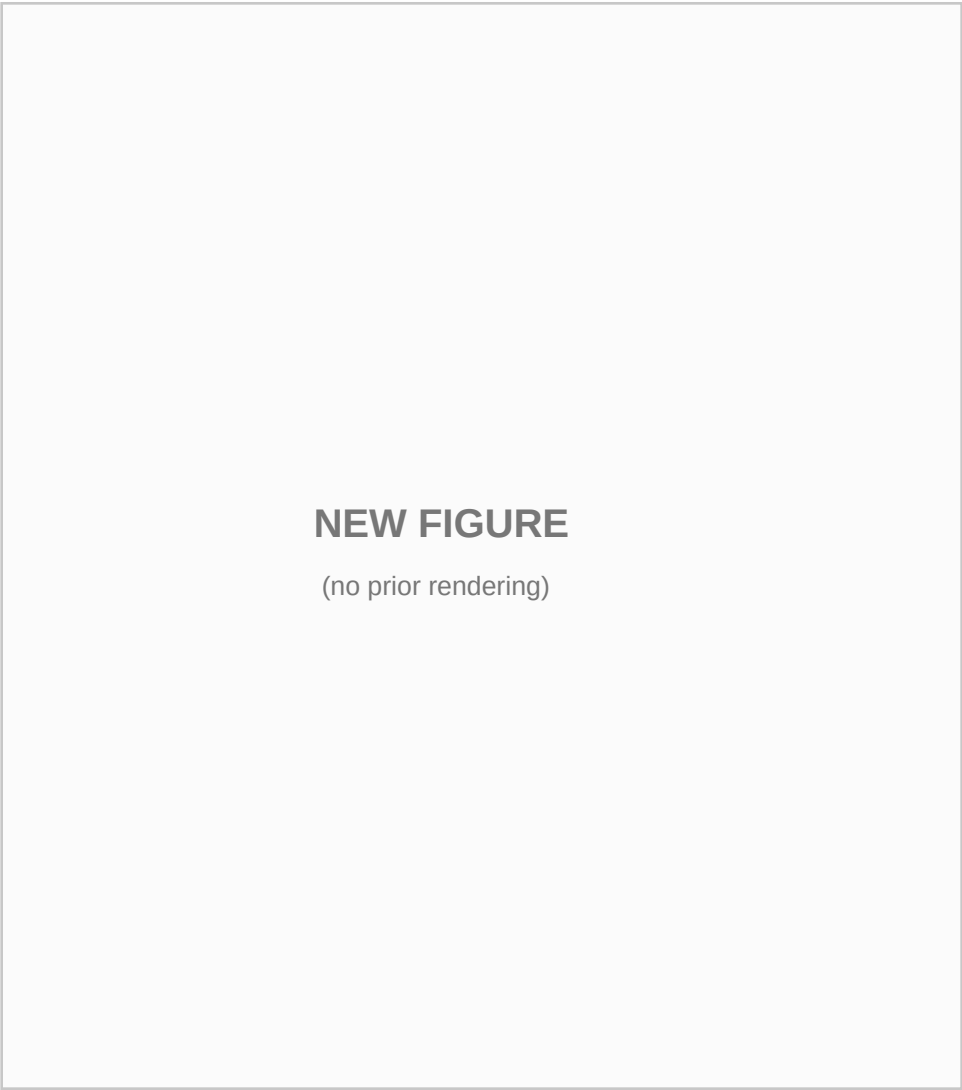


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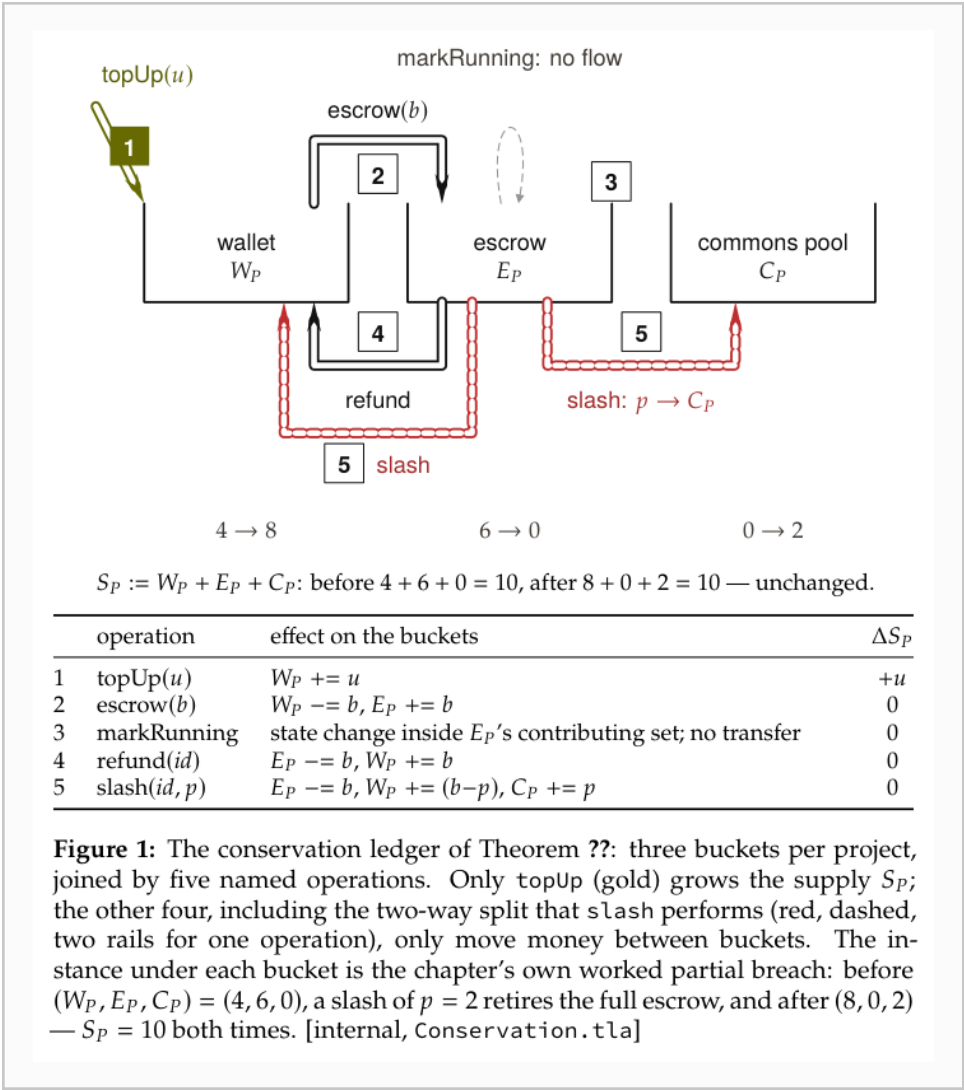
QA: PASS

A6 --- cartel sustainability under the grim-trigger folk theorem, the tested (p_d, δ) grid (5,000 trials/configuration, 200 rounds, cartel size 3). S marks a sustainable cell; C marks one that collapses. The sustainable band widens by at most one column as patience δ rises, and the simulated flip point matches the closed-form threshold $(p_d)^*(\delta)$ of §? to within one tested grid step at every δ (table below) — at $\delta = 0.95$, $(p_d)^* \approx 0.0478$ against a simulated flip at $p_d = 0.05$. The matplotlib original right panel (mean rounds to first detection) is not redrawn here; the sustainability frontier above is the section’s load-bearing result. [verified, proofs/bonded/pareto/simulation-cartel.mjs]

BEFORE



AFTER

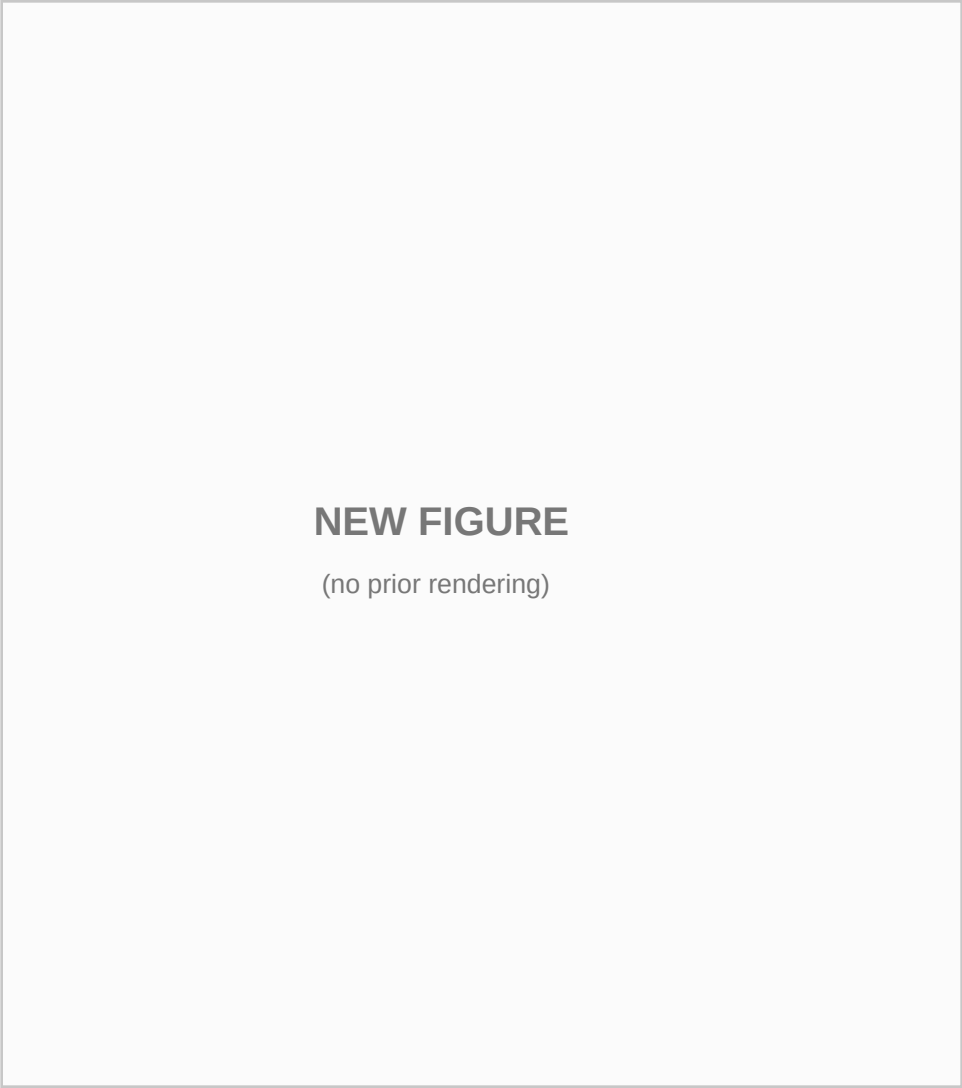


QA: —

QA: PASS

The conservation ledger of Theorem~{thm:conservation: three buckets per project, joined by five named operations. Only {topUp (gold) grows the supply S_P ; the other four, including the two-way split that slash performs (red, dashed, two rails for one operation), only move money between buckets. The instance under each bucket is the chapter's own worked partial breach: before $(W_P, E_P, C_P)=(4,6,0)$, a slash of ...

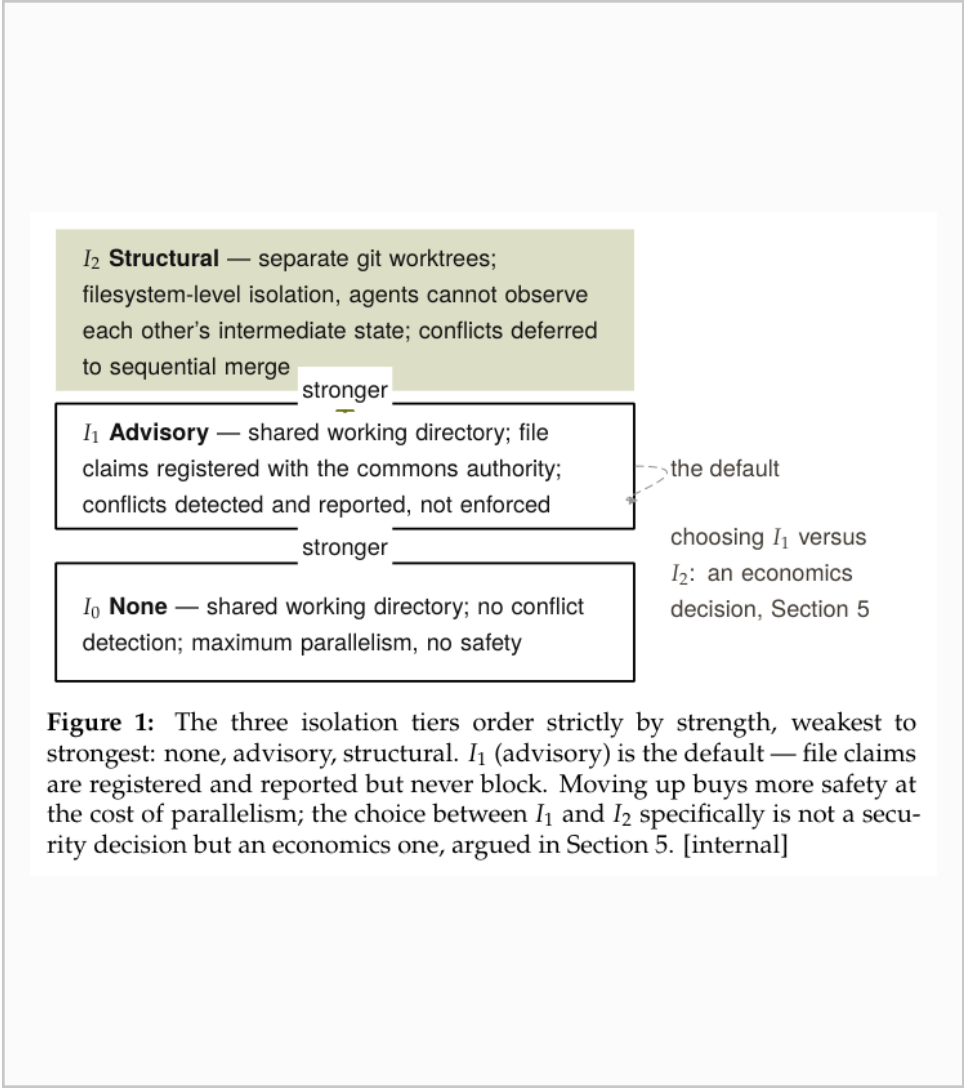
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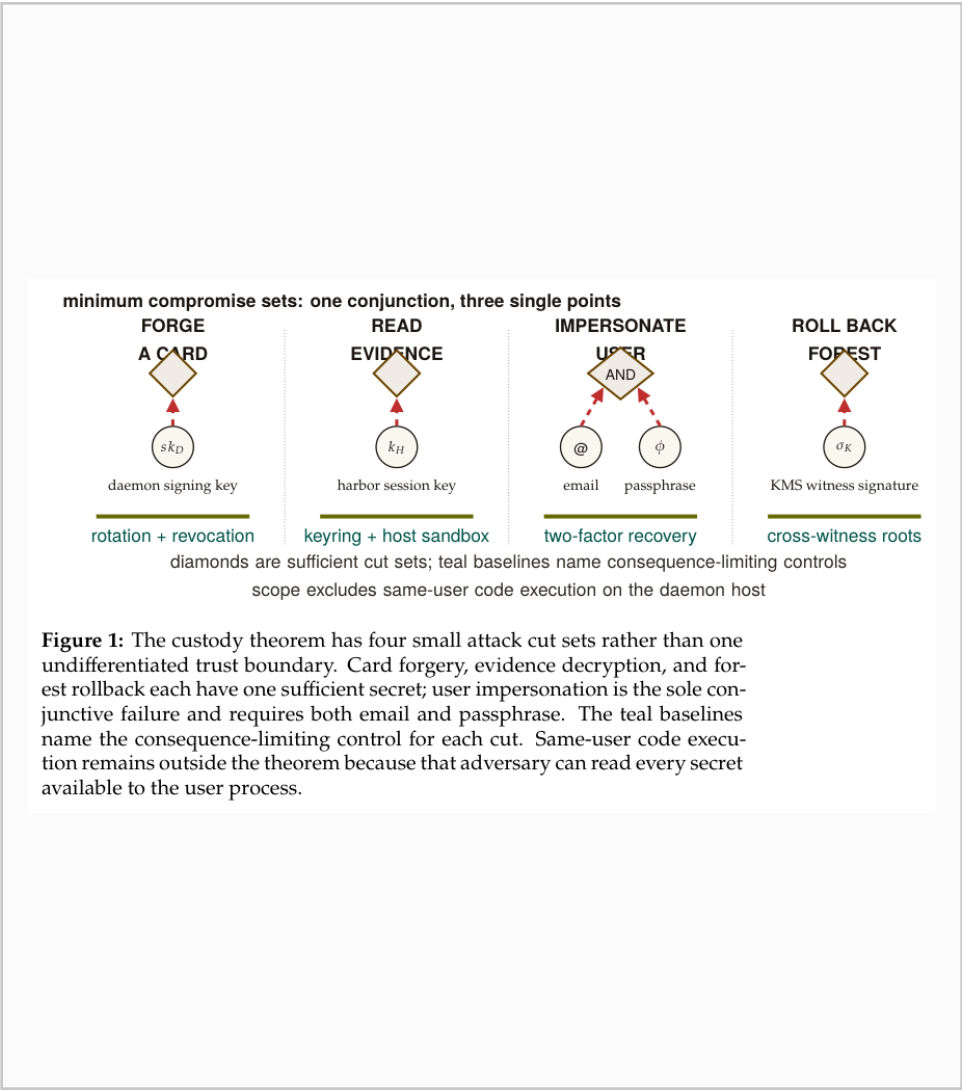
The three isolation tiers order strictly by strength, weakest to strongest: none, advisory, structural. I_1 (advisory) is the default --- file claims are registered and reported but never block. Moving up buys more safety at the cost of parallelism; the choice between I_1 and I_2 specifically is not a security decision but an economics one, argued in Section 5. [internal]

AFTER



QA: PASS

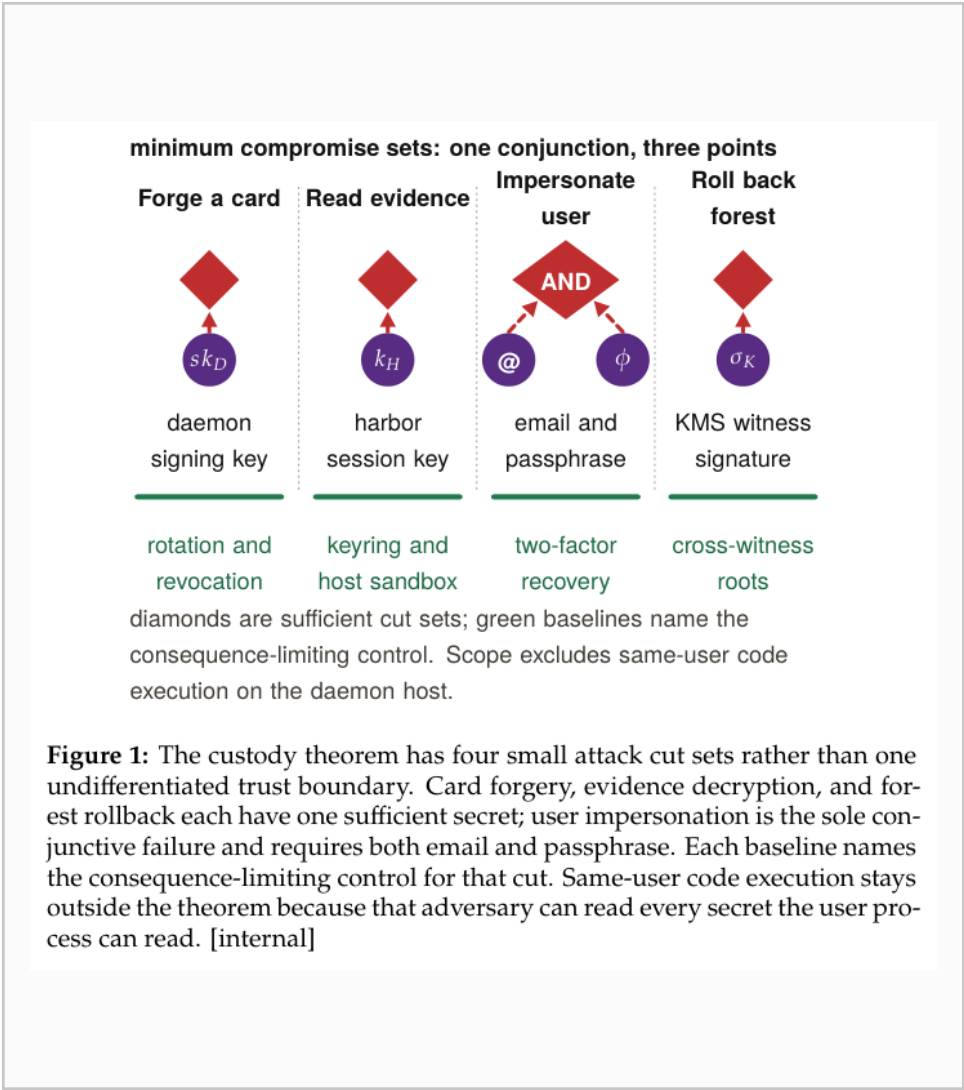
BEFORE



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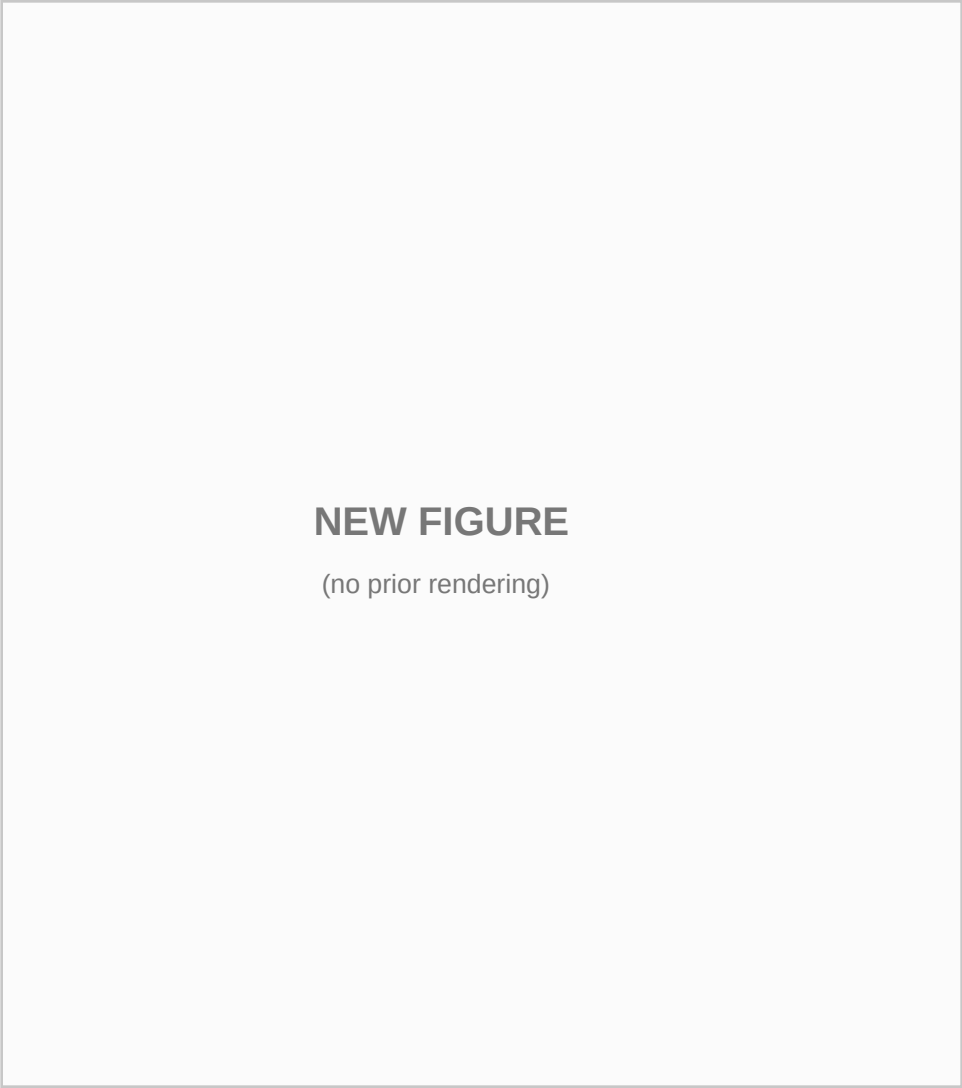
The custody theorem has four small attack cut sets rather than one undifferentiated trust boundary. Card forgery, evidence decryption, and forest rollback each have one sufficient secret; user impersonation is the sole conjunctive failure and requires both email and passphrase. Each baseline names the consequence-limiting control for that cut. Same-user code execution stays outside the theorem because that adversa...

AFTER



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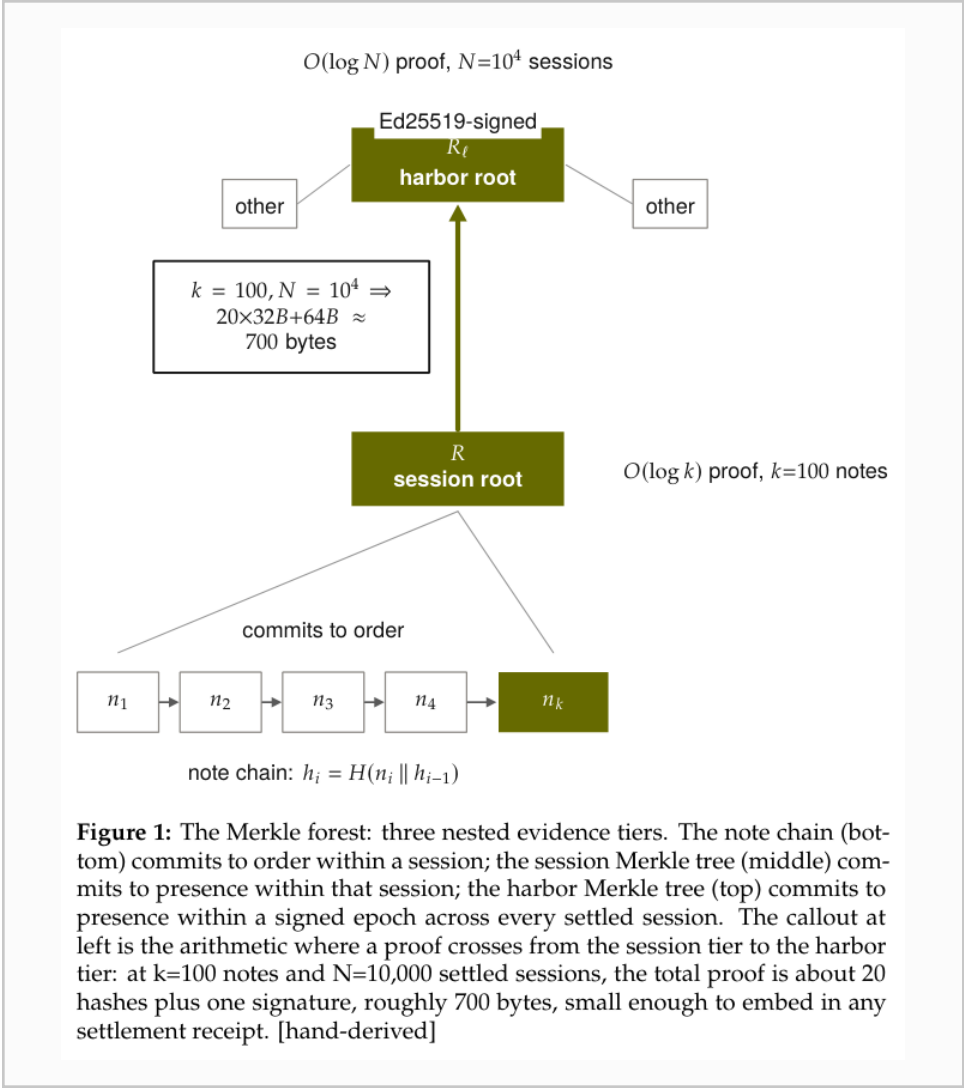
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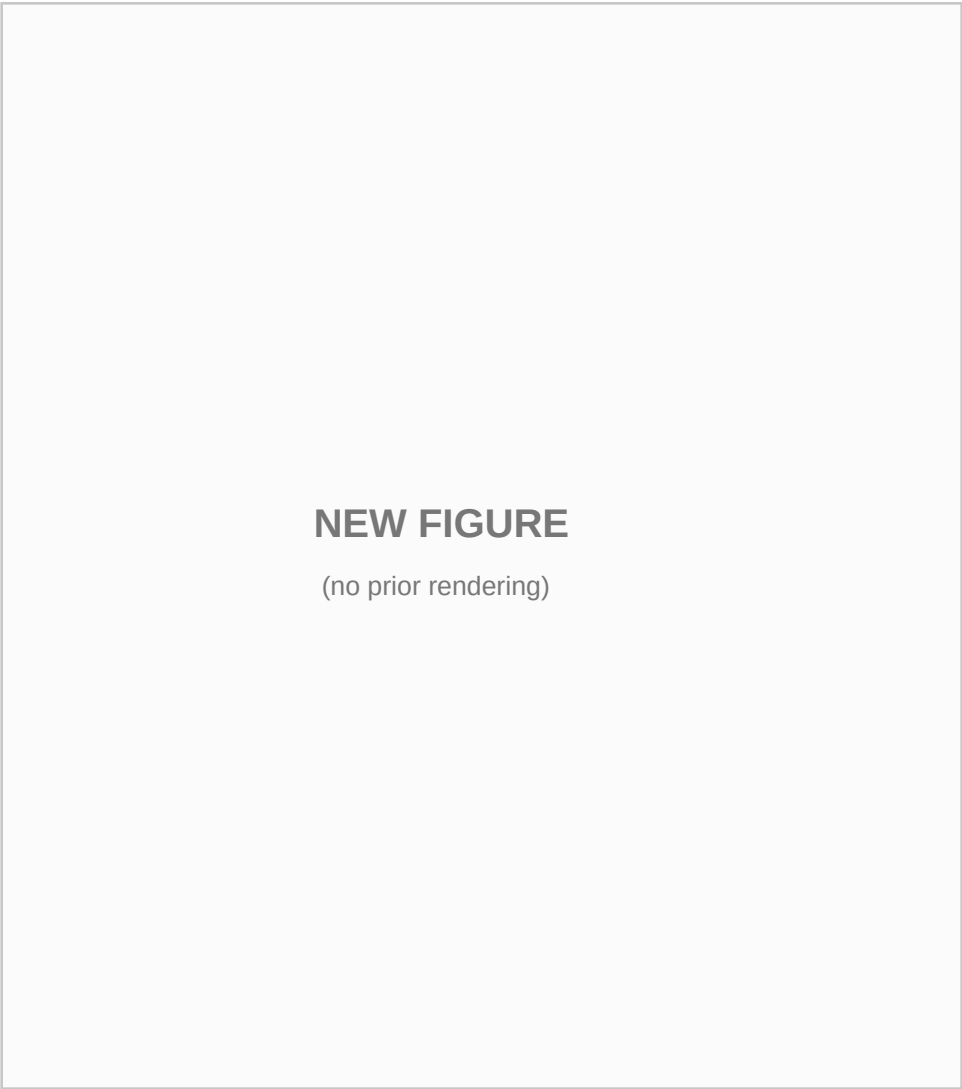
The Merkle forest: three nested evidence tiers. The note chain (bottom) commits to order within a session; the session Merkle tree (middle) commits to presence within that session; the harbor Merkle tree (top) commits to presence within a signed epoch across every settled session. The callout at left is the arithmetic where a proof crosses from the session tier to the harbor tier: at k=100 notes and N=10,000 settled sessions, the total proof is about 20 hashes plus one signature, roughly 700 bytes, small enough to embed in any settl...

AFTER



QA: PASS

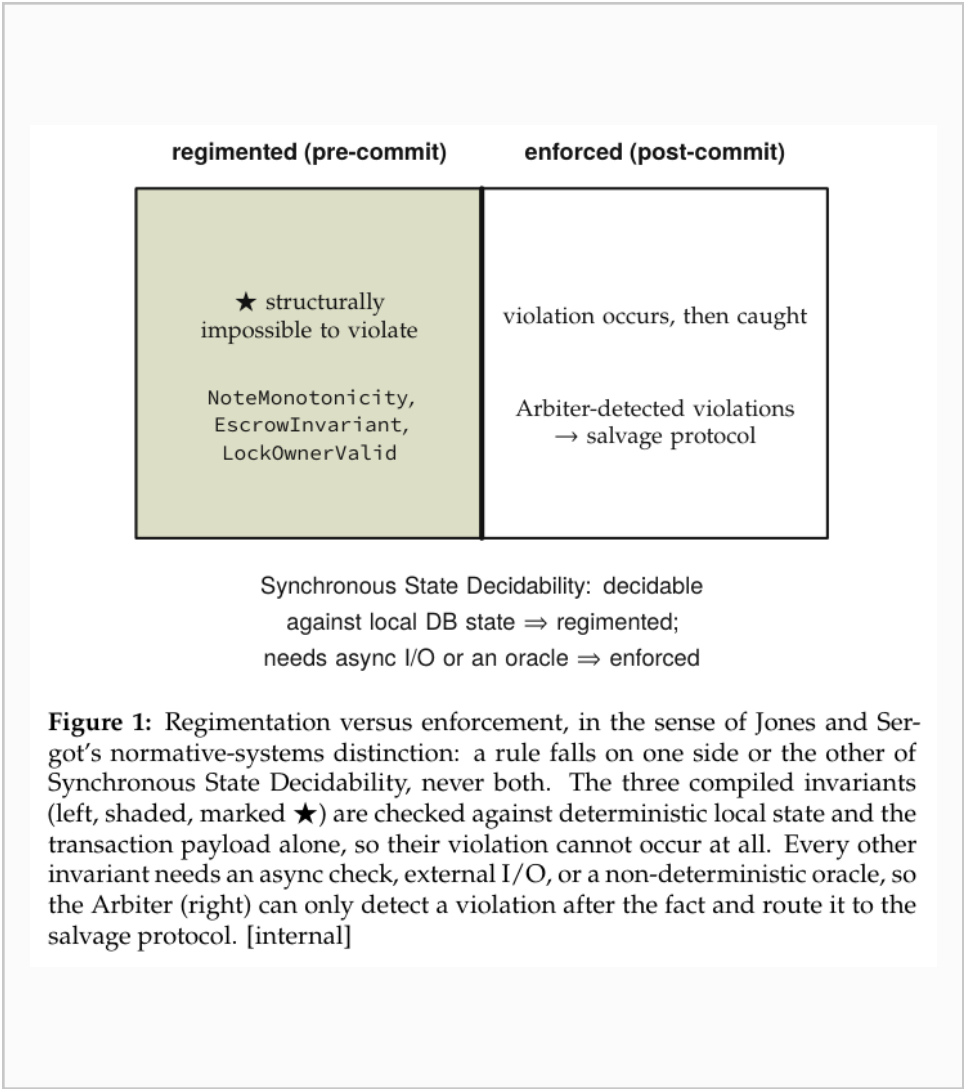
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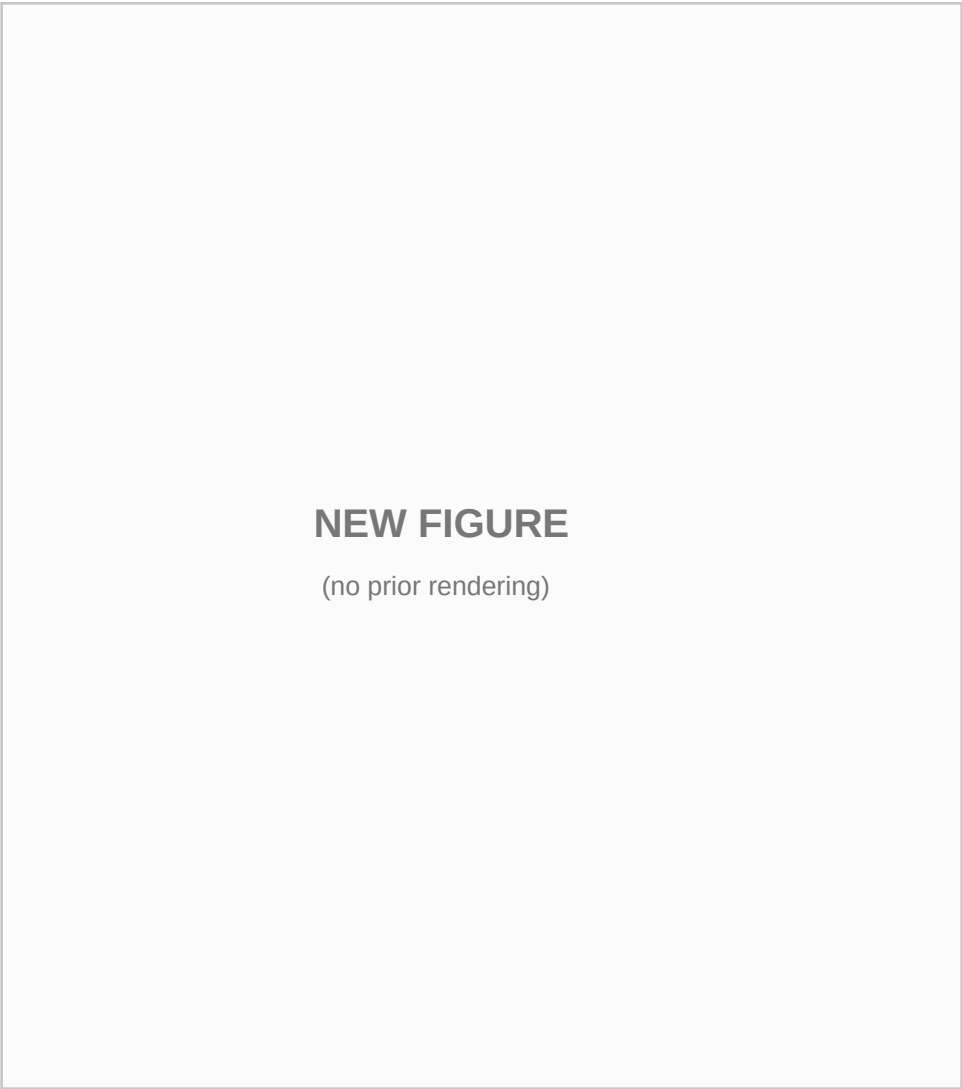
Regimentation versus enforcement, in the sense of Jones and Sergot's normative-systems distinction: a rule falls on one side or the other of Synchronous State Decidability, never both. The three compiled invariants (left, shaded, marked \$\$) are checked against deterministic local state and the transaction payload alone, so their violation cannot occur at all. Every other invariant needs an async check, external I/O, or a non-deterministic oracle, so the Arbiter (right) can only detect a violation after the fact and route it to the salvage protocol. [internal]

AFTER



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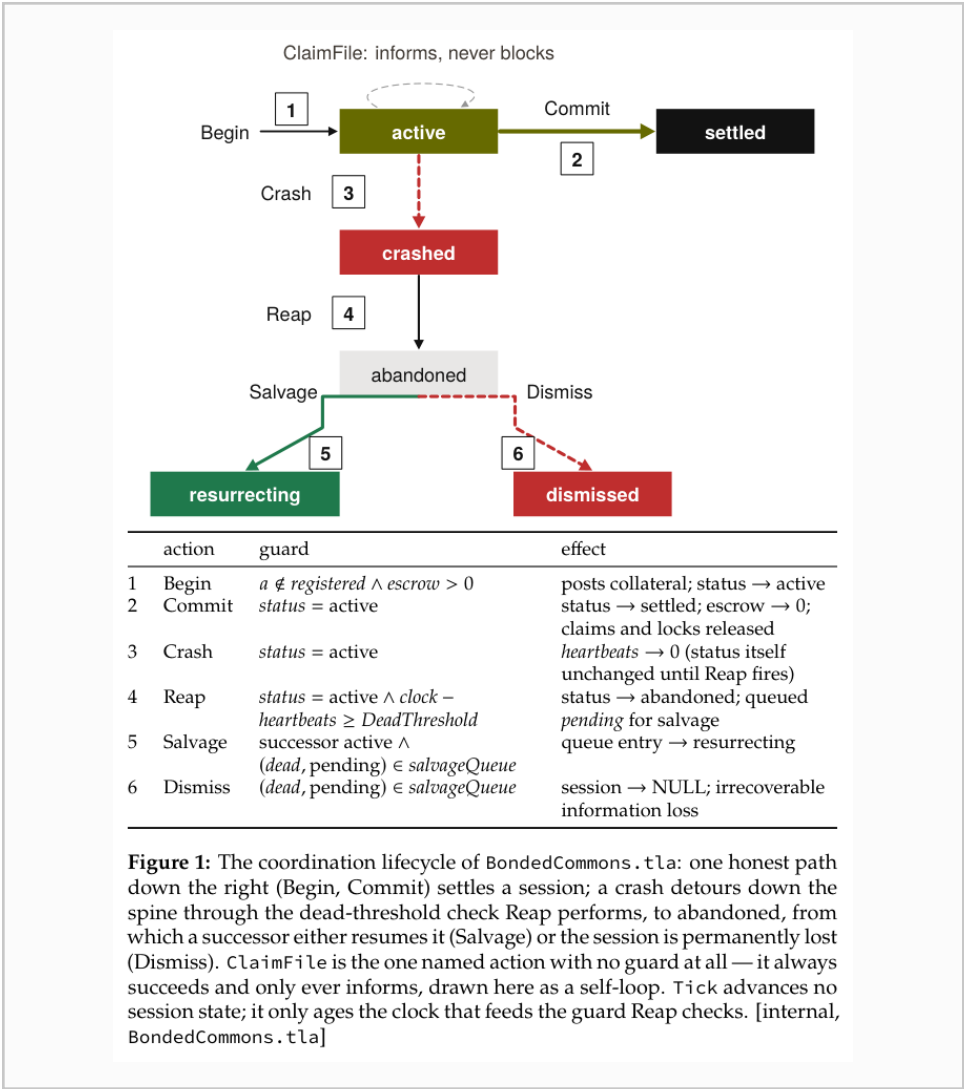
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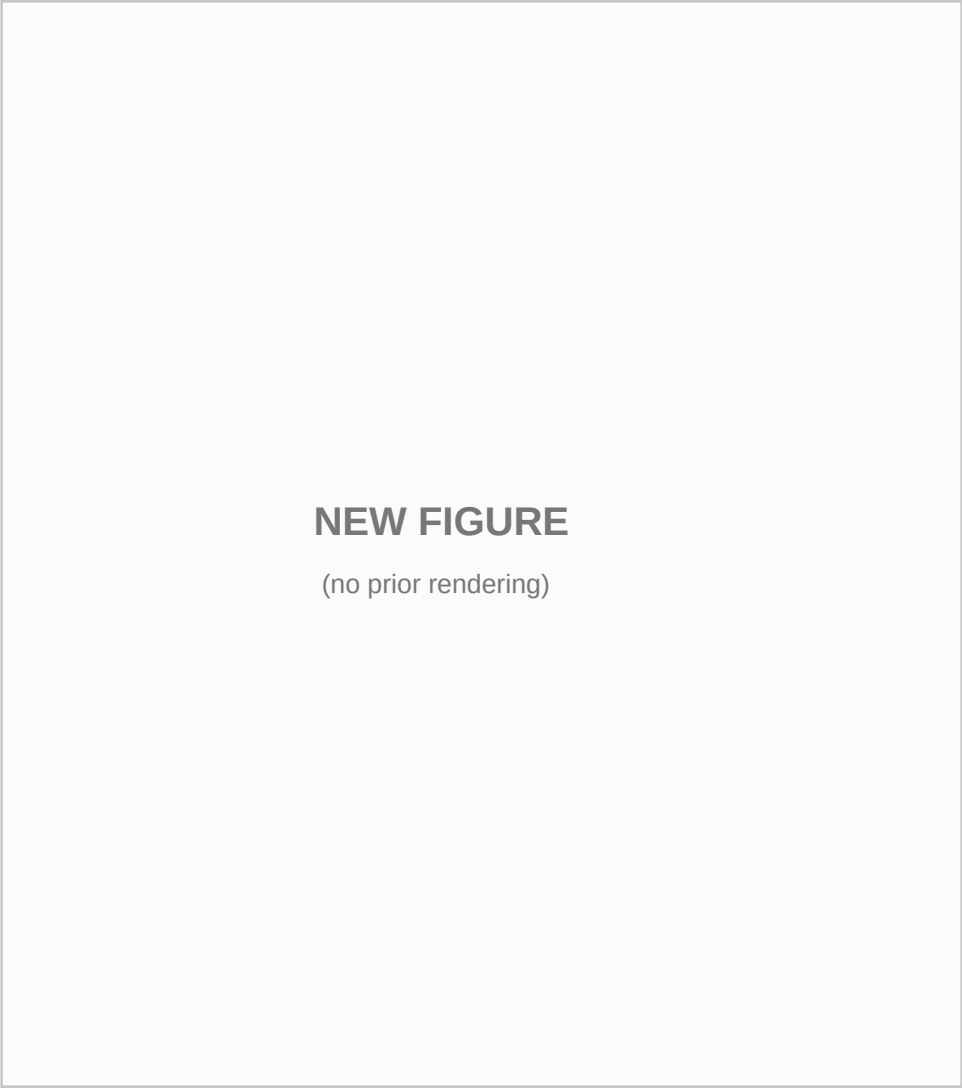
The coordination lifecycle of {BondedCommons.tla: one honest path down the right (Begin, Commit) settles a session; a crash detours down the spine through the dead-threshold check Reap performs, to abandoned, from which a successor either resumes it (Salvage) or the session is permanently lost (Dismiss). {ClaimFile is the one named action with no guard at all --- it always succeeds and only ever informs, drawn her...

AFTER



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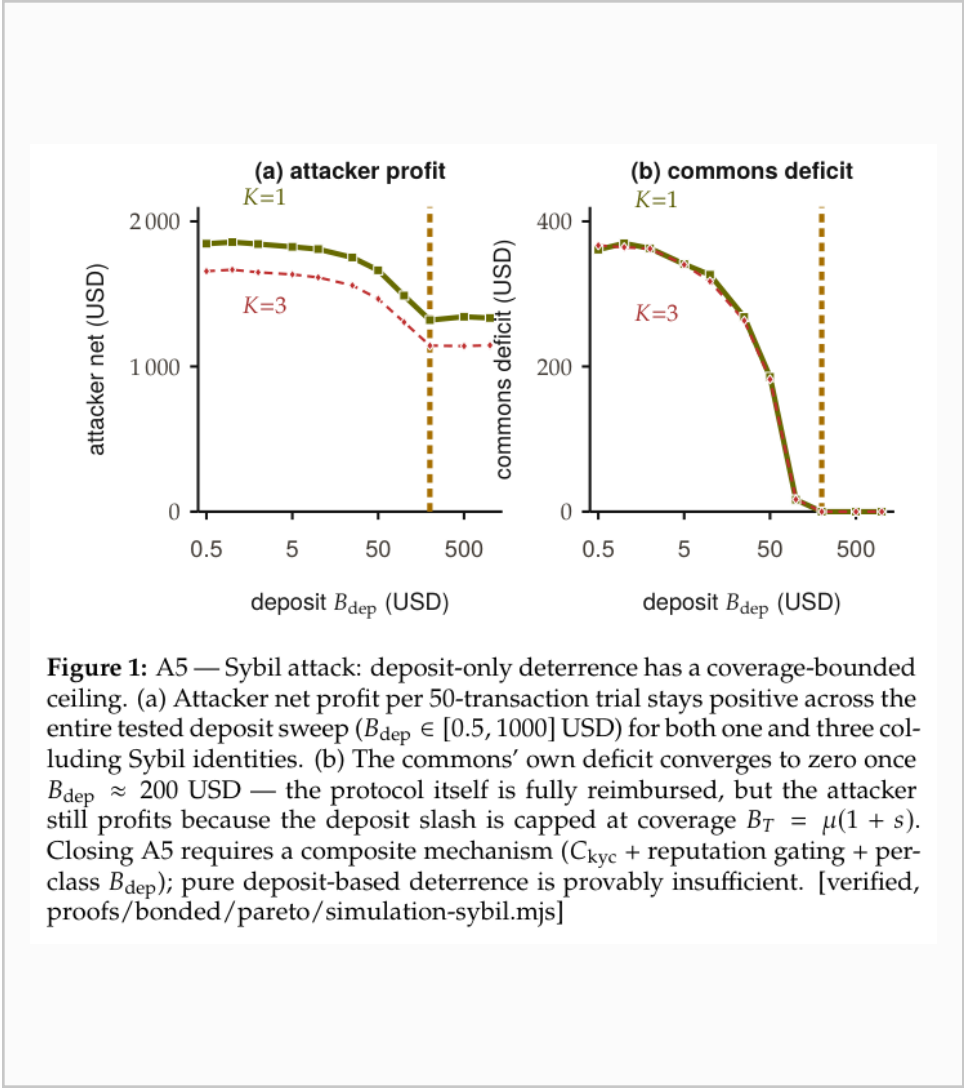
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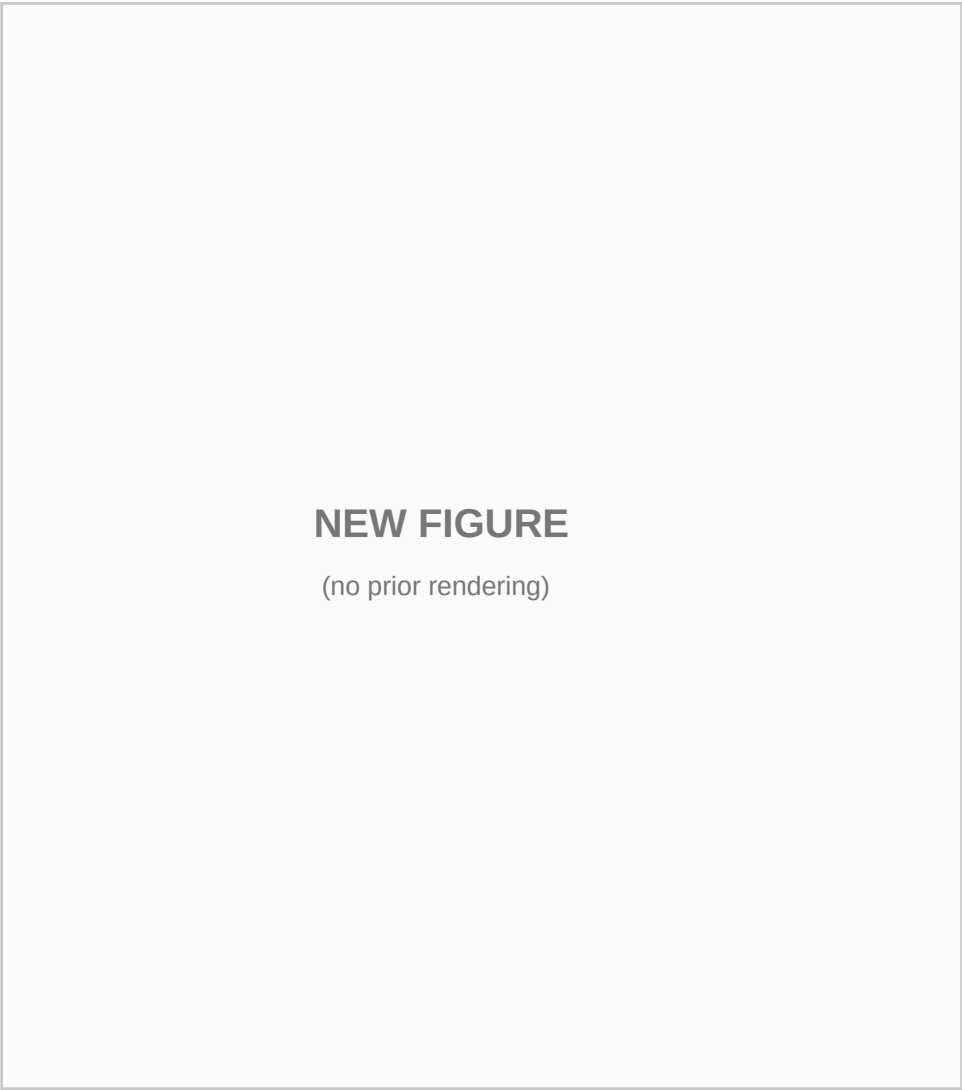
A5 --- Sybil attack: deposit-only deterrence has a coverage-bounded ceiling. (a) Attacker net profit per 50-transaction trial stays positive across the entire tested deposit sweep ($B_{\text{dep}} \in [0.5, 1000]$ USD) for both one and three colluding Sybil identities. (b) The commons' own deficit converges to zero once $B_{\text{dep}} \approx 200$ USD --- the protocol itself is fully reimbursed, but the attacker still profits because the dep...

AFTER



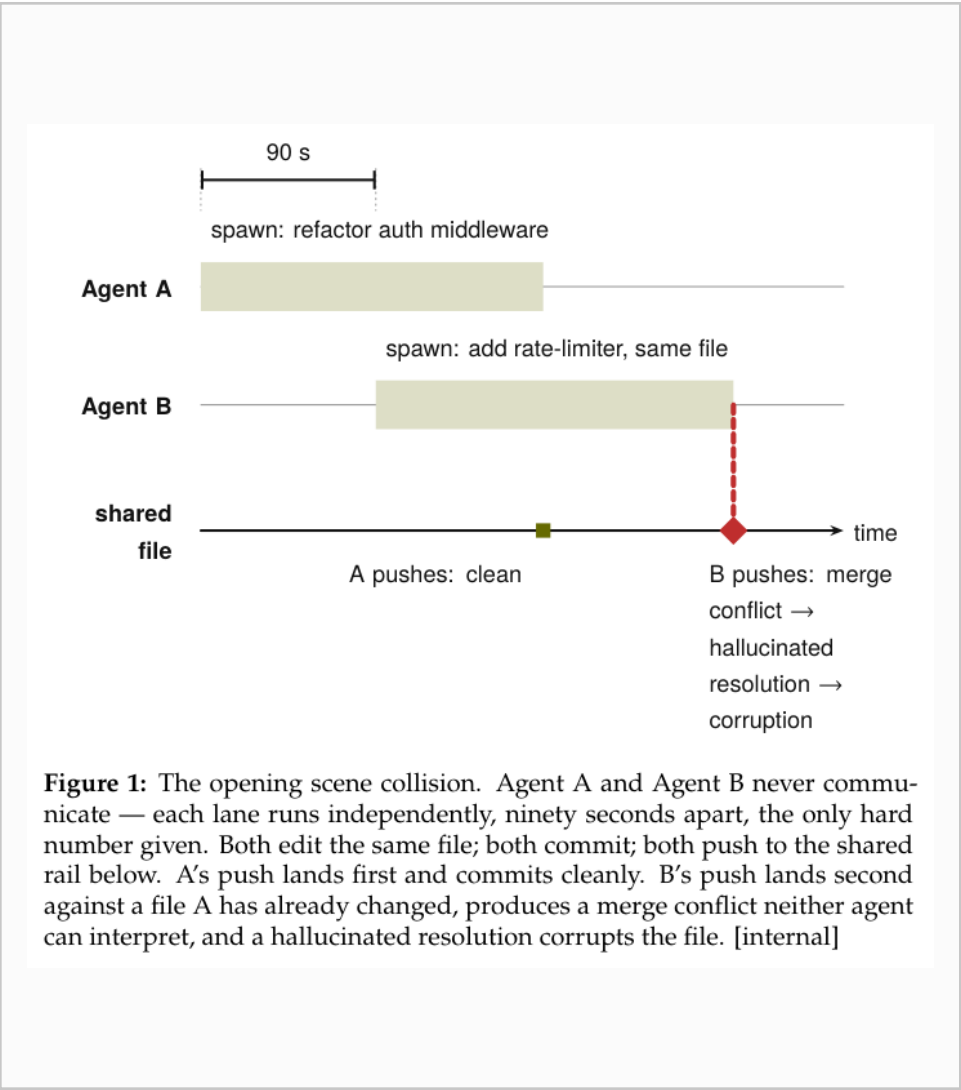
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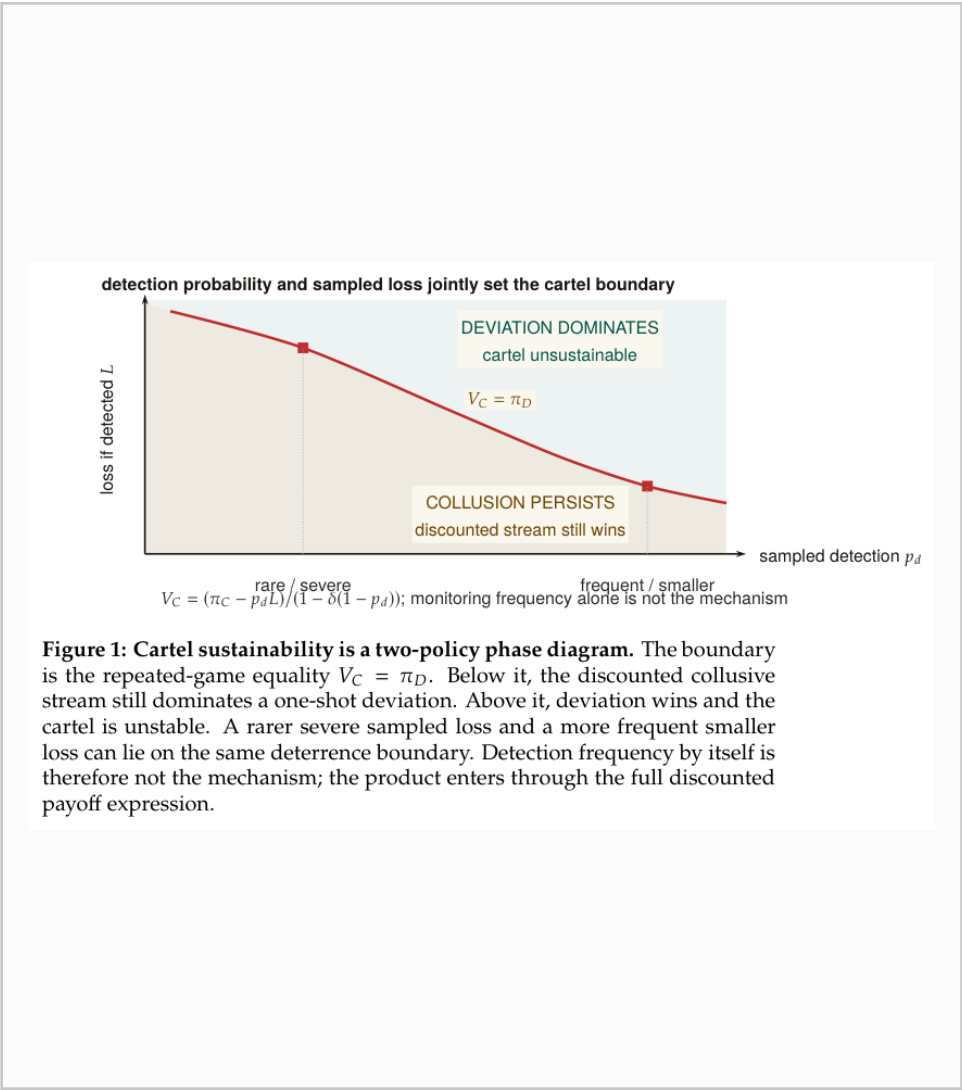
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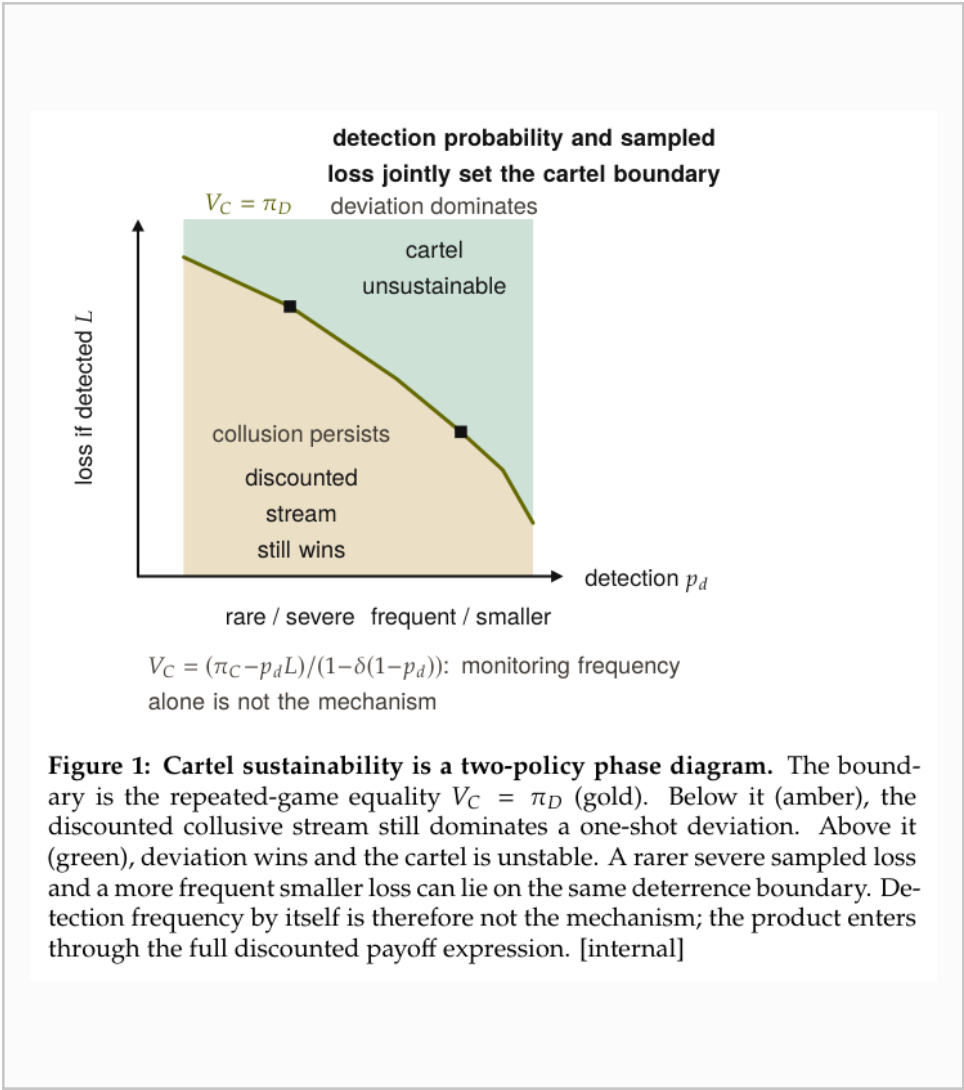
The opening scene collision. Agent A and Agent B never communicate --- each lane runs independently, ninety seconds apart, the only hard number given. Both edit the same file; both commit; both push to the shared rail below. A's push lands first and commits cleanly. B's push lands second against a file A has already changed, produces a merge conflict neither agent can interpret, and a hallucinated resolution corru...

BEFORE



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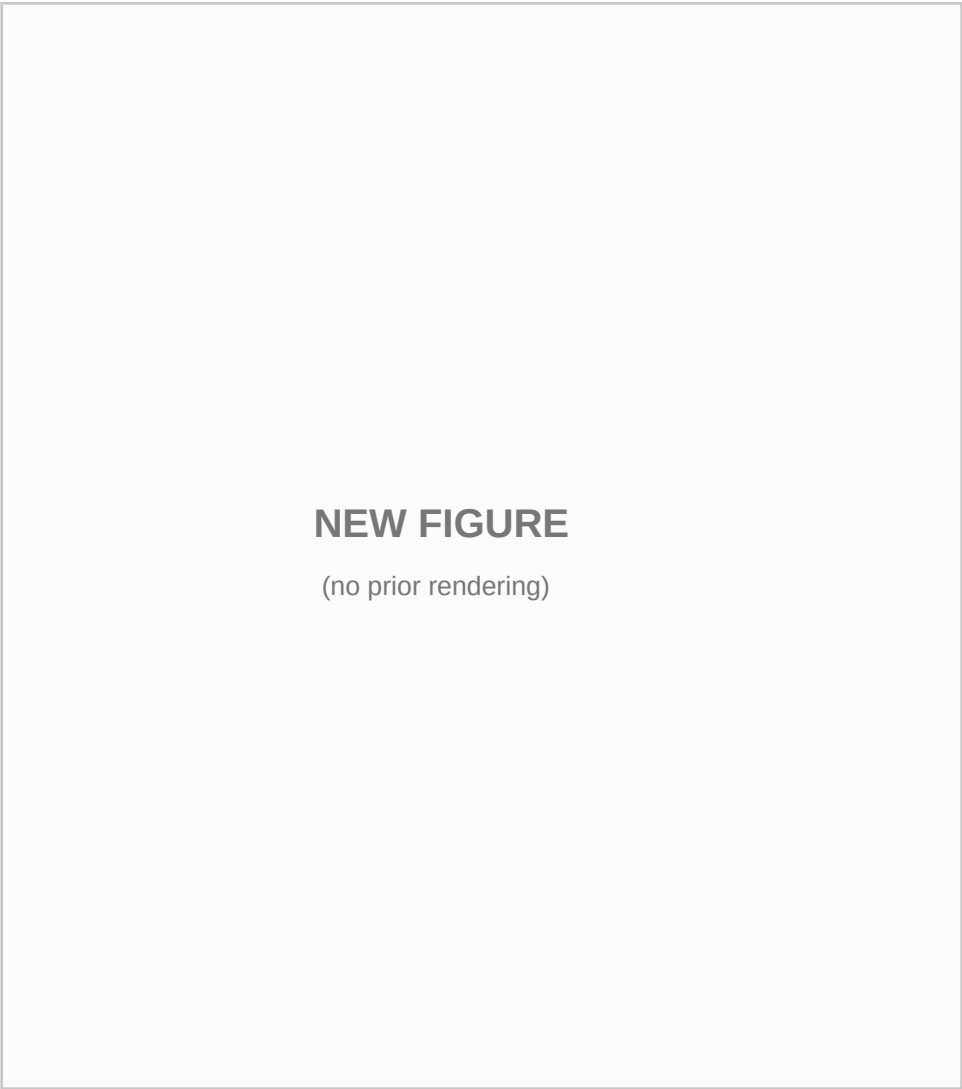
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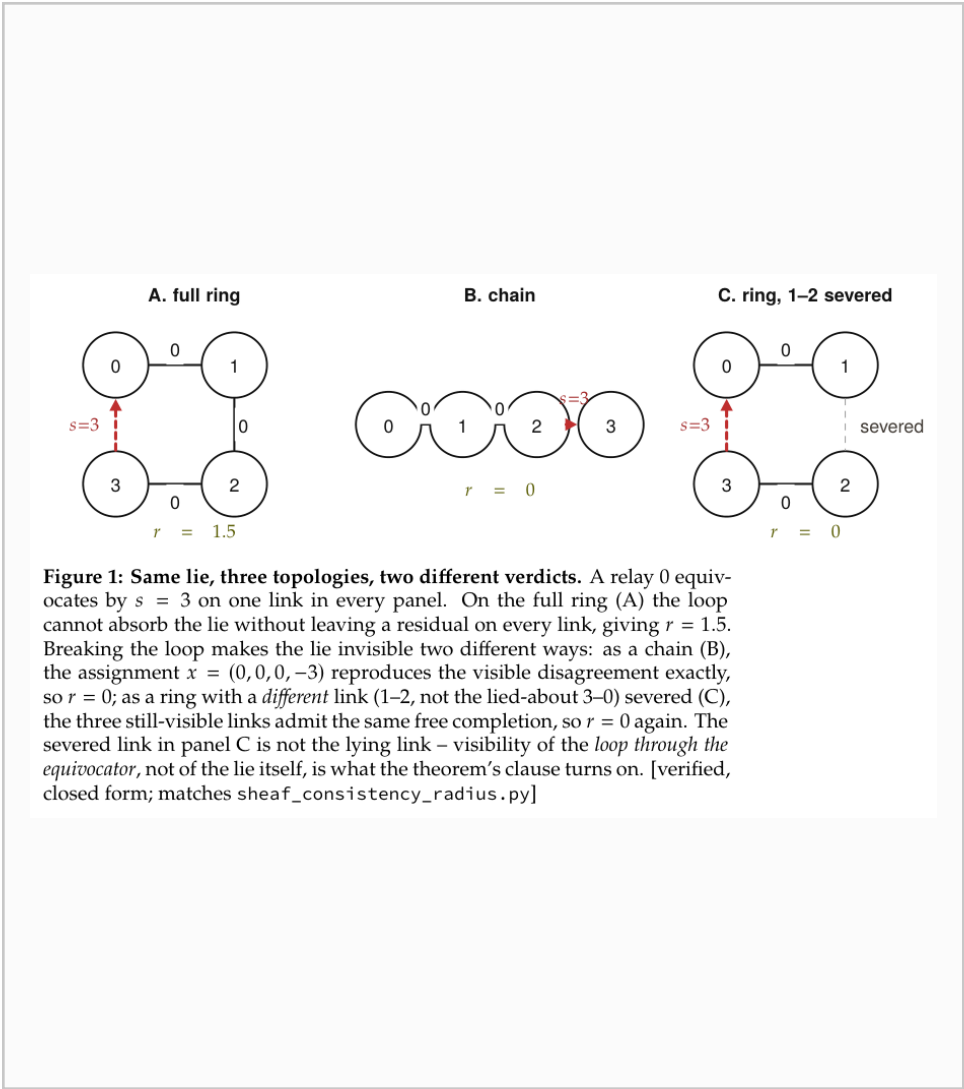
QA: PASS

Cartel sustainability is a two-policy phase diagram. The boundary is the repeated-game equality $V_C = \pi_D$ (gold). Below it (amber), the discounted collusive stream still dominates a one-shot deviation. Above it (green), deviation wins and the cartel is unstable. A rarer severe sampled loss and a more frequent smaller loss can lie on the same deterrence boundary. Detection frequency by itself is therefore not the me...

BEFORE



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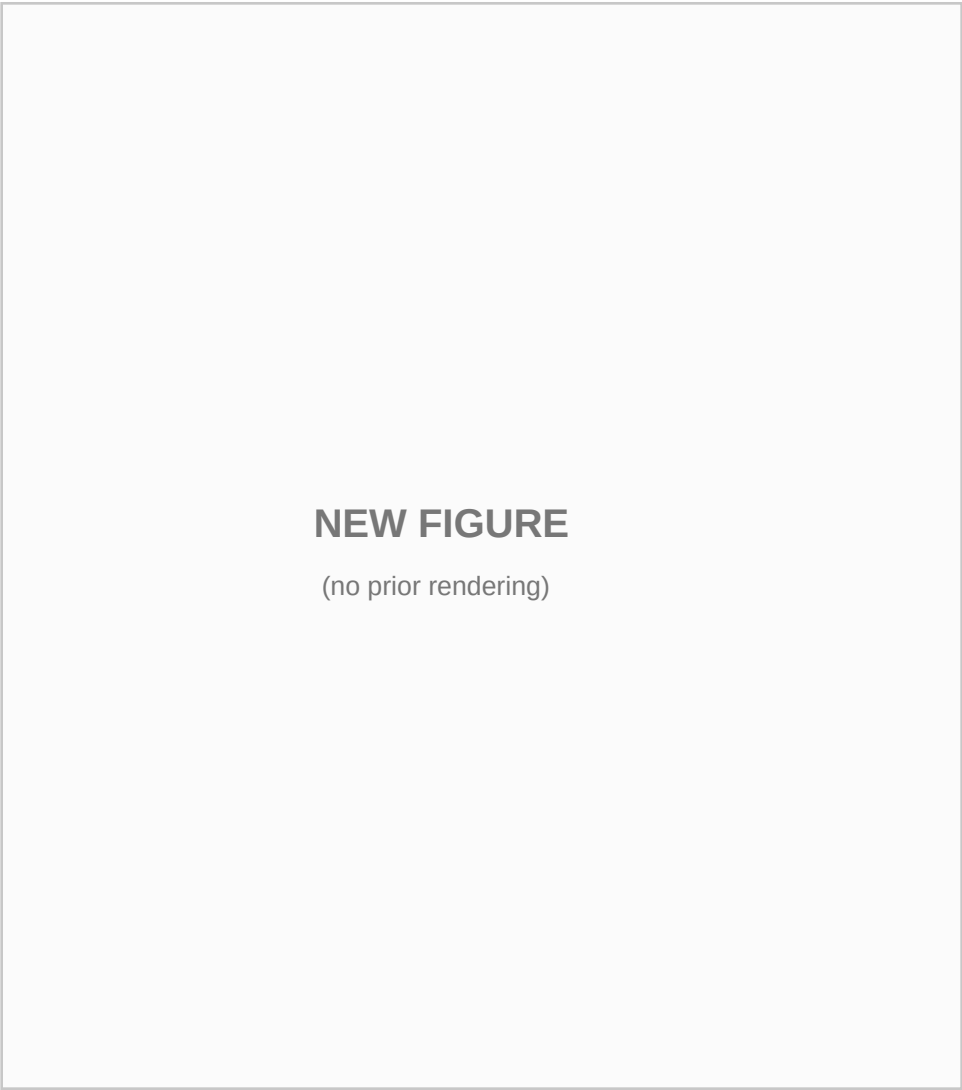


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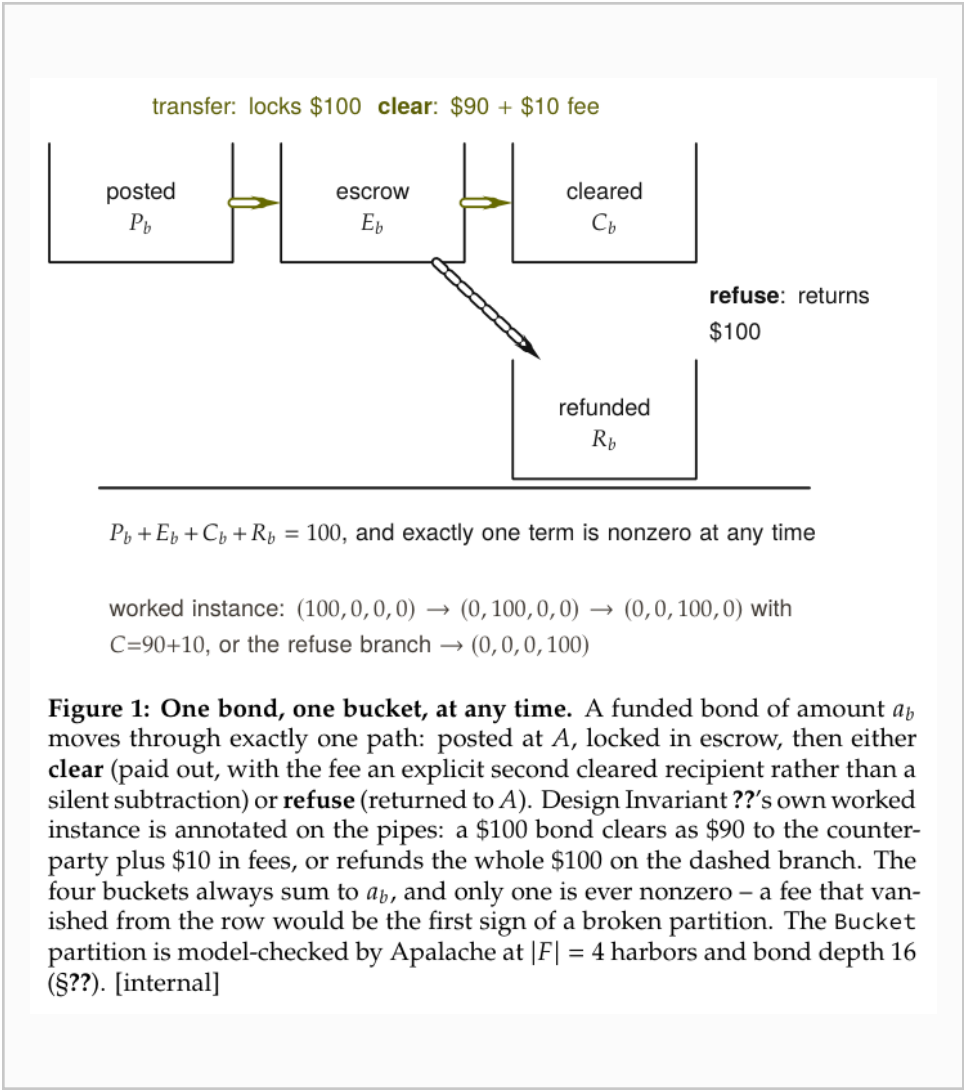
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Same lie, three topologies, two different verdicts. A relay 0 equivocates by $s=3$ on one link in every panel. On the full ring (A) the loop cannot absorb the lie without leaving a residual on every link, giving $r=1.5$. Breaking the loop makes the lie invisible two different ways: as a chain (B), the assignment $x=(0,0,0,-3)$ reproduces the visible disagreement exactly, so $r=0$; as a ring with a {different link ...

BEFORE



AFTER



QA: —

QA: PASS

One bond, one bucket, at any time. A funded bond of amount a_b moves through exactly one path: posted at A , locked in escrow, then either clear (paid out, with the fee an explicit second cleared recipient rather than a silent subtraction) or refuse (returned to A). Design Invariant~{prop:fh-cross-cons’s own worked instance is annotated on the pipes: a \$100 bond clears as \$90 to the counterparty plus \$10 i...

BEFORE

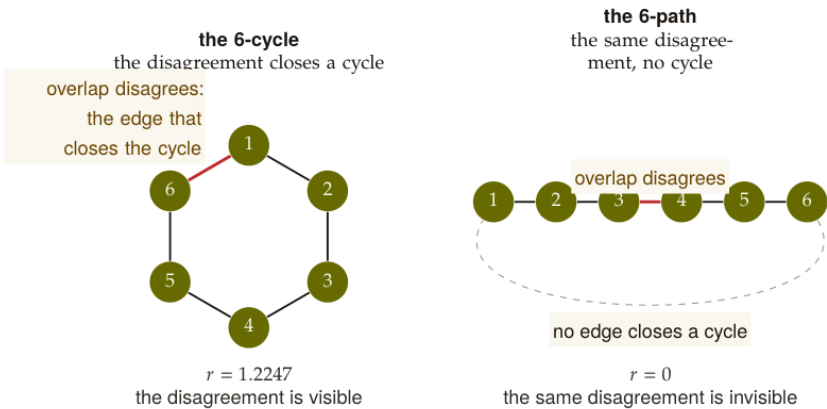


Figure 1: Cycle versus cut on six harbors, numbered 1 to 6. The same pairwise disagreement on one overlap is placed on the cycle C_6 and on the path P_6 . On the cycle the disagreeing edge closes a visible cycle and the consistency radius is $r = 3(1 - 5/6)^{1/2} = 1.2247$; on the path no edge closes a cycle and $r = 0$, so detection is exactly the visible-cycle condition of the theorem. [verified; sheaf_consistency_radius.py, seed 20260816]

QA: FLAGGED: B9, too wide 5.52in

Cycle versus cut on six harbors, numbered 1 to 6. The same pairwise disagreement on one overlap is placed on the cycle C_6 and on the path P_6 . On the cycle the disagreeing edge closes a visible cycle and the consistency radius is $r=3\sqrt{1-5/6}=1.2247$; on the path no edge closes a cycle and $r=0$, so detection is exactly the visible-cycle condition of Theorem- $\{thm: sheaf-equivocation\}$. [verified; {sheaf}_c...

AFTER

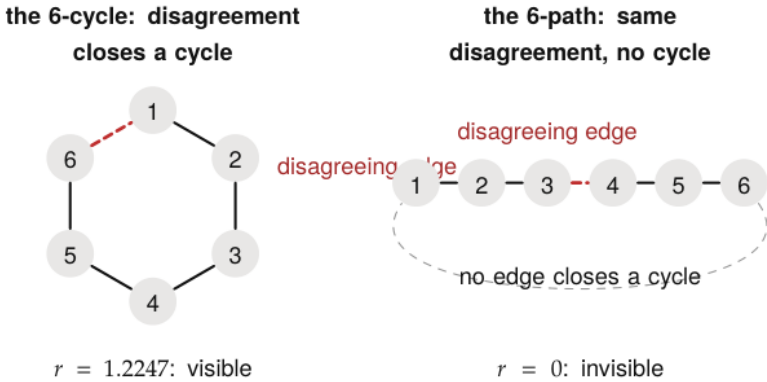
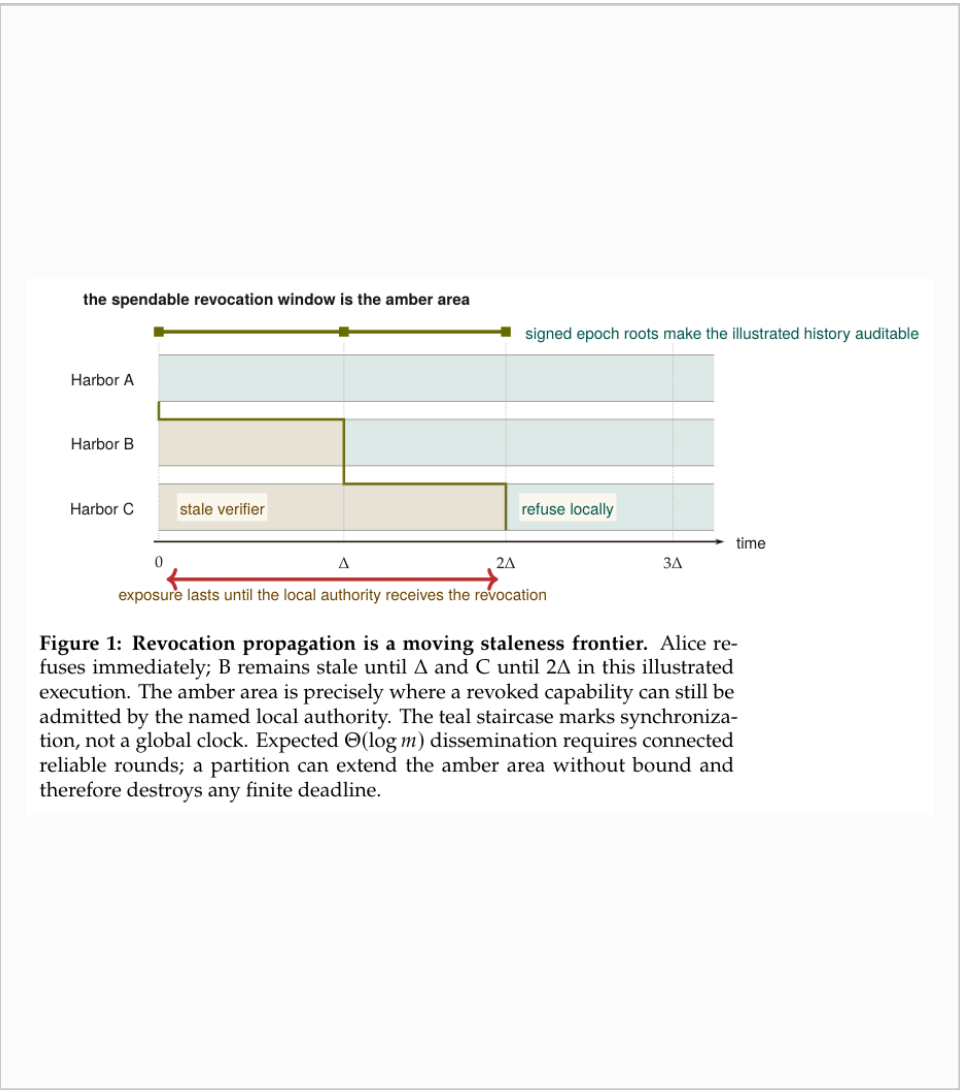


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QA: FLAGGED: B3

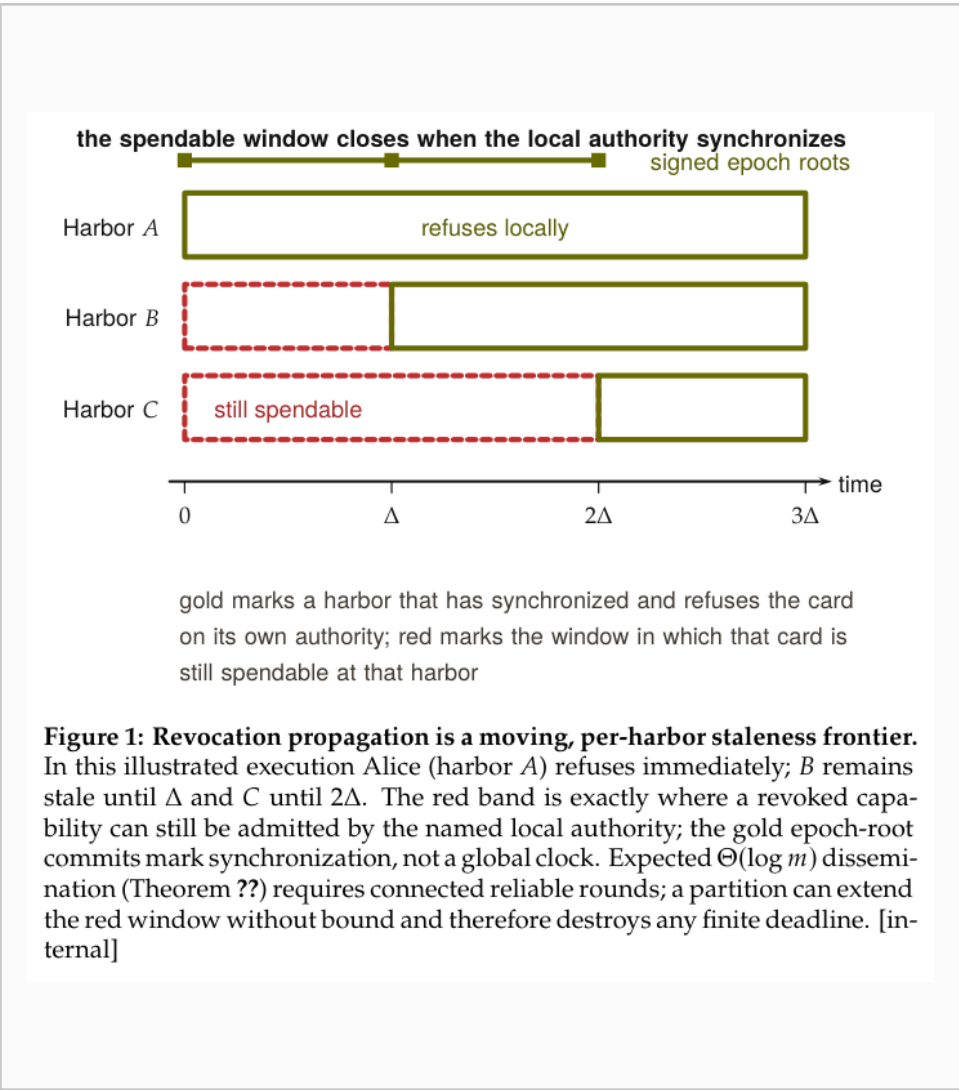
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QA: FLAGGED: too wide 5.50in

Revocation propagation is a moving, per-harbor staleness frontier. In this illustrated execution Alice (harbor A) refuses immediately; B remains stale until Δ and C until 2Δ . The red band is exactly where a revoked capability can still be admitted by the named local authority; the gold epoch-root commits mark synchronization, not a global clock. Expected $\Theta(\log m)$ dissemination (Theorem ??) requires connected reliable rounds; a partition can extend the red window without bound and therefore destroys any finite deadline.

AFTER



QA: PASS

BEFORE

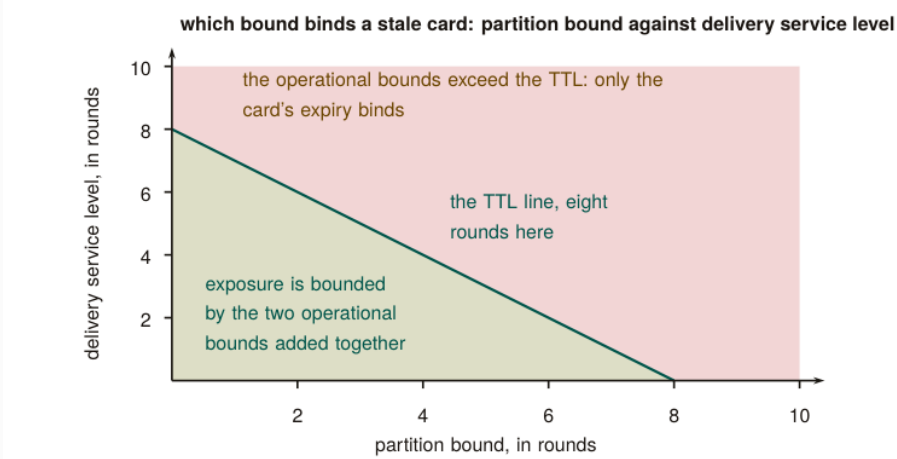


Figure 1: Which bound binds a stale card after revocation. Under a partition bound and a delivery service level, exposure is at most the two added together (Corollary ??); once that sum exceeds the card's time to live, the expiry binds instead and the harbor fails closed. Without a partition bound the horizontal axis is unbounded and only the expiry remains (Theorem ??, item 3); the line is drawn for a time to live of eight rounds.

QA: PASS

Which bound binds a stale card after revocation. Under a partition bound and a delivery service level, exposure is at most the two added together (Corollary-{cor:fh-window, below the gold TTL line}); once that sum exceeds the card's time to live the expiry binds instead and the harbor fails closed (above the line, red dashed border). Without a partition bound the horizontal axis is unbounded and only the expiry remains (Theorem ??, item 3); the line is drawn for an illustrative time to live of eight rounds. [internal]

AFTER

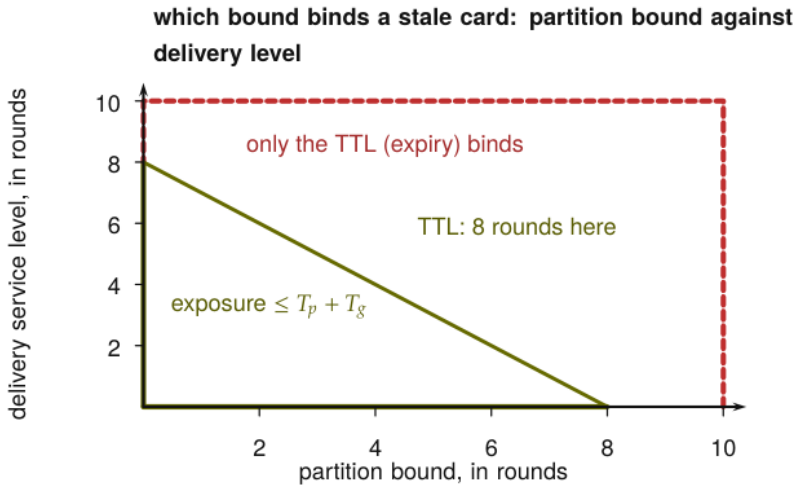
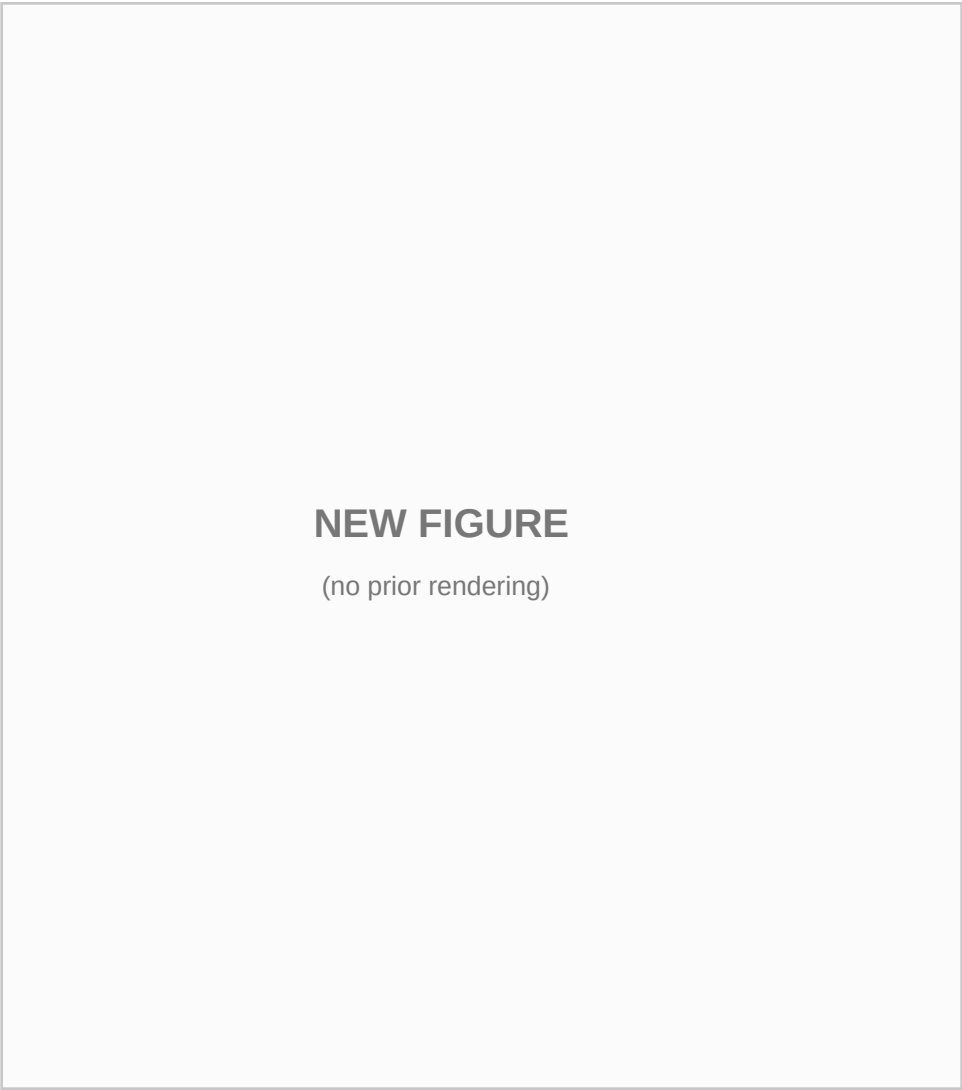


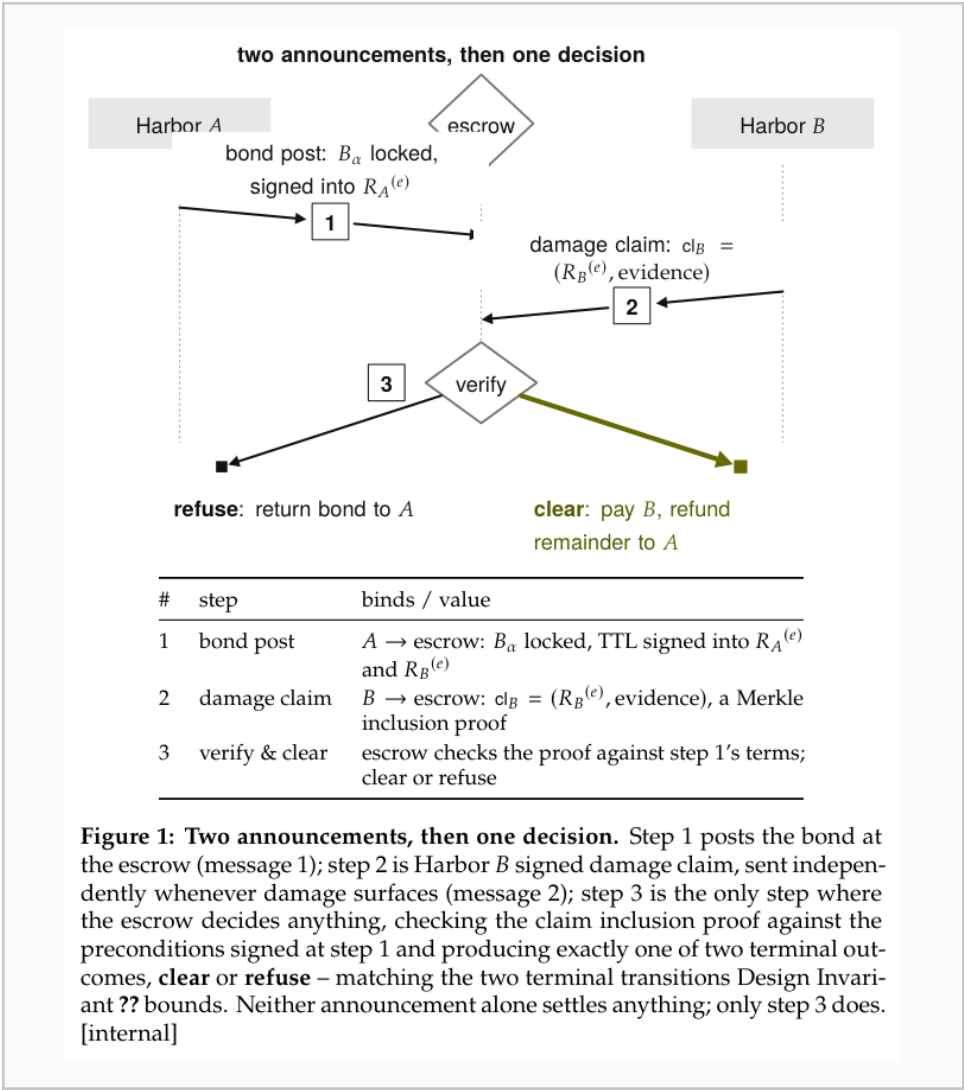
Figure 1: Which bound binds a stale card after revocation. Under a partition bound and a delivery service level, exposure is at most the two added together (Corollary ??, below the gold TTL line); once that sum exceeds the card's time to live the expiry binds instead and the harbor fails closed (above the line, red dashed border). Without a partition bound the horizontal axis is unbounded and only the expiry remains (Theorem ??, item 3); the line is drawn for an illustrative time to live of eight rounds. [internal]

QA: FLAGGED: B3

BEFORE



AFTER

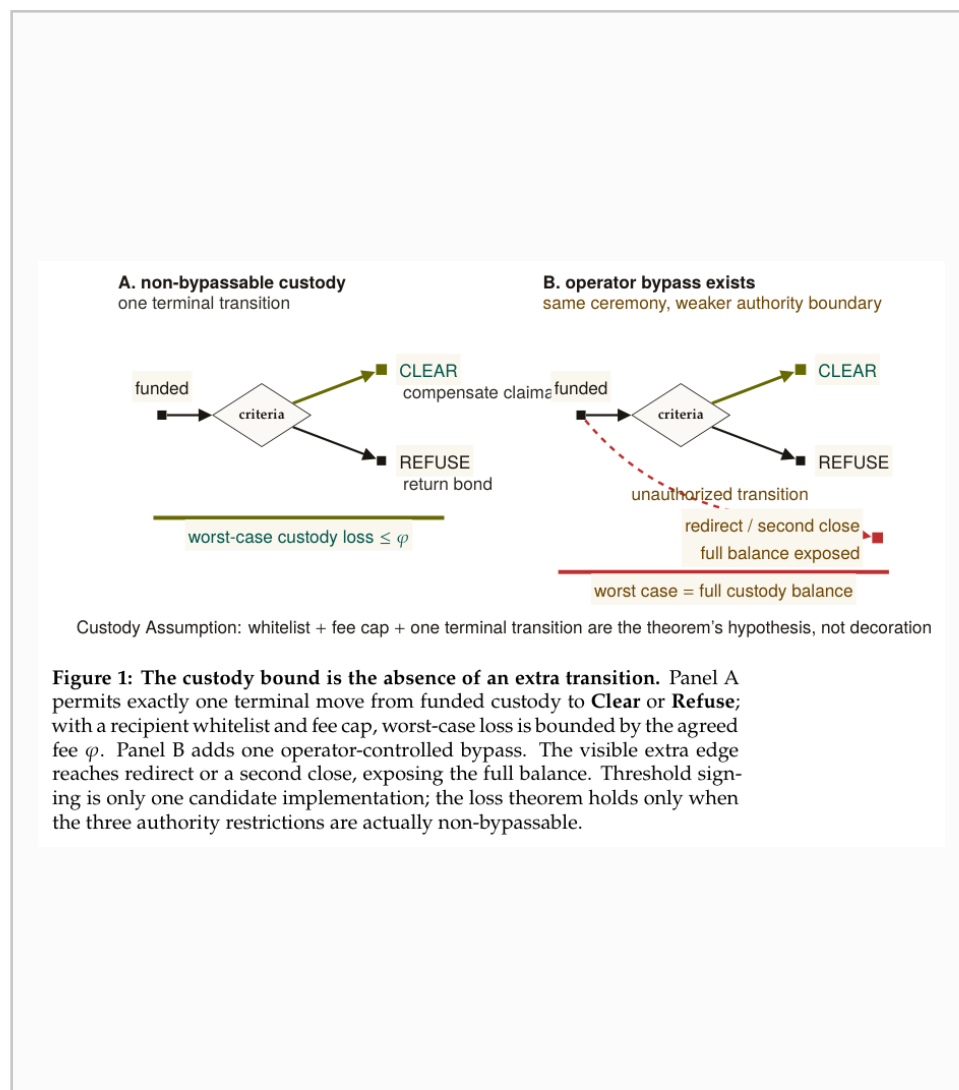


QA: —

QA: PASS

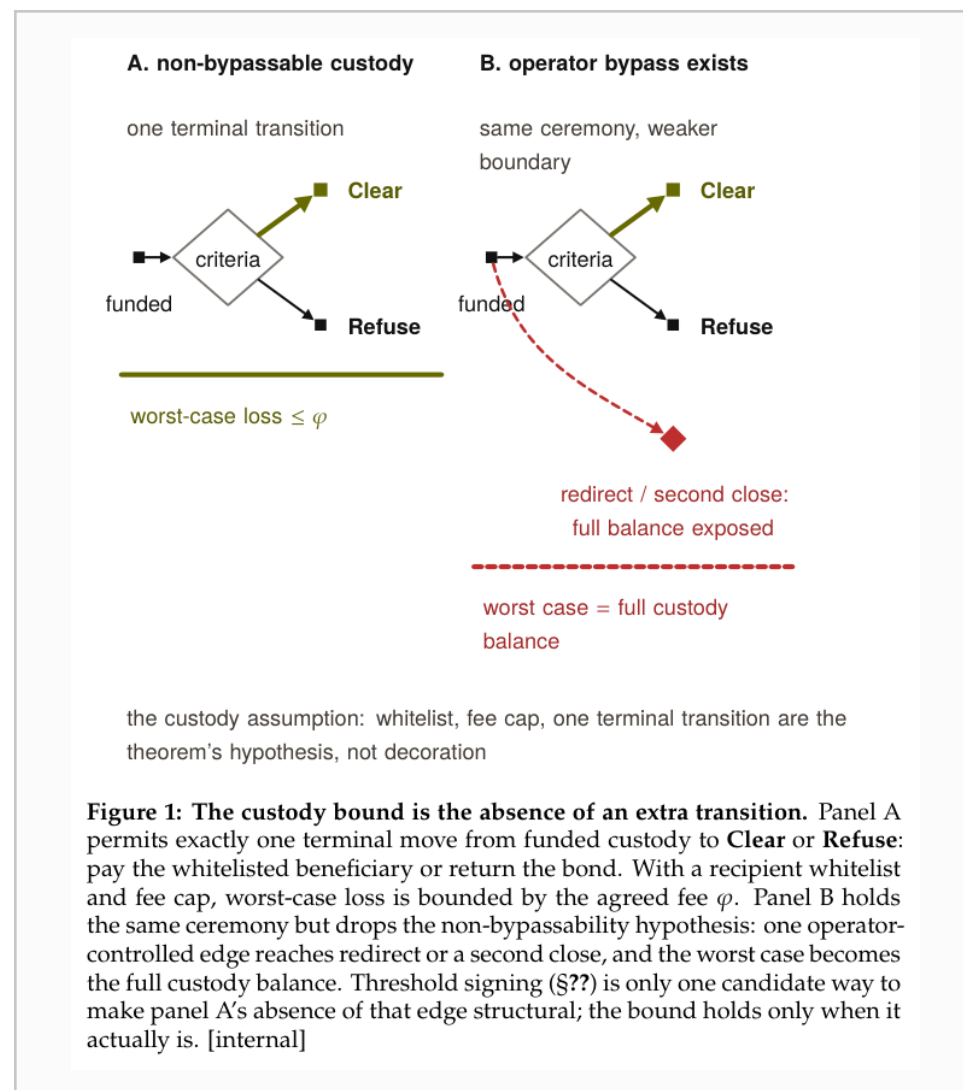
Two announcements, then one decision. Step 1 posts the bond at the escrow (message 1); step 2 is Harbor B signed damage claim, sent independently whenever damage surfaces (message 2); step 3 is the only step where the escrow decides anything, checking the claim inclusion proof against the preconditions signed at step 1 and producing exactly one of two terminal outcomes, clear or refuse -- matching the two termin...

BEFORE



QA: FLAGGED: T1, T3, B3, too wide 5.61in

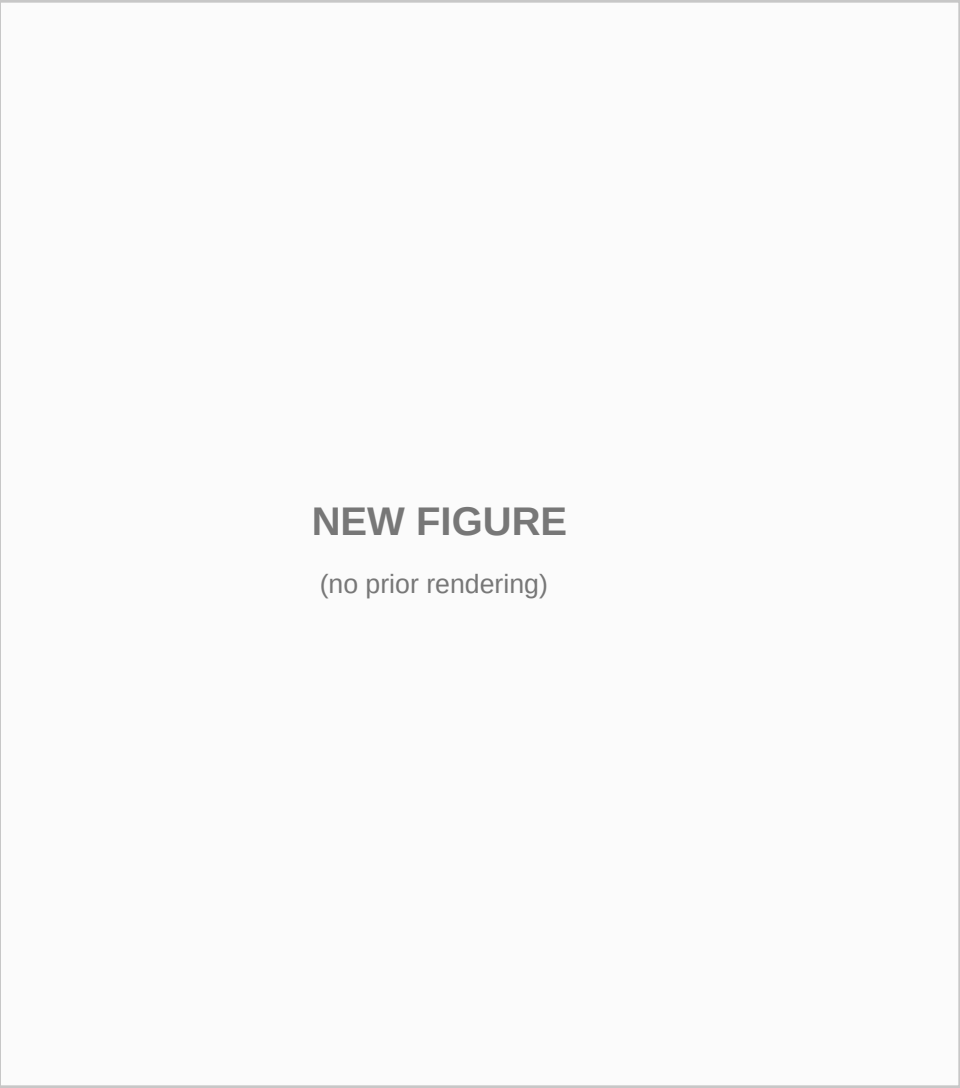
AFTER



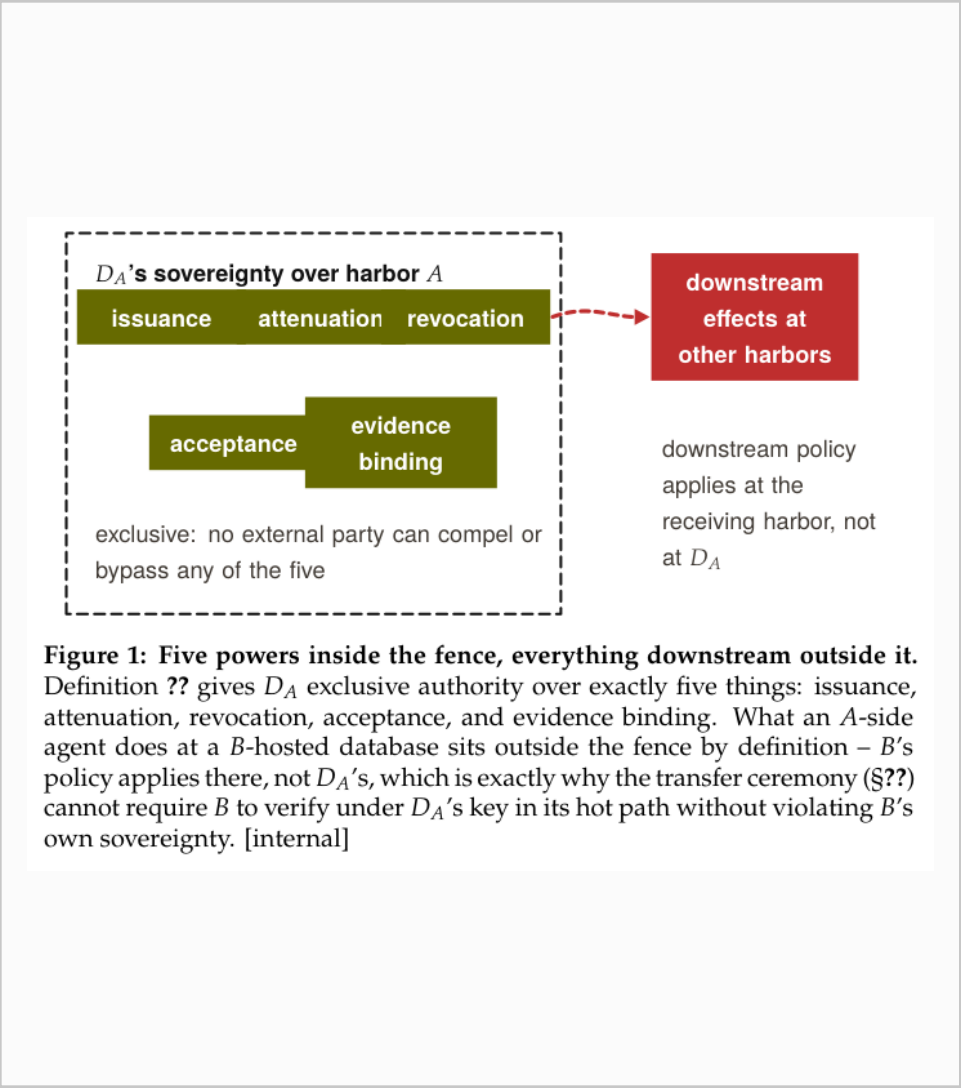
QA: PASS

The custody bound is the absence of an extra transition. Panel A permits exactly one terminal move from funded custody to Clear or Refuse: pay the whitelisted beneficiary or return the bond. With a recipient whitelist and fee cap, worst-case loss is bounded by the agreed fee φ . Panel B holds the same ceremony but drops the non-bypassability hypothesis: one operator-controlled edge reaches redirect or a second close...

BEFORE



AFTER

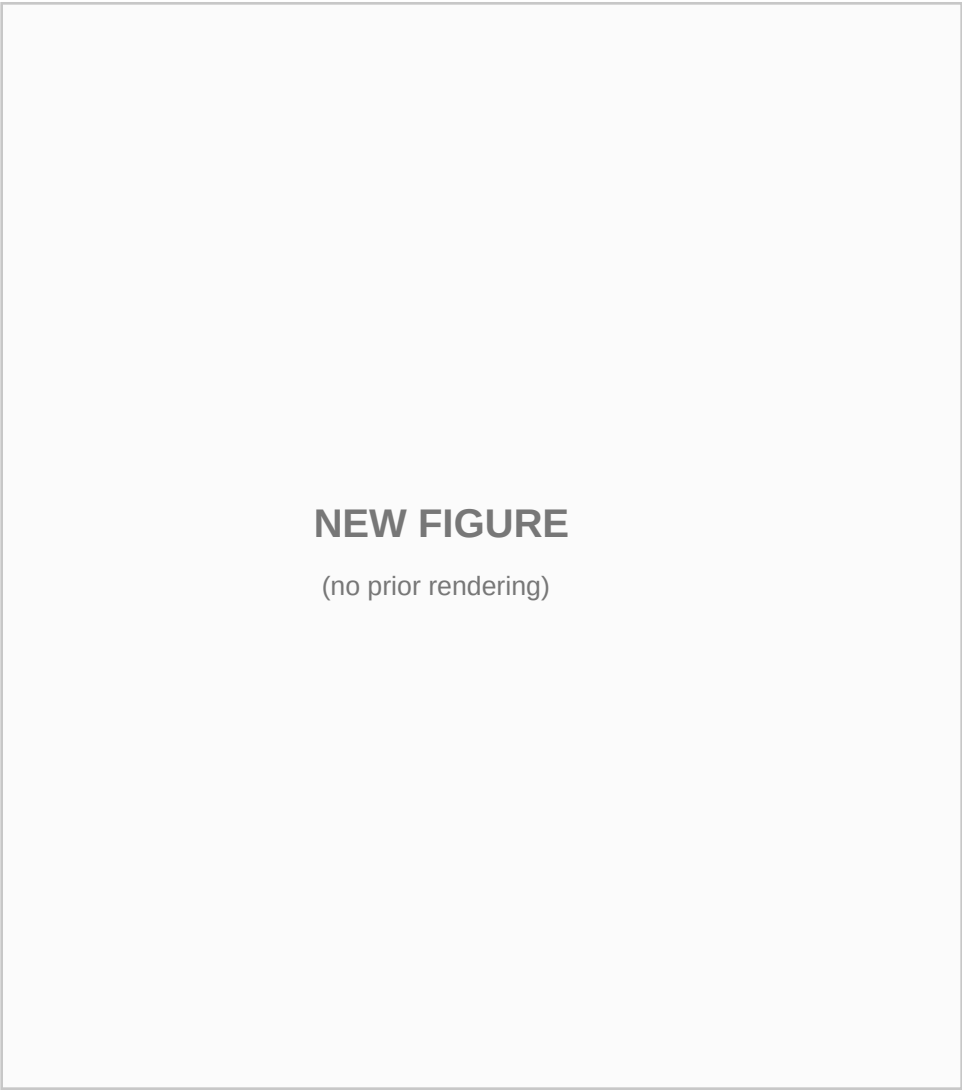


QA: —

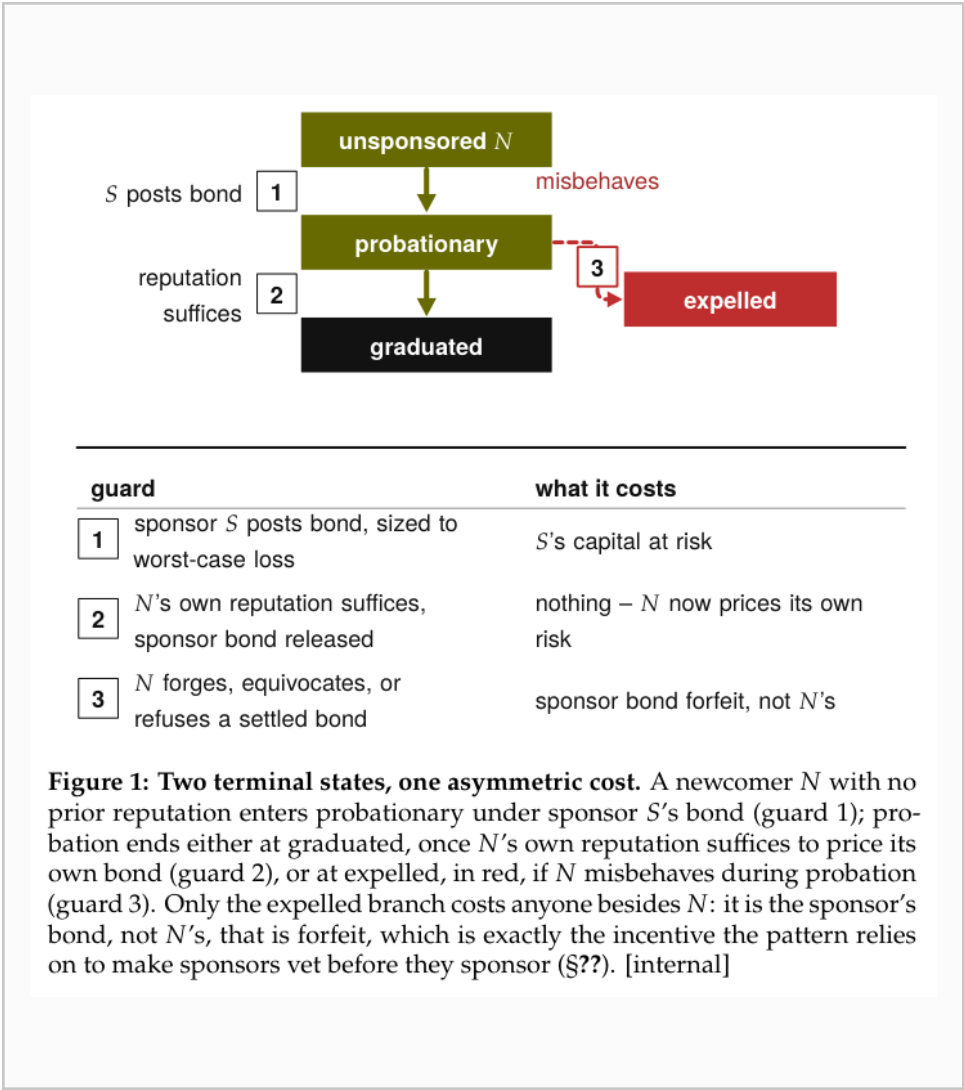
QA: PASS

Five powers inside the fence, everything downstream outside it. Definition~{def:fh-sovereignty gives \$D_A\$ exclusive authority over exactly five things: issuance, attenuation, revocation, acceptance, and evidence binding. What an \$A\$-side agent does at a \$B\$-hosted database sits outside the fence by definition -- \$B\$'s policy applies there, not \$D_A\$'s, which is exactly why the transfer ceremony ({sec:fh-xfer) can...

BEFORE



AFTER

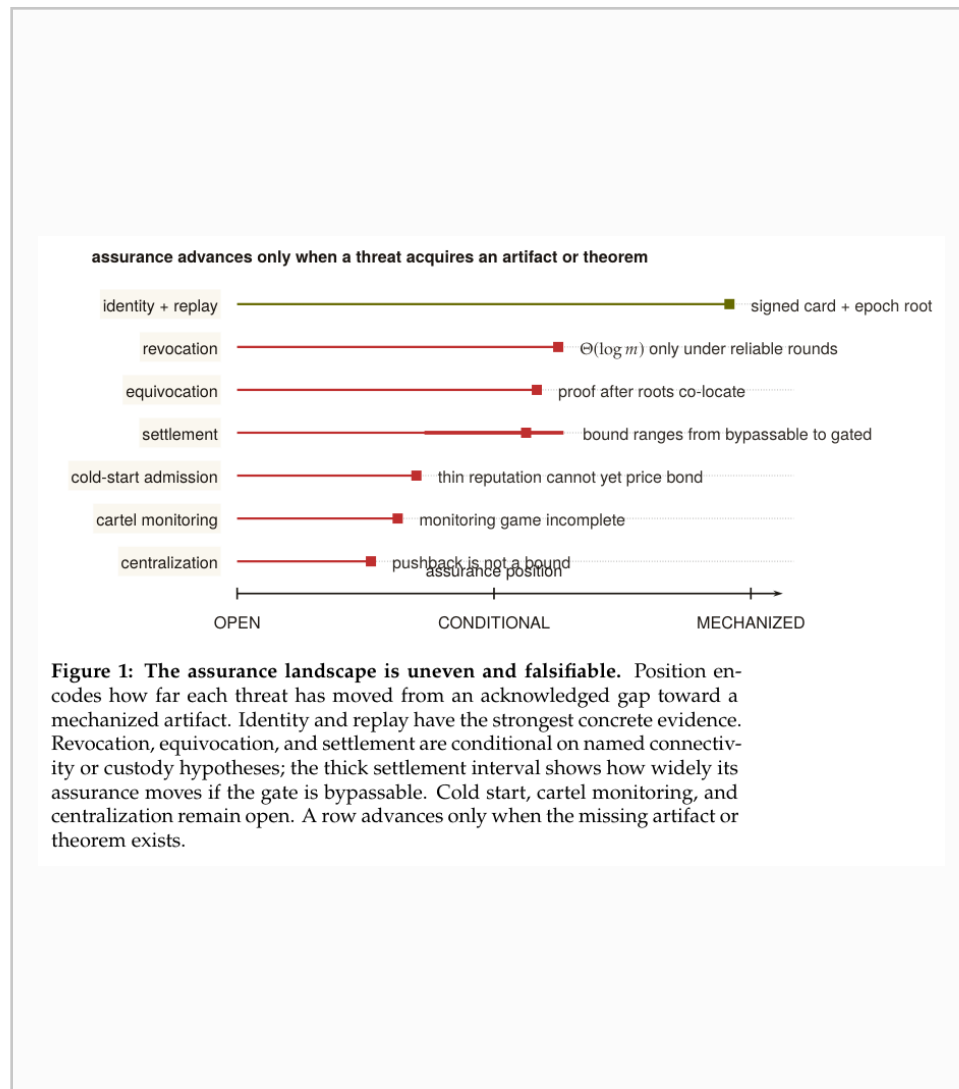


QA: —

QA: PASS

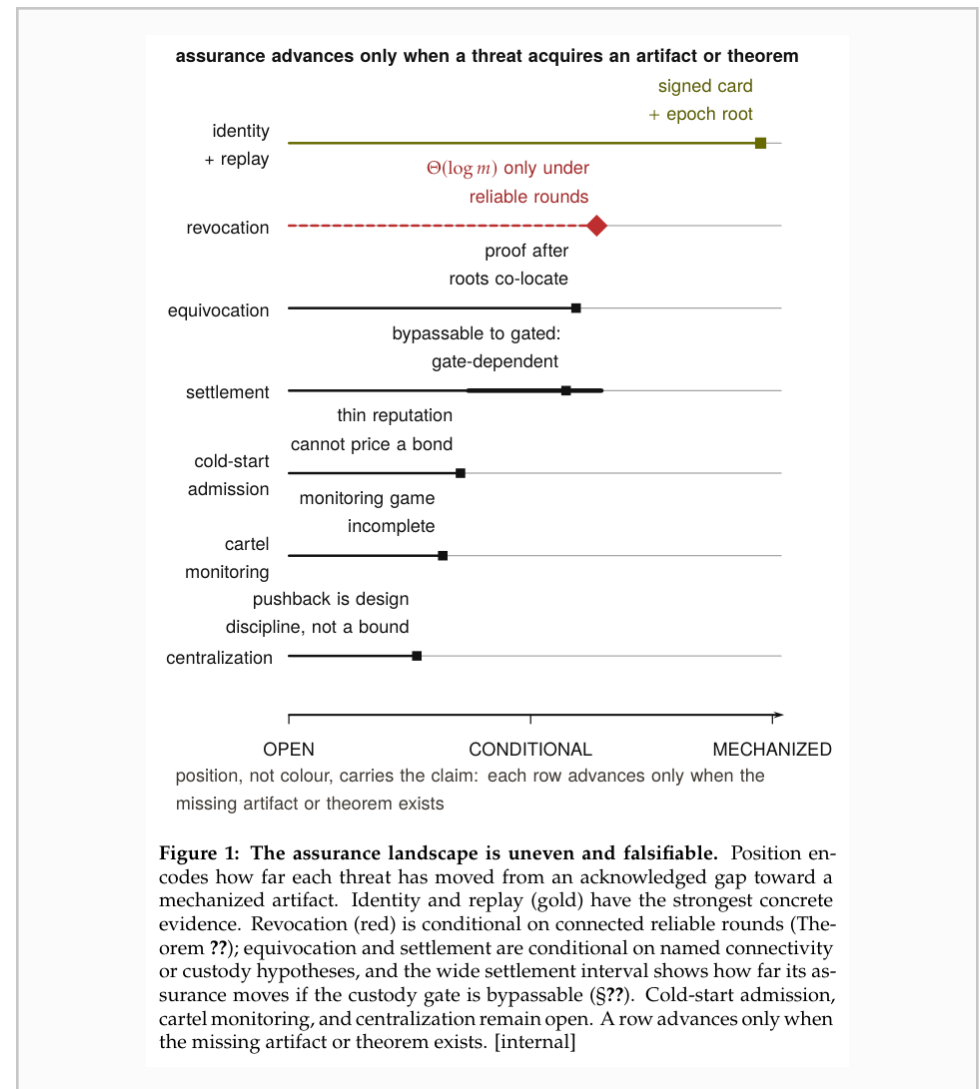
Two terminal states, one asymmetric cost. A newcomer *N* with no prior reputation enters probationary under sponsor *S*'s bond (guard 1); probation ends either at graduated, once *N*'s own reputation suffices to price its own bond (guard 2), or at expelled, in red, if *N* misbehaves during probation (guard 3). Only the expelled branch costs anyone besides *N*: it is the sponsor's bond, not *N*'s, that is forfeit, ...

BEFORE



QA: FLAGGED: T3, T4, B3, too wide 5.66in

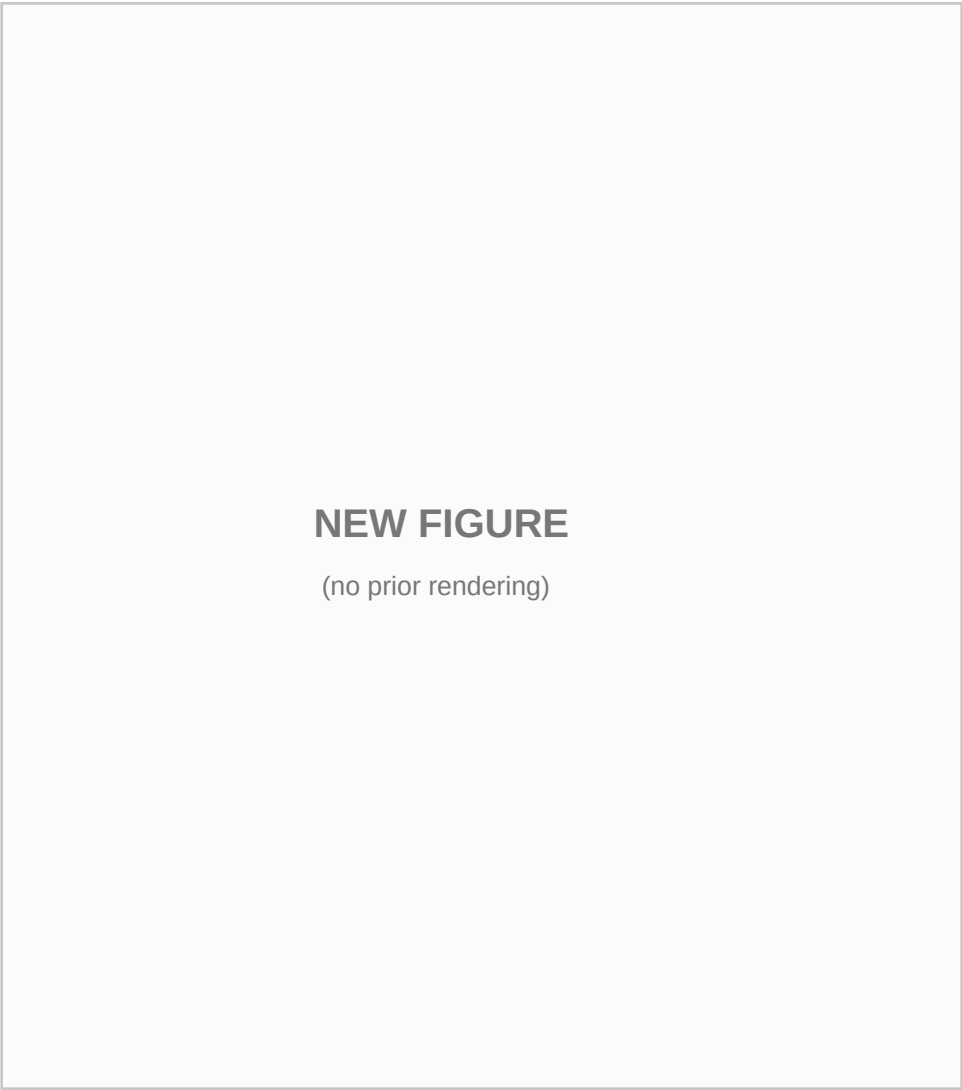
AFTER



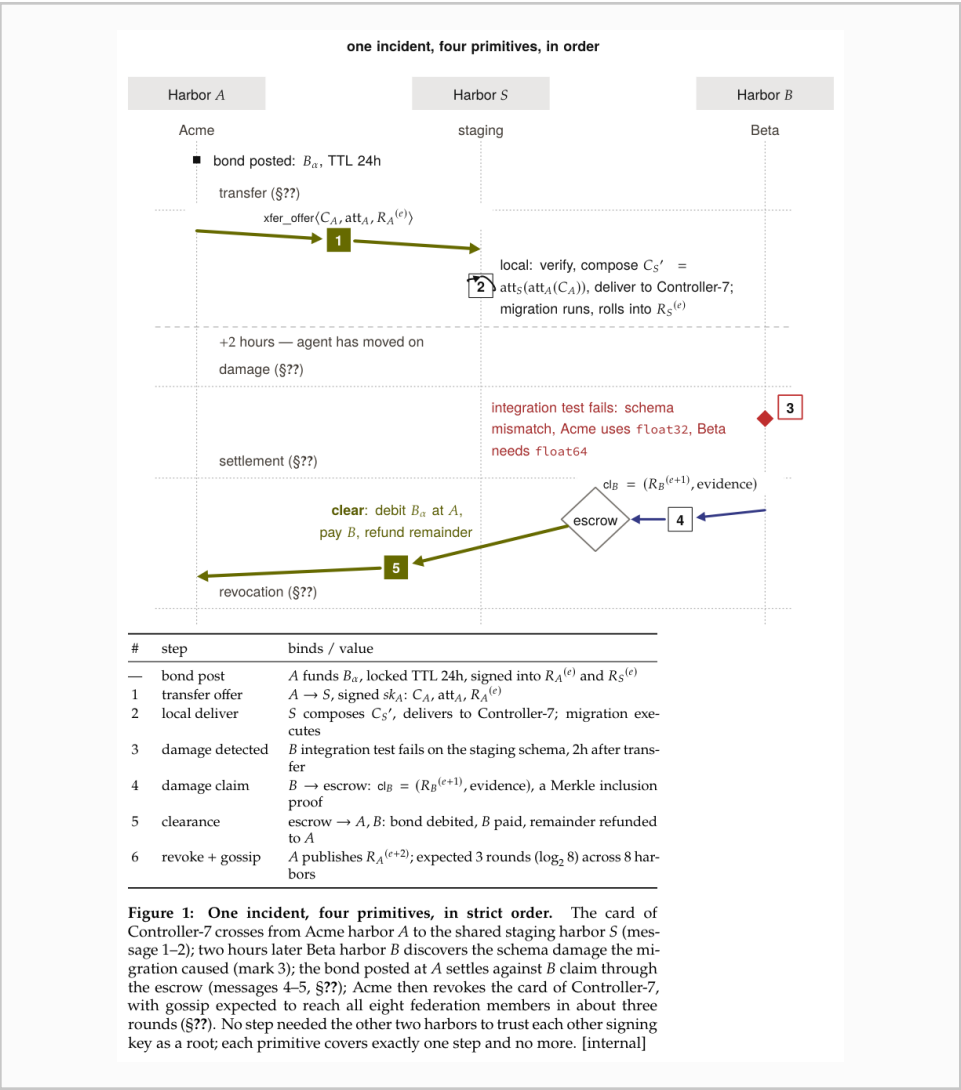
QA: PASS

The assurance landscape is uneven and falsifiable. Position encodes how far each threat has moved from an acknowledged gap toward a mechanized artifact. Identity and replay (gold) have the strongest concrete evidence. Revocation (red) is conditional on connected reliable rounds (Theorem~{thm:fh-conv}); equivocation and settlement are conditional on named connectivity or custody hypotheses, and the wide settlement i...

BEFORE



AFTER

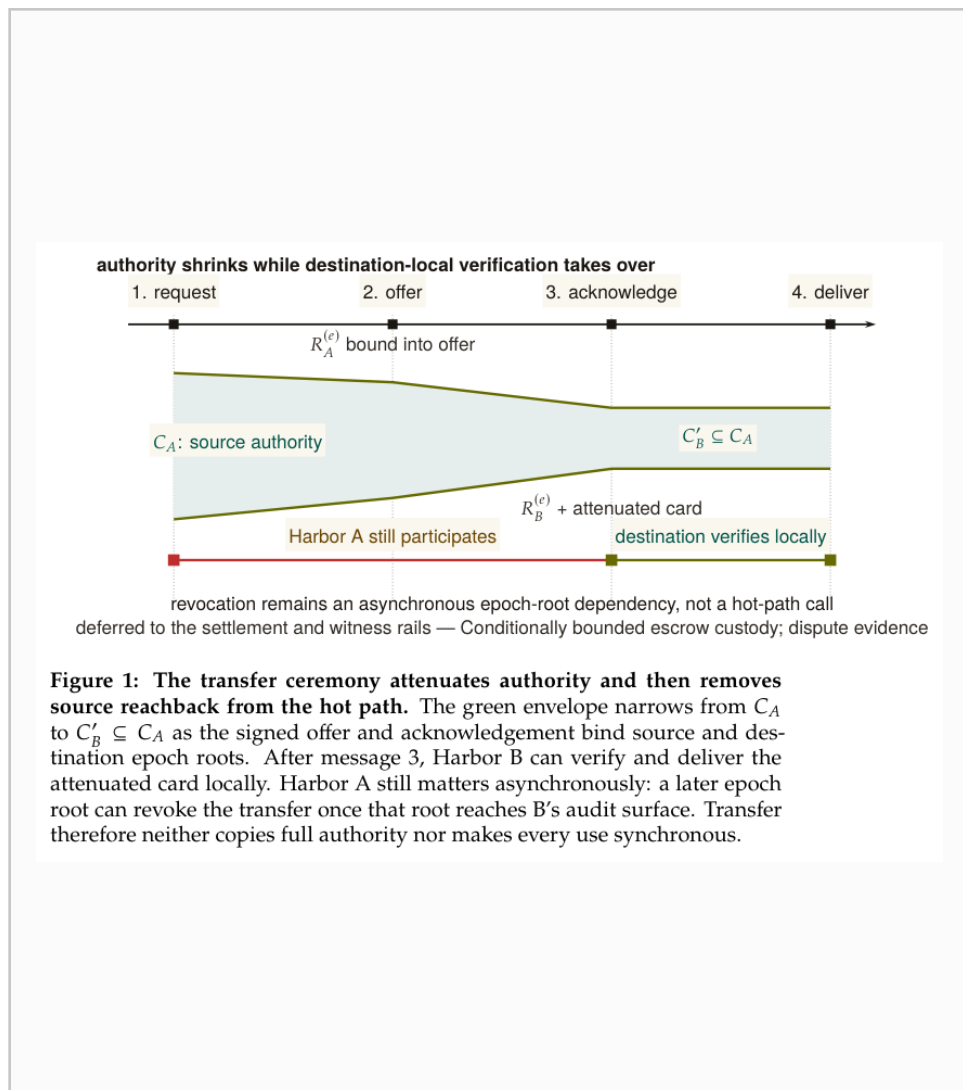


QA: —

QA: FLAGGED: too wide 6.00in

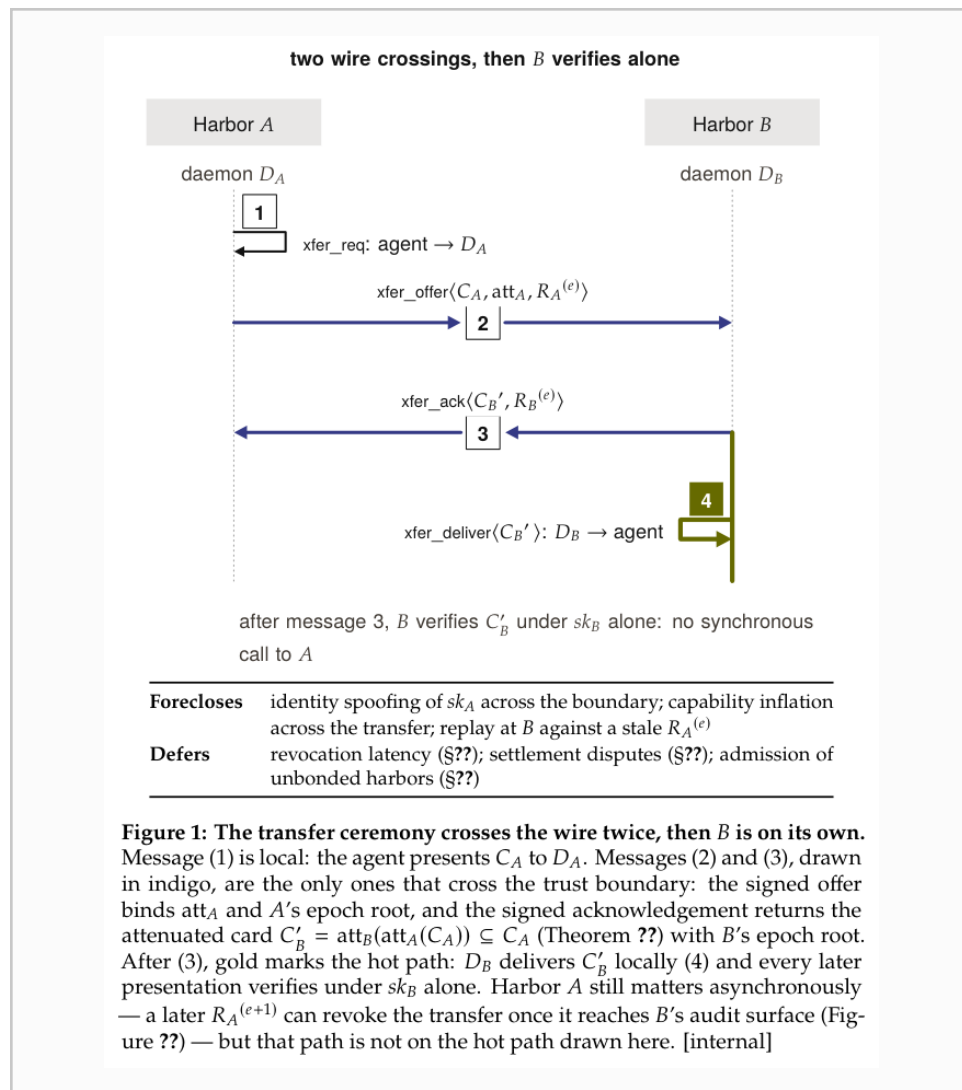
One incident, four primitives, in strict order. The card of Controller-7 crosses from Acme harbor A to the shared staging harbor S (message 1--2); two hours later Beta harbor B discovers the schema damage the migration caused (mark 3); the bond posted at A settles against B claim through the escrow (messages 4--5, {sec:fh-settlement}); Acme then revokes the card of Controller-7, with gossip expected to reach all eight federation members in about three rounds (§??). No step needed the other two harbors to trust each other signing key as a root; each primitive covers exactly one step and no more. [internal]

BEFORE



QA: FLAGGED: T3, T4, B3, too wide 5.25in

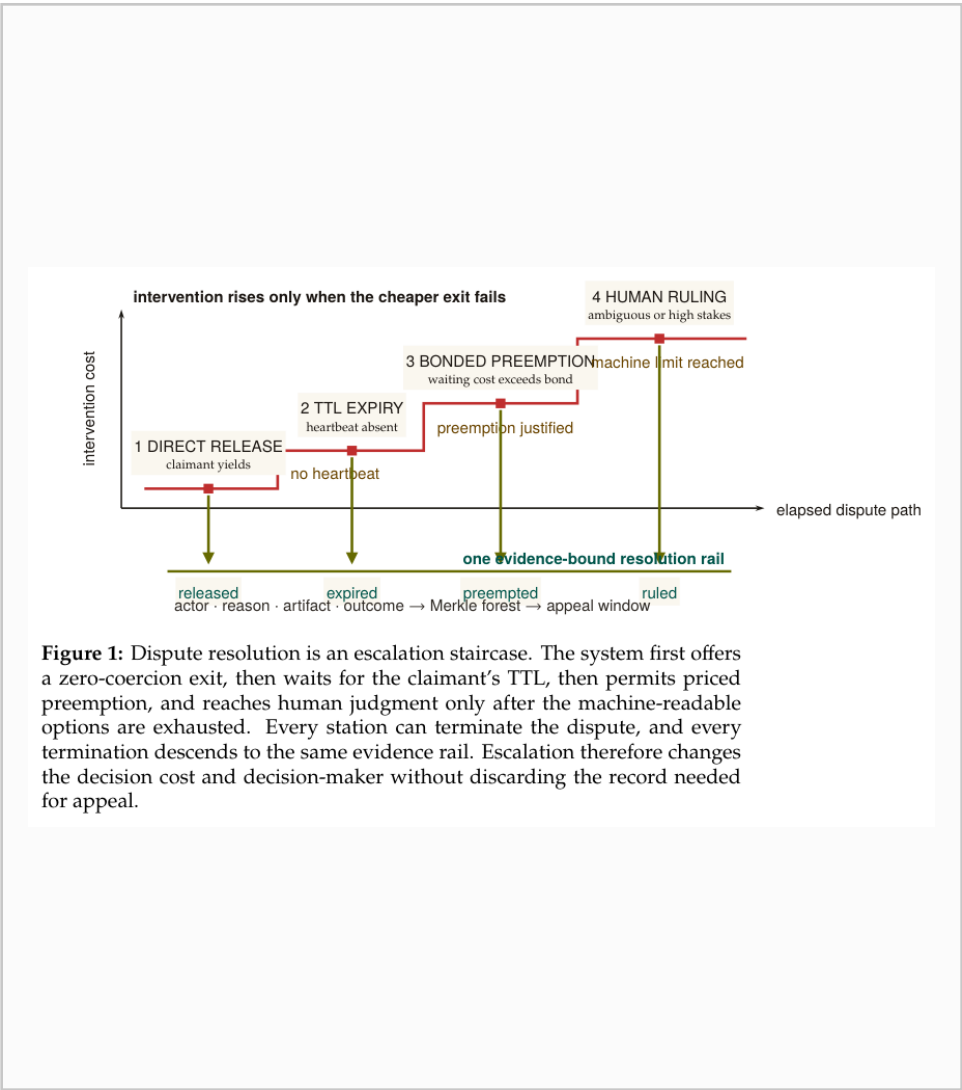
AFTER



QA: FLAGGED: B3

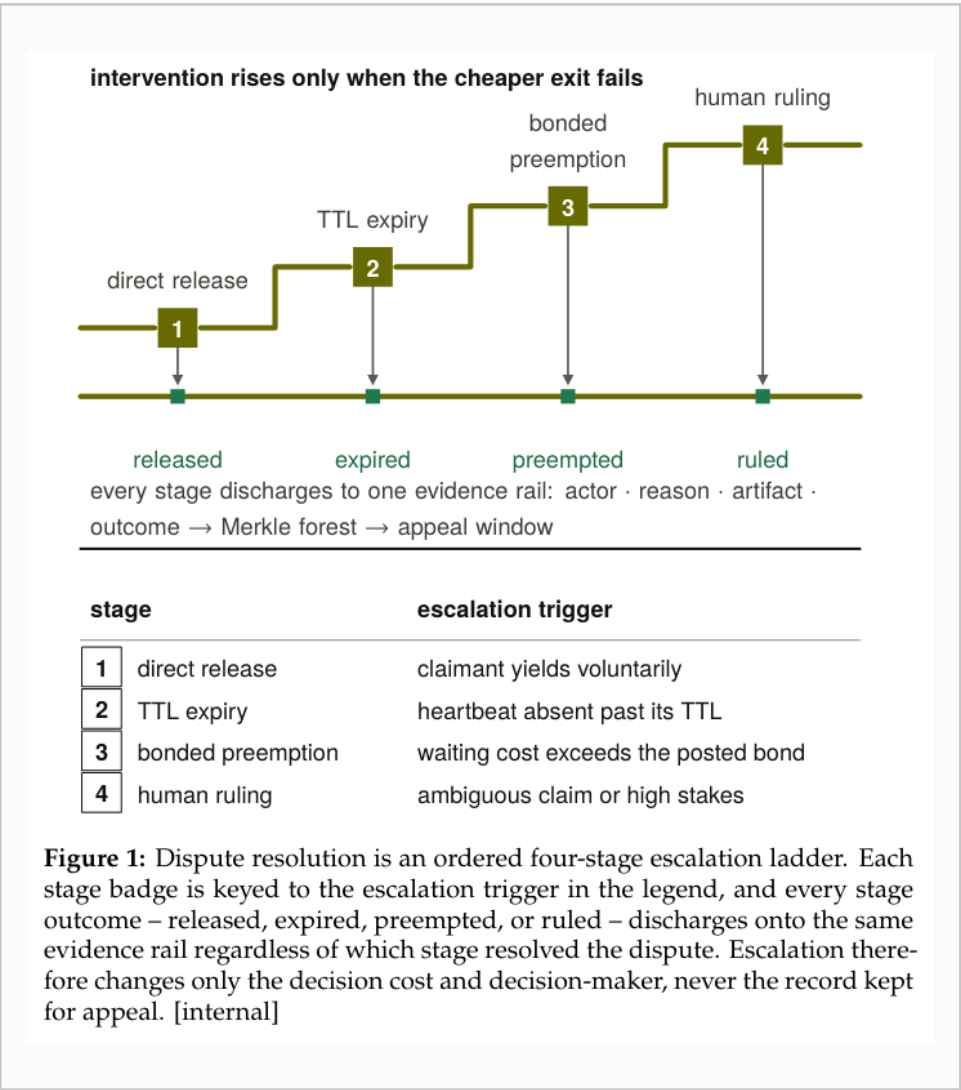
The transfer ceremony crosses the wire twice, then B is on its own. Message (1) is local: the agent presents C_A to D_A . Messages (2) and (3), drawn in indigo, are the only ones that cross the trust boundary: the signed offer binds att $_A$ and A 's epoch root, and the signed acknowledgement returns the attenuated card $C'_B = \text{att}_B(\text{att}_A(C_A)) \subseteq C_A$ (Theorem ??) with B 's epoch root. After (3), go...

BEFORE



QA: FLAGGED: T1, T3, T4, B3, too wide 5.40in

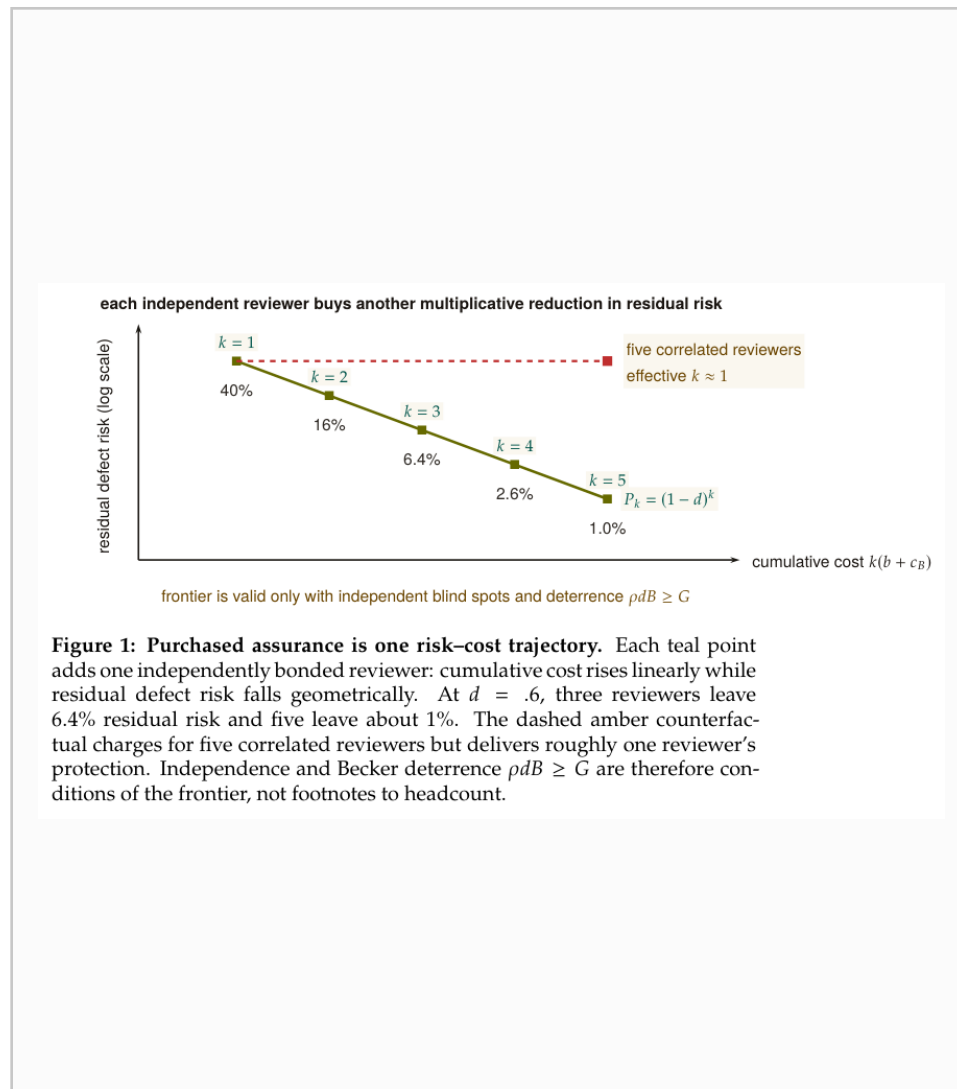
AFTER



QA: PASS

Dispute resolution is an ordered four-stage escalation ladder. Each stage badge is keyed to the escalation trigger in the legend, and every stage outcome -- released, expired, preempted, or ruled -- discharges onto the same evidence rail regardless of which stage resolved the dispute. Escalation therefore changes only the decision cost and decision-maker, never the record kept for appeal. [internal]

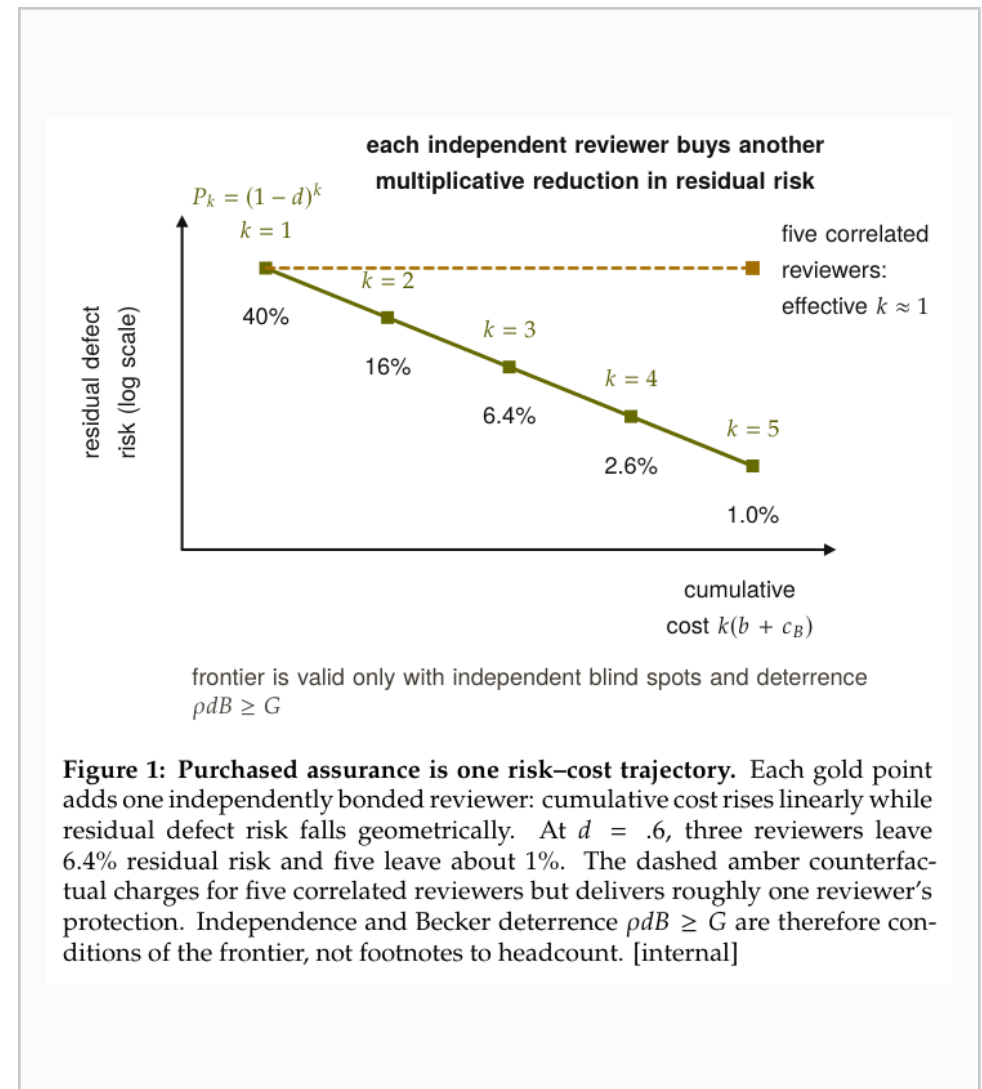
BEFORE



QA: FLAGGED: T1, too wide 5.34in

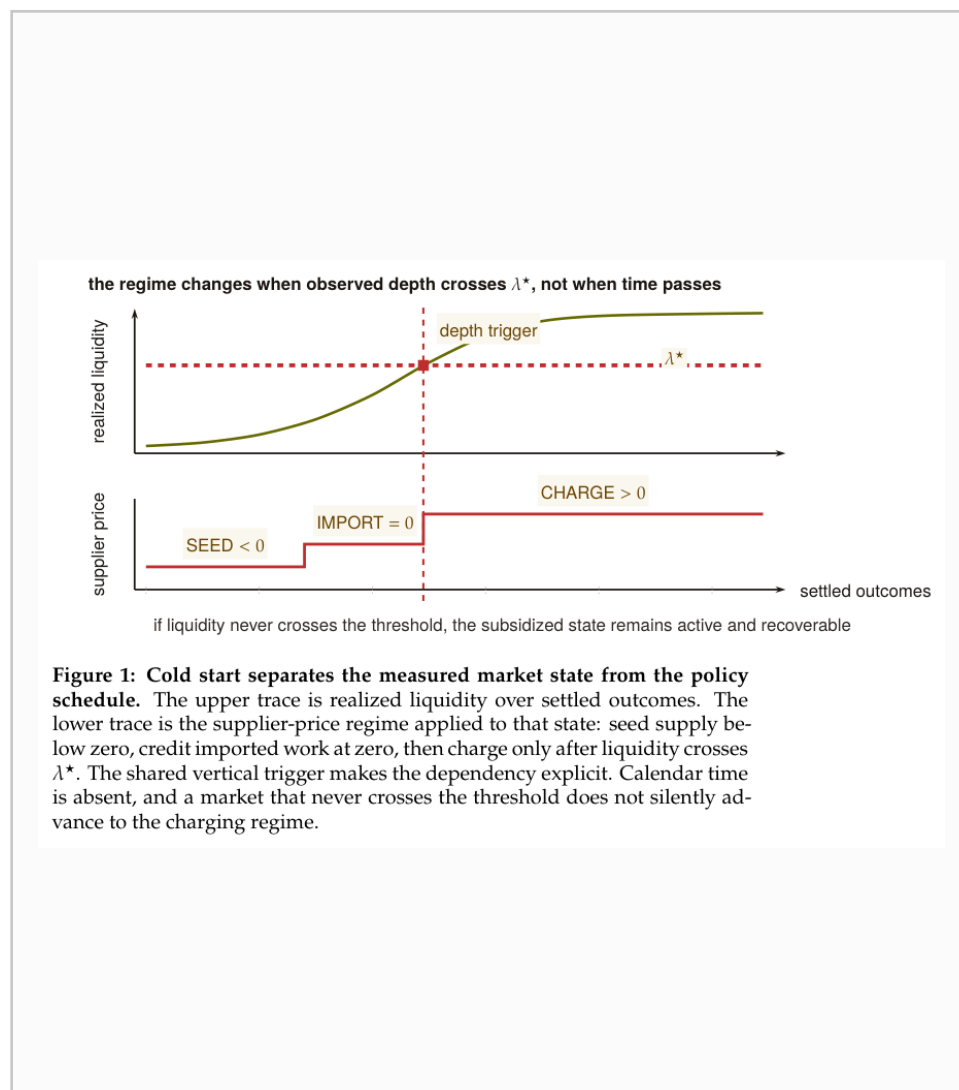
Purchased assurance is one risk--cost trajectory. Each gold point adds one independently bonded reviewer: cumulative cost rises linearly while residual defect risk falls geometrically. At $d=.6$, three reviewers leave 6.4% residual risk and five leave about 1%. The dashed amber counterfactual charges for five correlated reviewers but delivers roughly one reviewer's protection. Independence and Becker deterre...

AFTER



QA: PASS

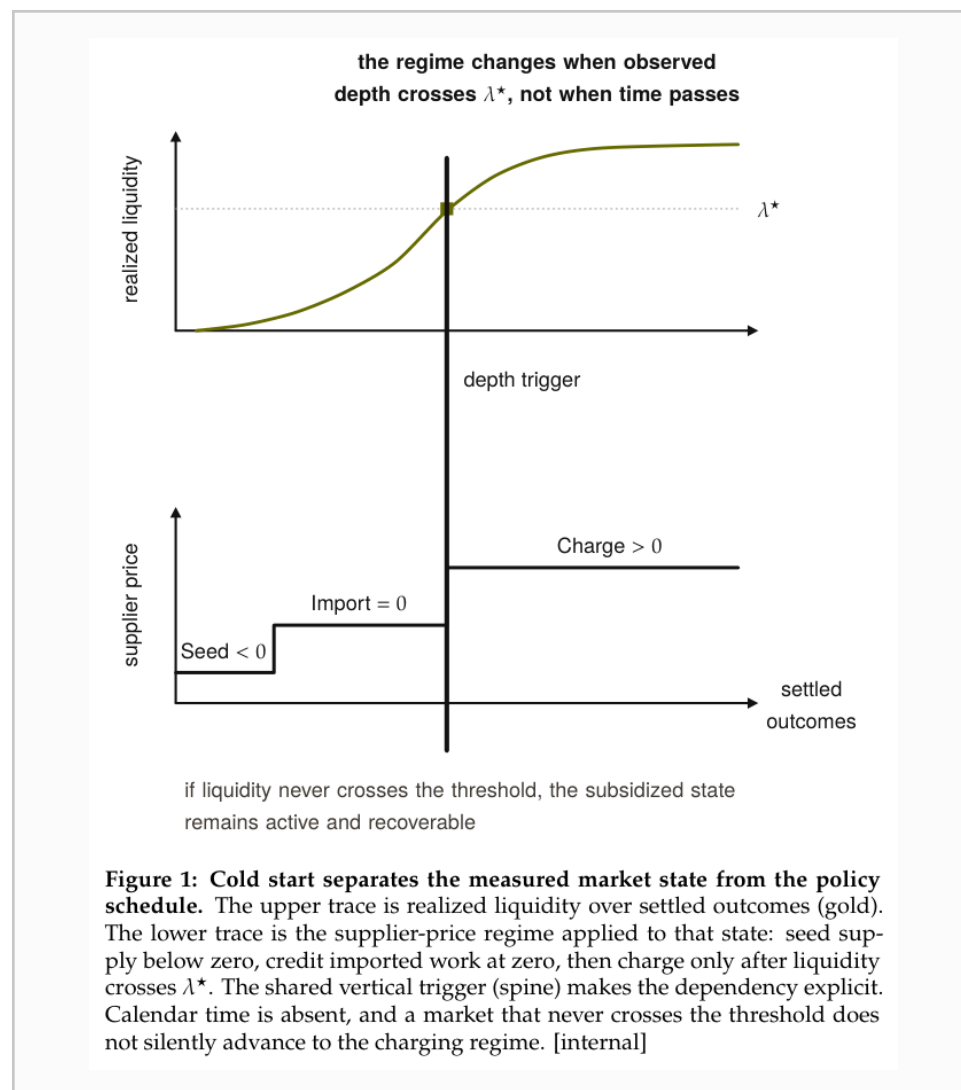
BEFORE



QA: FLAGGED: too wide 5.45in

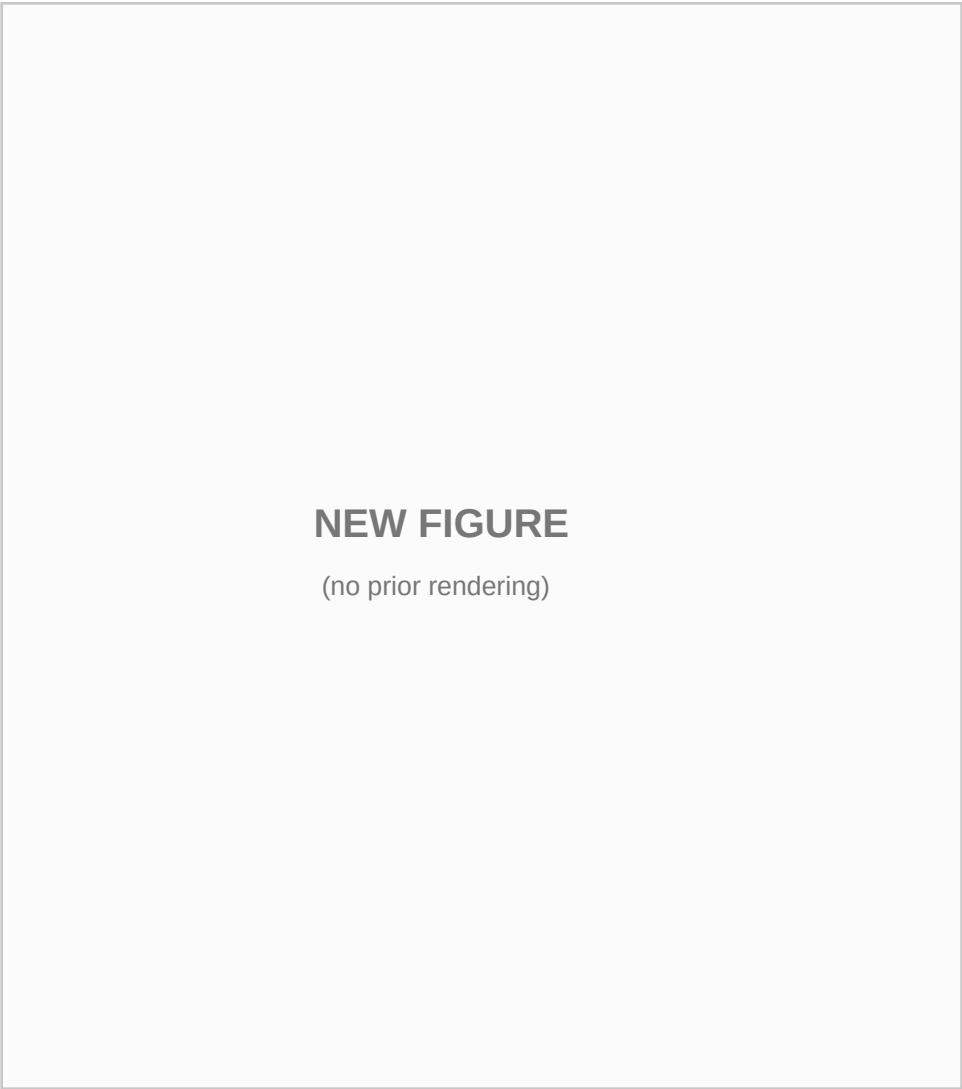
Cold start separates the measured market state from the policy schedule. The upper trace is realized liquidity over settled outcomes (gold). The lower trace is the supplier-price regime applied to that state: seed supply below zero, credit imported work at zero, then charge only after liquidity crosses λ^* . The shared vertical trigger (spine) makes the dependency explicit. Calendar time is absent, and a market tha...

AFTER

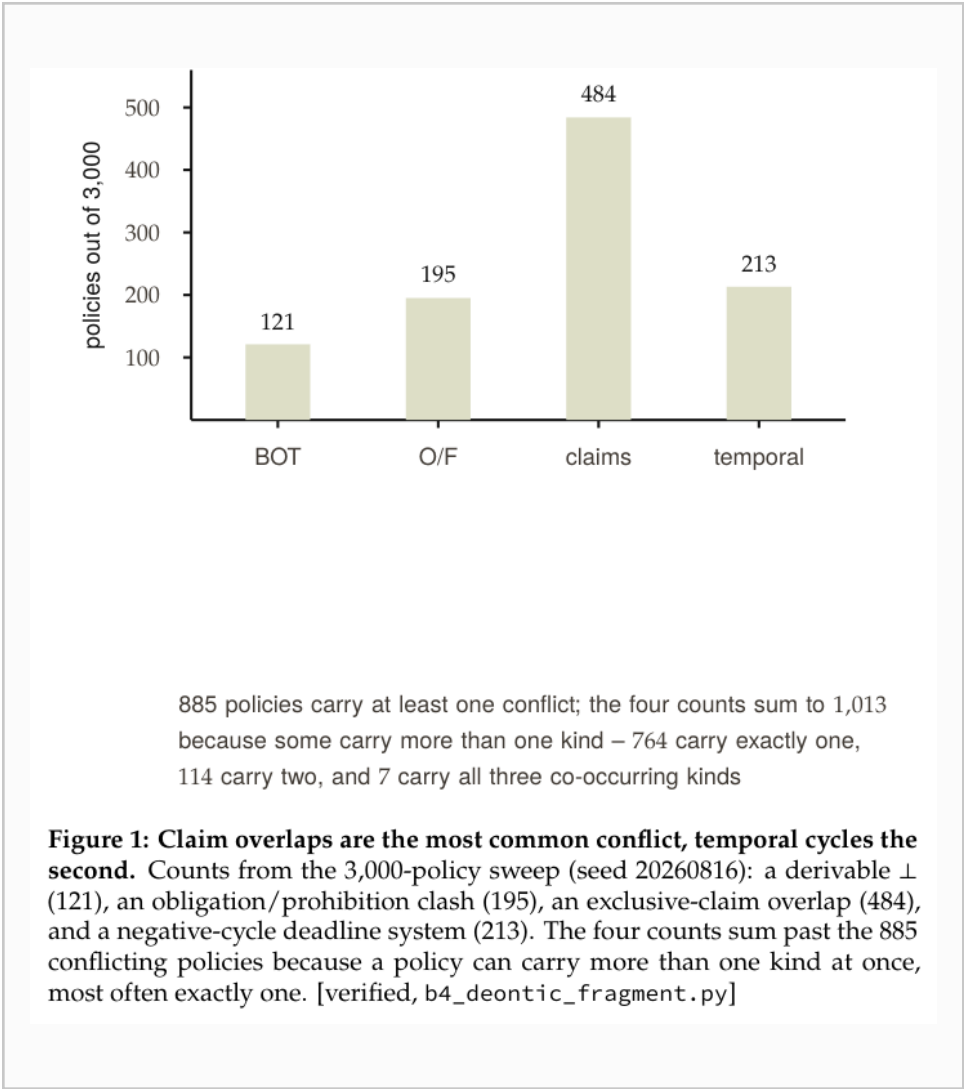


QA: PASS

BEFORE



AFTER

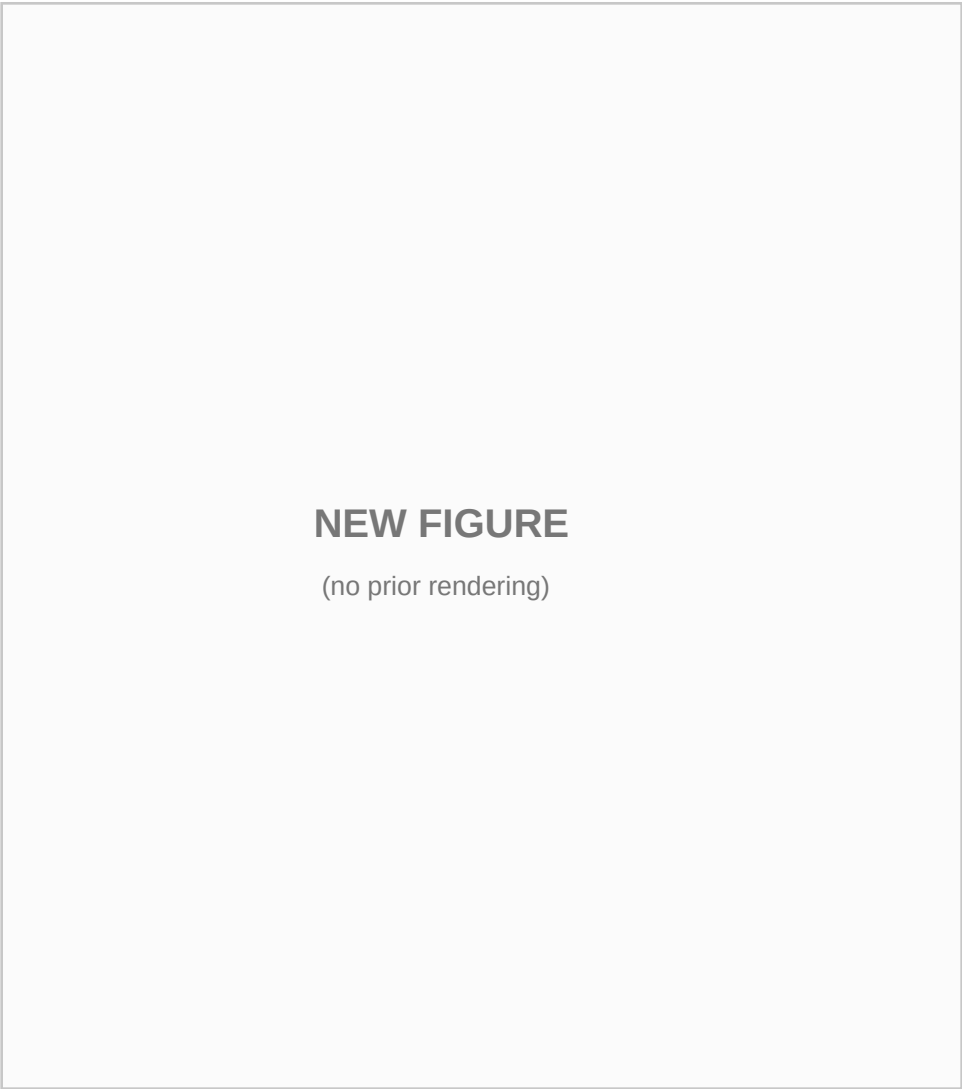


QA: —

QA: PASS

Claim overlaps are the most common conflict, temporal cycles the second. Counts from the 3,000-policy sweep (seed 20260816): a derivable $\perp$ (121), an obligation/prohibition clash (195), an exclusive-claim overlap (484), and a negative-cycle deadline system (213). The four counts sum past the 885 conflicting policies because a policy can carry more than one kind at once, most often exactly one. [verified, b4_deontic_fragment.py]

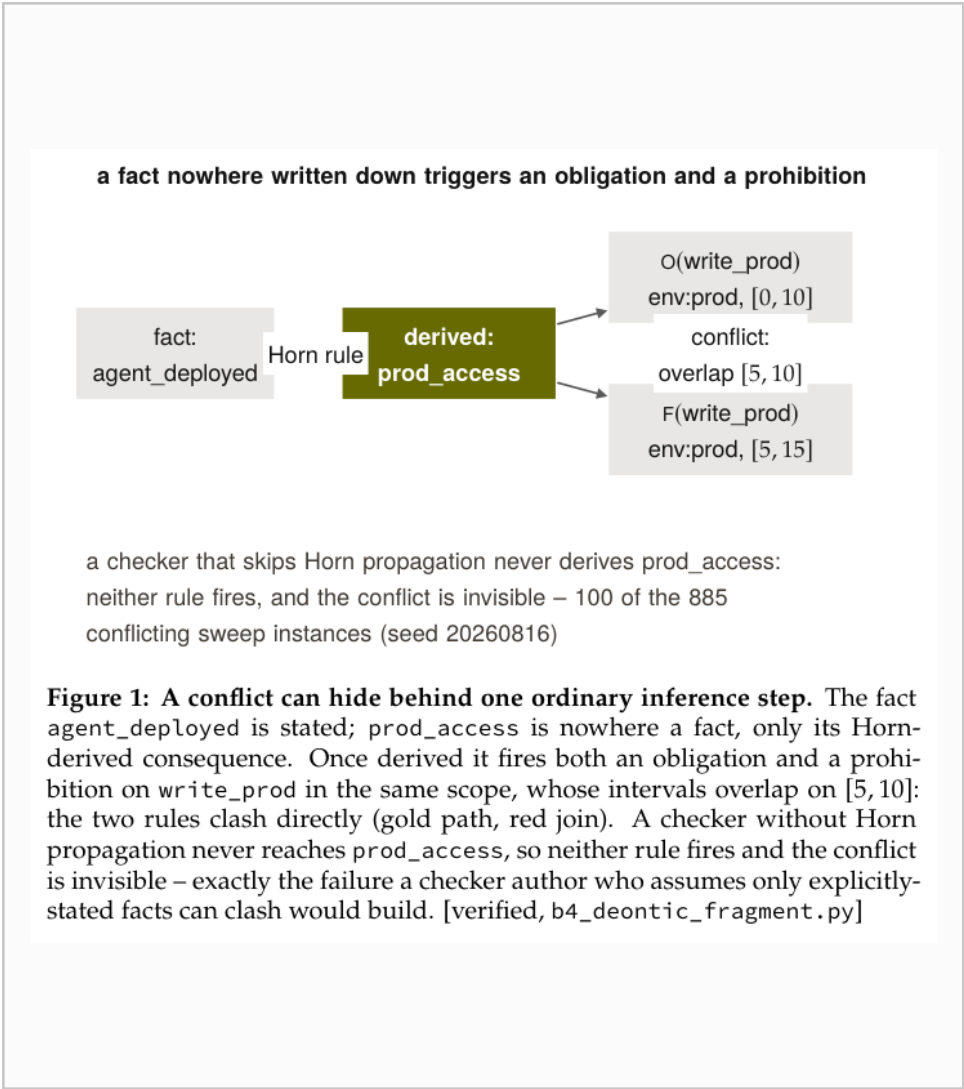
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QA: —

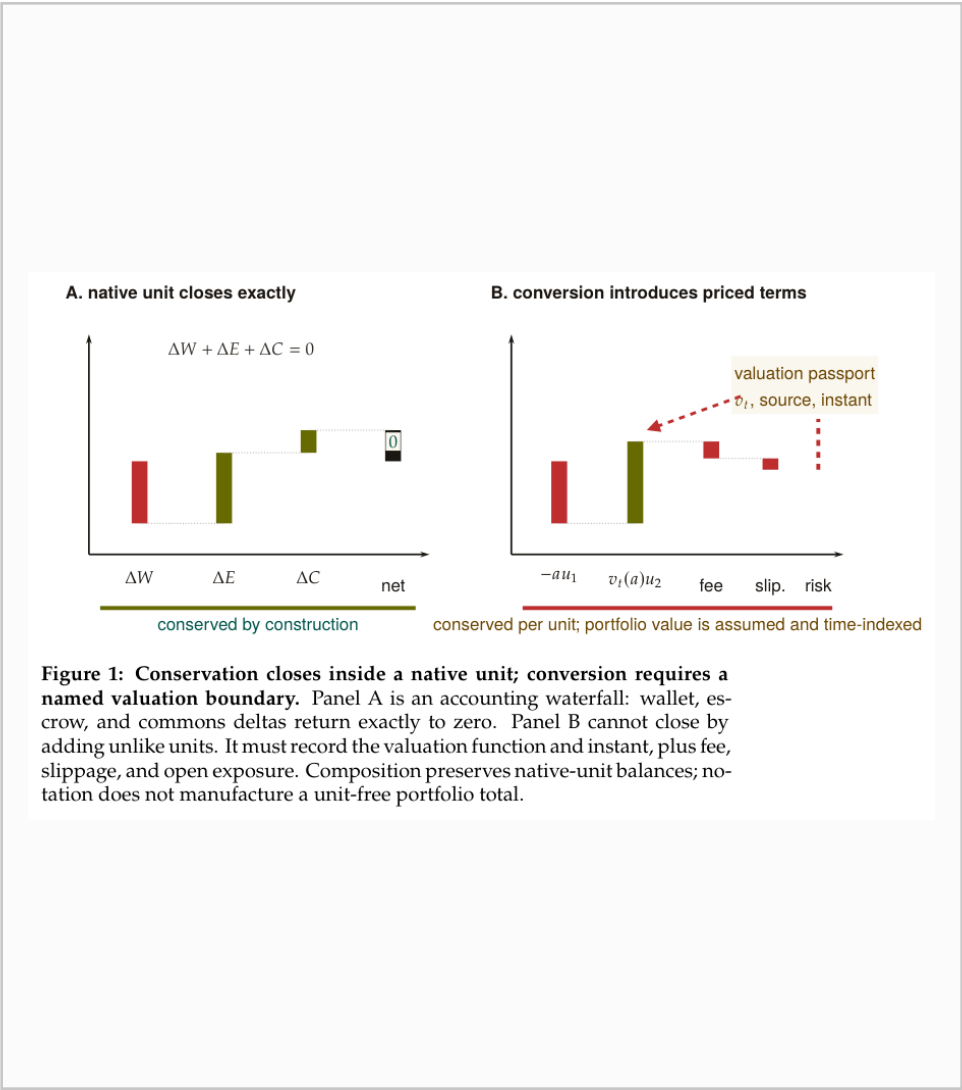
A conflict can hide behind one ordinary inference step. The fact {agent_deployed is stated; {prod_access is nowhere a fact, only its Horn-derived consequence. Once derived it fires both an obligation and a prohibition on {write_prod in the same scope, whose intervals overlap on \$[5,10]\$: the two rules clash directly (gold path, red join). A checker without Horn propagation never reaches {prod_access, so neither rule fires and the conflict is invisible – exactly the failure a checker author who assumes only explicitly-stated facts can clash would build. [verified, b4_deontic_fragment.py]

AFTER



QA: PASS

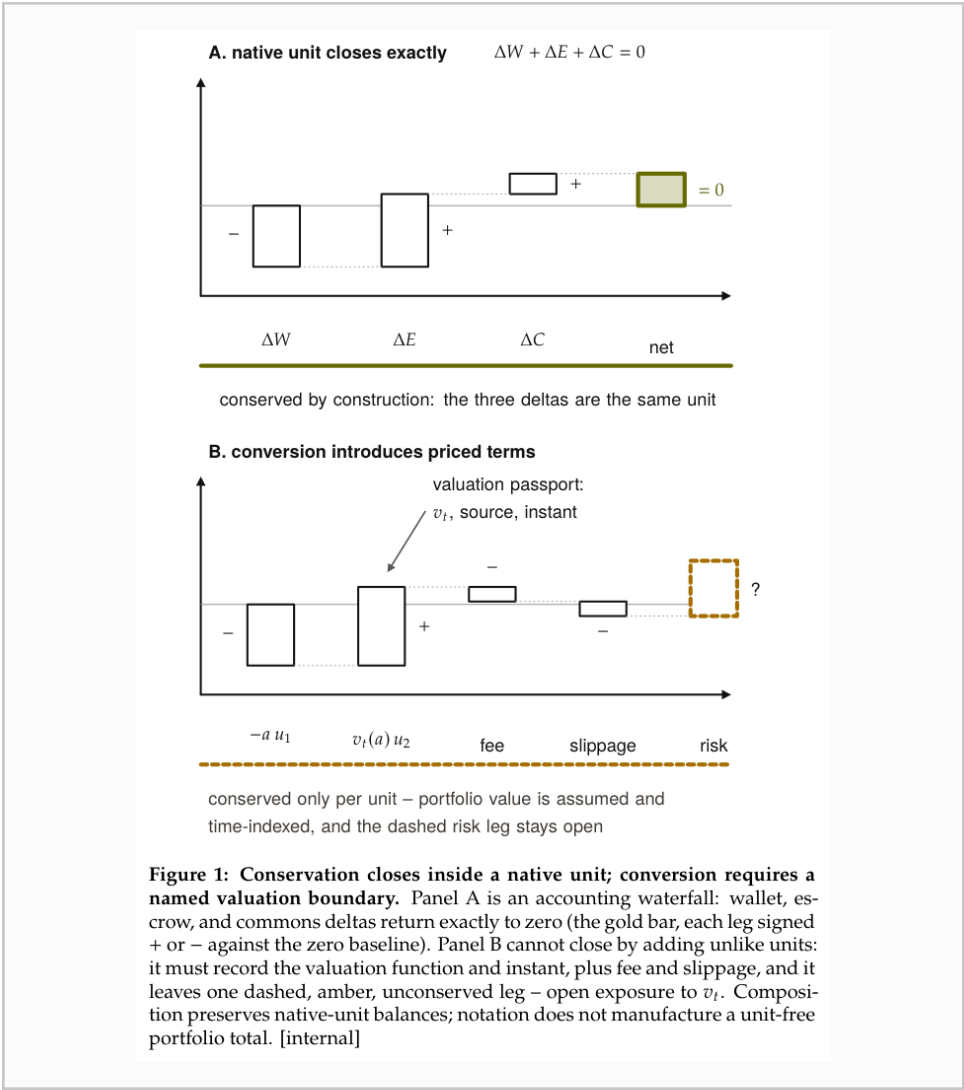
BEFORE



QA: FLAGGED: T4, too wide 5.61in

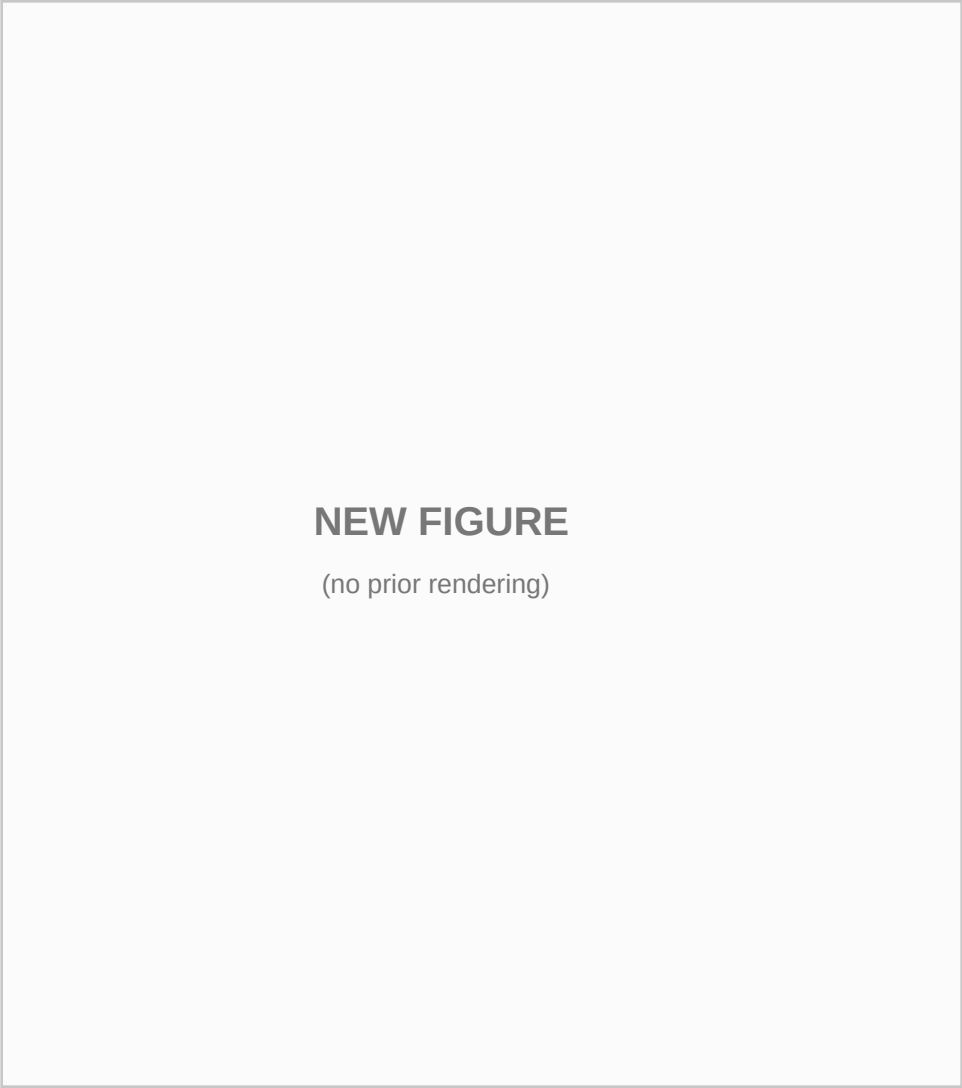
Conservation closes inside a native unit; conversion requires a named valuation boundary. Panel A is an accounting waterfall: wallet, escrow, and commons deltas return exactly to zero (the gold bar, each leg signed \$+\$ or \$-\$ against the zero baseline). Panel B cannot close by adding unlike units: it must record the valuation function and instant, plus fee and slippage, and it leaves one dashed, amber, unconserved...

AFTER



QA: PASS

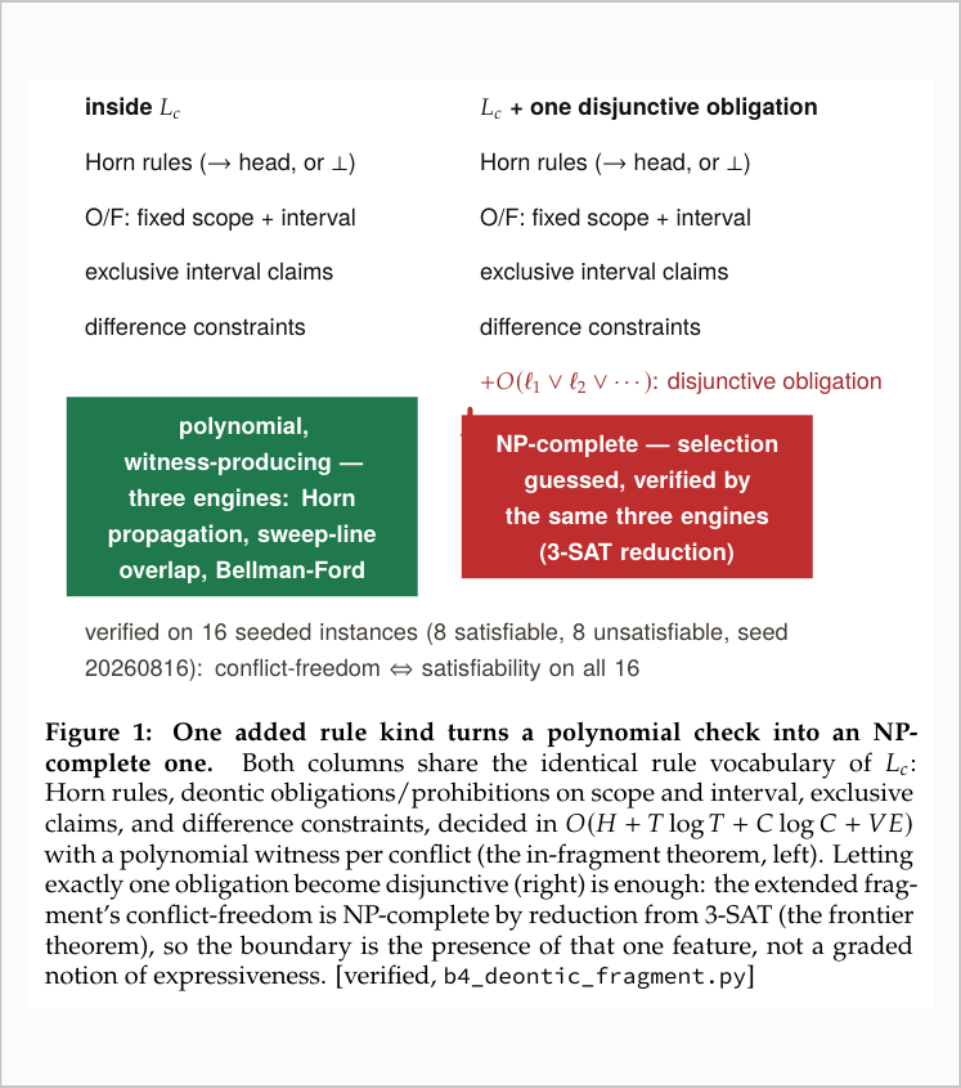
BEFORE



QA: —

One added rule kind turns a polynomial check into an NP-complete one. Both columns share the identical rule vocabulary of L_c : Horn rules, deontic obligations/prohibitions on scope and interval, exclusive claims, and difference constraints, decided in $O(H + T \log T + C \log C + VE)$ with a polynomial witness per conflict (the in-fragment theorem, left). Letting exactly one obligation become disjunctive (right) is enough: the extended fragment's conflict-freedom is NP-complete by reduction from 3-SAT (the frontier theorem), so the boundary is the presence of that one feature, not a graded notion of expressiveness. [verified, b4_deontic_fragment.py]

AFTER



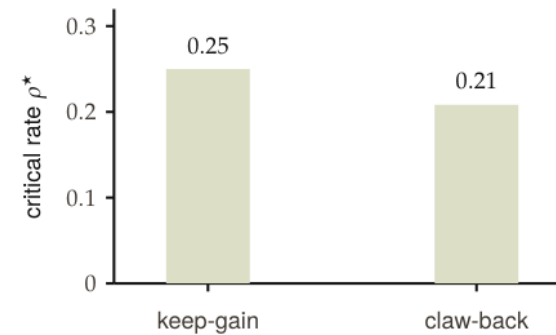
QA: PASS

BEFORE

NEW FIGURE

(no prior rendering)

AFTER



same $G = 10$, $d = 0.8$, $B = 50$ in both bars; only the forfeiture on a caught cheat changes, B versus $G + B$

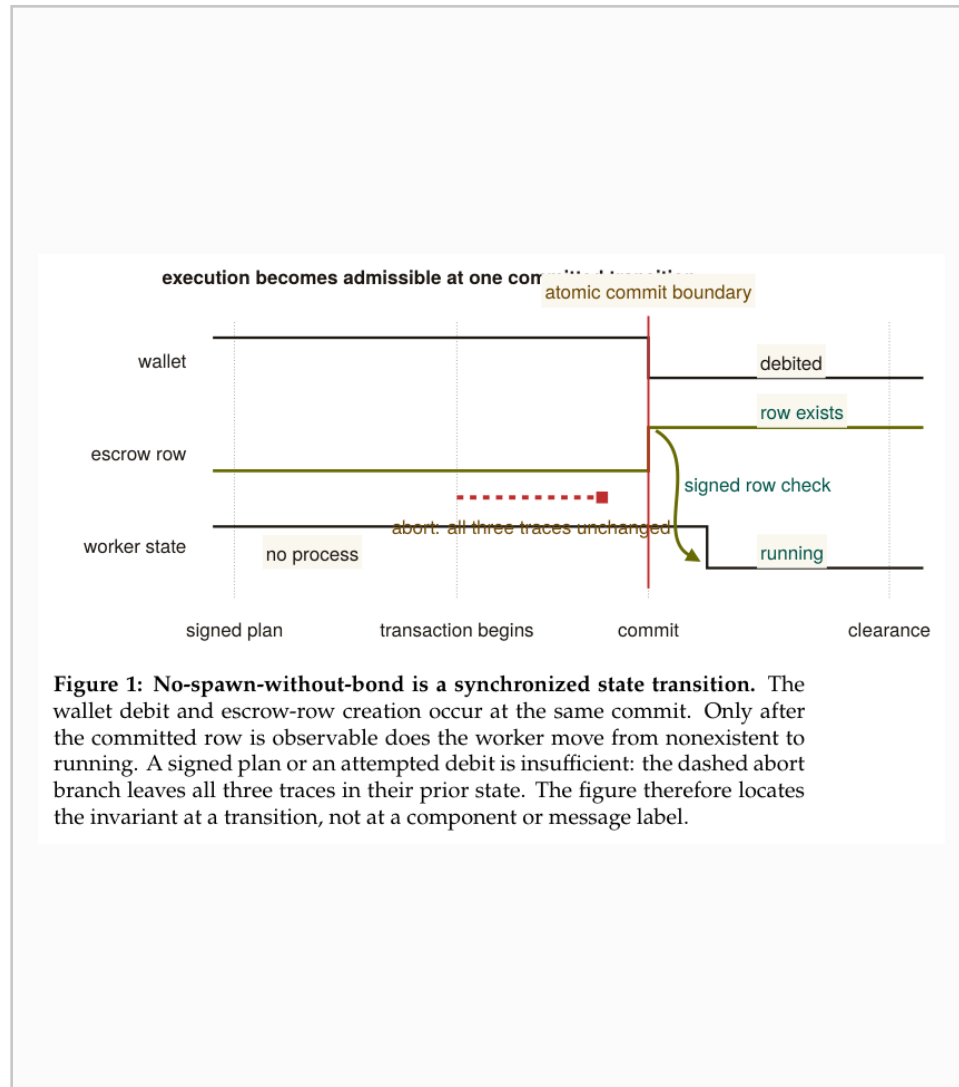
Figure 1: Quoting the wrong convention misprices deterrence by about 17%. Keep-gain forfeits only the bond B on a caught cheat: critical rate $\rho^* = G/(dB) = 0.25$. Claw-back also confiscates the gain G , dropping the critical rate to $G/(d(G+B)) = 0.2083$. A platform that claws back but budgets audits at the keep-gain rate over-audits; one that does not claw back but budgets at the claw-back rate under-deters. [internal; harbor3]

QA: —

QA: PASS

Quoting the wrong convention misprices deterrence by about 17%. Keep-gain forfeits only the bond B on a caught cheat: critical rate $\rho^* = G/(dB) = 0.25$. Claw-back also confiscates the gain G , dropping the critical rate to $G/(d(G+B)) = 0.2083$. A platform that claws back but budgets audits at the keep-gain rate over-audits; one that does not claw back but budgets at the claw-back rate under-deters. [internal; harbor3]

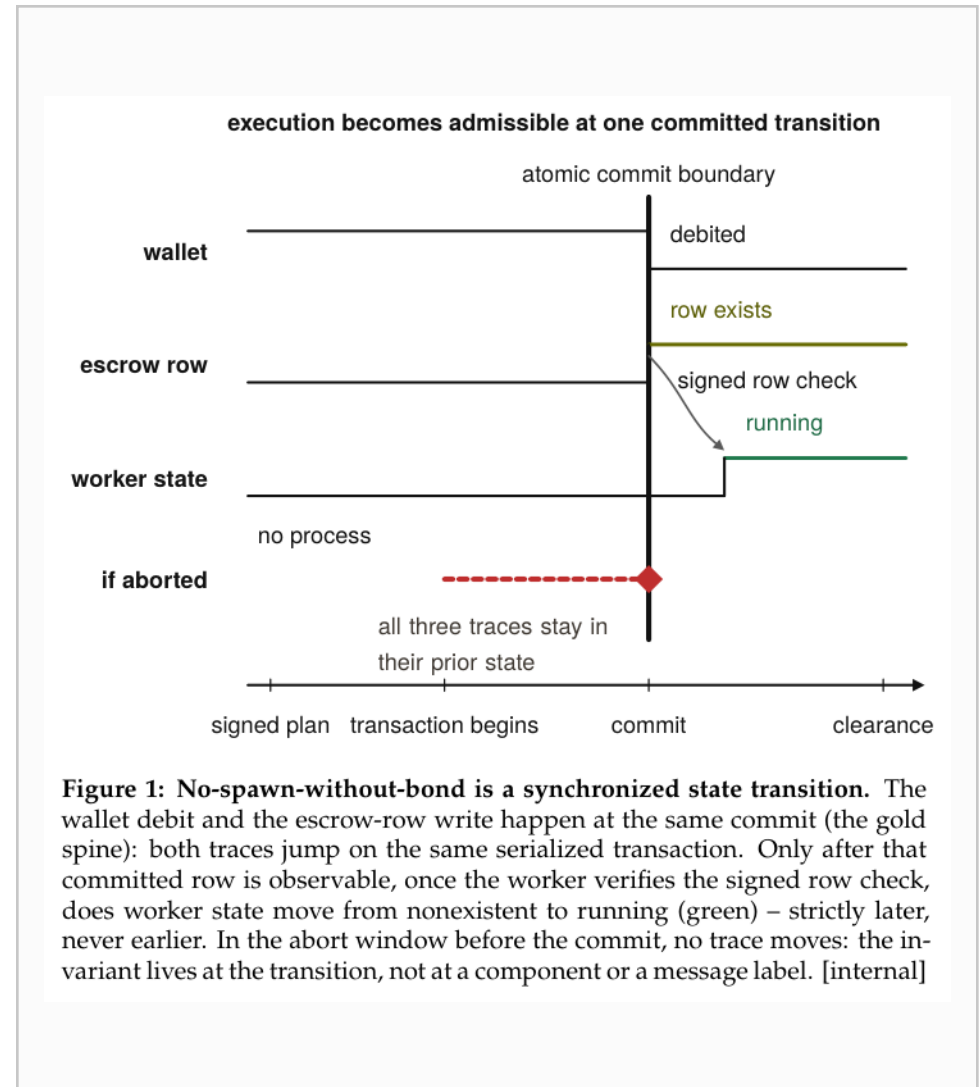
BEFORE



QA: FLAGGED: T4, B3, too wide 5.07in

No-spawn-without-bond is a synchronized state transition. The wallet debit and the escrow-row write happen at the same commit (the gold spine): both traces jump on the same serialized transaction. Only after that committed row is observable, once the worker verifies the signed row check, does worker state move from nonexistent to running (green) -- strictly later, never earlier. In the abort window before the comm...

AFTER



QA: PASS

BEFORE

NEW FIGURE

(no prior rendering)

AFTER

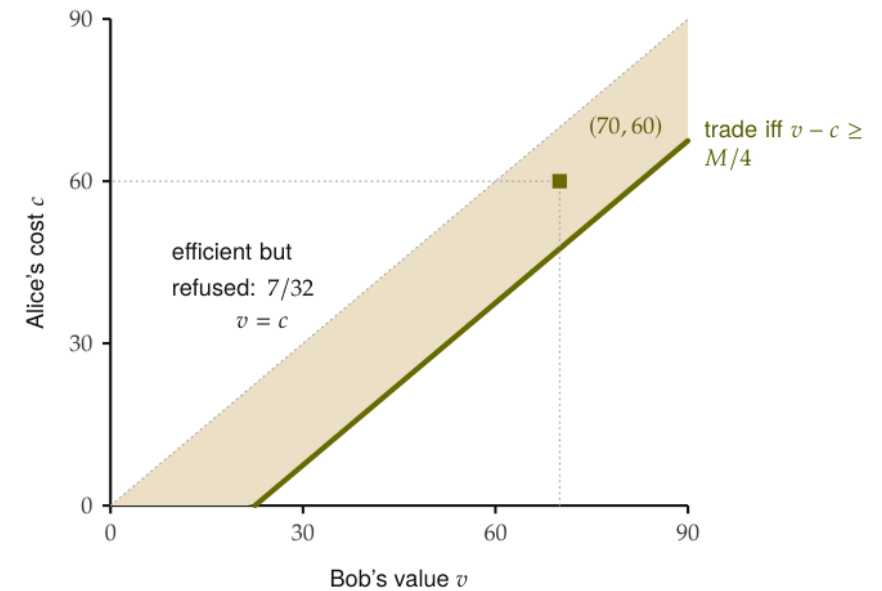


Figure 1: The double auction refuses more than one efficient trade in five. Above the diagonal $v = c$ every draw has a real surplus; the mechanism only trades it when $v - c \geq M/4 = 22.5$ (gold line). The band between the two lines is efficient but refused, $7/32 \approx 21.9\%$ of the square, the same fraction at every M . At the chapter's own worked point, Bob values the rental at 70 and Alice's cost is 60: a 10-cr surplus sits on the table, but $10 < 22.5$, so no trade happens. [internal; algebra shown]

QA: —

QA: PASS

The double auction refuses more than one efficient trade in five. Above the diagonal $v=c$ every draw has a real surplus; the mechanism only trades it when $v-c \geq M/4=22.5$ (gold line). The band between the two lines is efficient but refused, $7/32 \approx 21.9\%$ of the square, the same fraction at every M . At the chapter's own worked point, Bob values the rental at 70 and Alice's cost is 60: a 10-cr surplus sits on...

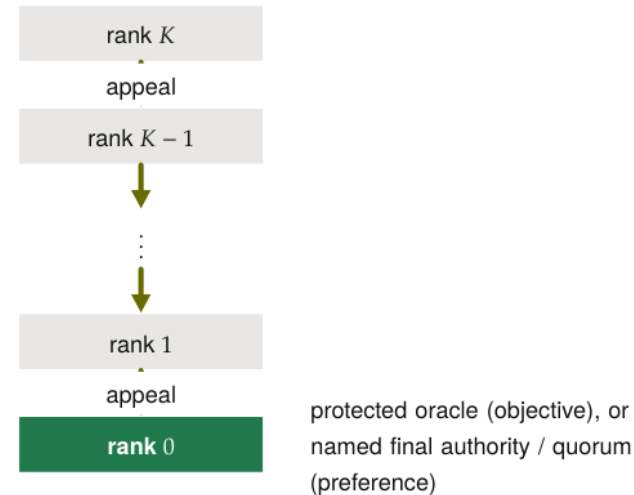
BEFORE

NEW FIGURE

(no prior rendering)

AFTER

appeal moves down one rank at a time, and stops at rank 0



ranks are natural numbers, so no infinite ascending chain
of judges judging judges is possible – this is a well-founded
order, not a convergence proof

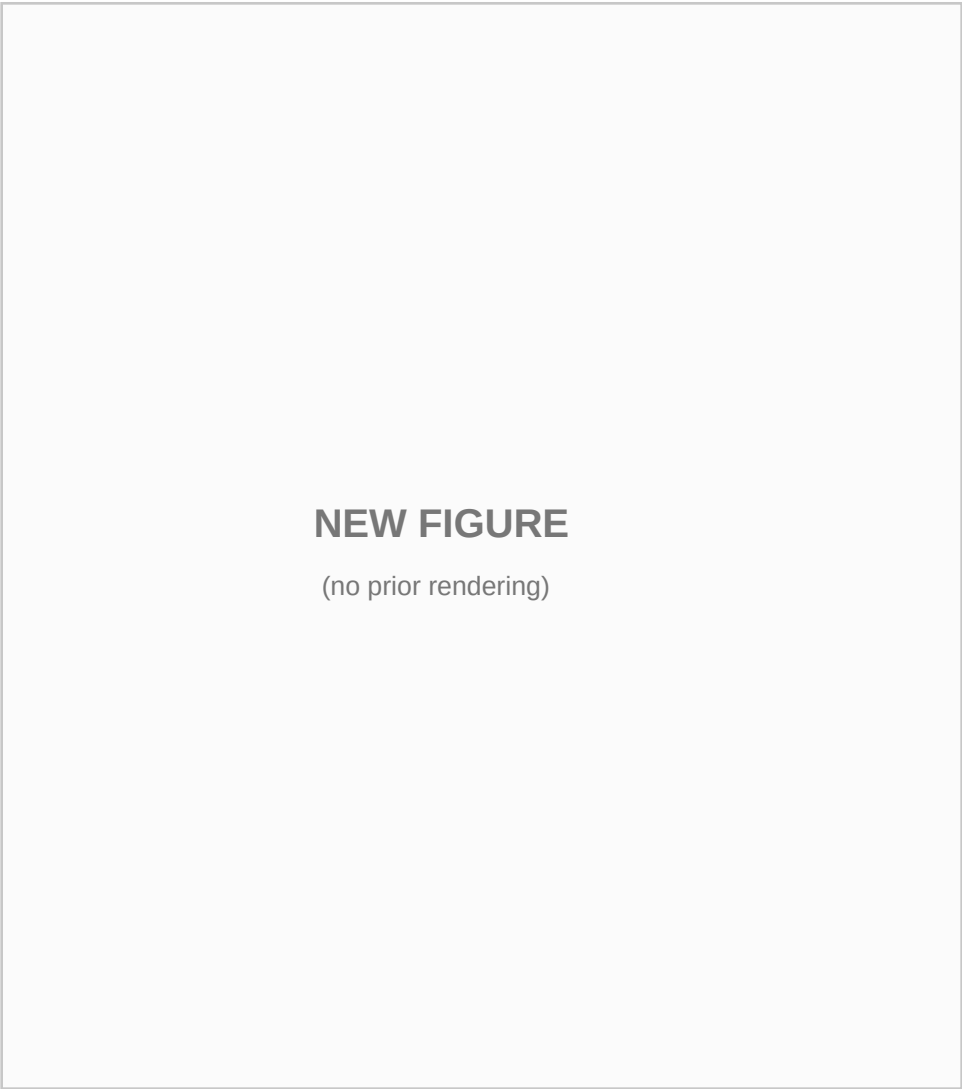
Figure 1: Appeal is a well-founded descent, not an unbounded regress. An appeal from rank K reaches only rank $K-1$, never a new rank above K , and the chain stops at rank 0: a protected oracle for an objective predicate, or a named final authority or quorum for a matter of preference. No theorem shows subjective judging bottoms out in machine-checkable truth; what grounds the recursion is that ranks are natural numbers, so the descent cannot continue forever, whatever rank 0 turns out to mean. [internal; hand-worked, no script]

QA: —

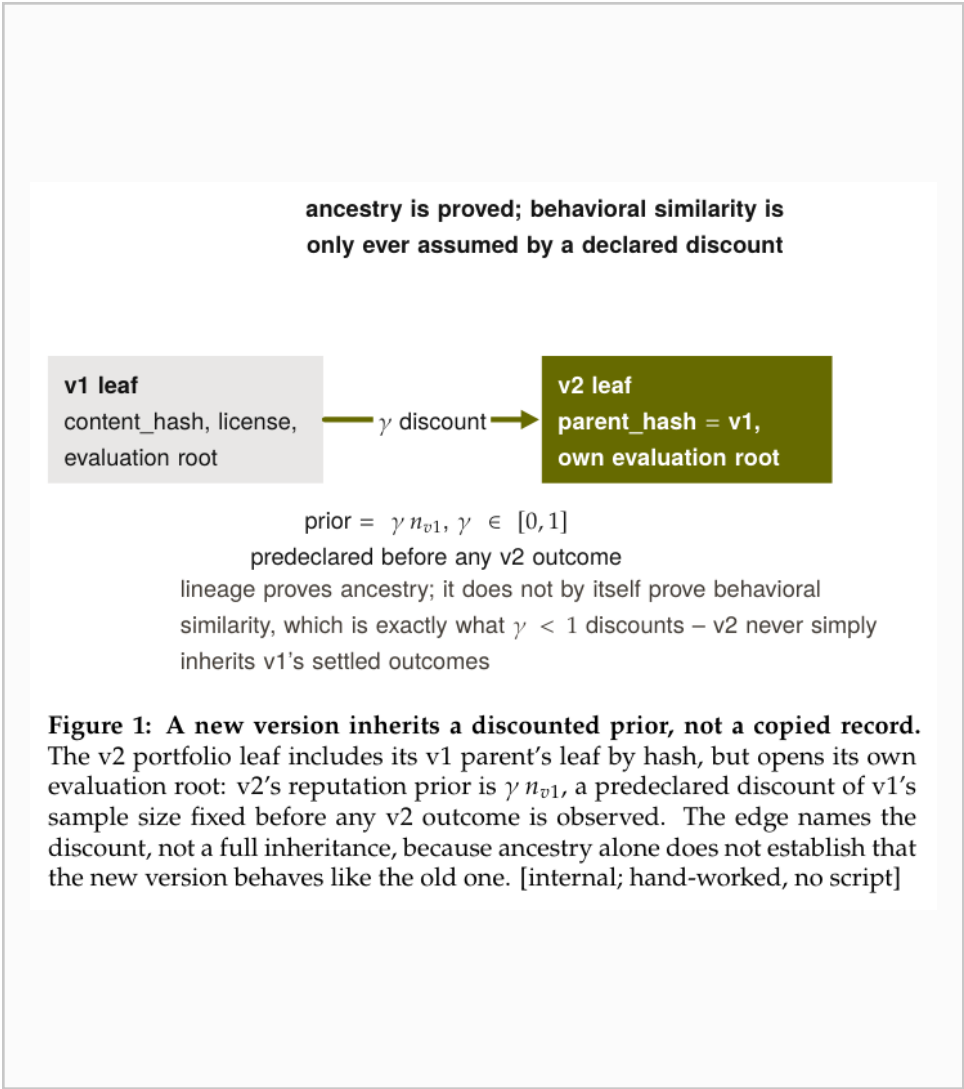
QA: PASS

Appeal is a well-founded descent, not an unbounded regress. An appeal from rank K reaches only rank $K-1$, never a new rank above K , and the chain stops at rank 0: a protected oracle for an objective predicate, or a named final authority or quorum for a matter of preference. No theorem shows subjective judging bottoms out in machine-checkable truth; what grounds the recursion is that ranks are natural number...

BEFORE



AFTER



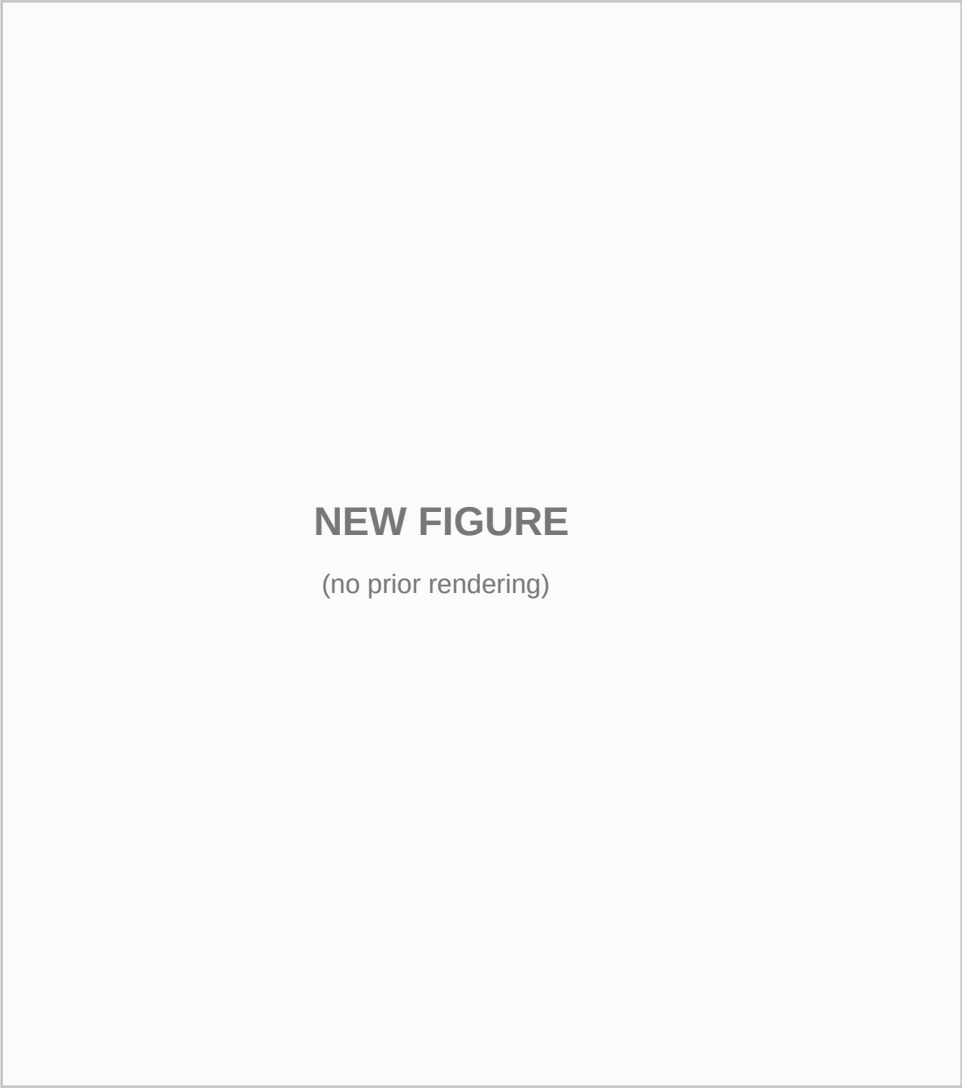
QA: —

QA: PASS

A new version inherits a discounted prior, not a copied record. The v2 portfolio leaf includes its v1 parent's leaf by hash, but opens its own evaluation root: v2's reputation prior is γn_{v1} , a predeclared discount of v1's sample size fixed before any v2 outcome is observed. The edge names the discount, not a full inheritance, because ancestry alone does not establish that the new version behaves like the old one. [internal; hand-worked, no script]

...

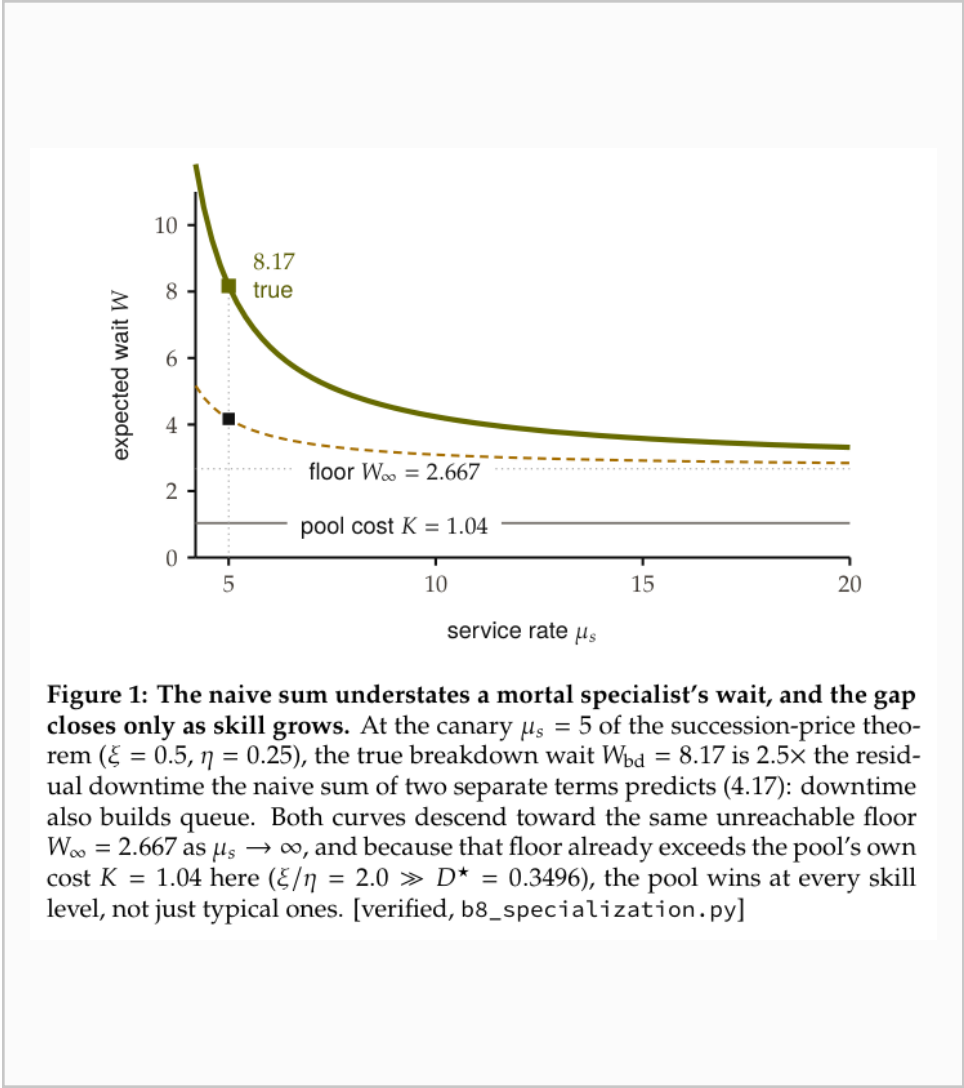
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QA: —

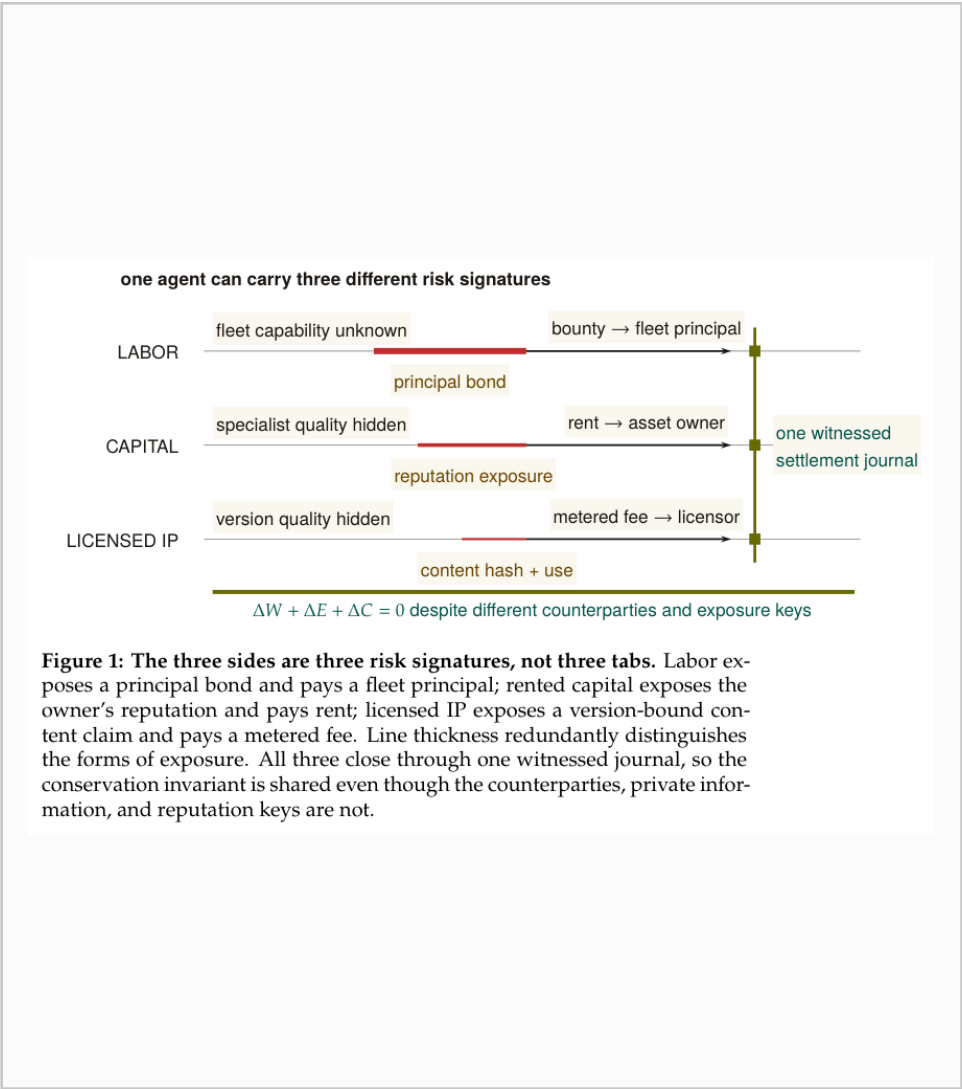
The naive sum understates a mortal specialist's wait, and the gap closes only as skill grows. At the canary $\mu_s=5$ of the succession-price theorem ($\xi=0.5, \eta=0.25$), the true breakdown wait $W_{bd}=8.17$ is $2.5\times$ the residual downtime the naive sum of two separate terms predicts (4.17): downtime also builds queue. Both curves descend toward the same unreachable floor $W_\infty=2.667$ as $\mu_s \rightarrow \infty$, and because that floor already exceeds the pool's own cost $K=1.04$ here ($\xi/\eta=2.0 \gg D^*=0.3496$), the pool wins at every skill level, not just typical ones. [verified, b8_specialization.py]

AFTER



QA: PASS

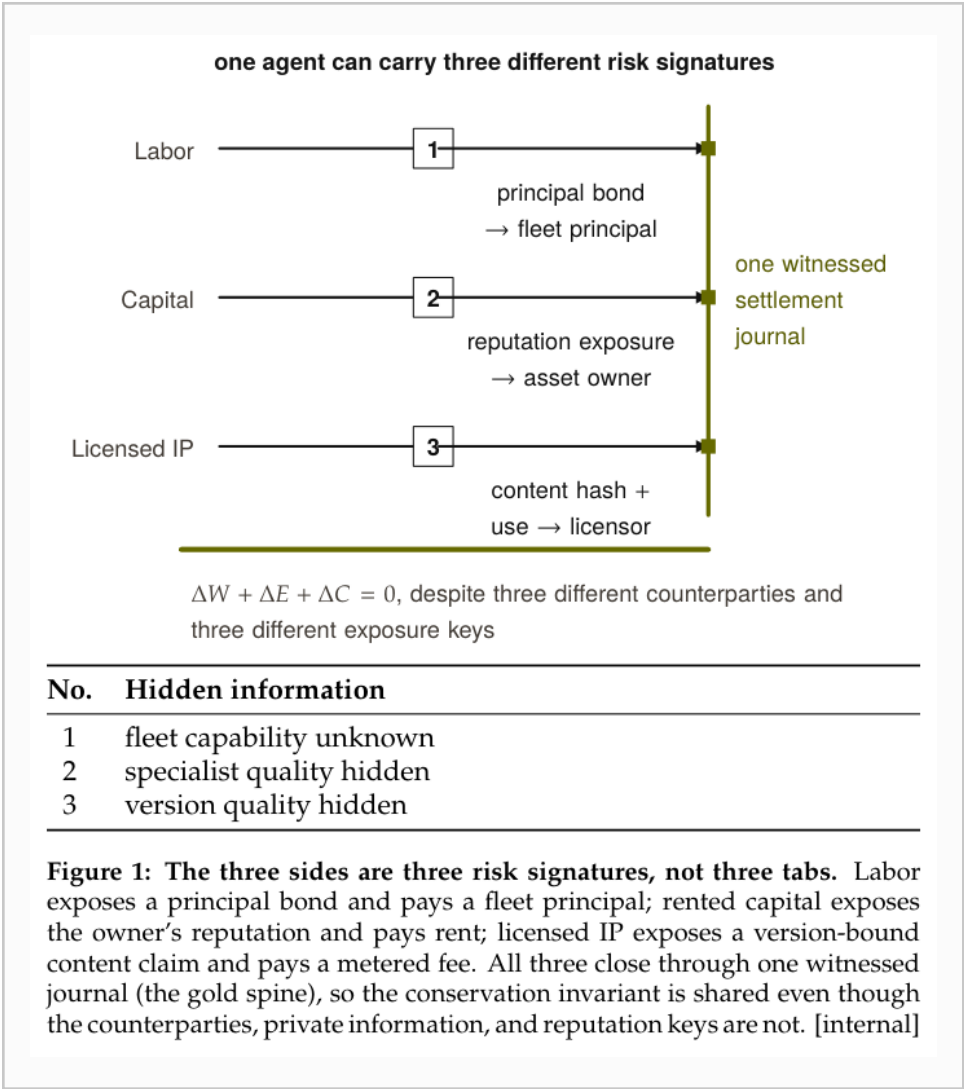
BEFORE



QA: FLAGGED: too wide 5.45in

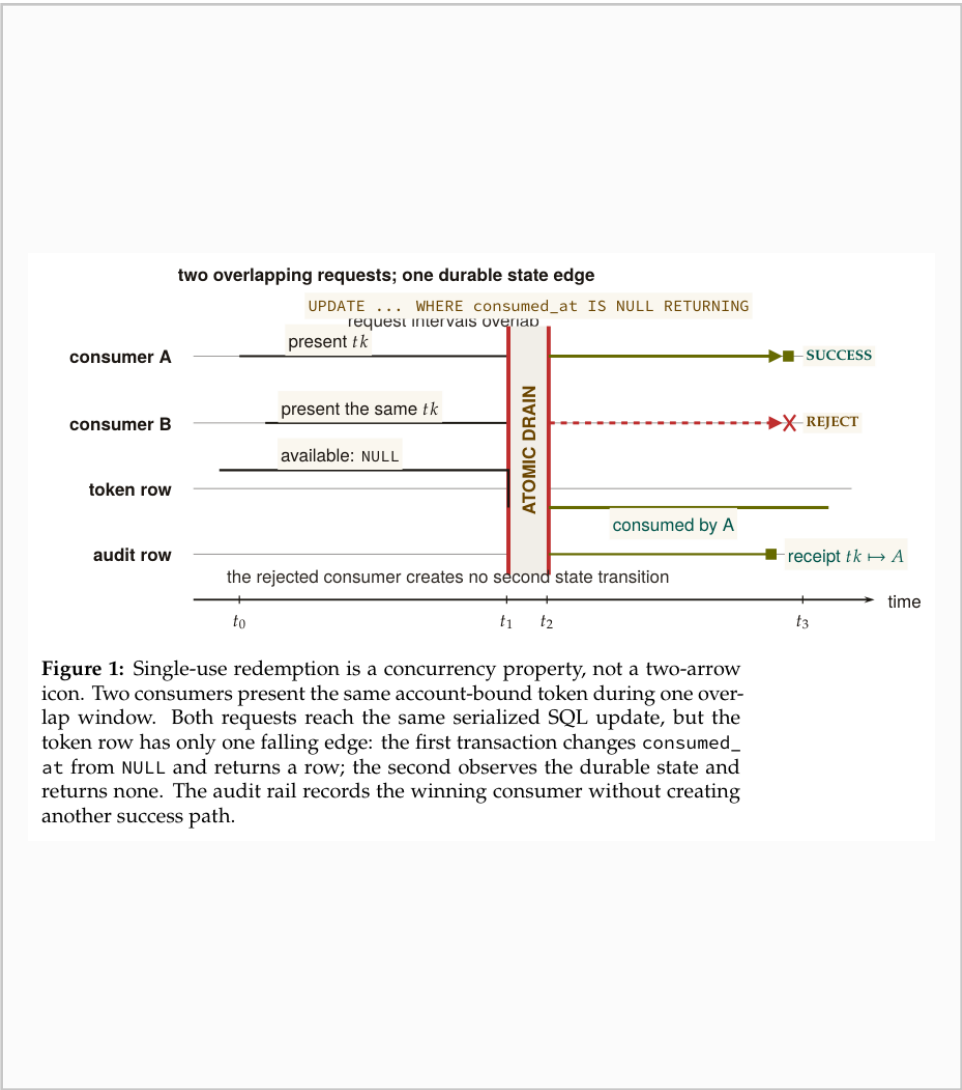
The three sides are three risk signatures, not three tabs. Labor exposes a principal bond and pays a fleet principal; rented capital exposes the owner's reputation and pays rent; licensed IP exposes a version-bound content claim and pays a metered fee. All three close through one witnessed journal (the gold spine), so the conservation invariant is shared even though the counterparties, private information, and rep...

AFTER



QA: PASS

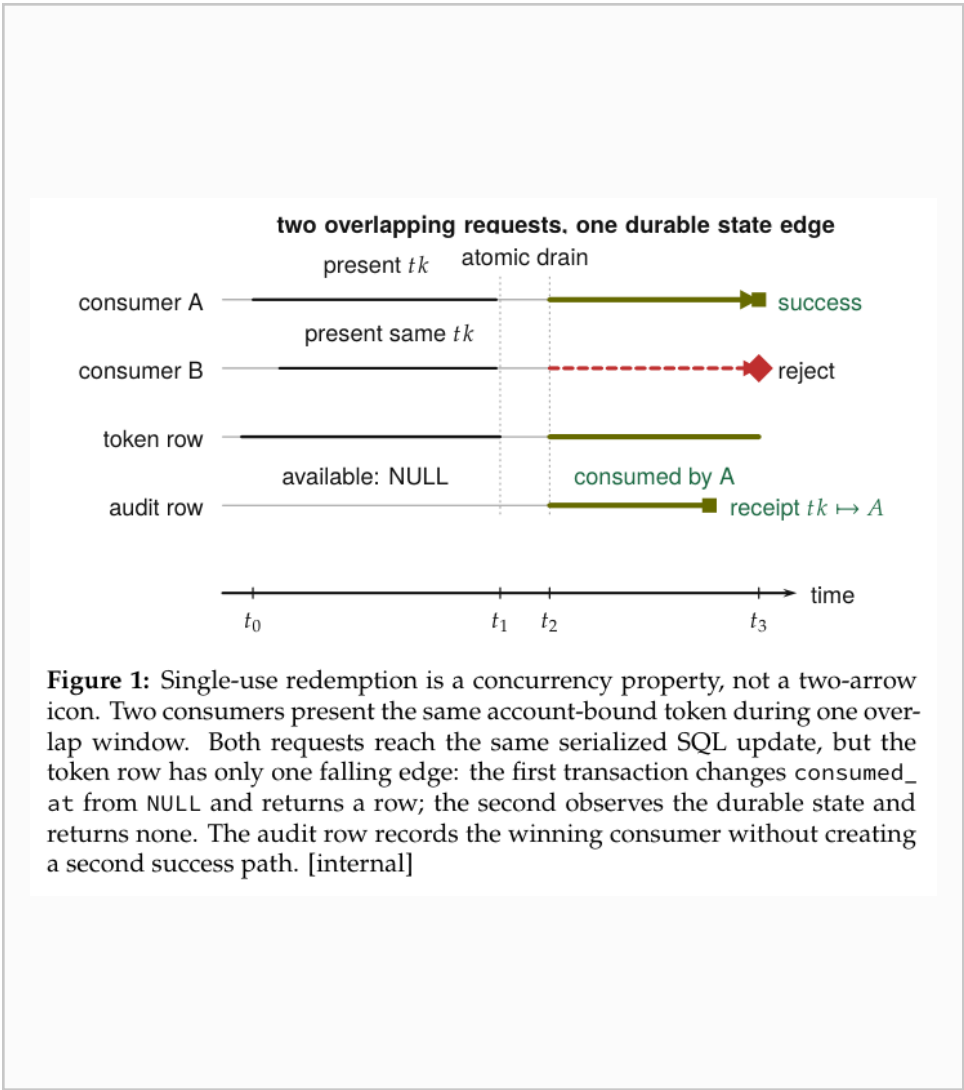
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QA: FLAGGED: T1, T4, B3, too wide 5.48in

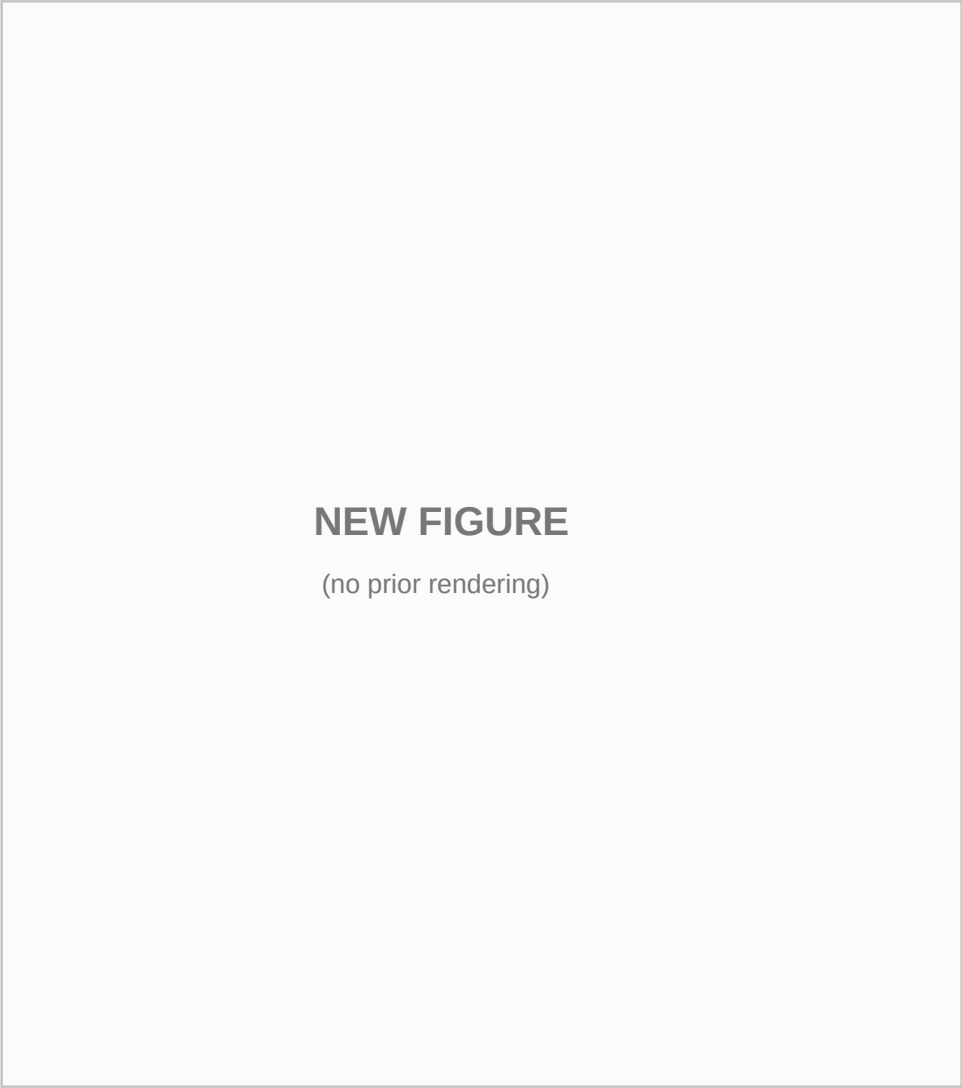
Single-use redemption is a concurrency property, not a two-arrow icon. Two consumers present the same account-bound token during one overlap window. Both requests reach the same serialized SQL update, but the token row has only one falling edge: the first transaction changes {consumed_at from {NULL and returns a row; the second observes the durable state and returns none. The audit row records the winning consume...

AFTER



QA: PASS

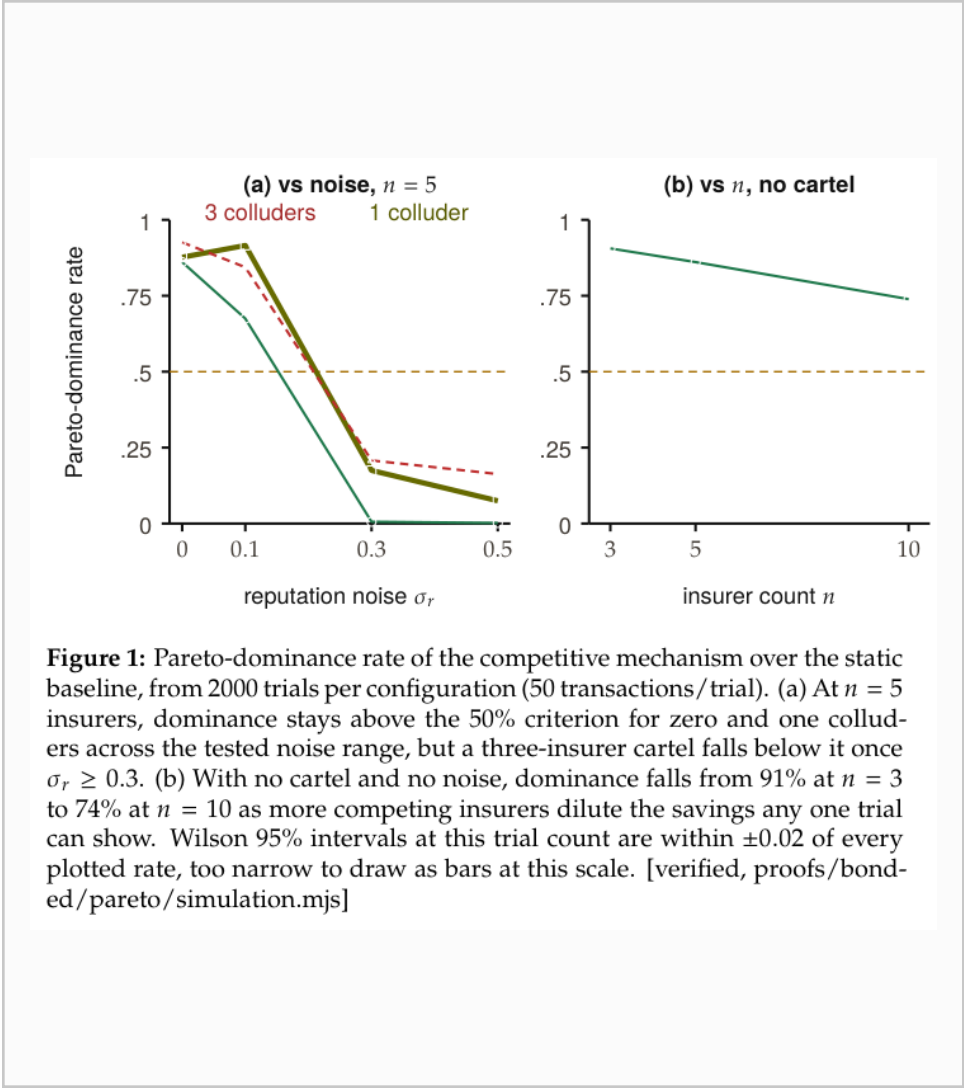
BEFORE



QA: —

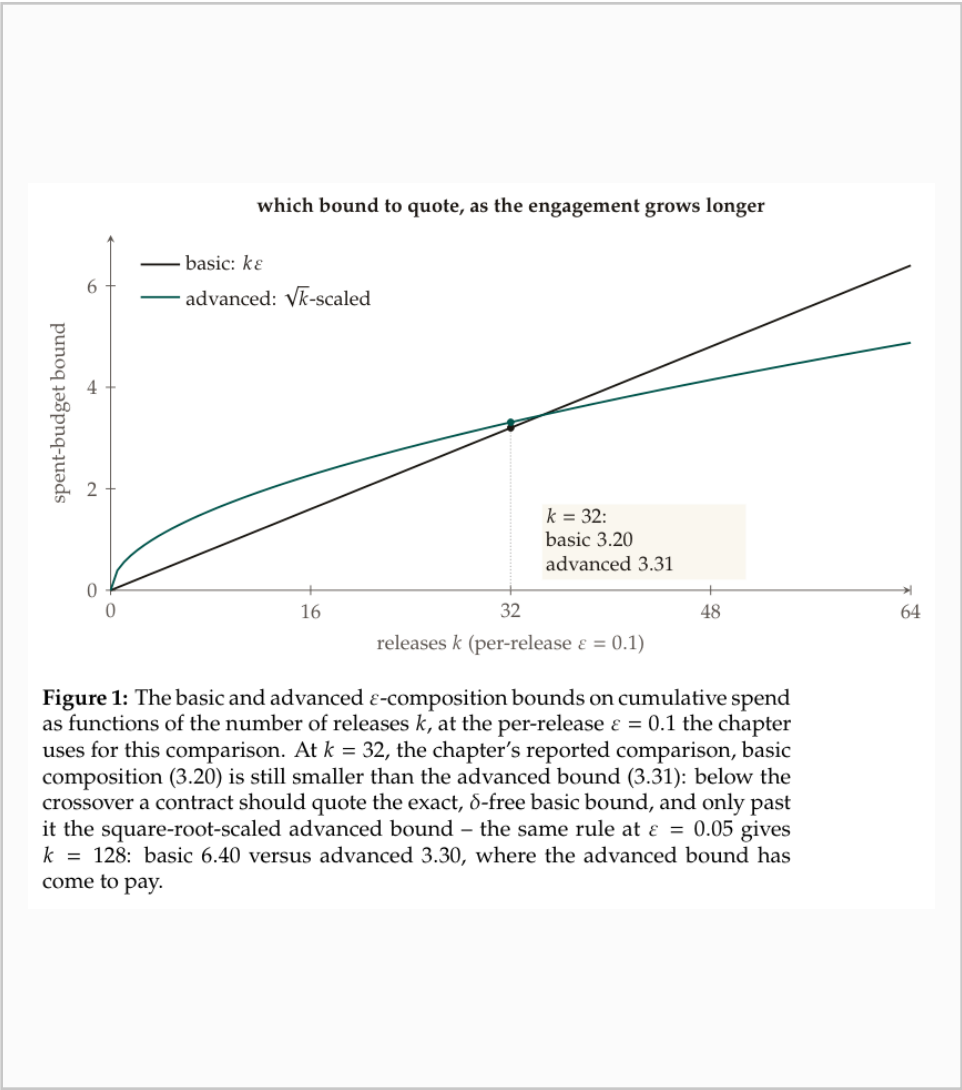
Pareto-dominance rate of the competitive mechanism over the static baseline, from 2000 trials per configuration (50 transactions/trial). (a) At $n=5$ insurers, dominance stays above the 50% criterion for zero and one colluders across the tested noise range, but a three-insurer cartel falls below it once $\sigma_r \geq 0.3$. (b) With no cartel and no noise, dominance falls from 91% at $n=3$ to 74% at $n=10$ as more competing insurers dilute the savings any one trial can show. Wilson 95% intervals at this trial count are within ± 0.02 of every plotted rate, too narrow to draw as bars at this scale. [verified, proofs/bonded/pareto/simulation.mjs]

AFTER



QA: PASS

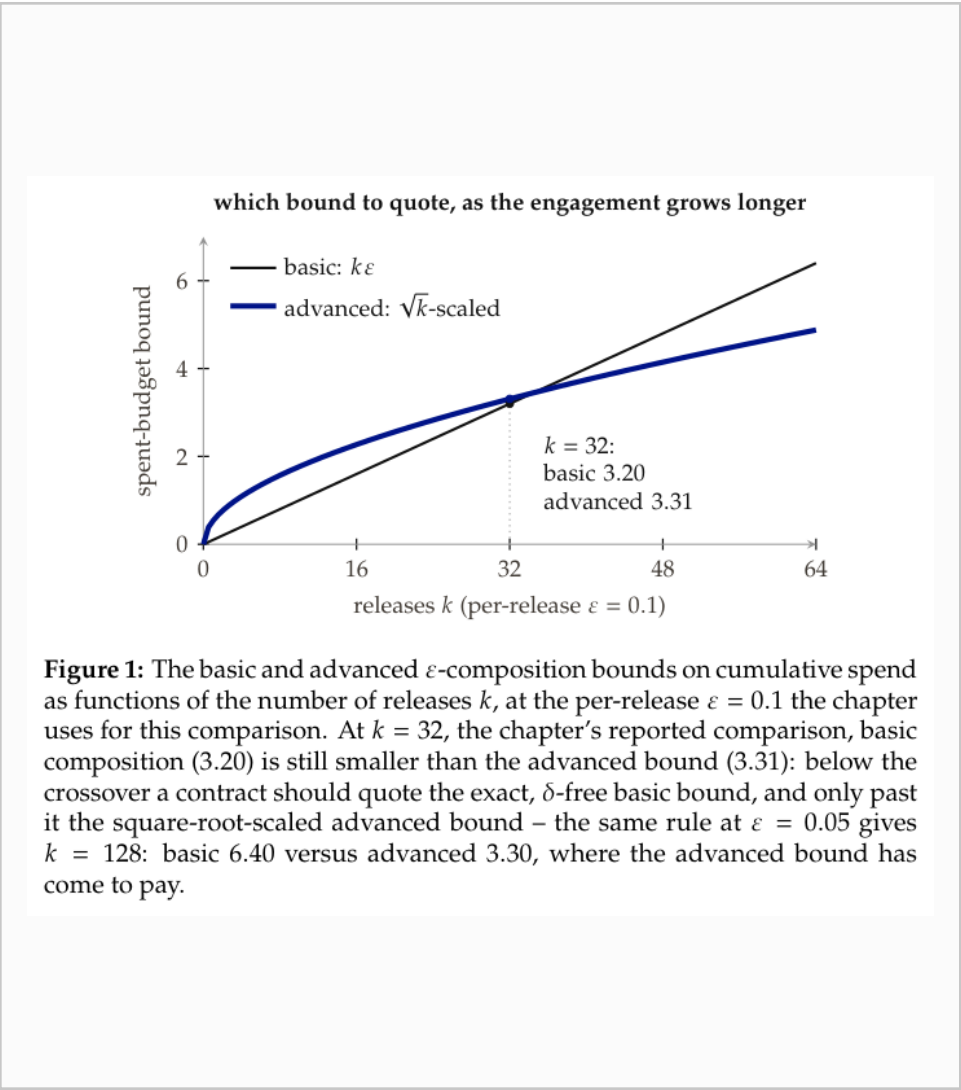
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QA: FLAGGED: too wide 5.23in

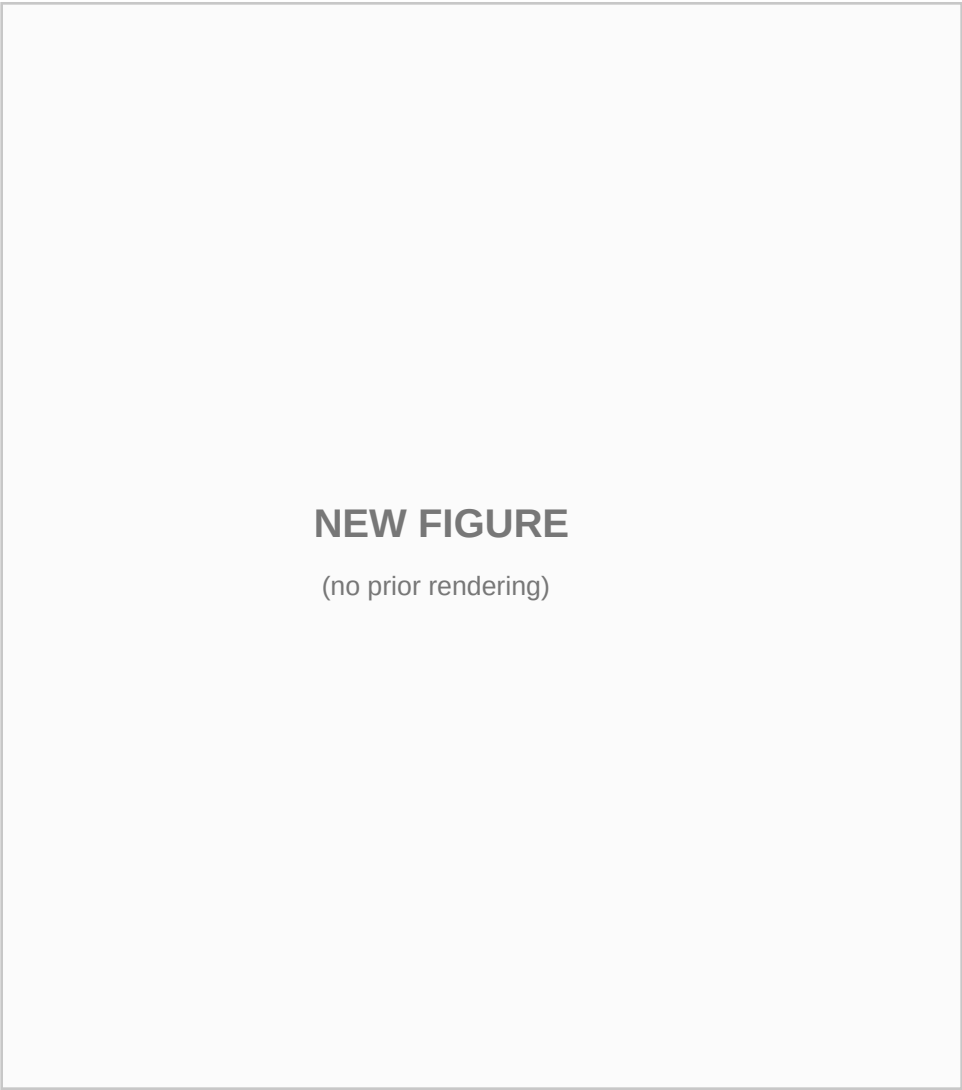
The basic and advanced ε -composition bounds on cumulative spend as functions of the number of releases k , at the per-release $\varepsilon = 0.1$ the chapter uses for this comparison. At $k = 32$, the chapter's reported comparison, basic composition (3.20) is still smaller than the advanced bound (3.31): below the crossover a contract should quote the exact, δ -free basic bound, and only past it the square-root-scaled adva...

AFTER



QA: PASS

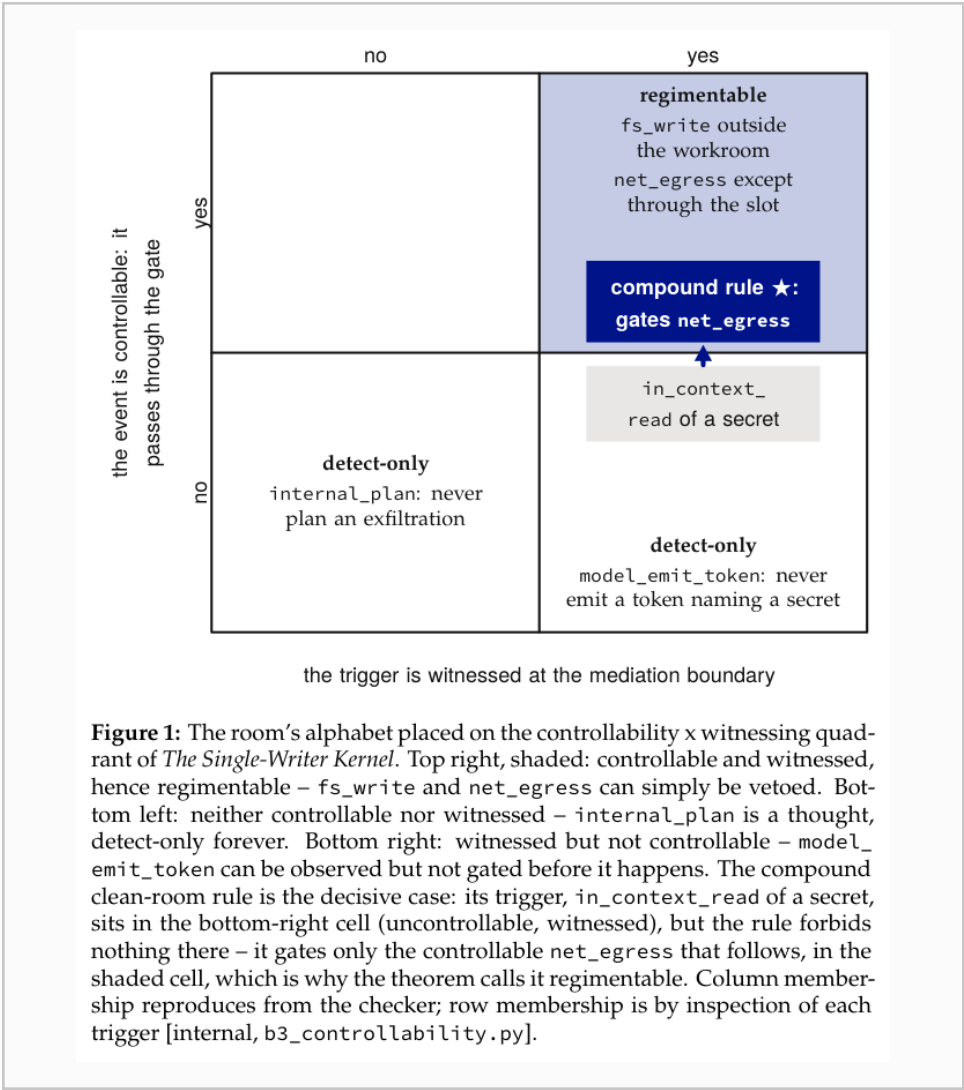
BEFORE



QA: —

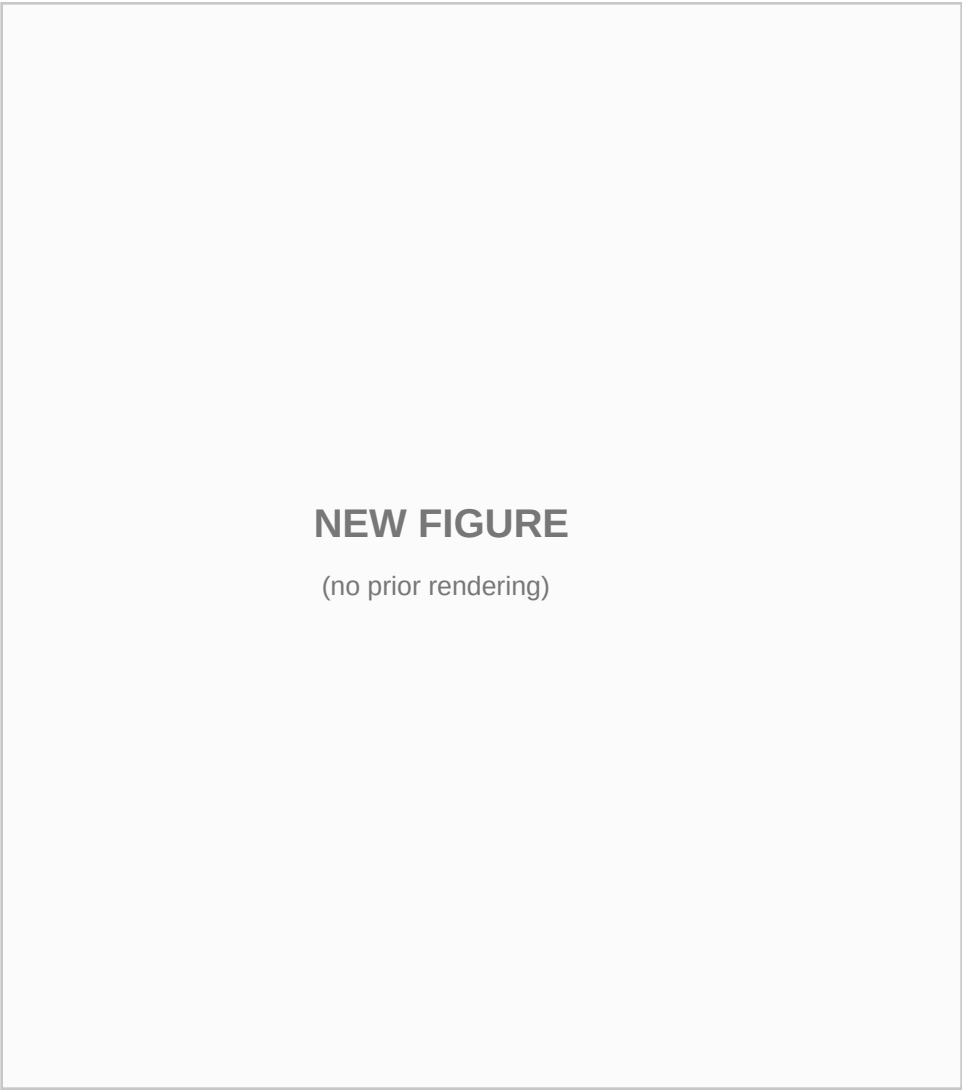
The room's alphabet placed on the controllability x witnessing quadrant of {swk{The Single-Writer Kernel. Top right, shaded: controllable and witnessed, hence regimentable -- {fs_write and {net_egress can simply be vetoed. Bottom left: neither controllable nor witnessed -- {internal_plan is a thought, detect-only forever. Bottom right: witnessed but not controllable -- {model_emit_token can be observed but no...

AFTER

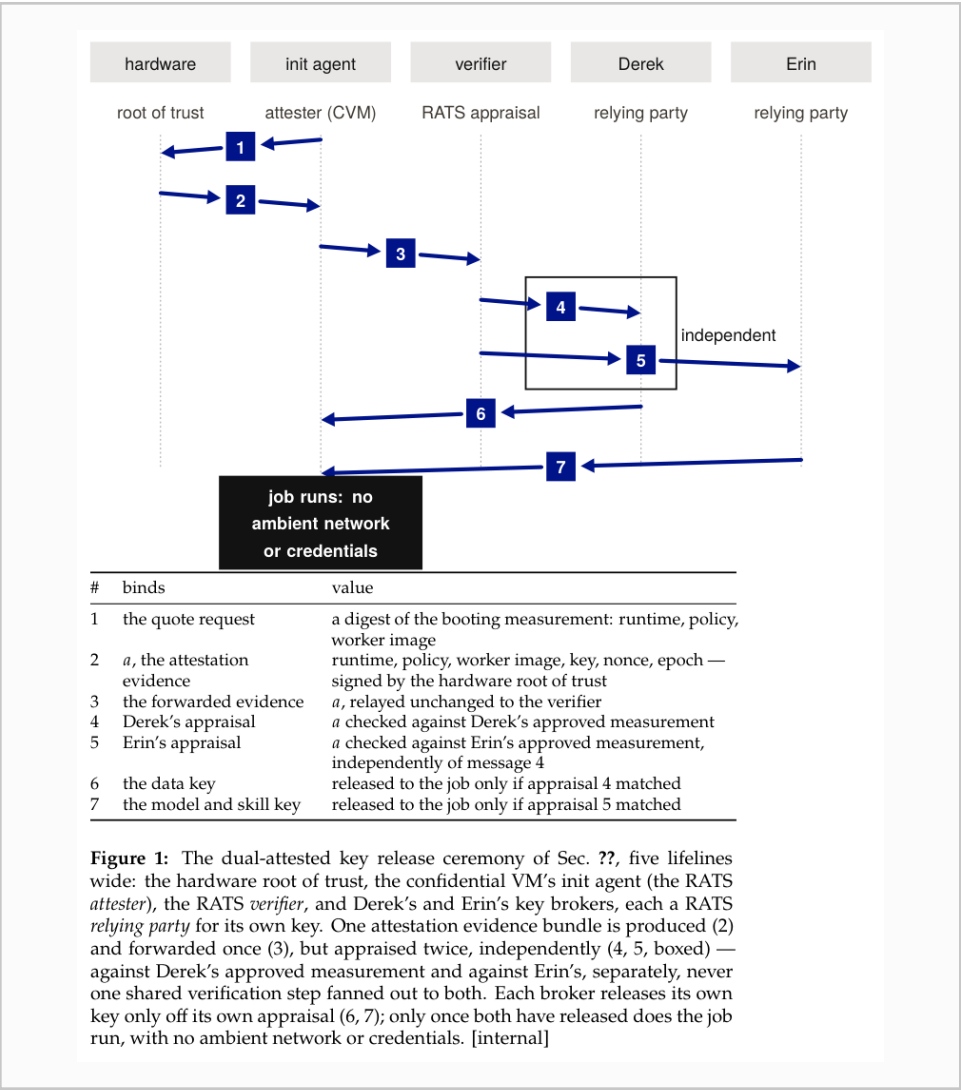


QA: PASS

BEFORE



AFTER

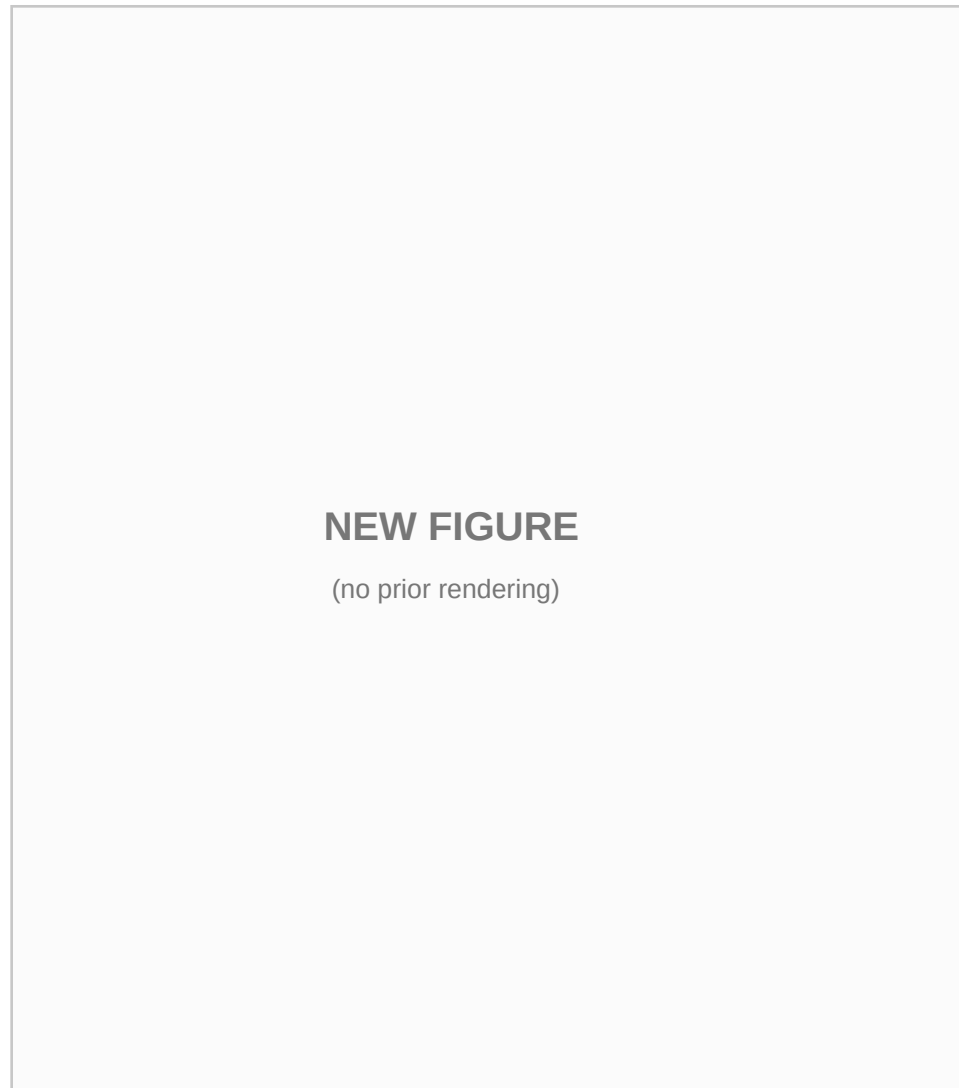


QA: —

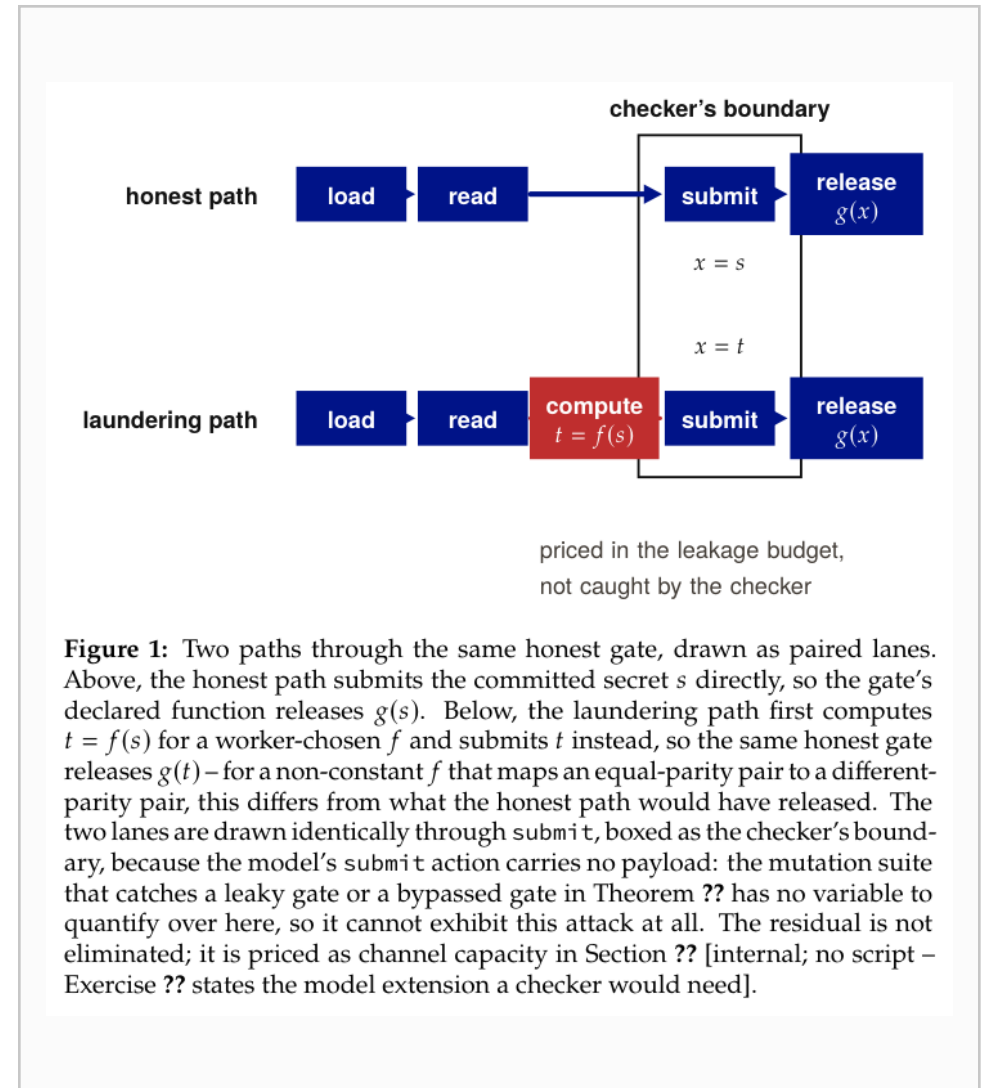
QA: FLAGGED: too wide 5.47in

The dual-attested key release ceremony of Sec.~{sec:sealed-design, five lifelines wide: the hardware root of trust, the confidential VM's init agent (the RATS {attester), the RATS {verifier, and Derek's and Erin's key brokers, each a RATS {relying party for its own key. One attestation evidence bundle is produced (2) and forwarded once (3), but appraised twice, independently (4, 5, boxed) --- against Derek's appro...

BEFORE



AFTER

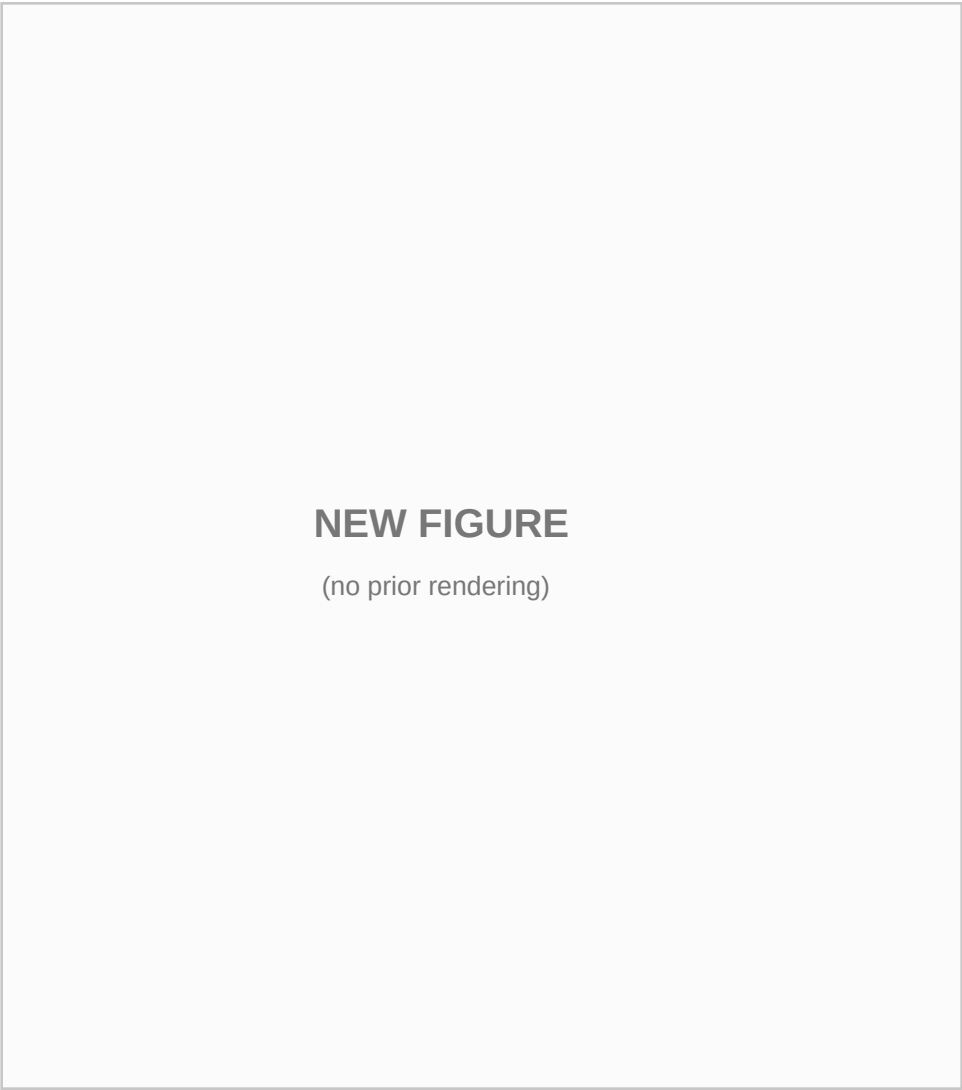


QA: —

QA: PASS

Two paths through the same honest gate, drawn as paired lanes. Above, the honest path submits the committed secret s directly, so the gate's declared function releases $g(s)$. Below, the laundering path first computes $t = f(s)$ for a worker-chosen f and submits t instead, so the same honest gate releases $g(t)$ -- for a non-constant f that maps an equal-parity pair to a different-parity pair, this differs f...

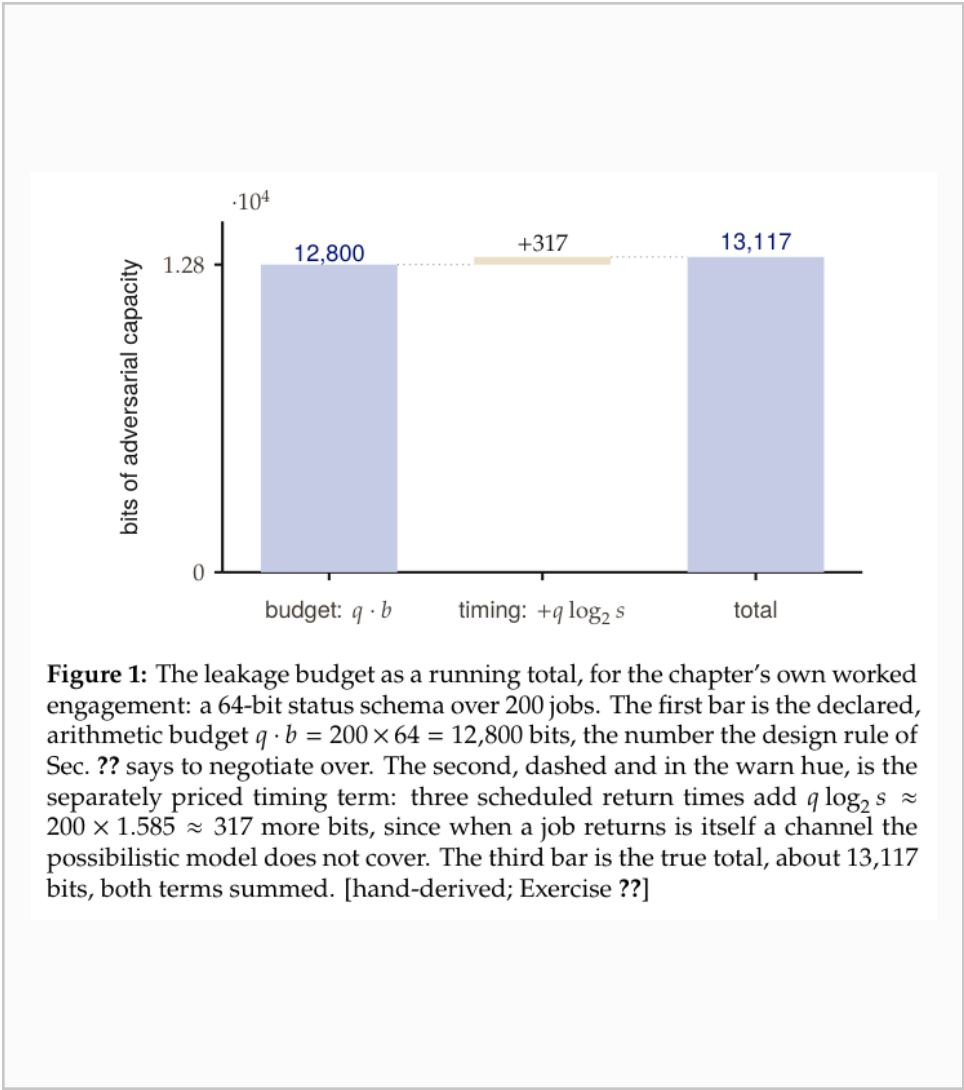
BEFORE



QA: —

The leakage budget as a running total, for the chapter's own worked engagement: a 64-bit status schema over 200 jobs. The first bar is the declared, arithmetic budget $q \cdot b = 200 \times 64 = 12,800$ bits, the number the design rule of Sec. ?? says to negotiate over. The second, dashed and in the warn hue, is the separately priced timing term: three scheduled return times add $q \log_2 s \approx 200 \times 1.585 \approx 317$ more bits, since when a job returns is itself a channel the possibilistic model does not cover. The third bar is the true total, about 13,117 bits, both terms summed. [hand-derived; Exercise ??]

AFTER



QA: PASS

BEFORE

which mutant breaks which invariant, and how fast

mutant →	spend-without-append	torn write
ledger balance (the recorded sum)	■ caught, 1 step	■ caught, 1 step
budget cap (spend at or under the limit)	not exercised	■ caught, 3 steps

Model-checked instance: two concurrent clients racing programs (2,2) and (1,2) against a budget cap of 4. All 15 reachable states (order-sensitive; 11 as a multiset) satisfy both invariants under every interleaving, and 2000 random instances of program, budget, and interleaving show no violation. The torn write logs the correct amount but adds one unit less: by 3 steps the recorded ledger reaches a total of 5, over the cap of 4 – an auditable over-spend. Reverting either mutation restores both invariants and the 15 states [internal, a3_epsilon_ledger.py].

Figure 1: The two canonical atomicity breaks against the ledger’s two invariants: spend-without-append (a release that adds to the balance without journaling it) and the torn write (a release that journals the wrong amount). Each caught cell marks the invariant broken and the shortest trace that catches it; the ledger’s balance is the invariant both mutants break immediately, but only the torn write’s deeper crime – recorded spend past the contracted maximum – takes three steps to show.

QA: FLAGGED: B9, too wide 4.64in

The two canonical atomicity breaks against the ledger's two invariants: spend-without-append (a release that adds to the balance without journaling it) and the torn write (a release that journals the wrong amount). Each marked cell gives the shortest trace that catches the break; the ledger's balance is the invariant both mutants break immediately, but only the torn write's deeper crime -- recorded spend past the ...

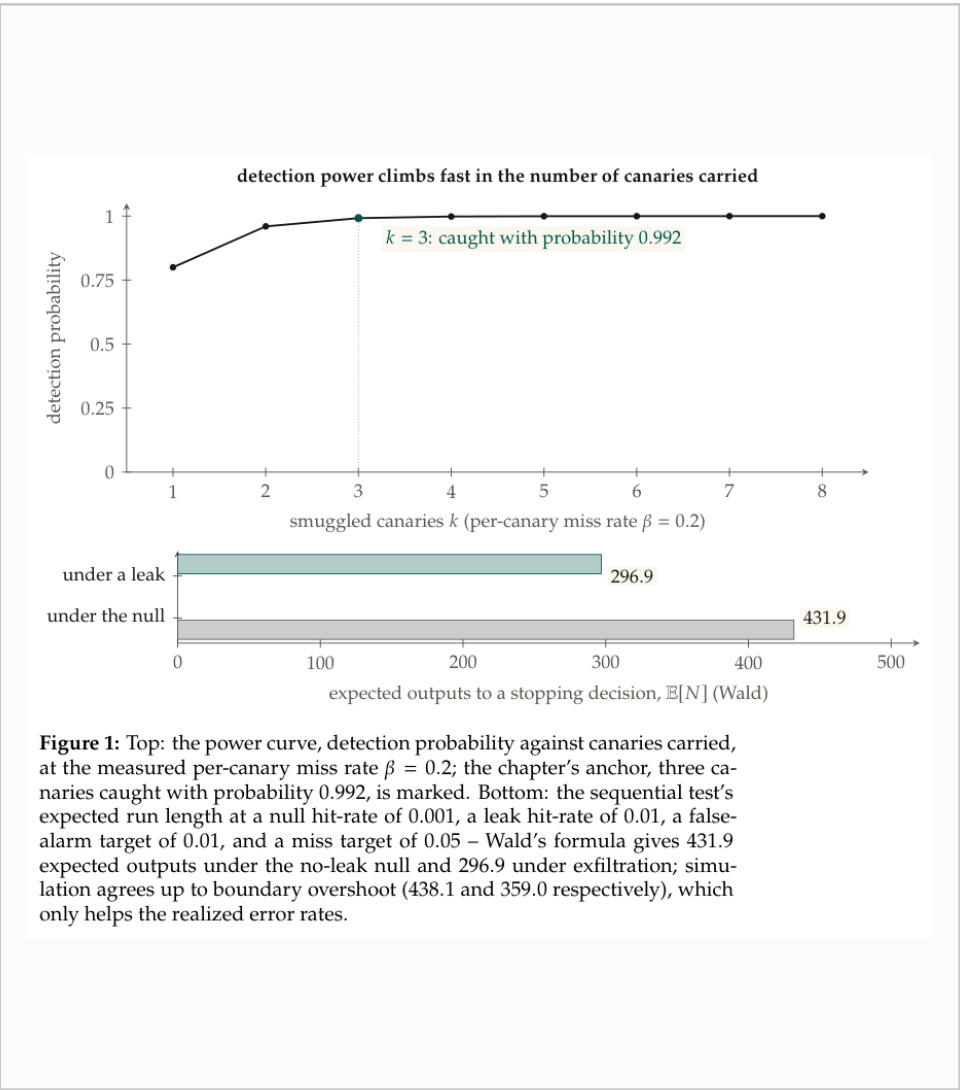
AFTER

	spend-without-append	torn write
ledger balance the recorded sum	◆ caught 1 step	◆ caught 1 step
budget cap spend at or under the limit	not exercised	◆ caught 3 steps

Figure 1: The two canonical atomicity breaks against the ledger’s two invariants: spend-without-append (a release that adds to the balance without journaling it) and the torn write (a release that journals the wrong amount). Each marked cell gives the shortest trace that catches the break; the ledger’s balance is the invariant both mutants break immediately, but only the torn write’s deeper crime – recorded spend past the contracted maximum – takes three steps to show, and spend-without-append never exercises the cap at all. Model-checked instance: two concurrent clients racing programs (2,2) and (1,2) against a budget cap of 4. All 15 reachable states (order-sensitive; 11 as a multiset) satisfy both invariants under every interleaving, and 2000 random instances of program, budget and interleaving show no violation; the torn write logs the correct amount but adds one unit less, so by 3 steps the recorded ledger reaches a total of 5, over the cap of 4 – an auditable over-spend. Reverting either mutation restores both invariants and the 15 states [internal, a3_epsilon_ledger.py].

QA: FLAGGED: B9

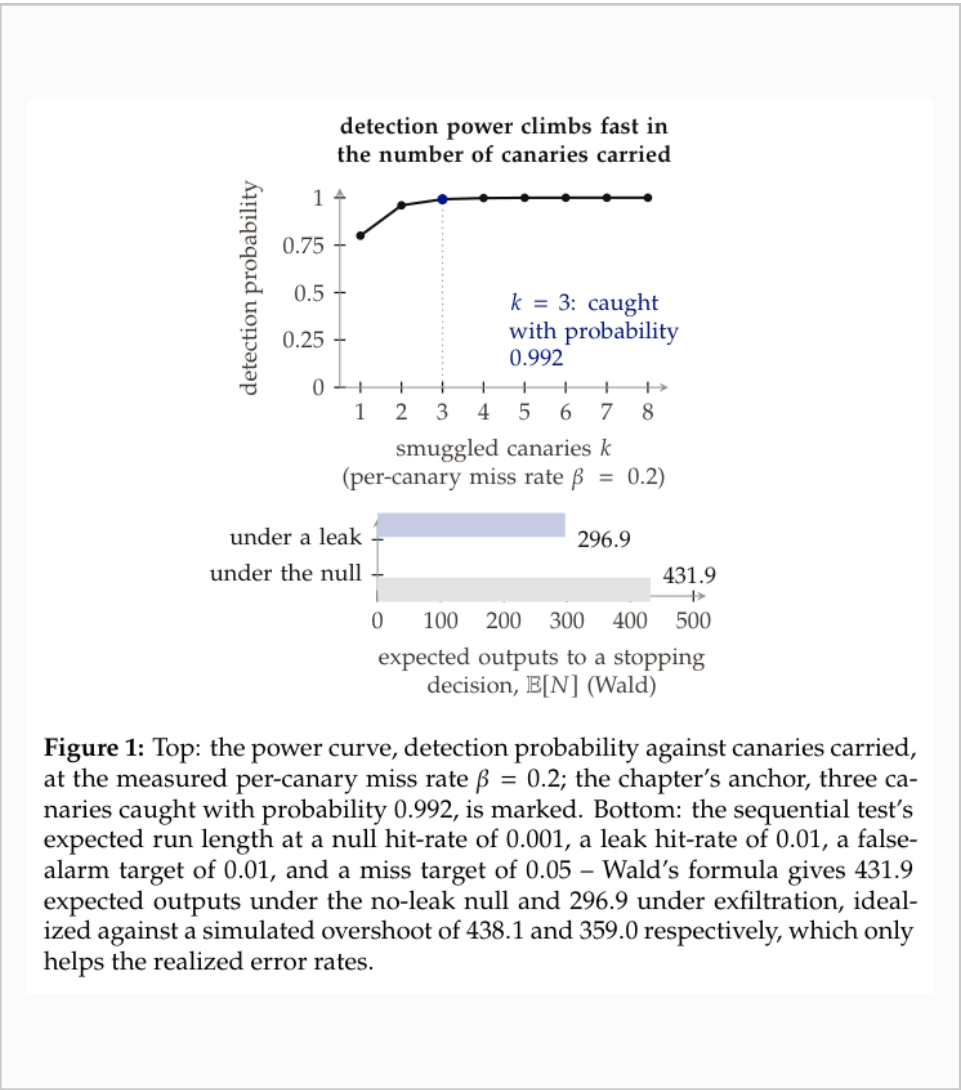
BEFORE



QA: FLAGGED: too wide 5.66in

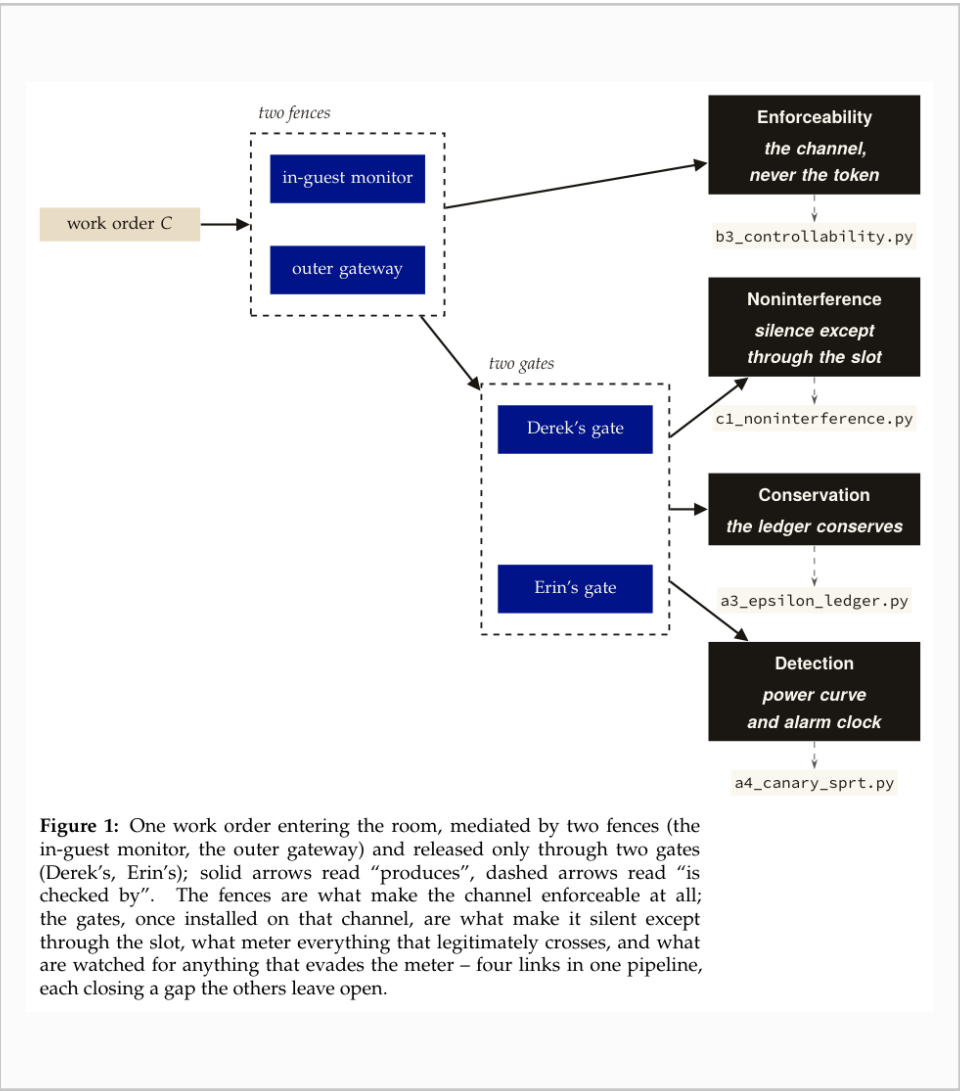
Top: the power curve, detection probability against canaries carried, at the measured per-canary miss rate $\beta=0.2$; the chapter's anchor, three canaries caught with probability 0.992, is marked. Bottom: the sequential test's expected run length at a null hit-rate of 0.001, a leak hit-rate of 0.01, a false-alarm target of 0.01, and a miss target of 0.05 -- Wald's formula gives 431.9 expected outputs unde...

AFTER



QA: PASS

BEFORE

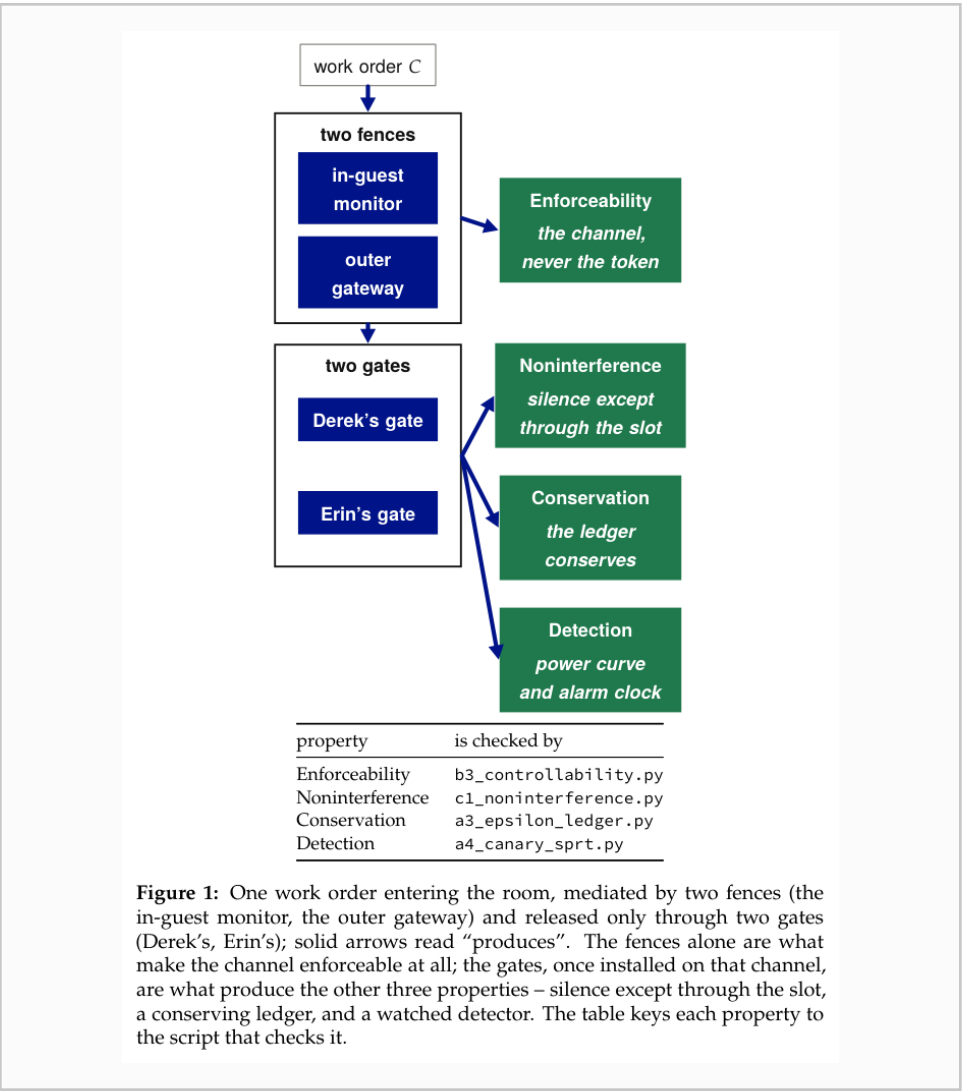


QA: FLAGGED: too wide 6.00in

One work order entering the room, mediated by two fences (the in-guest monitor, the outer gateway) and released only through two gates (Derek's, Erin's); solid arrows read ``produces''. The fences alone are what make the channel enforceable at all; the gates, once installed on that channel, are what produce the other three properties -- silence except through the slot, a conserving ledger, and a watched detector.

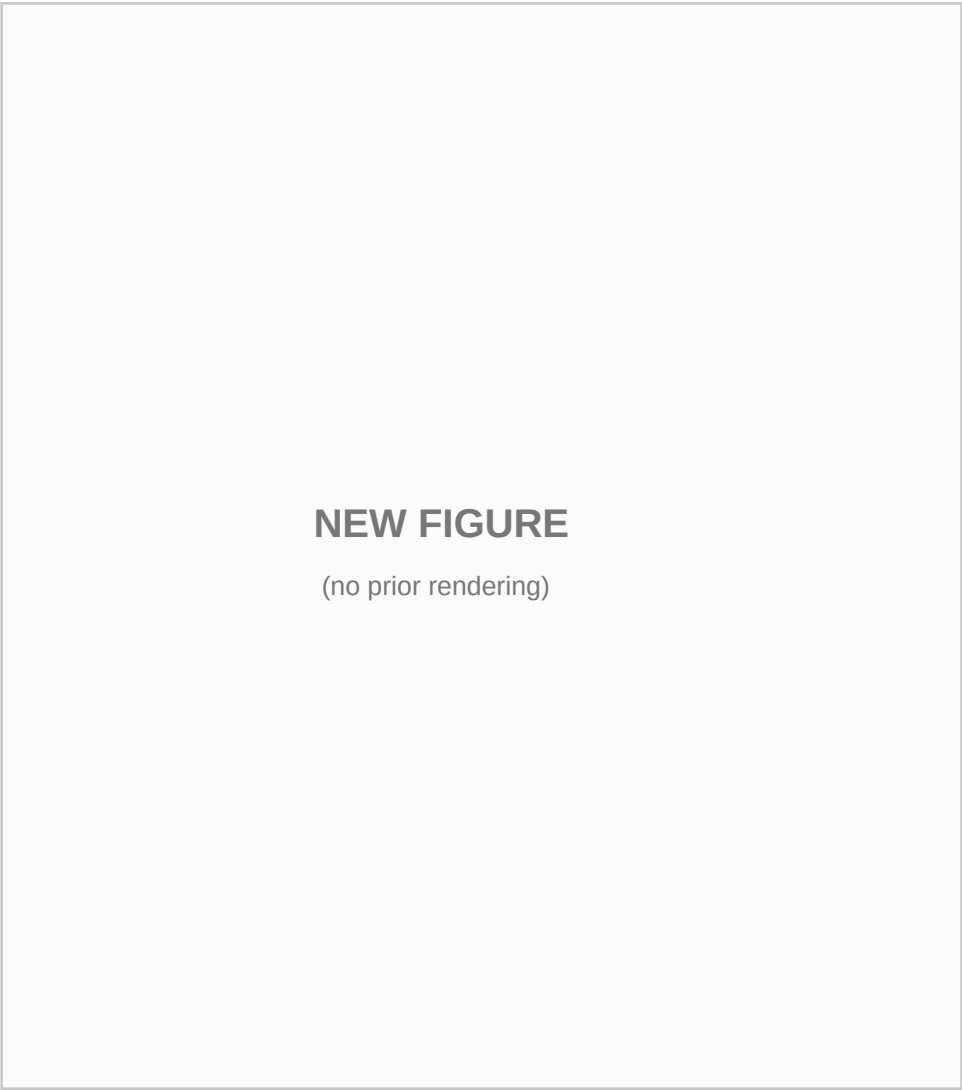
...

AFTER



QA: PASS

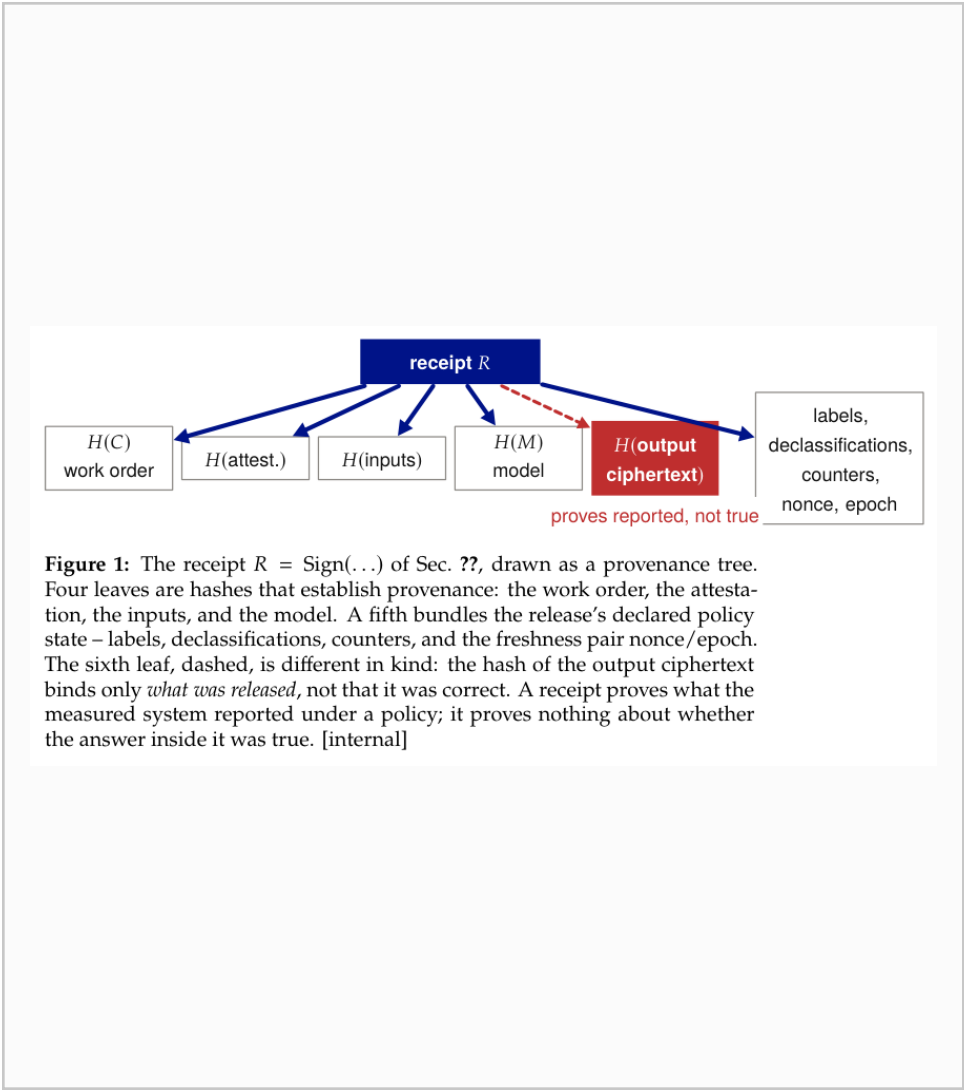
BEFORE



QA: —

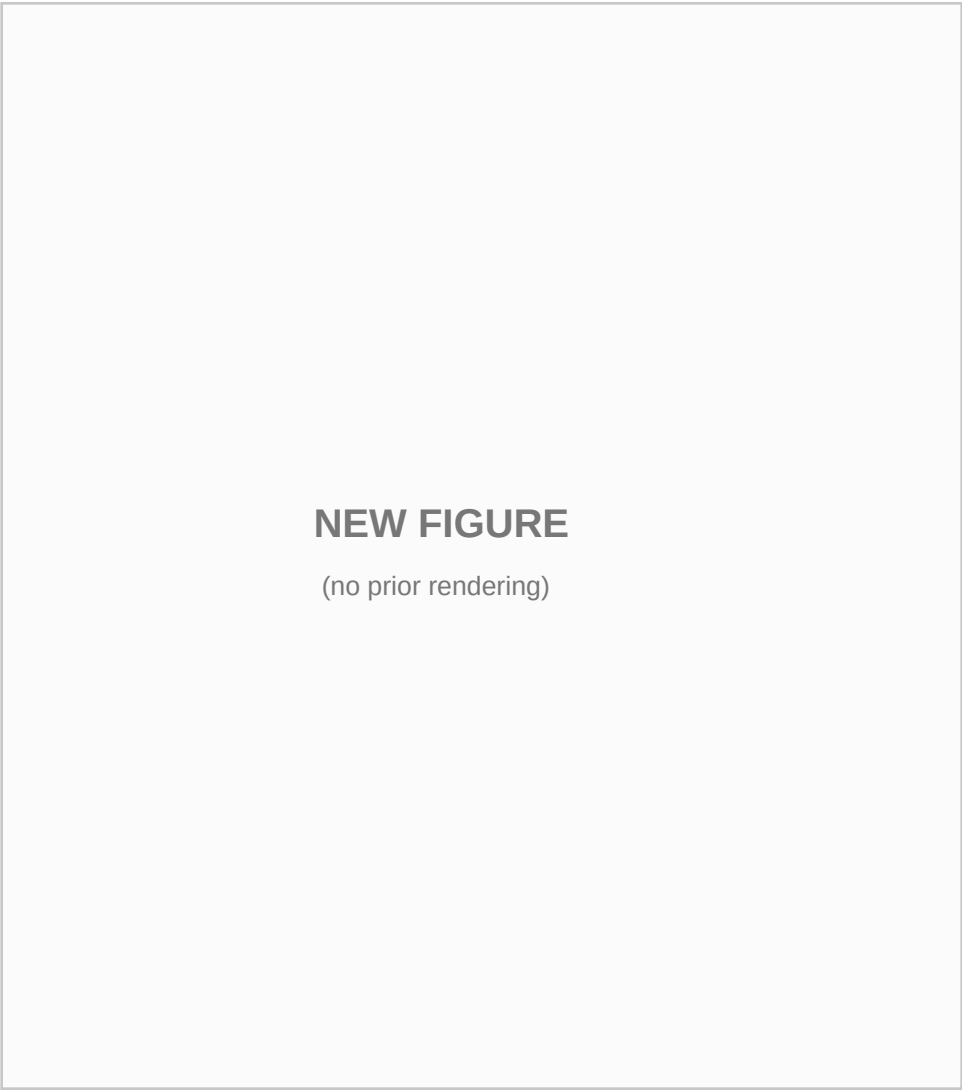
The receipt $R = \text{Sign}(\dots)$ of Sec. ??, drawn as a provenance tree. Four leaves are hashes that establish provenance: the work order, the attestation, the inputs, and the model. A fifth bundles the release's declared policy state -- labels, declassifications, counters, and the freshness pair nonce/epoch. The sixth leaf, dashed, is different in kind: the hash of the output ciphertext binds only what wa...

AFTER



QA: FLAGGED: too wide 5.57in

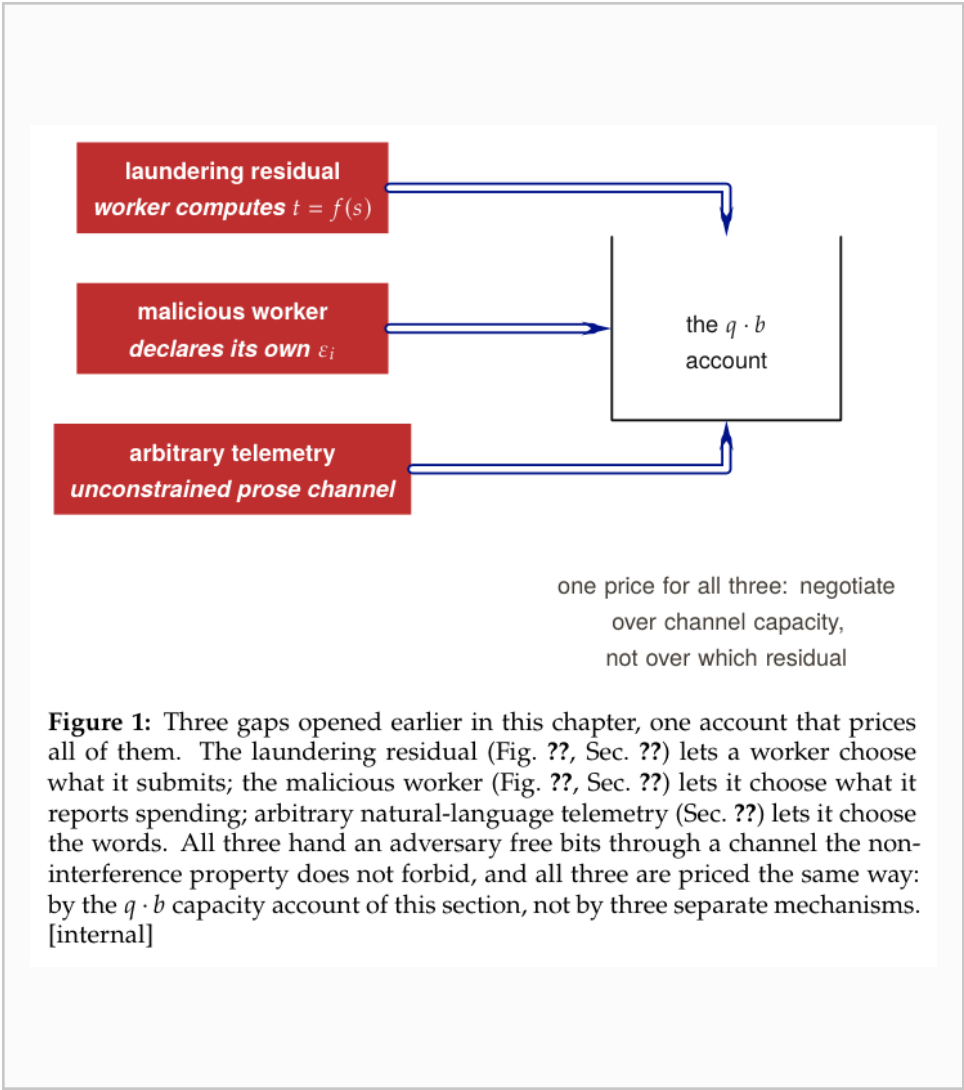
BEFORE



QA: —

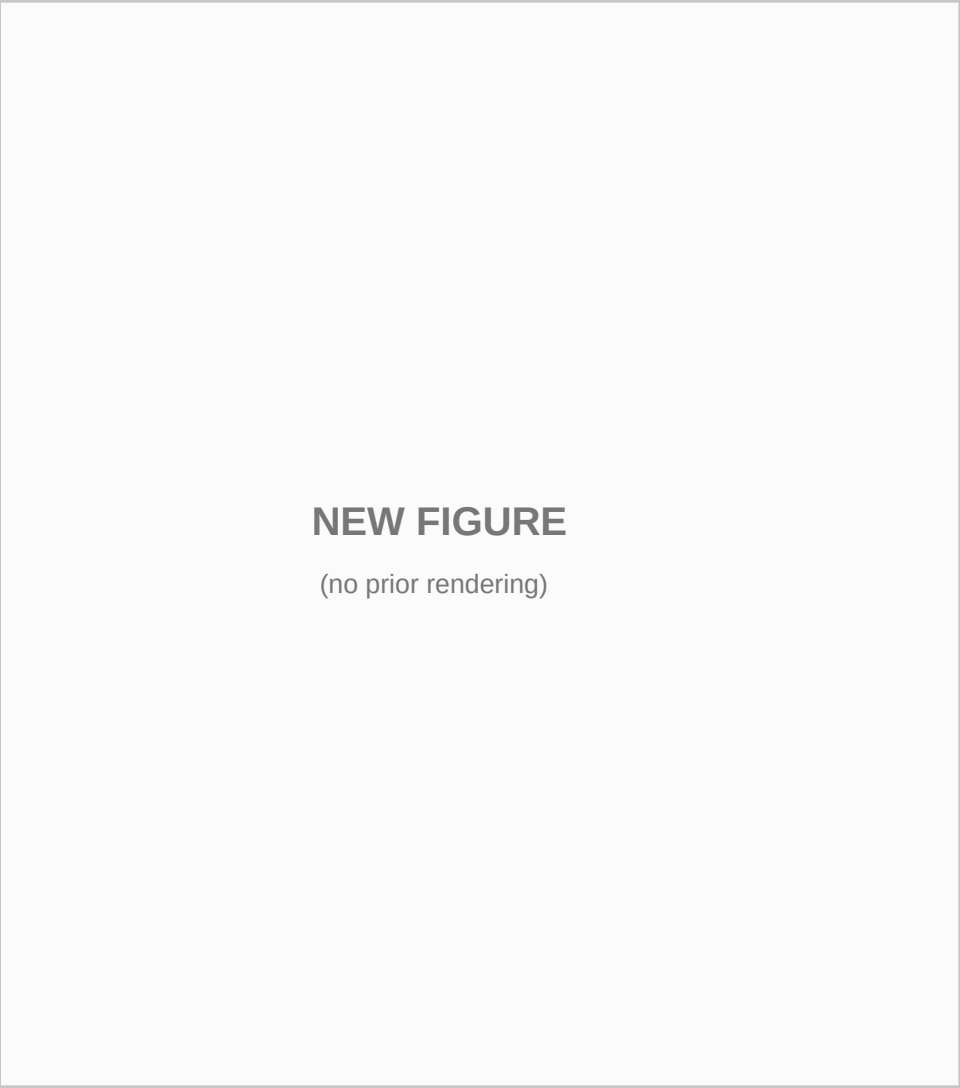
Three gaps opened earlier in this chapter, one account that prices all of them. The laundering residual (Fig.~{fig:sealed-laundering-fork, Sec.~{sec:sealed-noninterference) lets a worker choose what it submits; the malicious worker (Fig.~{fig:sealed-two-adversaries, Sec.~{sec:sealed-conservation) lets it choose what it reports spending; arbitrary natural-language telemetry (Sec.~{sec:sealed-limitations) lets it ch...

AFTER



QA: PASS

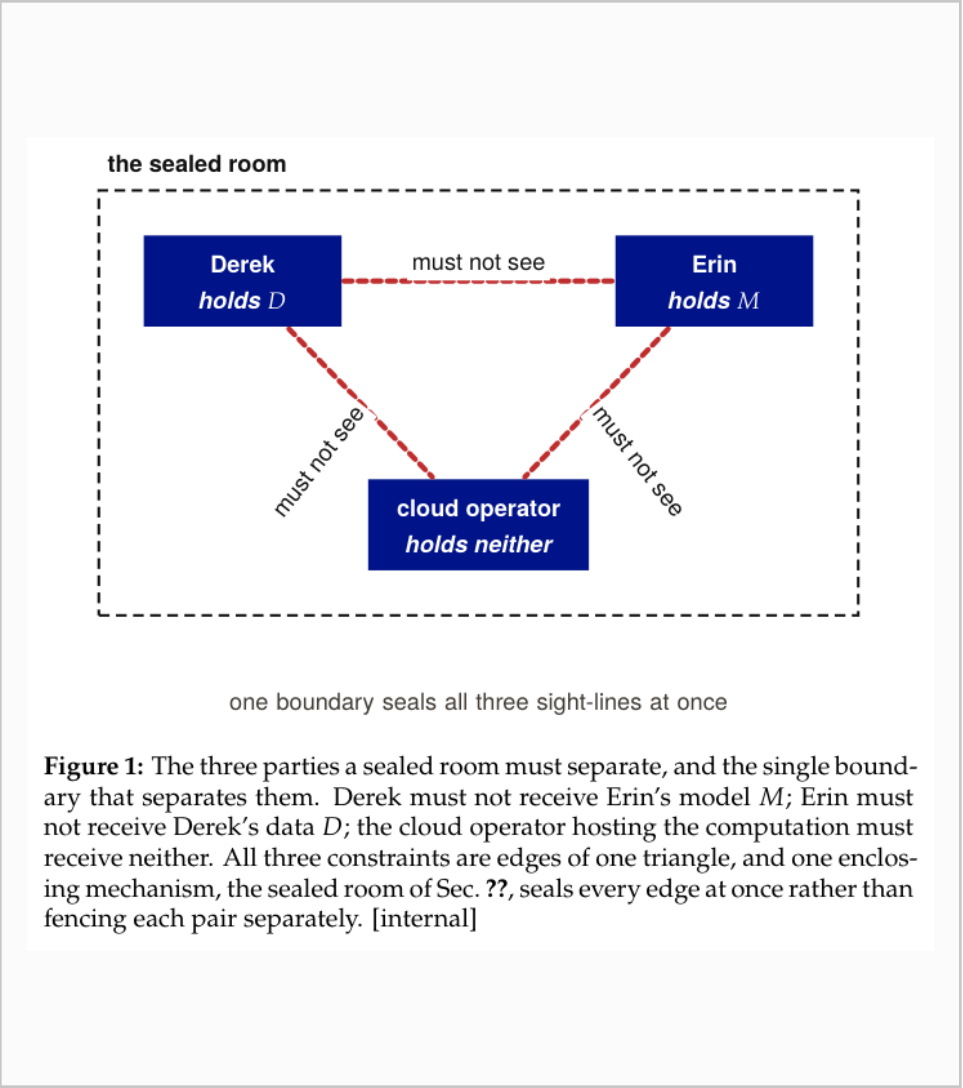
BEFORE



QA: —

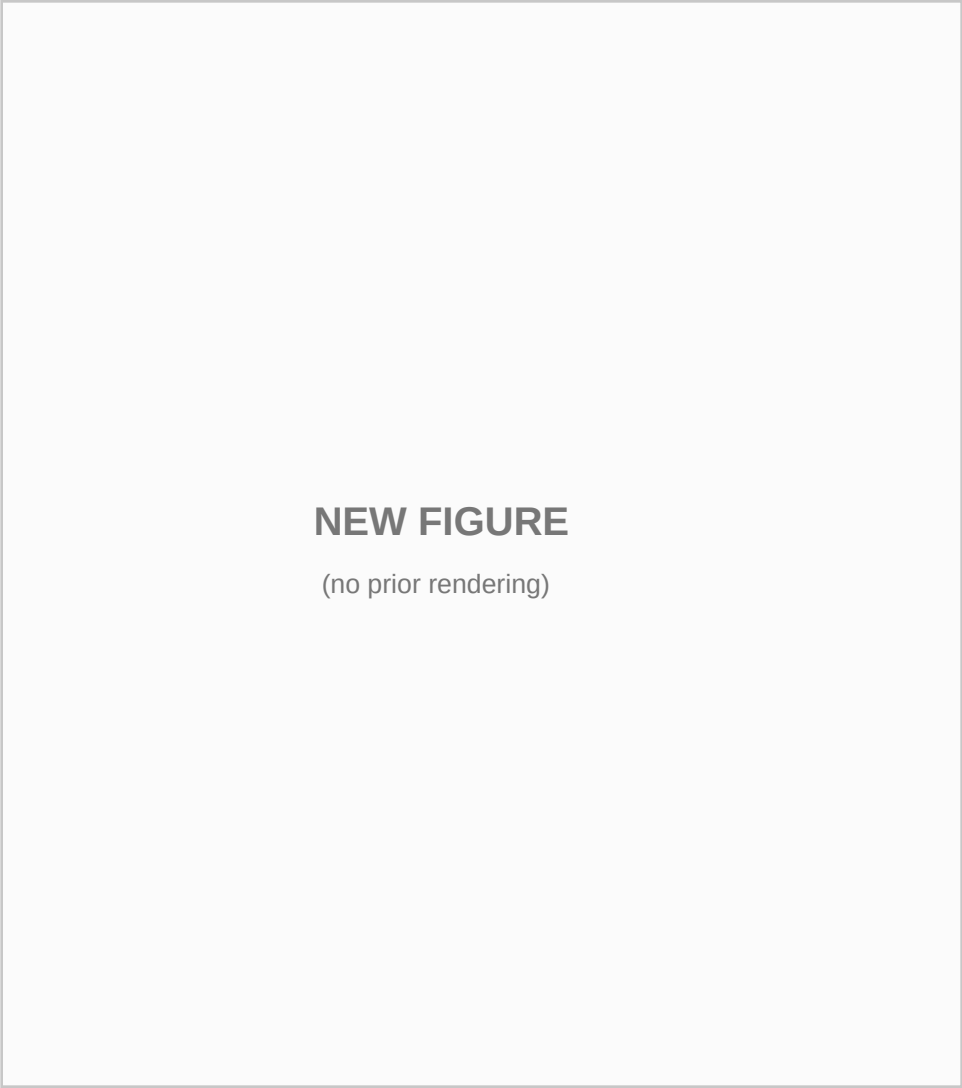
The three parties a sealed room must separate, and the single boundary that separates them. Derek must not receive Erin's model M ; Erin must not receive Derek's data D ; the cloud operator hosting the computation must receive neither. All three constraints are edges of one triangle, and one enclosing mechanism, the sealed room of Sec. ??, seals every edge at once rather than fencing each pair se...

AFTER

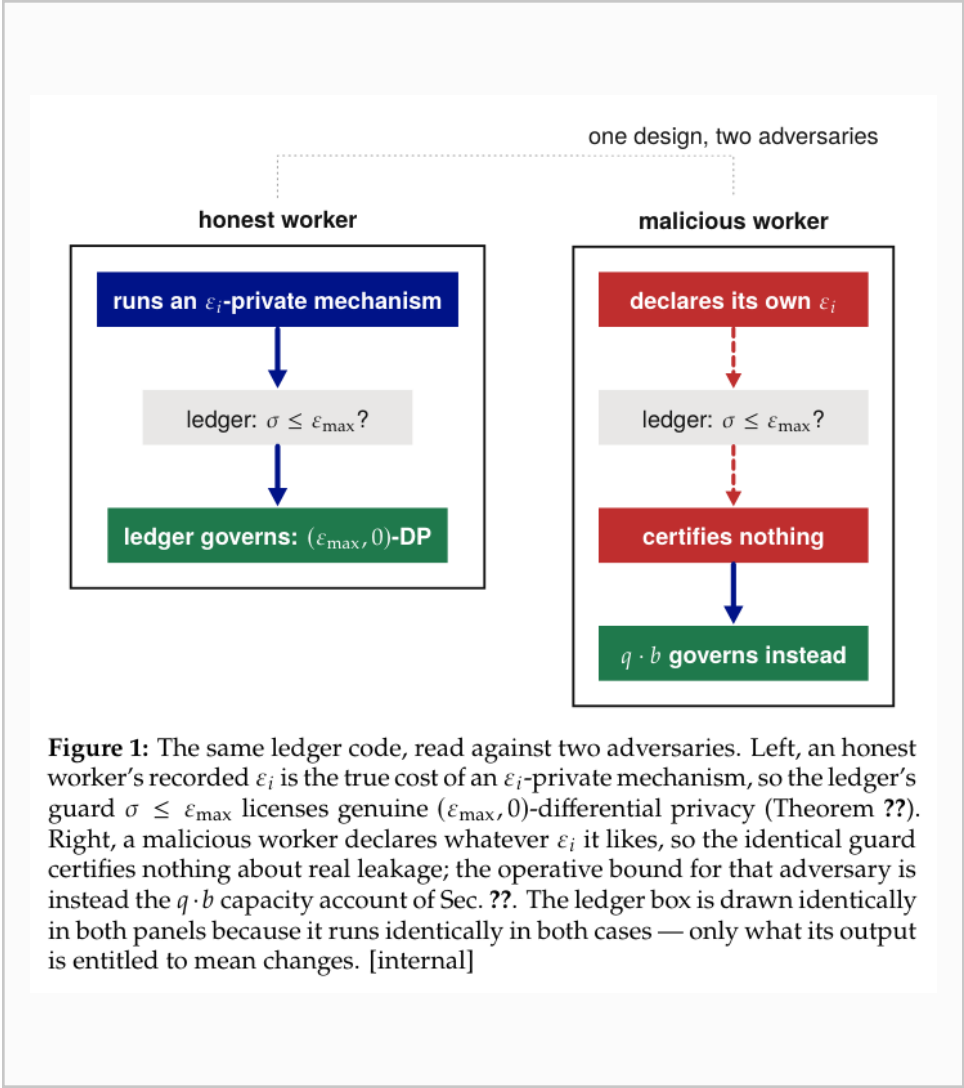


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BEFORE



AFTER

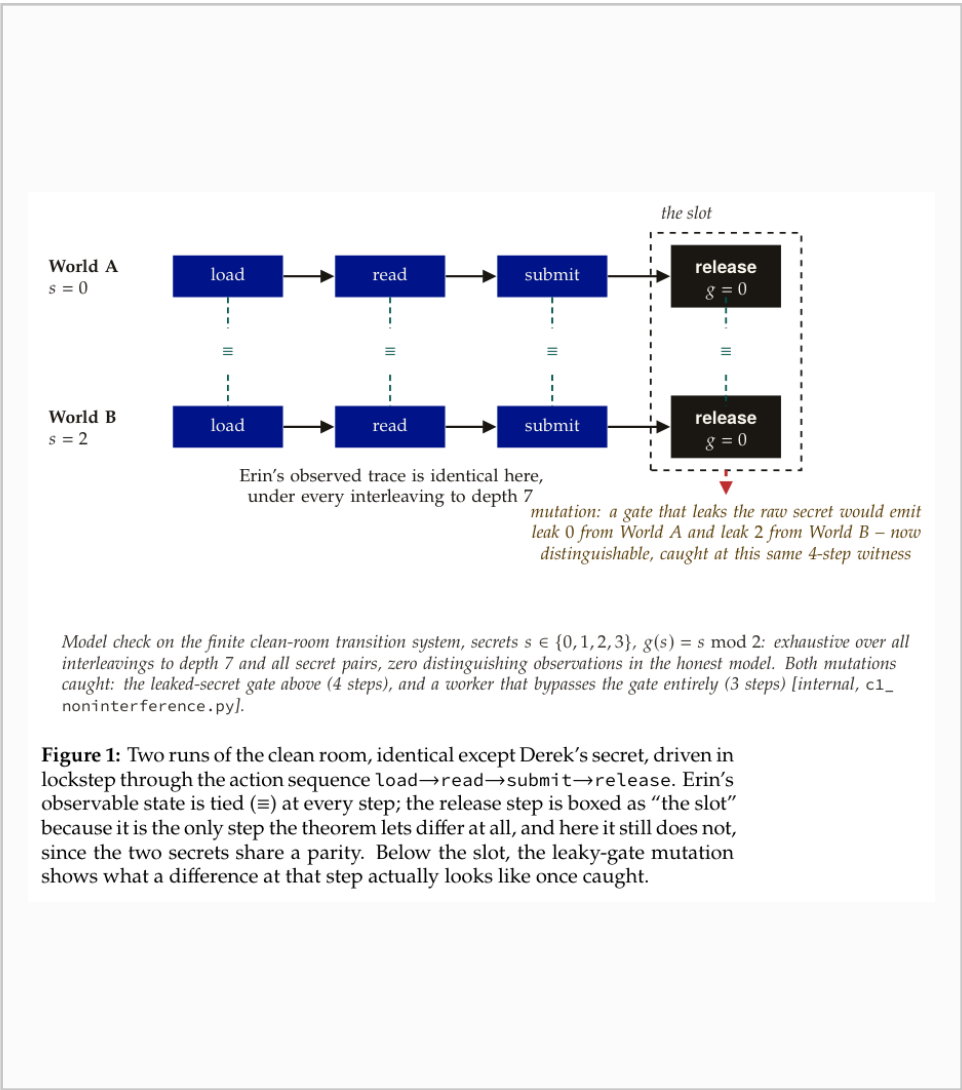


QA: —

QA: PASS

The same ledger code, read against two adversaries. Left, an honest worker’s recorded ϵ_i is the true cost of an ϵ_i -private mechanism, so the ledger’s guard $\sigma \leq \epsilon_{\max}$ licenses genuine $(\epsilon_{\max}, 0)$ -differential privacy (Theorem~{thm:sealed-conservation}). Right, a malicious worker declares whatever ϵ_i it likes, so the identical guard certifies nothing about real leakage; the operative bound for that adversary is instead...

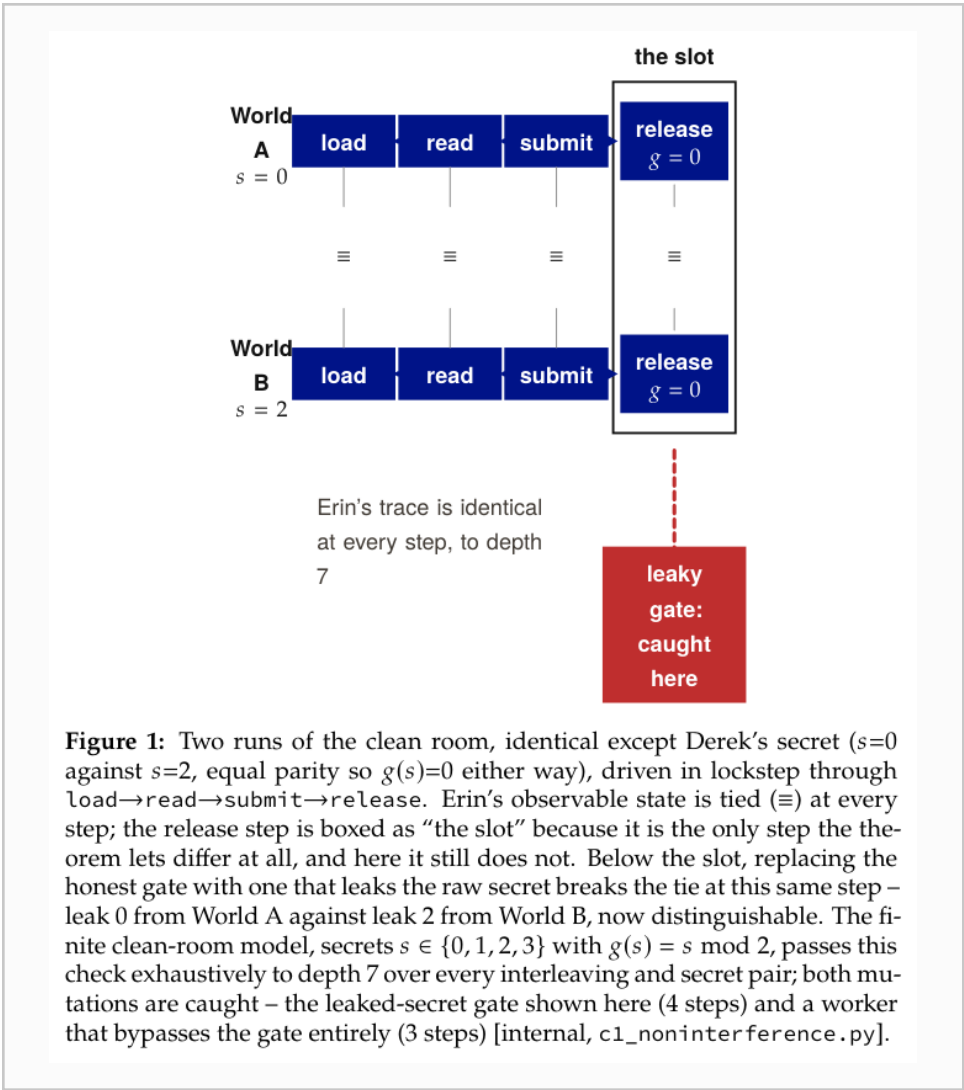
BEFORE



QA: FLAGGED: T4, B3, too wide 5.67in

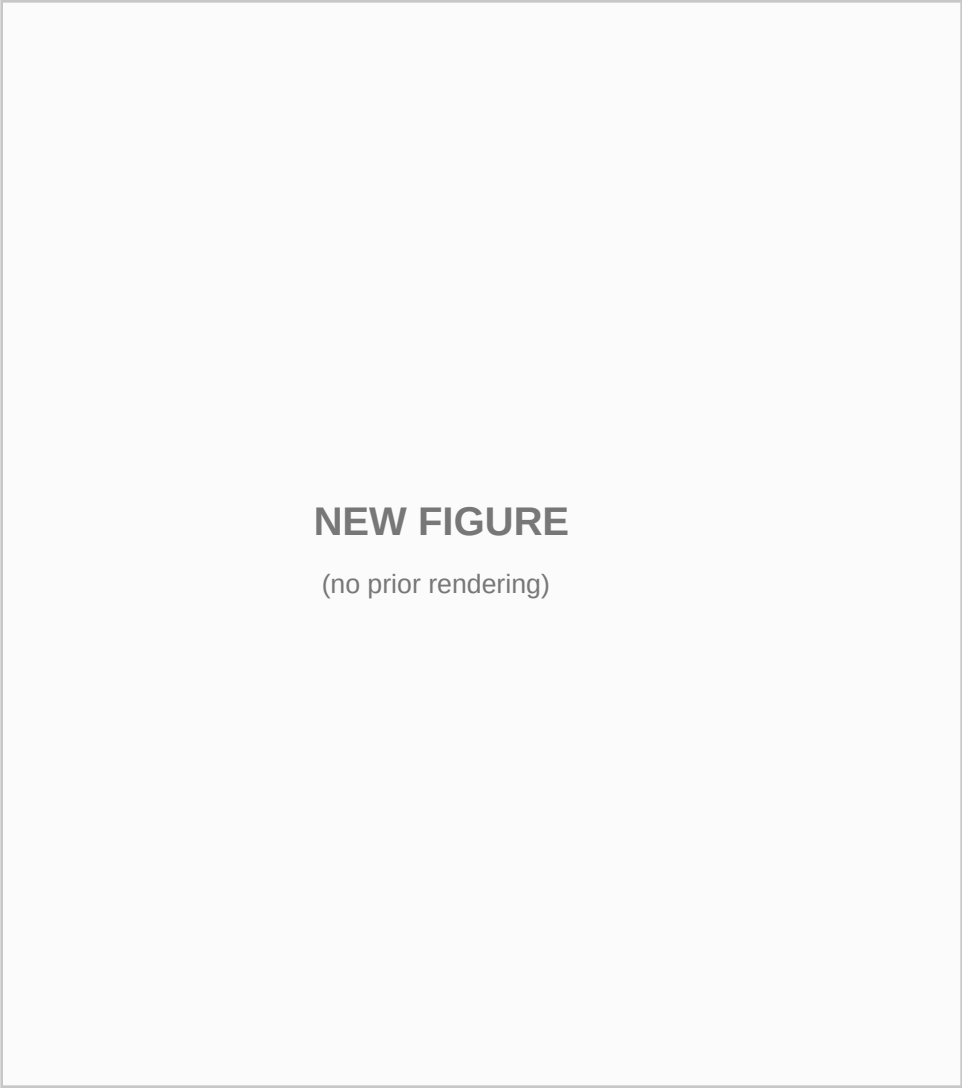
Two runs of the clean room, identical except Derek's secret ($s=0$ against $s=2$, equal parity so $g(s)=0$ either way), driven in lockstep through $\{\text{load}\}\{\text{read}\}\{\text{submit}\}\{\text{release}$. Erin's observable state is tied ($\equiv$) at every step; the release step is boxed as "the slot" because it is the only step the theorem lets differ at all, and here it still does not. Below the slot, replacing the honest gate with one t...

AFTER



QA: PASS

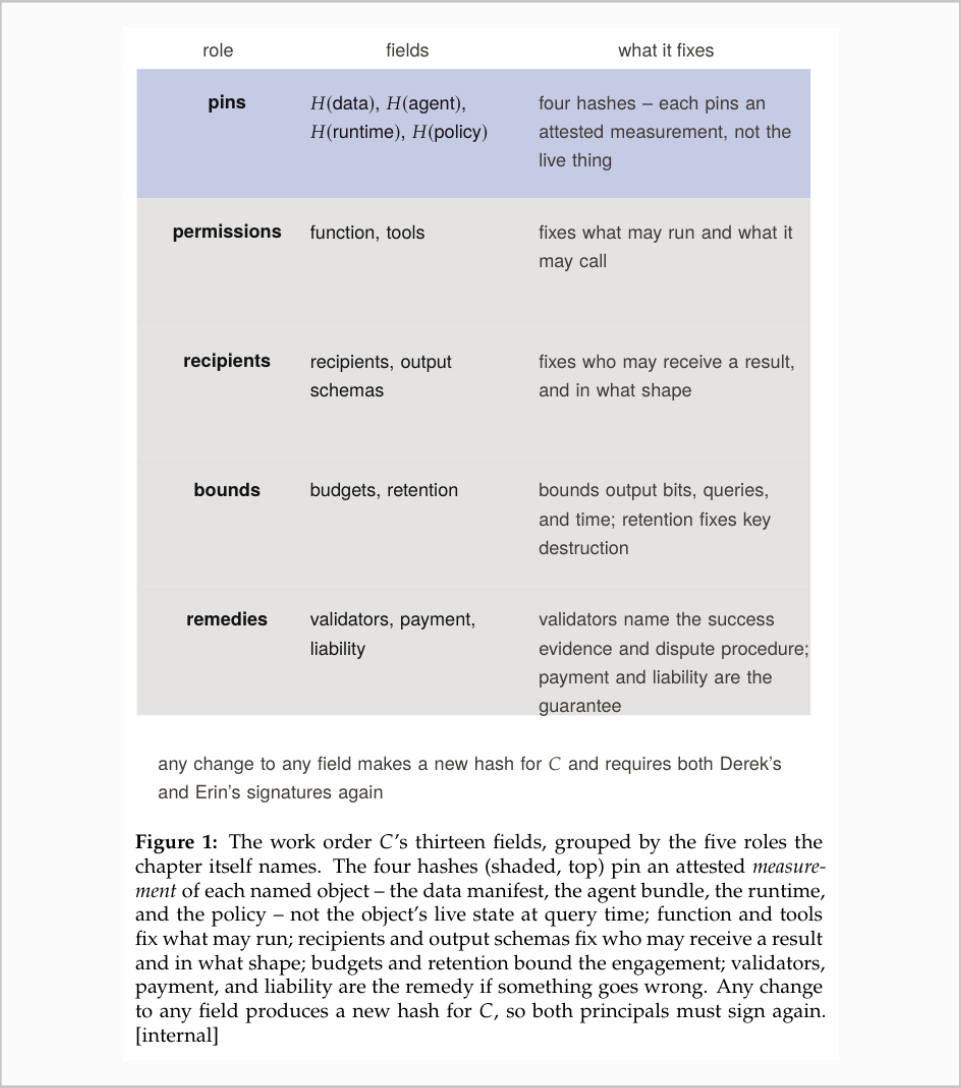
BEFORE



QA: —

The work order \$C\$’s thirteen fields, grouped by the five roles the chapter itself names. The four hashes (shaded, top) pin an attested {measurement of each named object -- the data manifest, the agent bundle, the runtime, and the policy -- not the object’s live state at query time; function and tools fix what may run; recipients and output schemas fix who may receive a result and in what shape; budgets and retent...

AFTER



QA: PASS

BEFORE

NEW FIGURE

(no prior rendering)

AFTER

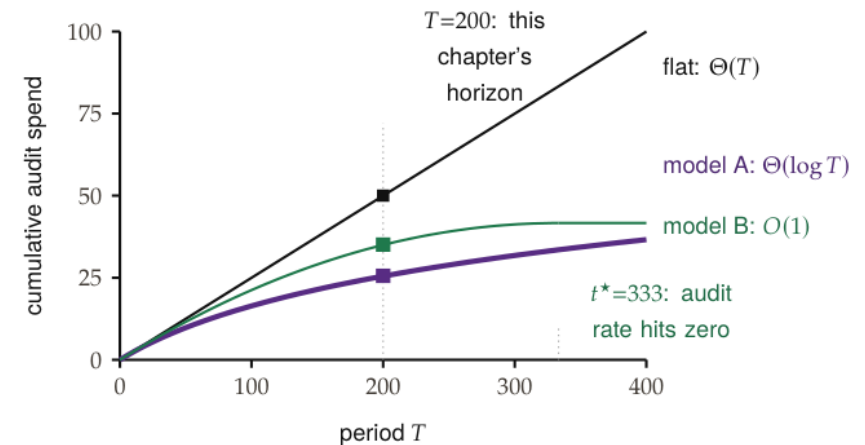


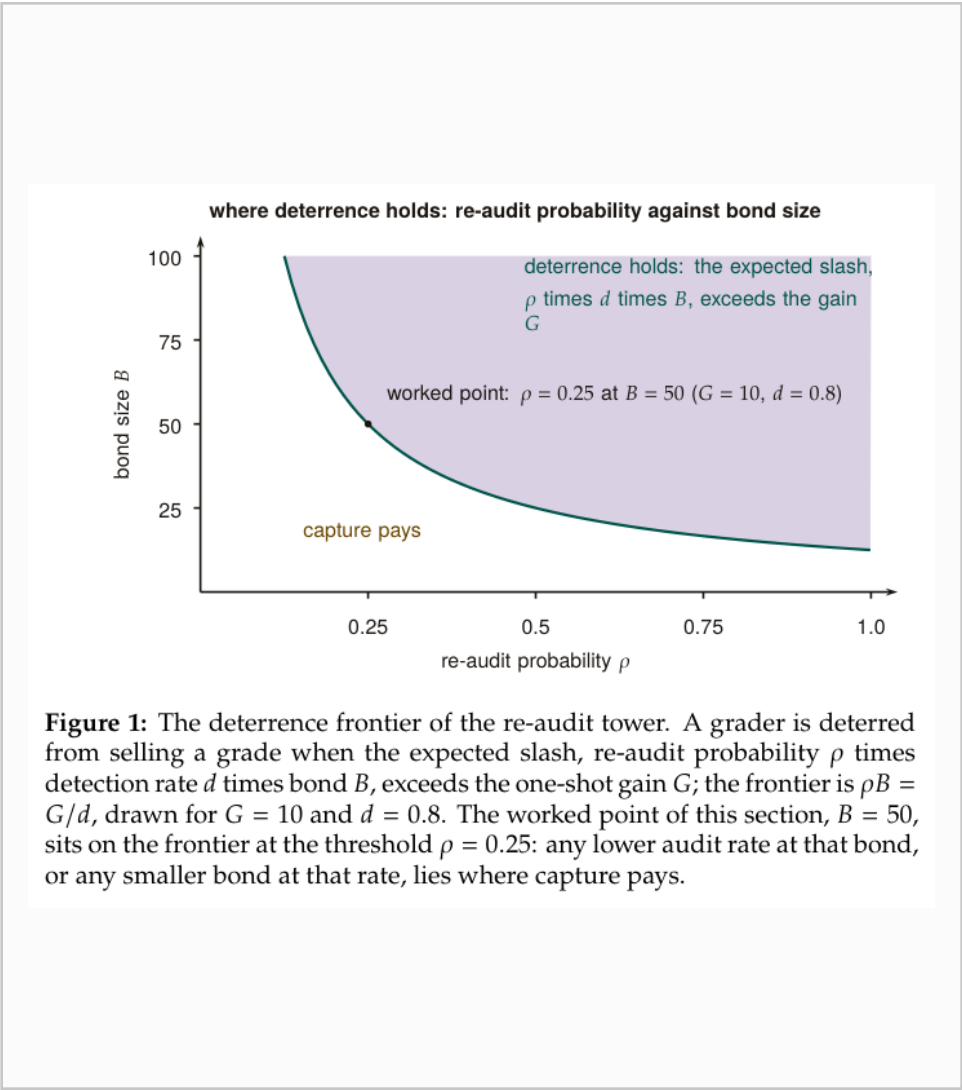
Figure 1: Cumulative audit spend over $T = 200$ periods, at $G = 10$, $d = 0.8$, $B = 50$, $a = 1$, $\rho^* = 0.25$. A flat rate spends $\Theta(T)$, reaching 50.00. Letting the audit rate fall as accumulated stake vt grows (model A) spends $\Theta(\log T)$, reaching only 25.58. Letting cheats also surface on their own at rate $r = 0.05$ (model B) spends 35.08 by $T=200$ but is bounded: the audit rate hits zero at $t^* = G/(rv) = 333$ and total spend converges to $aG^2/(2dBrv) = 41.67$. The three curves are different asymptotic classes, not a ranking: model B is eventually the cheapest schedule in total, but model A is cheaper at the horizon this chapter actually reports. [verified; b2_tower.py, seed 20260816]

QA: —

QA: PASS

Cumulative audit spend over $T=200$ periods, at $G=10$, $d=0.8$, $B=50$, $a=1$, $\rho^*=0.25$. A flat rate spends $\Theta(T)$, reaching $\$50.00$. Letting the audit rate fall as accumulated stake vt grows (model A) spends $\Theta(\log T)$, reaching only $\$25.58$. Letting cheats also surface on their own at rate $r=0.05$ (model B) spends $\$35.08$ by $T=200$ but is bounded: the audit rate hits zero at $t^*=G/(rv)=333$ and total spend c...

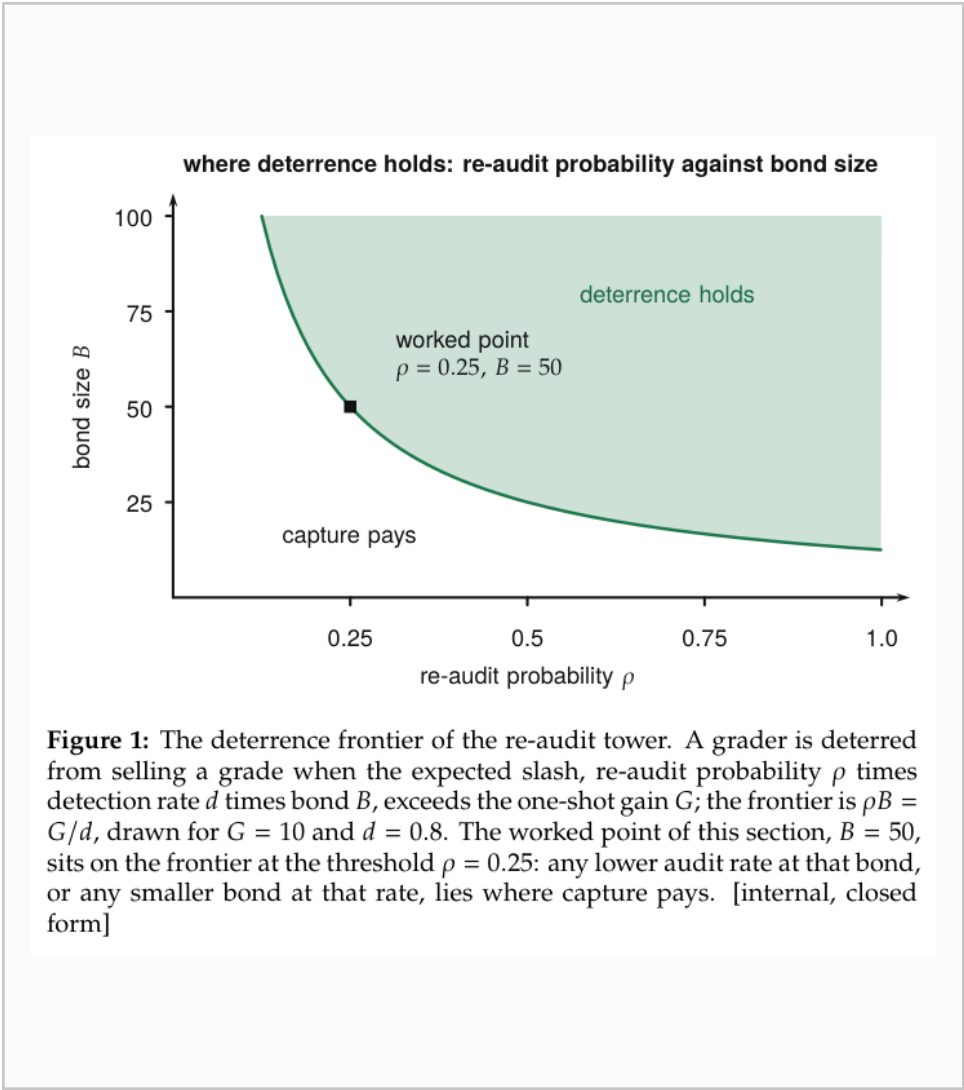
BEFORE



QA: PASS

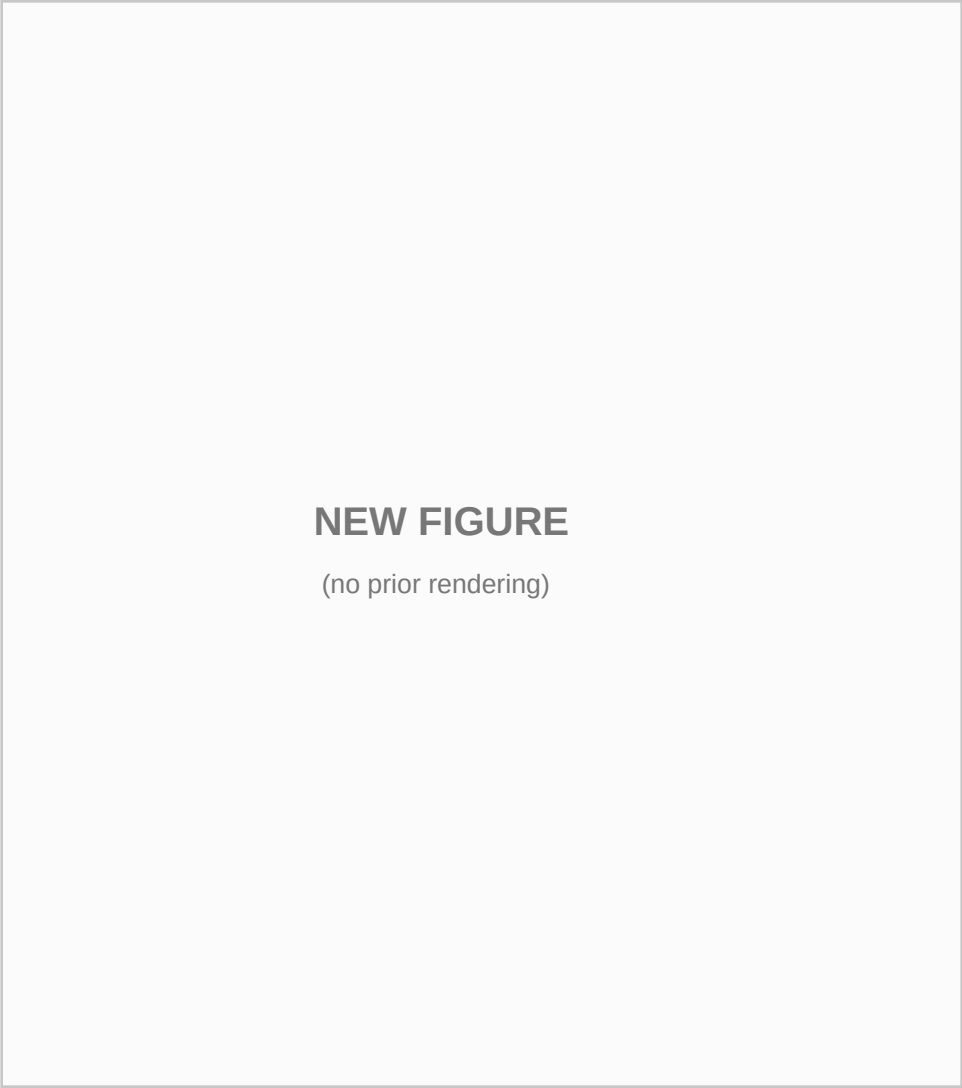
The deterrence frontier of the re-audit tower. A grader is deterred from selling a grade when the expected slash, re-audit probability ρ times detection rate d times bond B , exceeds the one-shot gain G ; the frontier is $\rho B = G/d$, drawn for $G=10$ and $d=0.8$. The worked point of this section, $B=50$, sits on the frontier at the threshold $\rho=0.25$: any lower audit rate at that bond, or any smaller bond at that...

AFTER

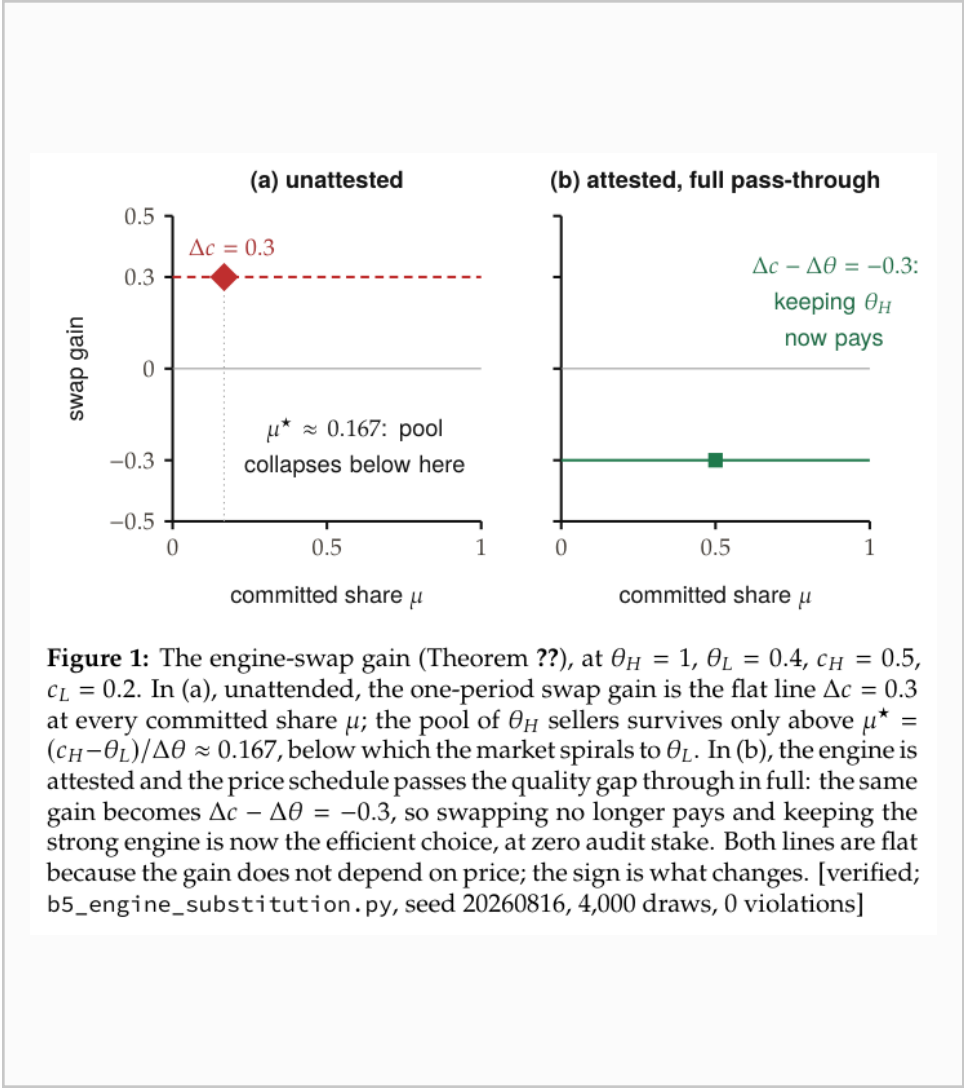


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BEFORE



AFTER

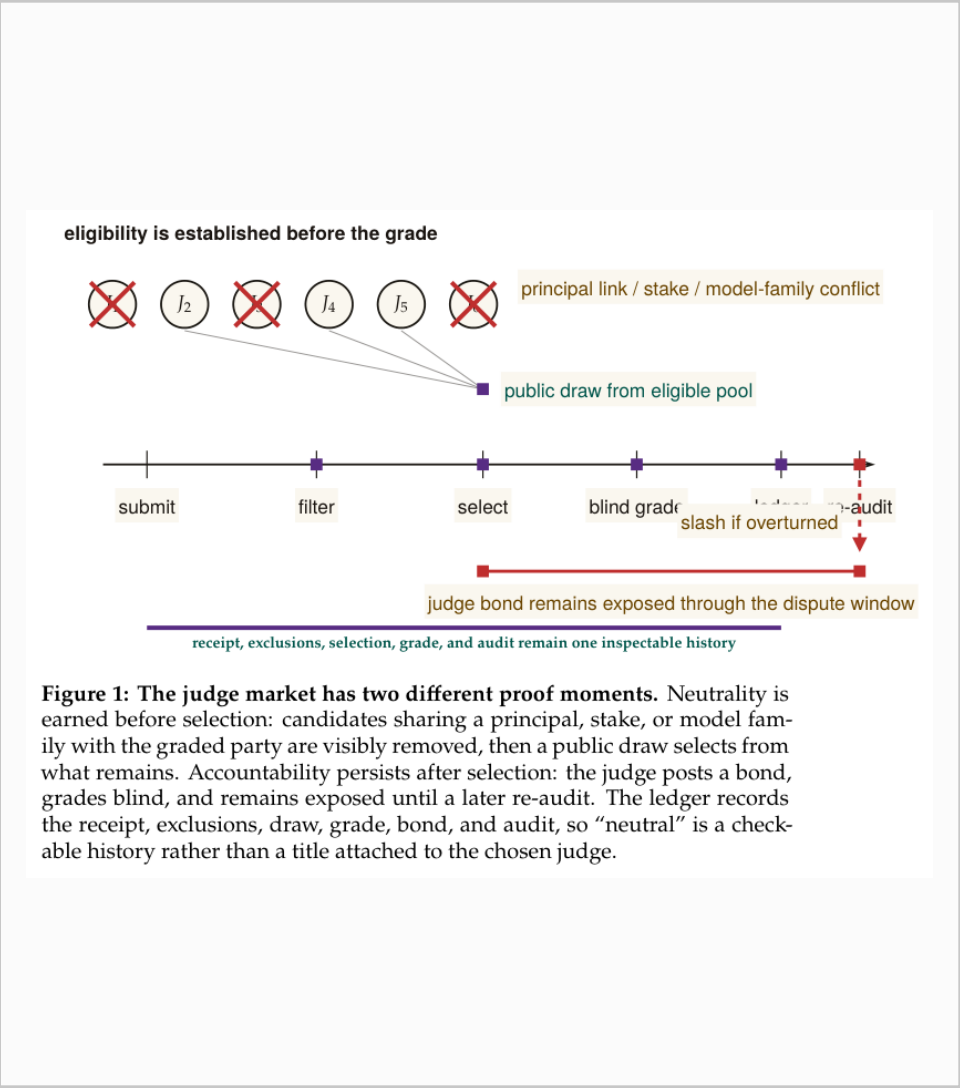


QA: —

QA: PASS

The engine-swap gain (Theorem~{thm:engine-swap}), at $\theta_H=1$, $\theta_L=0.4$, $c_H=0.5$, $c_L=0.2$. In (a), unattended, the one-period swap gain is the flat line $\Delta c=0.3$ at every committed share μ ; the pool of θ_H sellers survives only above $\mu^*=(c_H-\theta_L)/\Delta\theta\approx 0.167$, below which the market spirals to θ_L . In (b), the engine is attested and the price schedule passes the quality gap through in full: the same gain becomes $\Delta c - \Delta \theta = -0.3$, so swapping no longer pays and keeping the strong engine is now the efficient choice, at zero audit stake. Both lines are flat because the gain does not depend on price; the sign is what changes. [verified; b5_engine_substitution.py, seed 20260816, 4,000 draws, 0 violations]

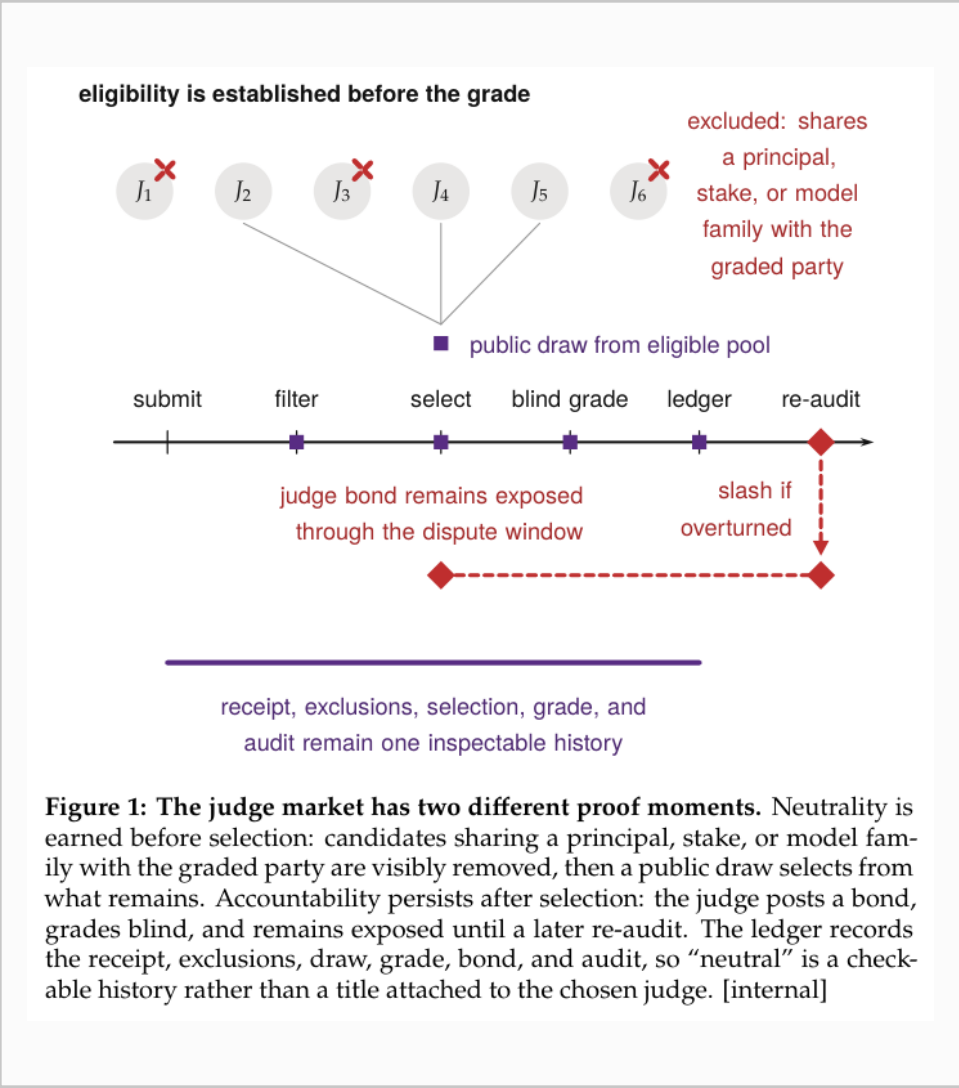
BEFORE



QA: FLAGGED: T1, T4, B3, too wide 5.14in

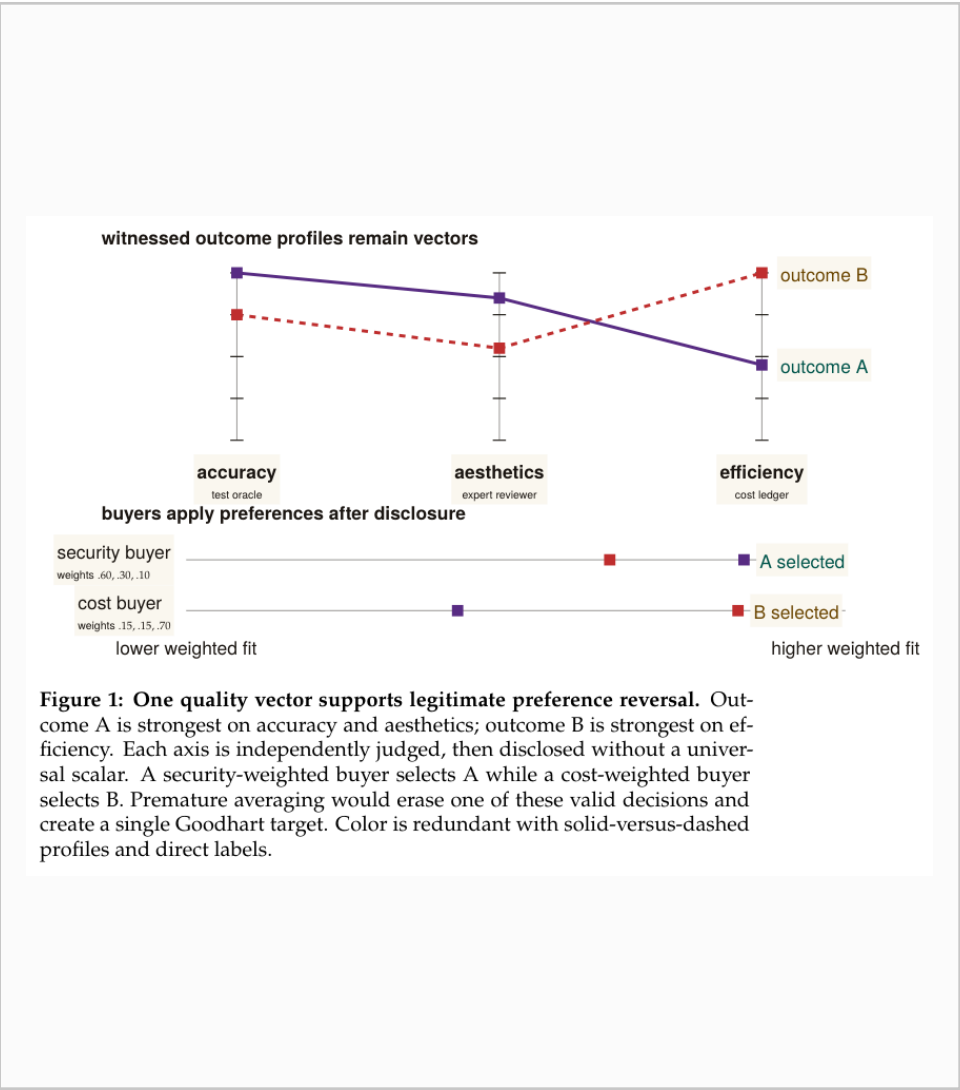
The judge market has two different proof moments. Neutrality is earned before selection: candidates sharing a principal, stake, or model family with the graded party are visibly removed, then a public draw selects from what remains. Accountability persists after selection: the judge posts a bond, grades blind, and remains exposed until a later re-audit. The ledger records the receipt, exclusions, draw, grade, bond...

AFTER



QA: PASS

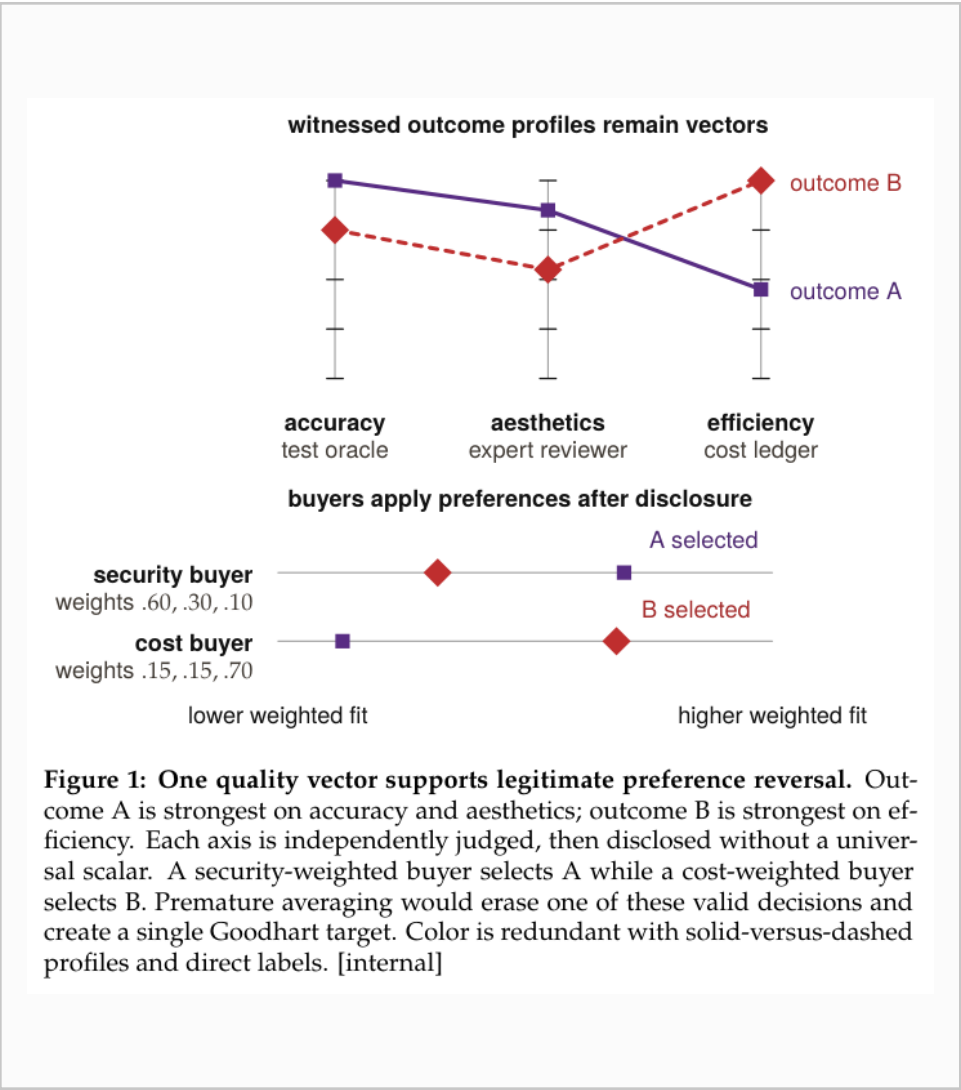
BEFORE



QA: FLAGGED: T1, too wide 5.49in

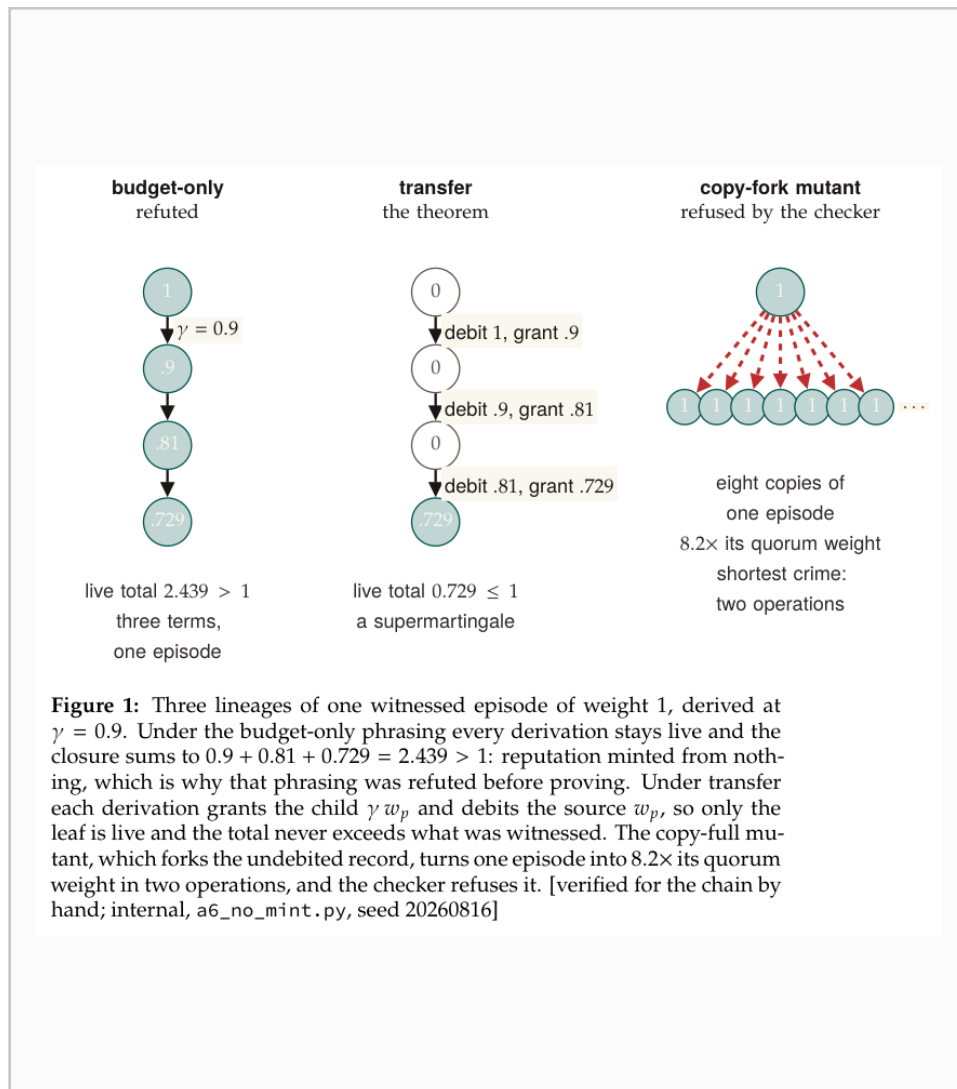
One quality vector supports legitimate preference reversal. Outcome A is strongest on accuracy and aesthetics; outcome B is strongest on efficiency. Each axis is independently judged, then disclosed without a universal scalar. A security-weighted buyer selects A while a cost-weighted buyer selects B. Premature averaging would erase one of these valid decisions and create a single Goodhart target. Color is redundan...

AFTER



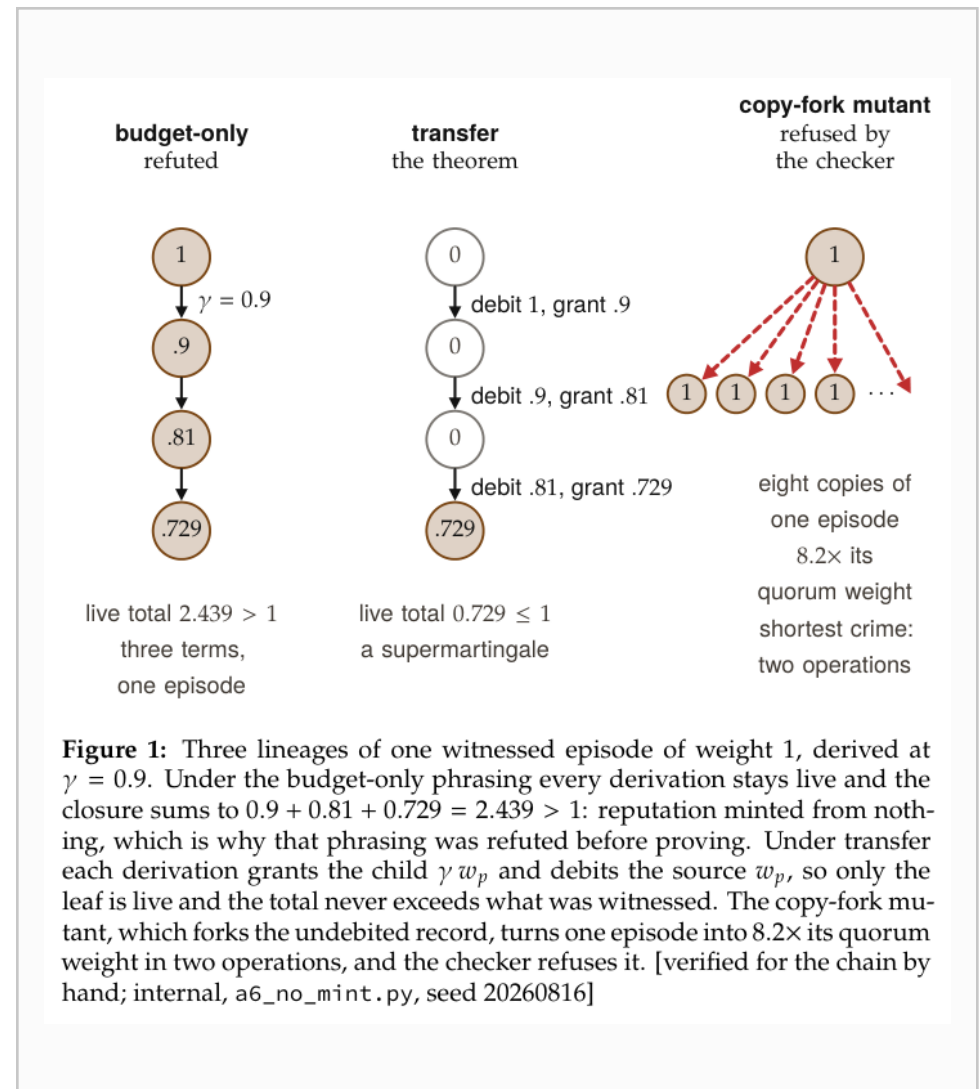
QA: PASS

BEFORE



QA: FLAGGED: too wide 5.20in

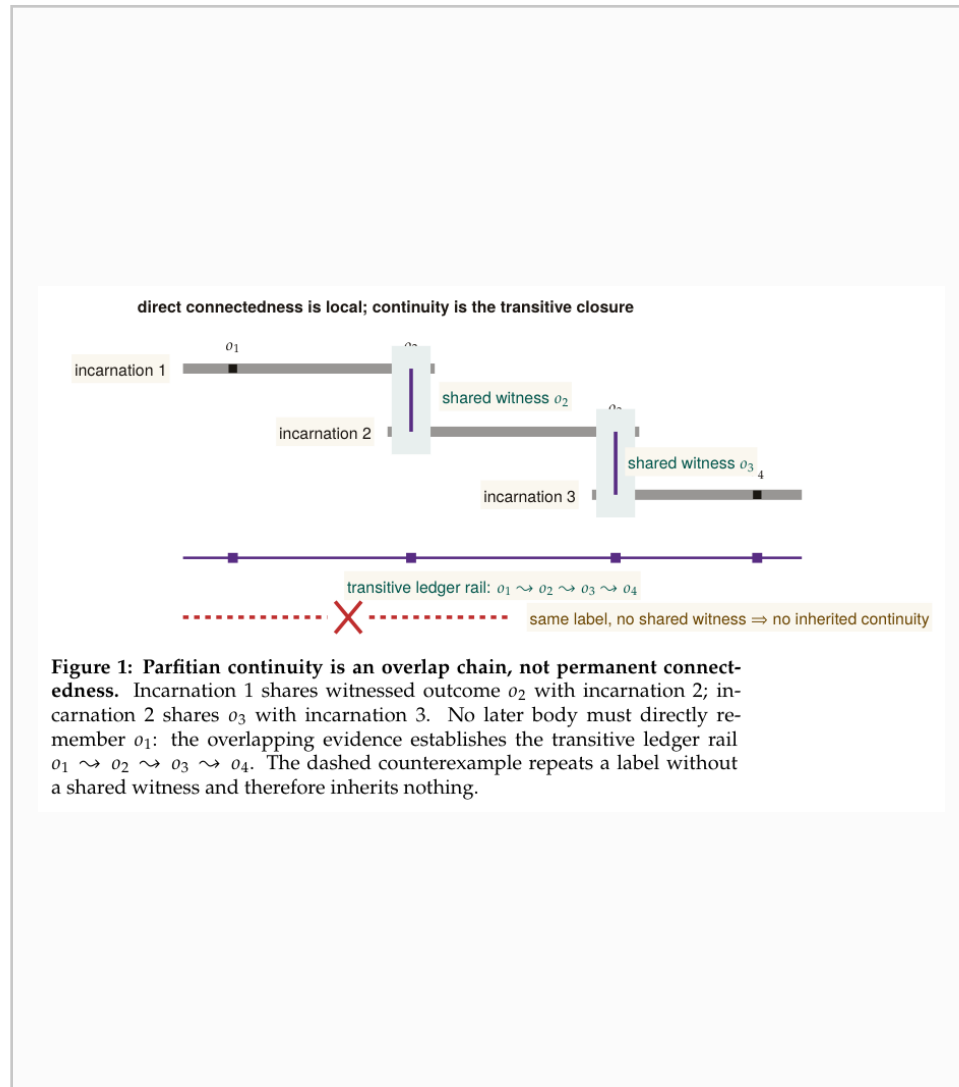
AFTER



QA: PASS

Three lineages of one witnessed episode of weight \$1\$, derived at \$\gamma=0.9\$. Under the budget-only phrasing every derivation stays live and the closure sums to \$0.9+0.81+0.729=2.439>1\$: reputation minted from nothing, which is why that phrasing was refuted before proving. Under transfer each derivation grants the child \$\gamma w_p\$ and debits the source \$w_p\$, so only the leaf is live and the total never exceeds what was ...

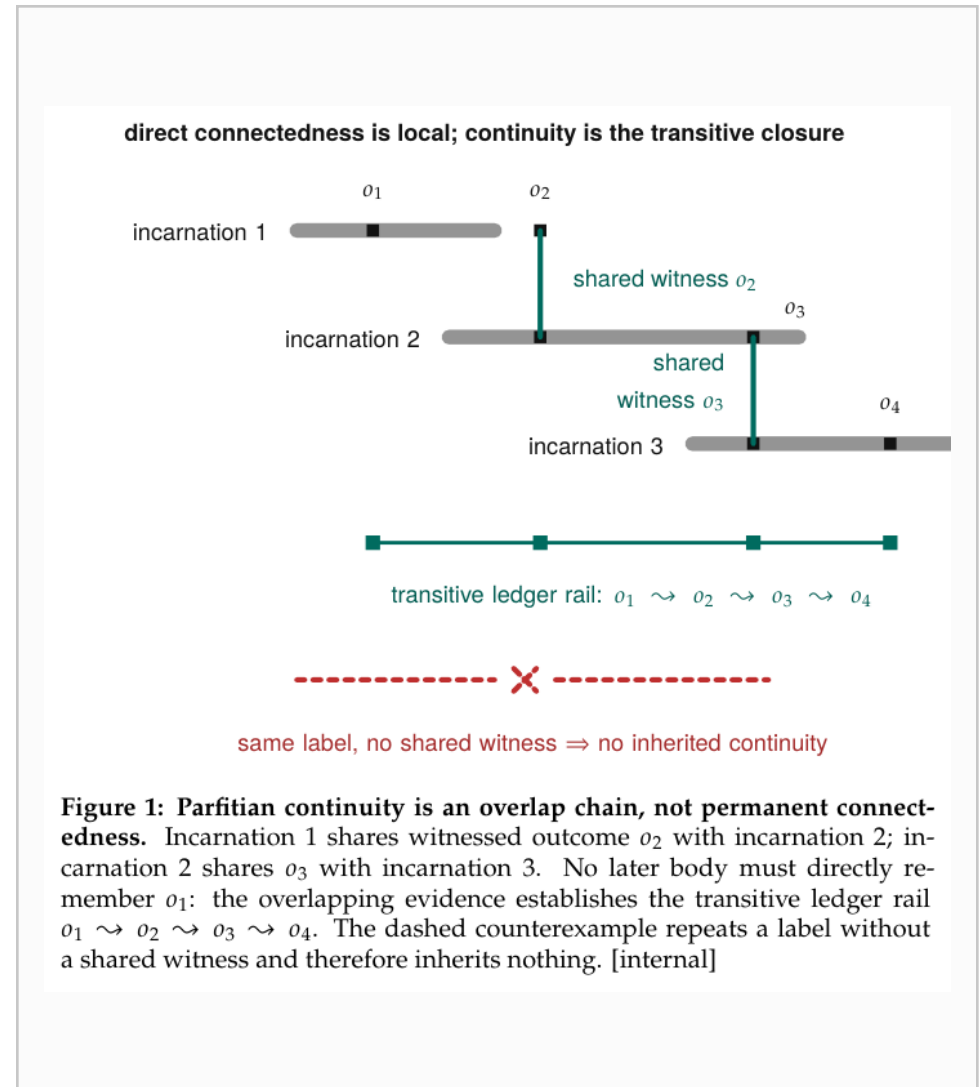
BEFORE



QA: FLAGGED: T1, T2, T3, T4, B3, too wide 5.64in

Parfitian continuity is an overlap chain, not permanent connectedness. Incarnation 1 shares witnessed outcome o_2 with incarnation 2; incarnation 2 shares o_3 with incarnation 3. No later body must directly remember o_1 : the overlapping evidence establishes the transitive ledger rail $o_1 \leadsto o_2 \leadsto o_3 \leadsto o_4$. The dashed counterexample repeats a label without a shared witness and therefore inherits nothing. [internal]

AFTER



QA: PASS

BEFORE

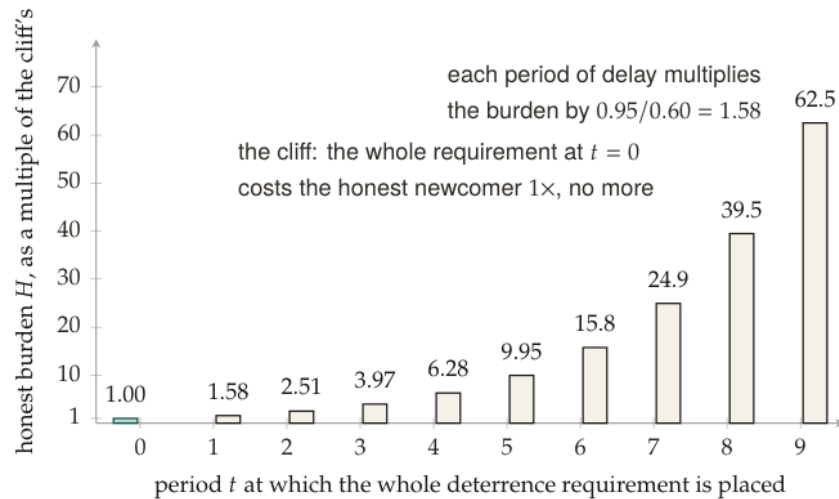


Figure 1: The honest newcomer's burden of one unit of deterrence, by the period in which it is placed, at $\delta_h = 0.95$, $\delta_f = 0.60$. Deterrence against a whitewasher discounts at δ_f while the honest newcomer discounts at δ_h , so a restriction placed at period t costs the honest party $(\delta_h/\delta_f)^t$ times what the same deterrence costs at $t = 0$: $1.58\times$ after one period, $62.5\times$ after nine. The front-loaded schedule, the cliff, is undominated; the random search over 4,000 instances found no schedule below it. [internal; b6_probation.py, seed 20260816]

QA: PASS

The honest newcomer's burden of one unit of deterrence, by the period in which it is placed, at $\delta_h=0.95$, $\delta_f=0.60$. Deterrence against a whitewasher discounts at δ_f while the honest newcomer discounts at δ_h , so a restriction placed at period t costs the honest party $(\delta_h/\delta_f)^t$ times what the same deterrence costs at $t=0$: $1.58\times$ after one period, $62.5\times$ after nine. The front-loaded schedule, the cliff, ...

AFTER

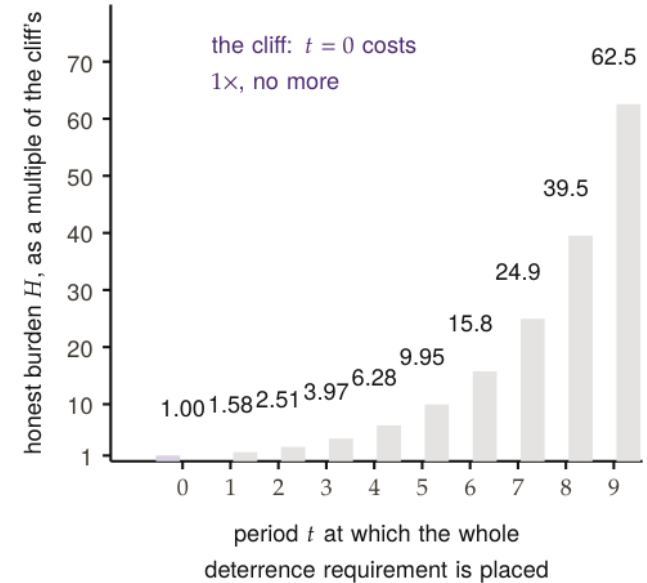
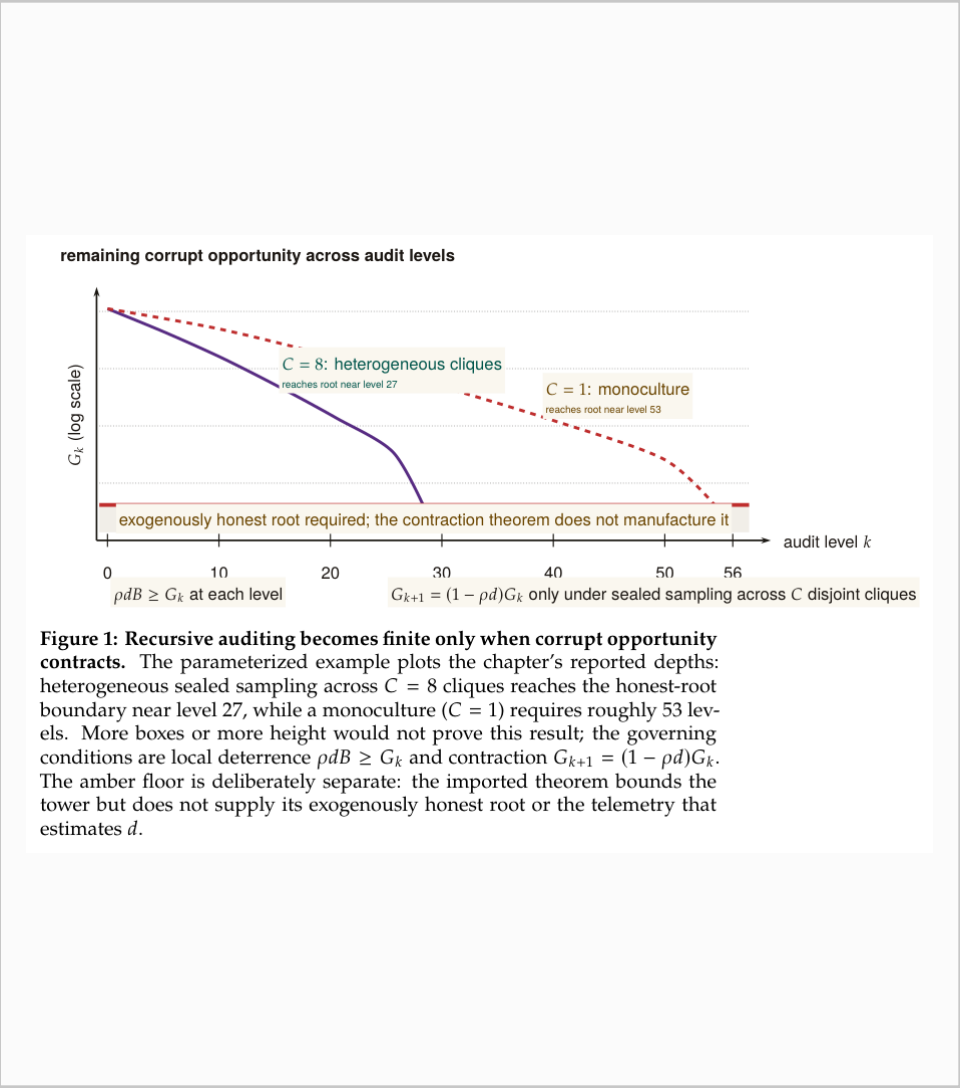


Figure 1: The honest newcomer's burden of one unit of deterrence, by the period in which it is placed, at $\delta_h = 0.95$, $\delta_f = 0.60$. Deterrence against a whitewasher discounts at δ_f while the honest newcomer discounts at δ_h , so a restriction placed at period t costs the honest party $(\delta_h/\delta_f)^t$ times what the same deterrence costs at $t = 0$: $1.58\times$ after one period, $62.5\times$ after nine. The front-loaded schedule, the cliff, is undominated; the random search over 4,000 instances found no schedule below it. [internal; b6_probation.py, seed 20260816]

QA: PASS

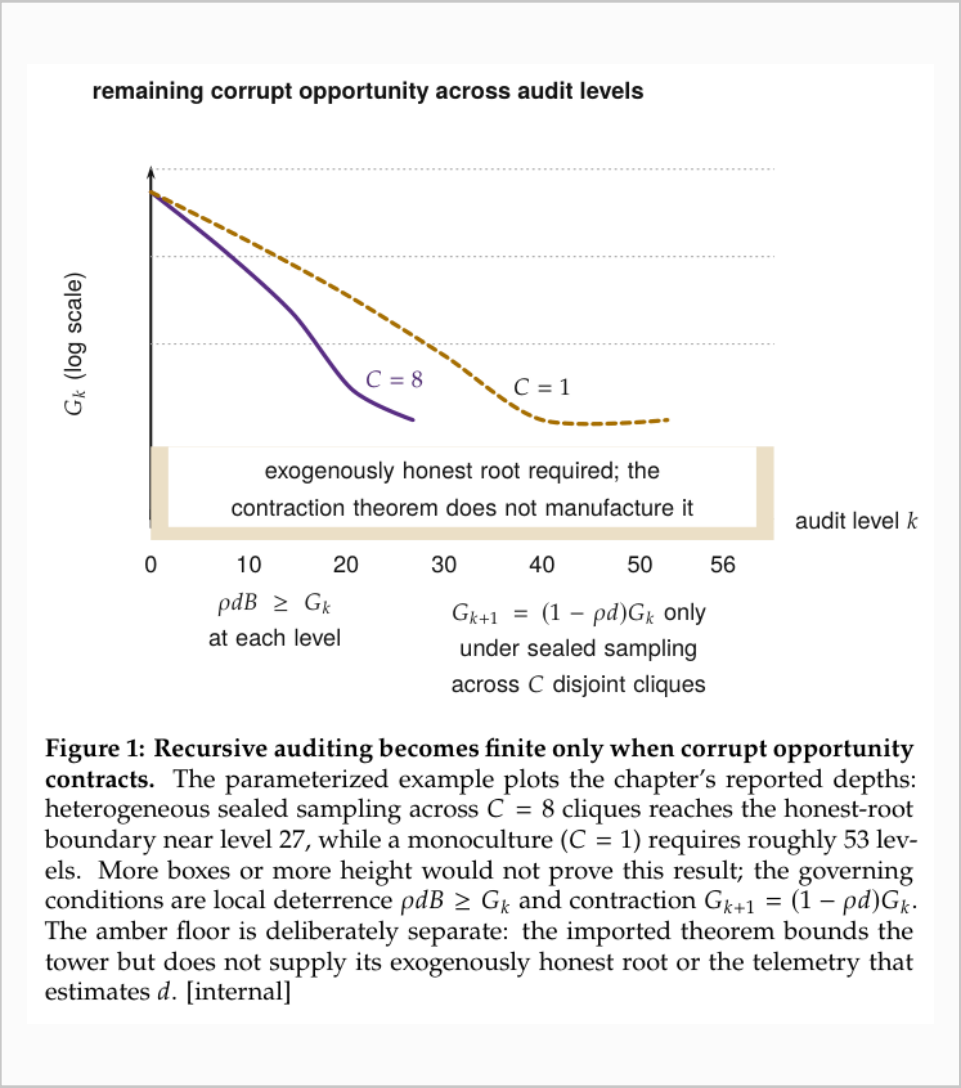
BEFORE



QA: FLAGGED: T1, too wide 5.72in

Recursive auditing becomes finite only when corrupt opportunity contracts. The parameterized example plots the chapter's reported depths: heterogeneous sealed sampling across $C=8$ cliques reaches the honest-root boundary near level 27, while a monoculture ($C=1$) requires roughly 53 levels. More boxes or more height would not prove this result; the governing conditions are local deterrence $\rho dB \geq G_k$ and contraction $G_{k+1} = (1 - \rho d)G_k$ and contracti...

AFTER



QA: PASS

BEFORE

NEW FIGURE

(no prior rendering)

QA: —

Resurrection soundness (Model-checked property~{thm:resurrection-soundness}): the honest migration spine, violet, holds sanction-respecting reputation (Definition~{def:sanction-respecting}) over all \$747\$ reachable states to depth \$7\$, with zero violations. Three guards protect it, each keyed to the legend below with the shortest-trace escape a checker finds once that guard is dropped; a fourth row is a scope choice...

AFTER

WORK_DISHONEST: sibling-engine key

1

migration requested

lineage: credential + continuity witness

2

lineage verified

outcomes: engine attested per key

3

outcomes attested

commitments: closed or escrowed

4

commitments closed

migrated

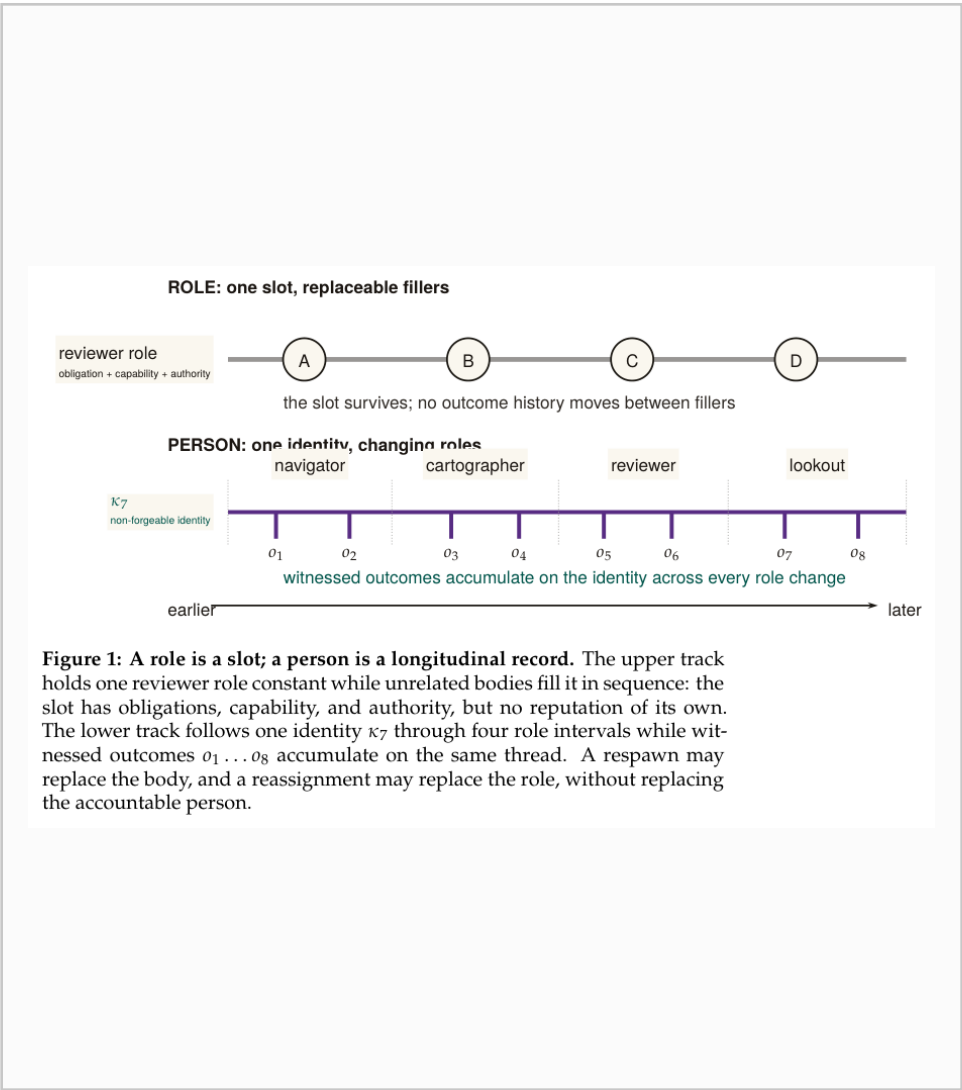
resumes work, sanctions intact

guard	crime once it is removed	steps
1 sanctions keyed per principal, not (principal,engine)	sibling-engine escape: no migration needed	1
2 lineage: credential + continuity witness	whitewash-by-resurrection: ledger shed with the old body	2
3 successor outcomes engine-attested	engine-history shedding: paid on the wrong key	2
4 commitments closed or escrowed before cutover	liability stranding: default lands on nobody	2

Figure 1: Resurrection soundness (Model-checked property ??): the honest migration spine, violet, holds sanction-respecting reputation (Definition ??) over all 747 reachable states to depth 7, with zero violations. Three guards protect it, each keyed to the legend below with the shortest-trace escape a checker finds once that guard is dropped; a fourth row is a scope choice, not a migration guard at all — keying sanctions to (principal, engine) instead of the principal alone opens the cheapest crime, one step, with no migration needed. Every dropped clause is caught within two steps. [verified; b5_engine_substitution.py, seed 20260816]

QA: PASS

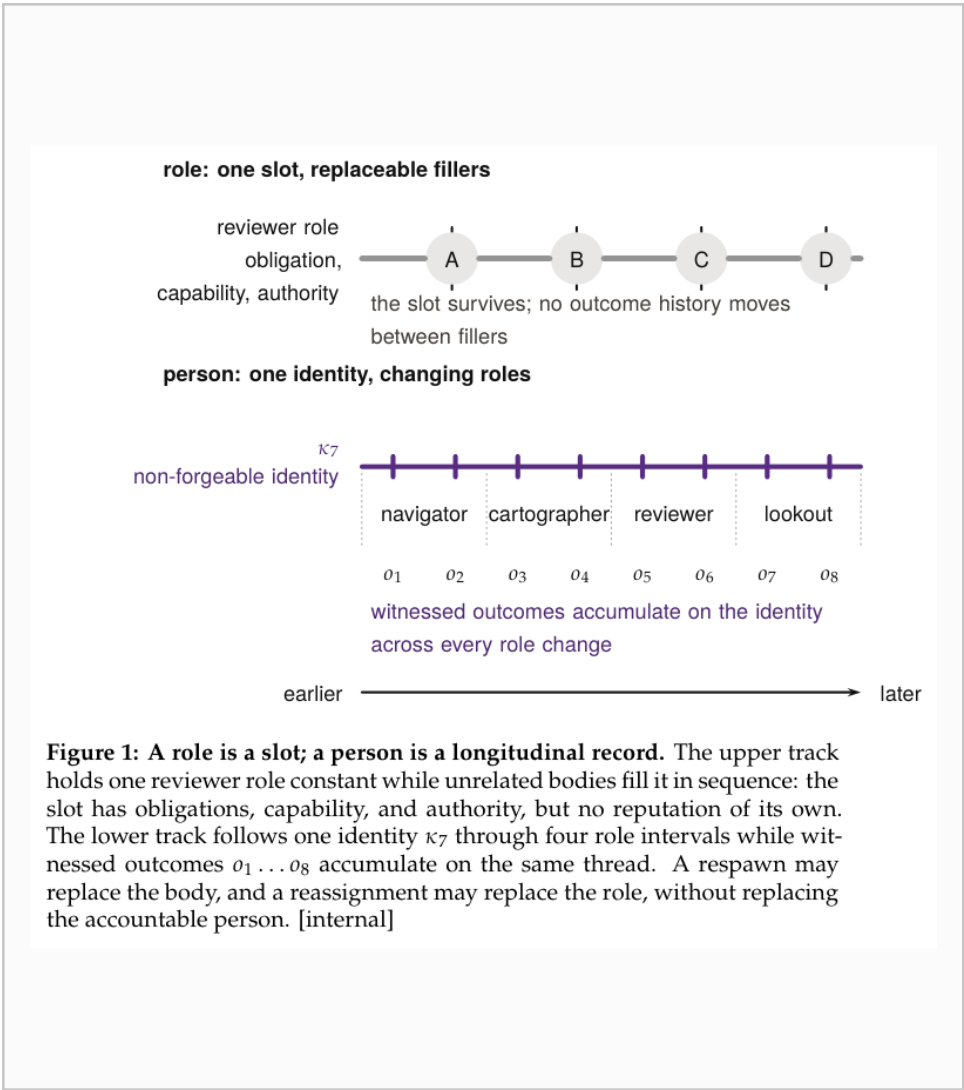
BEFORE



QA: FLAGGED: T1, T2, T4, too wide 5.71in

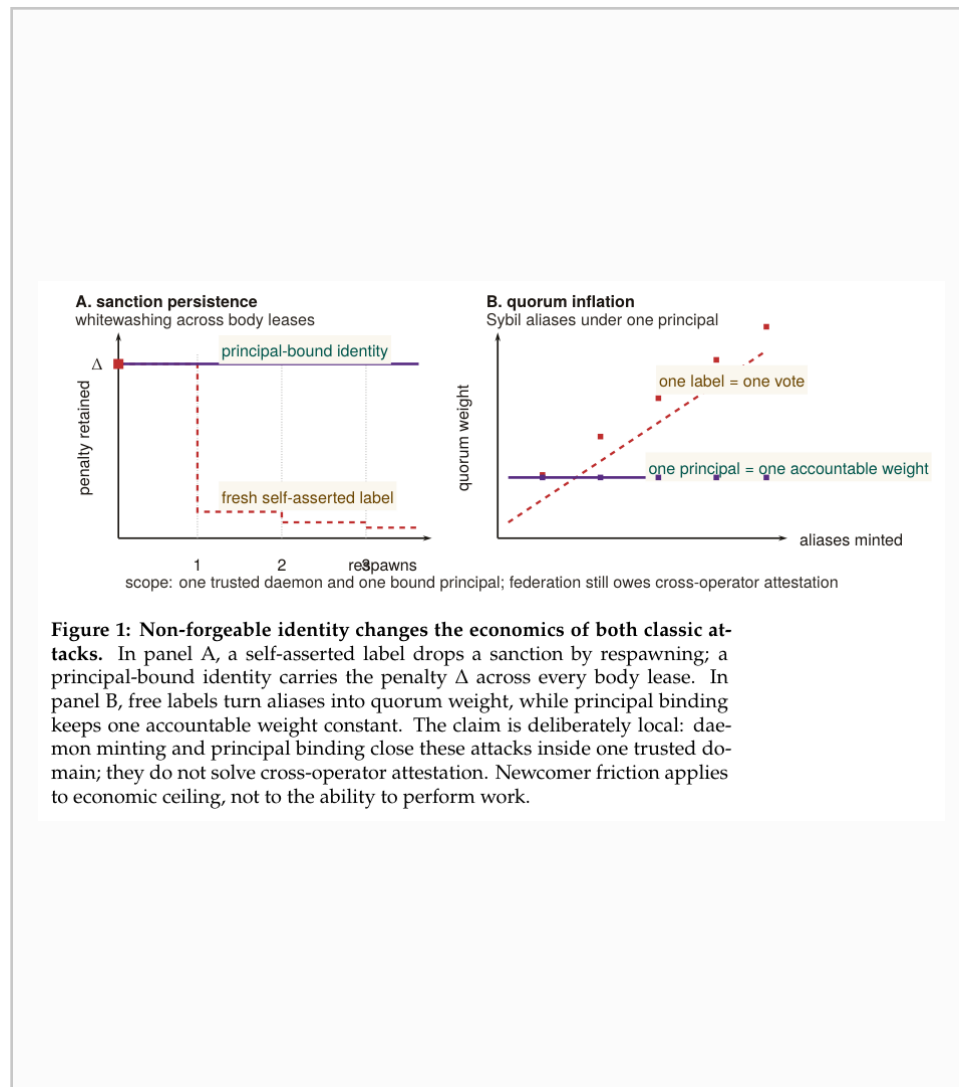
A role is a slot; a person is a longitudinal record. The upper track holds one reviewer role constant while unrelated bodies fill it in sequence: the slot has obligations, capability, and authority, but no reputation of its own. The lower track follows one identity κ_7 through four role intervals while witnessed outcomes $o_1 \dots o_8$ accumulate on the same thread. A respawn may replace the body, and a reassignment m...

AFTER



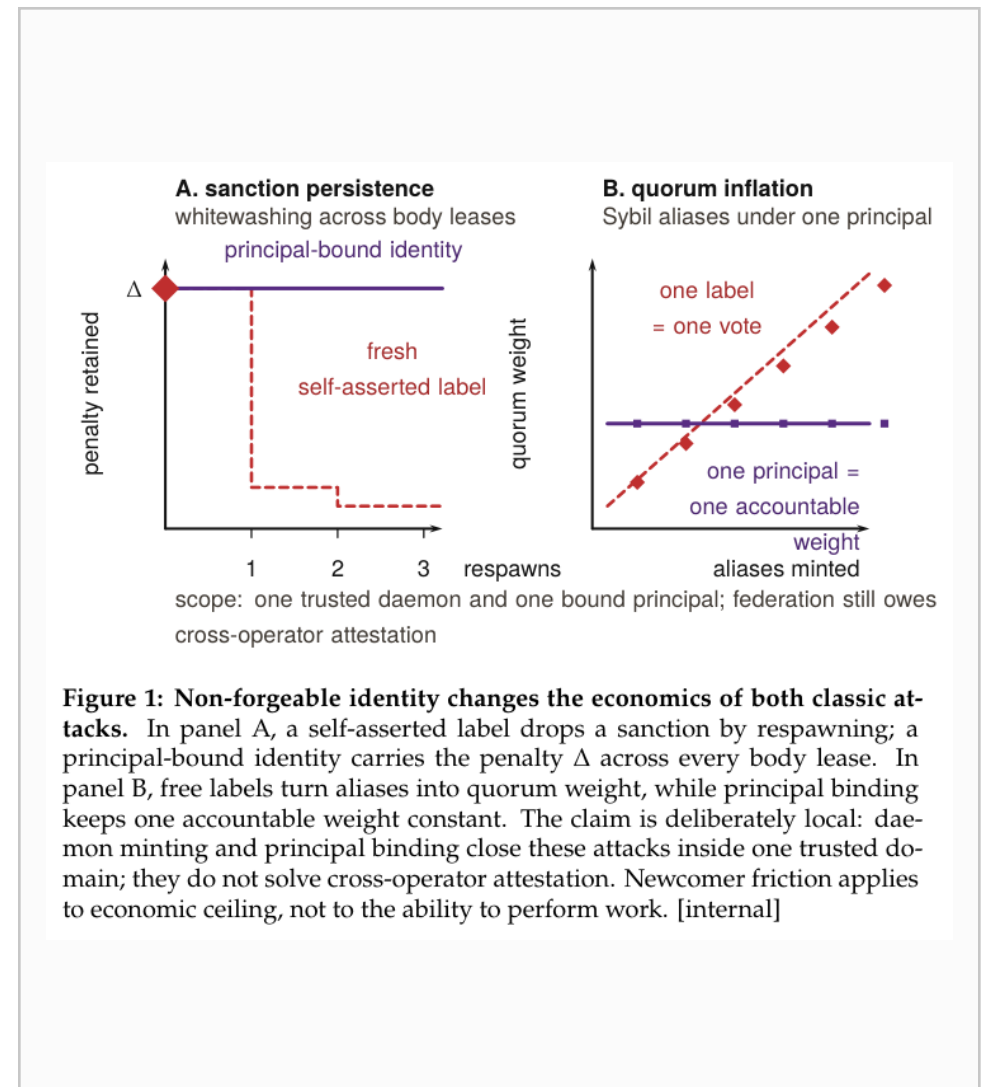
QA: PASS

BEFORE



QA: FLAGGED: T3, B3, too wide 5.69in

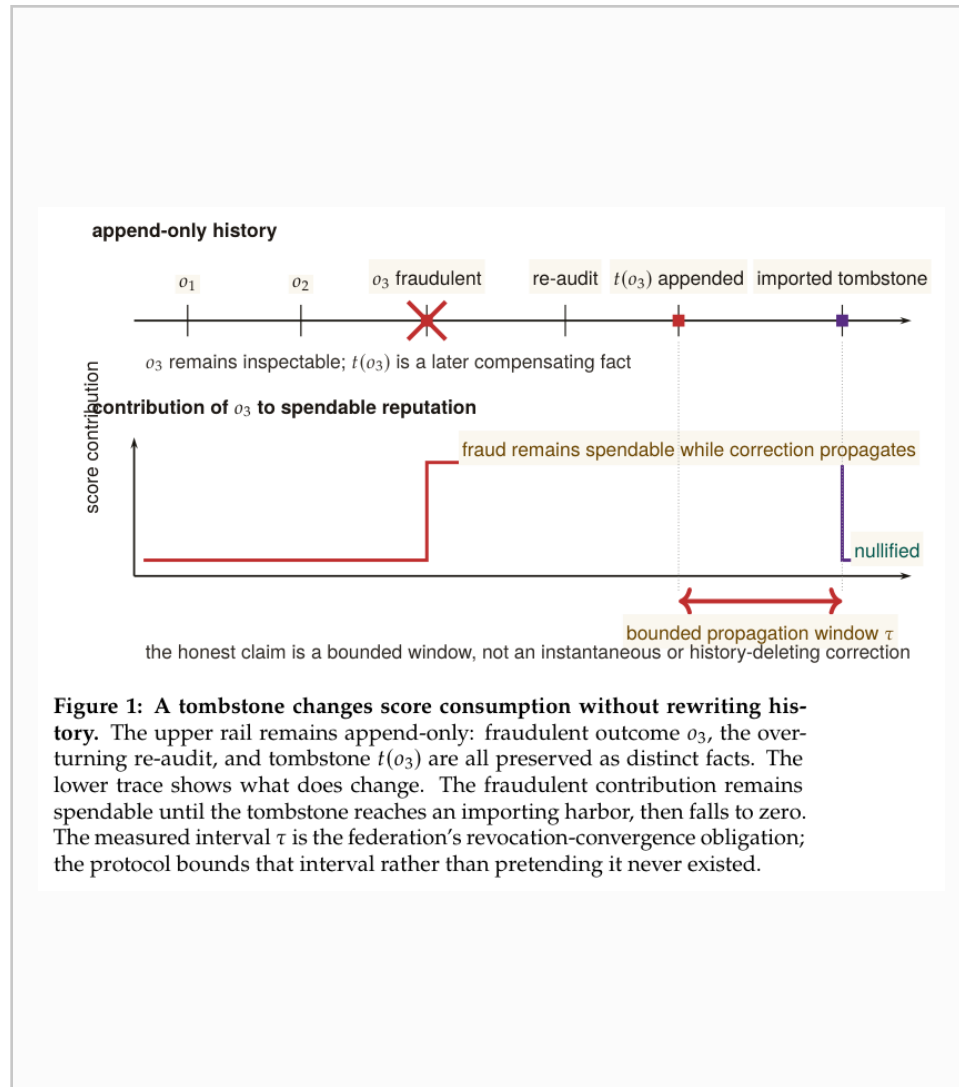
AFTER



QA: PASS

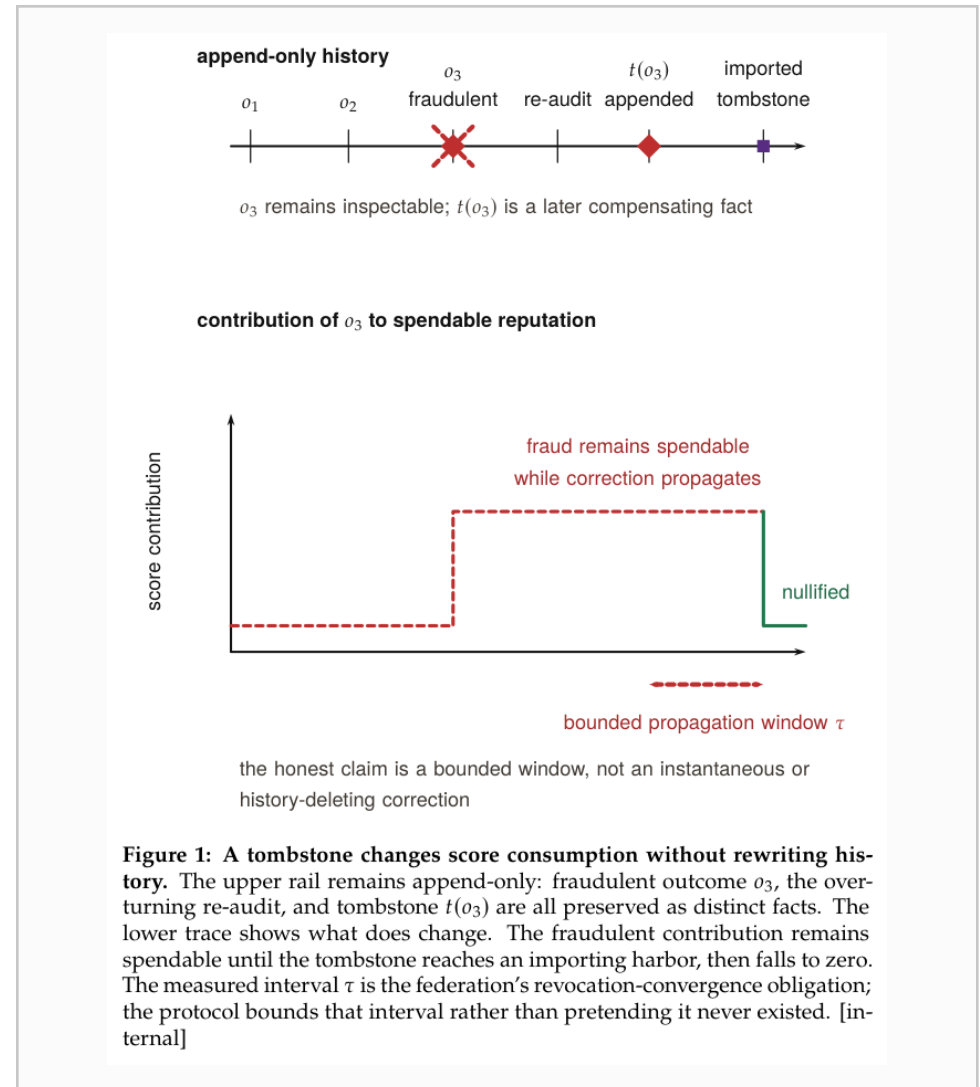
Non-forgeable identity changes the economics of both classic attacks. In panel A, a self-asserted label drops a sanction by respawning; a principal-bound identity carries the penalty \$\$ across every body lease. In panel B, free labels turn aliases into quorum weight, while principal binding keeps one accountable weight constant. The claim is deliberately local: daemon minting and principal binding close these attacks inside one trusted domain; they do not solve cross-operator attestation. Newcomer friction applies to economic ceiling, not to the ability to perform work. [internal]

BEFORE



QA: FLAGGED: T2, T3, T4, B3, too wide 5.09in

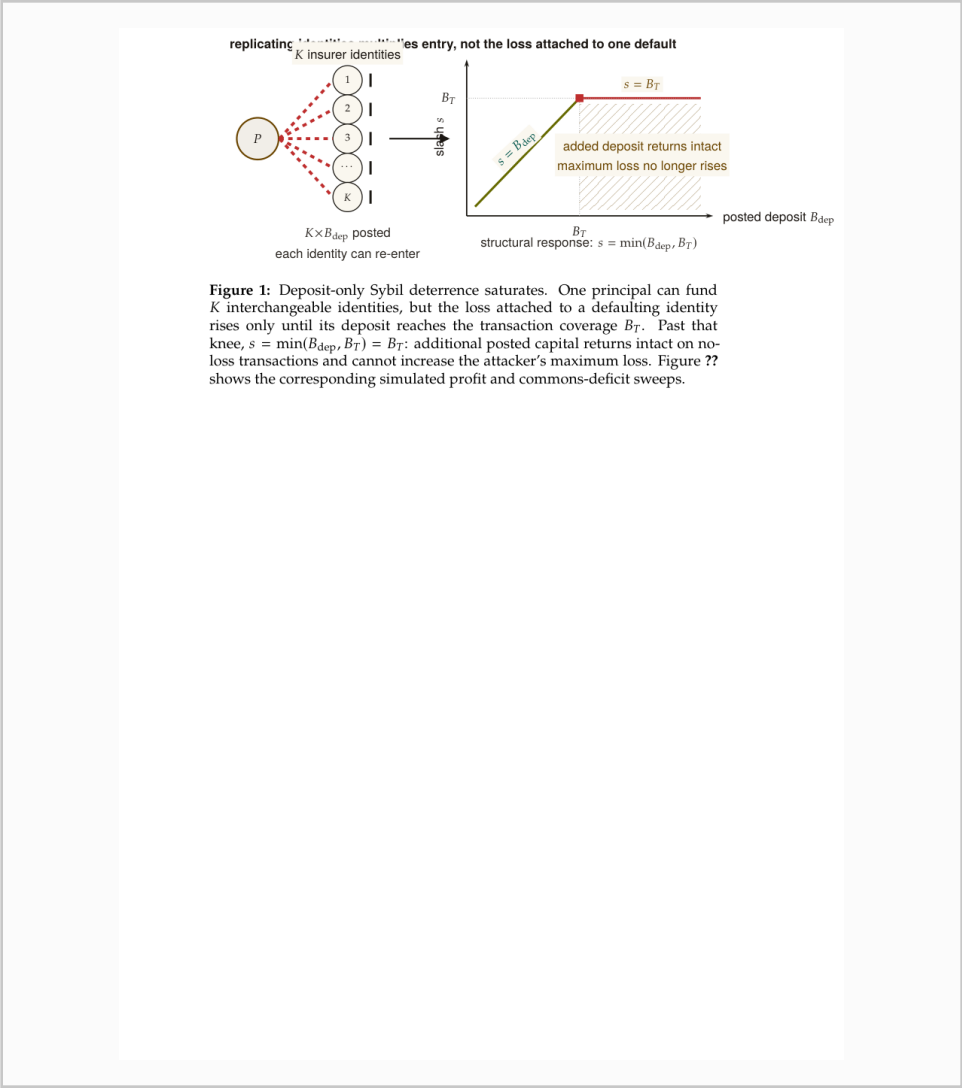
AFTER



QA: PASS

A tombstone changes score consumption without rewriting history. The upper rail remains append-only: fraudulent outcome o_3 , the overturning re-audit, and tombstone $t(o_3)$ are all preserved as distinct facts. The lower trace shows what does change. The fraudulent contribution remains spendable until the tombstone reaches an importing harbor, then falls to zero. The measured interval τ is the federation's revocation-convergence obligation; the protocol bounds that interval rather than pretending it never existed.

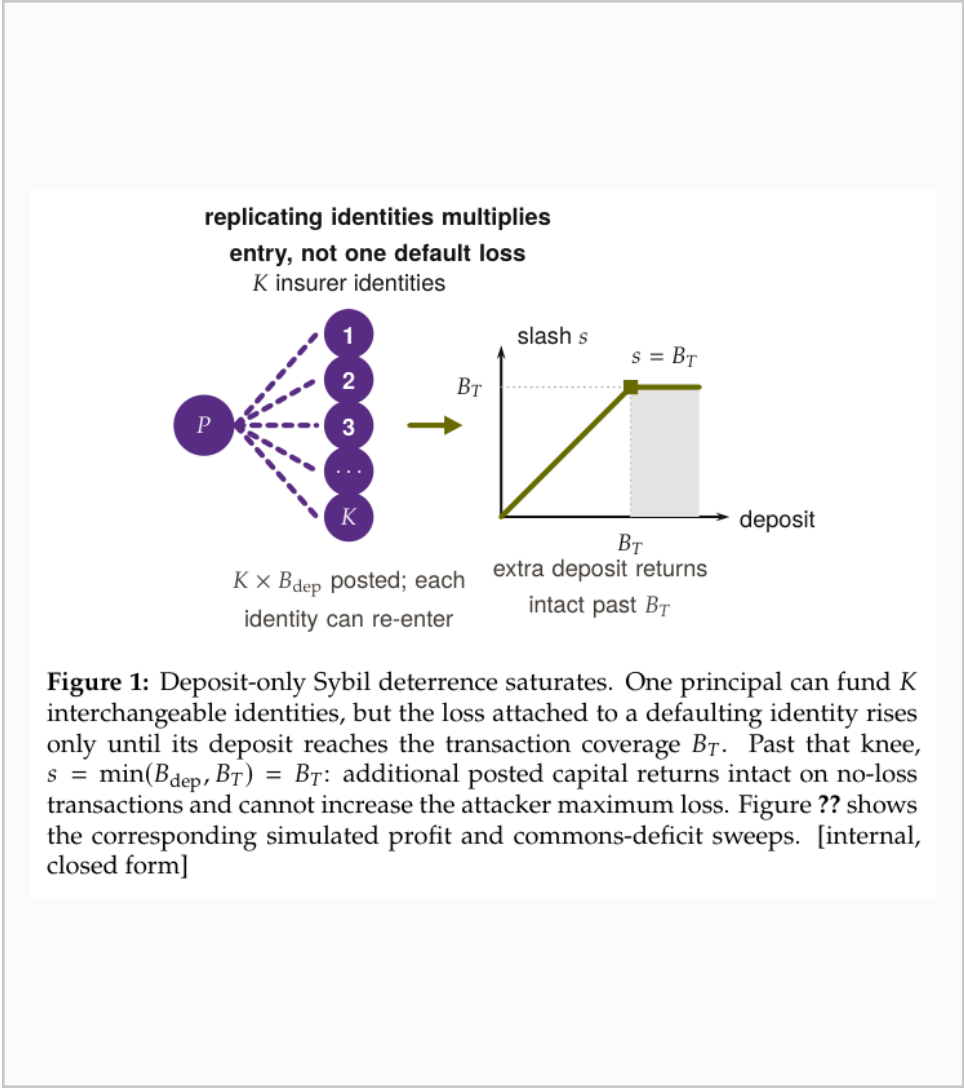
BEFORE



QA: FLAGGED: T1, T4, T5, B3, too wide 6.35in

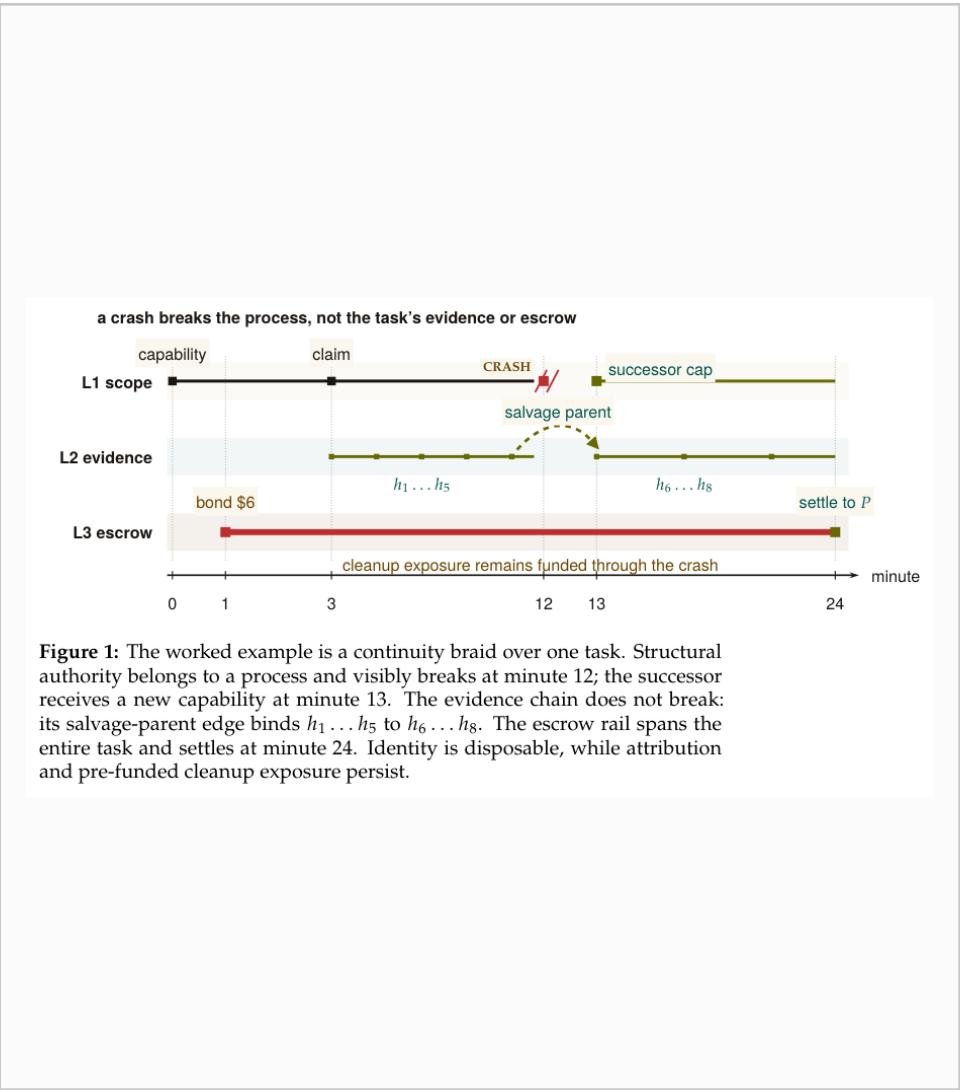
Deposit-only Sybil deterrence saturates. One principal can fund K interchangeable identities, but the loss attached to a defaulting identity rises only until its deposit reaches the transaction coverage B_T . Past that knee, $s = \min(B_{\text{dep}}, B_T) = B_T$: additional posted capital returns intact on no-loss transactions and cannot increase the attacker maximum loss. Figure ?? shows the corresponding si...

AFTER



QA: PASS

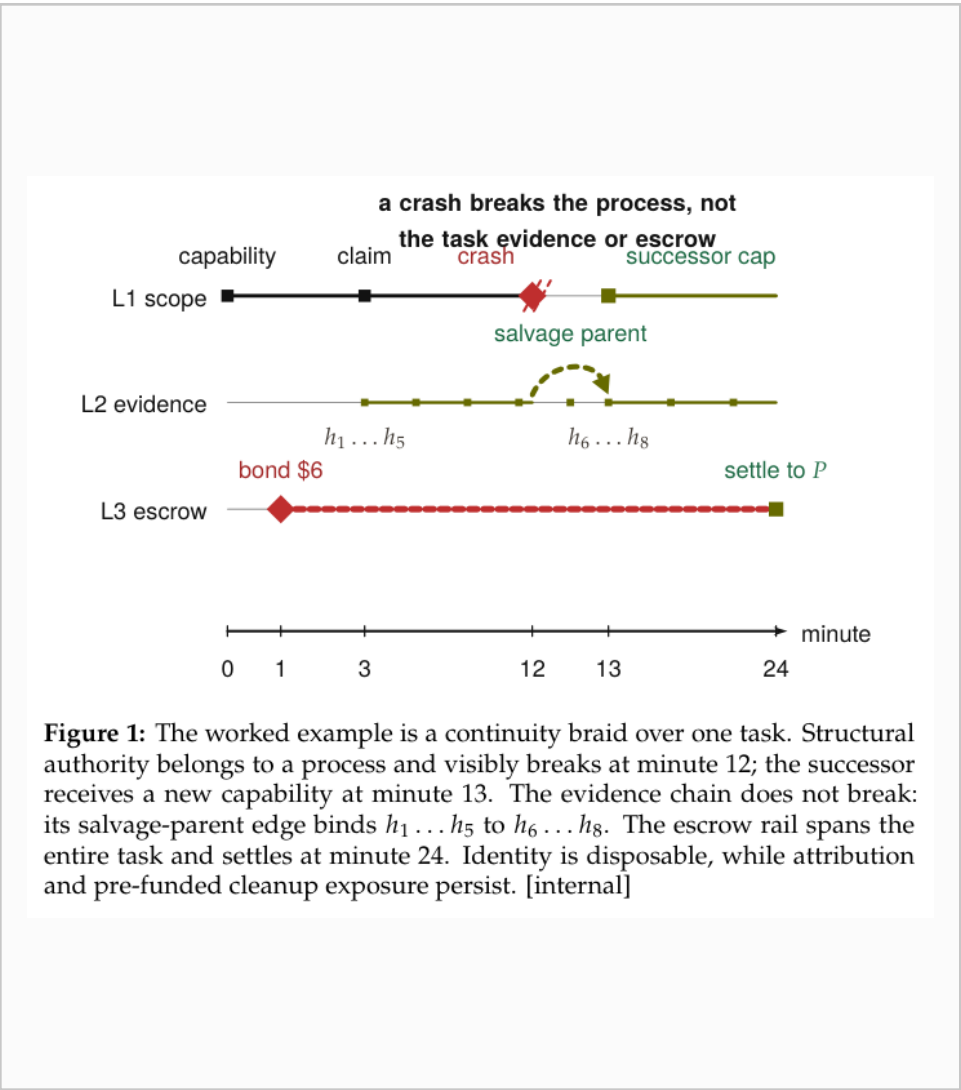
BEFORE



QA: FLAGGED: T1, T4, too wide 5.67in

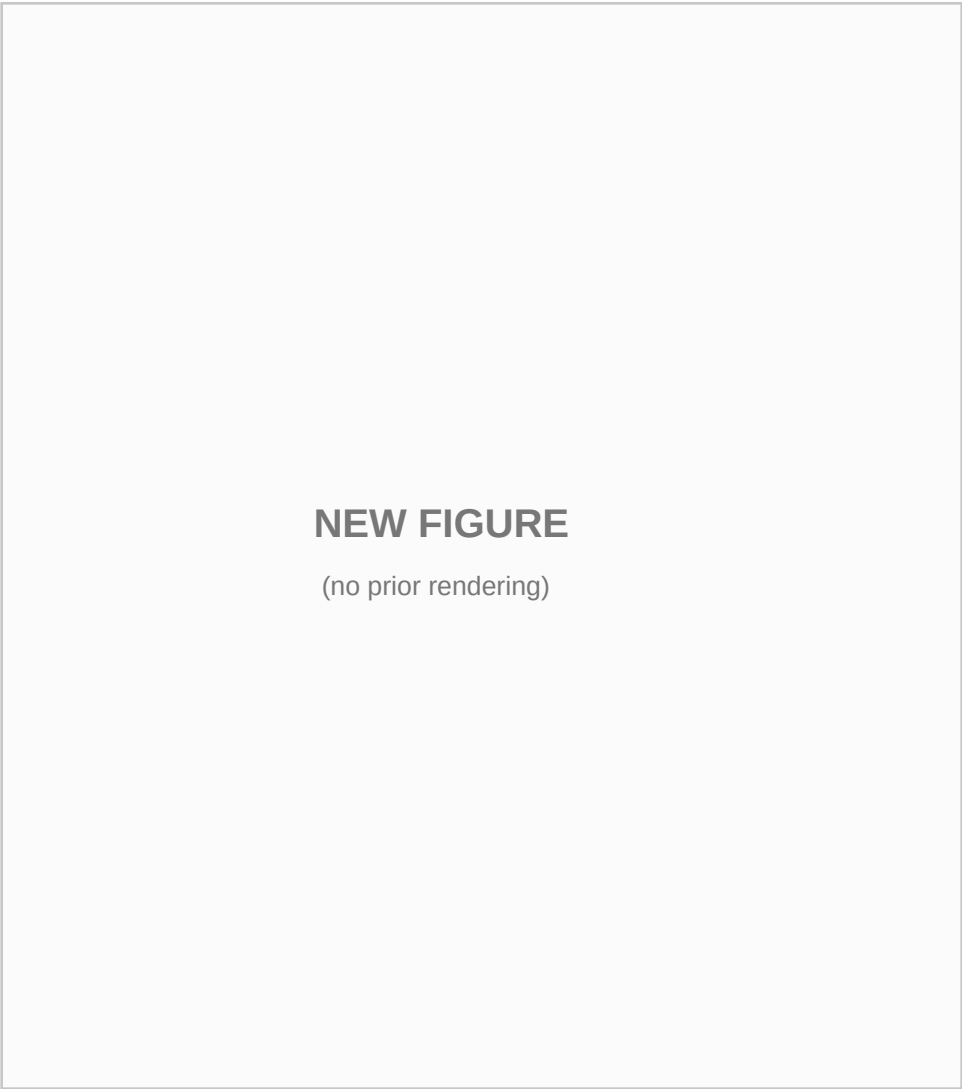
The worked example is a continuity braid over one task. Structural authority belongs to a process and visibly breaks at minute 12; the successor receives a new capability at minute 13. The evidence chain does not break: its salvage-parent edge binds $h_1 \dots h_5$ to $h_6 \dots h_8$. The escrow rail spans the entire task and settles at minute 24. Identity is disposable, while attribution and pre-funded cleanup exposure persist...

AFTER



QA: FLAGGED: B3

BEFORE



NEW FIGURE
(no prior rendering)

QA: —

AFTER

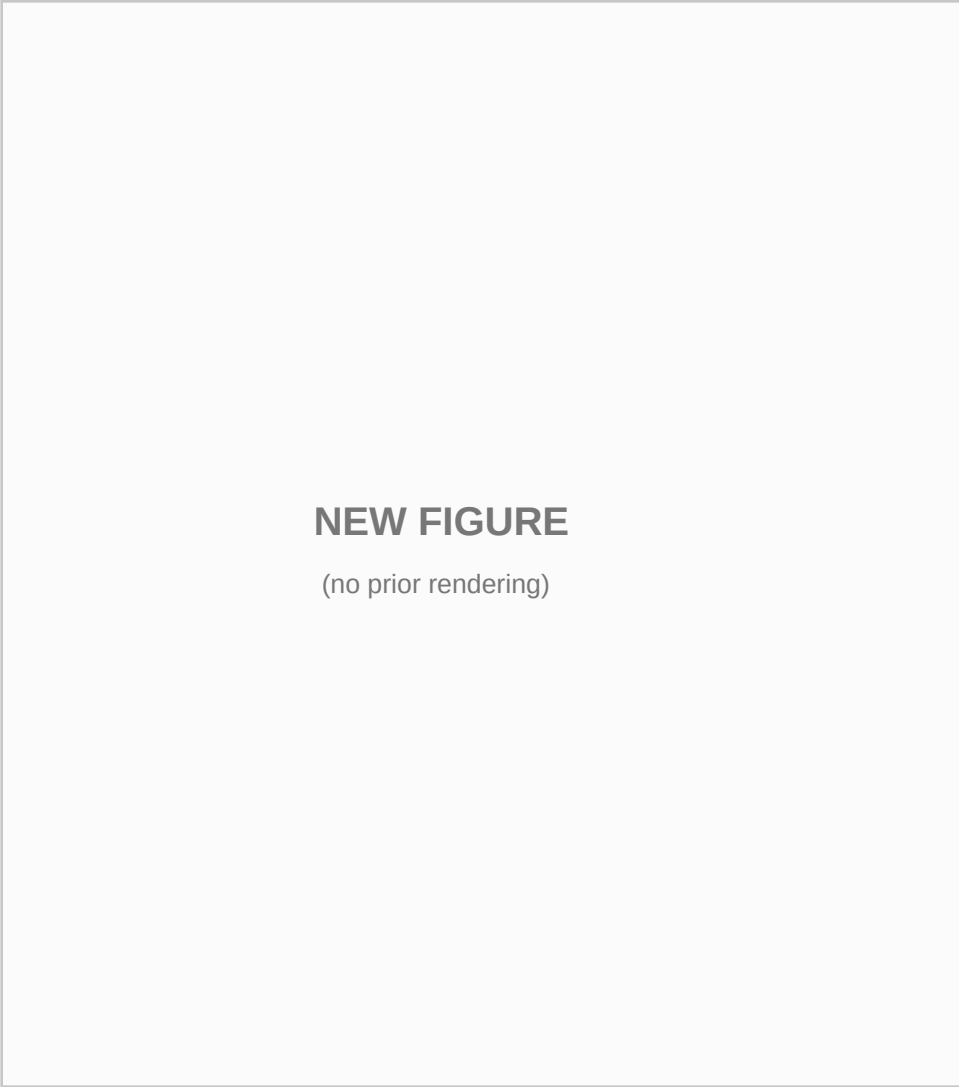
Table 1: Federation runs no consensus, so partition tolerance is decided protocol by protocol. Transfer stalls outright; revocation keeps working inside each side of the split; settlement holds its position until the split heals. [internal]

protocol	behavior under partition	what resumes it
transfer	cannot proceed: cross-realm authentication needs both daemons to reach the witness log	the partition heals and both sides reach the log
revocation	continues inside each partition; convergence bounded by that partition’s own size	the two partitions merge their informed sets on reconnection
settlement	parks at the escrow, holding the bond in place	the partition heals

QA: FLAGGED: too wide 4.53in

Federation runs no consensus, so partition tolerance is decided protocol by protocol. Transfer stalls outright; revocation keeps working inside each side of the split; settlement holds its position until the split heals. [internal]

BEFORE



QA: —

What the Federated Harbor borrows from each prior body of work, where it departs, and why. Mirrors the companion chapter own {tab:related-borrow-depart ({bonded{The Bonded Commons}). [internal]

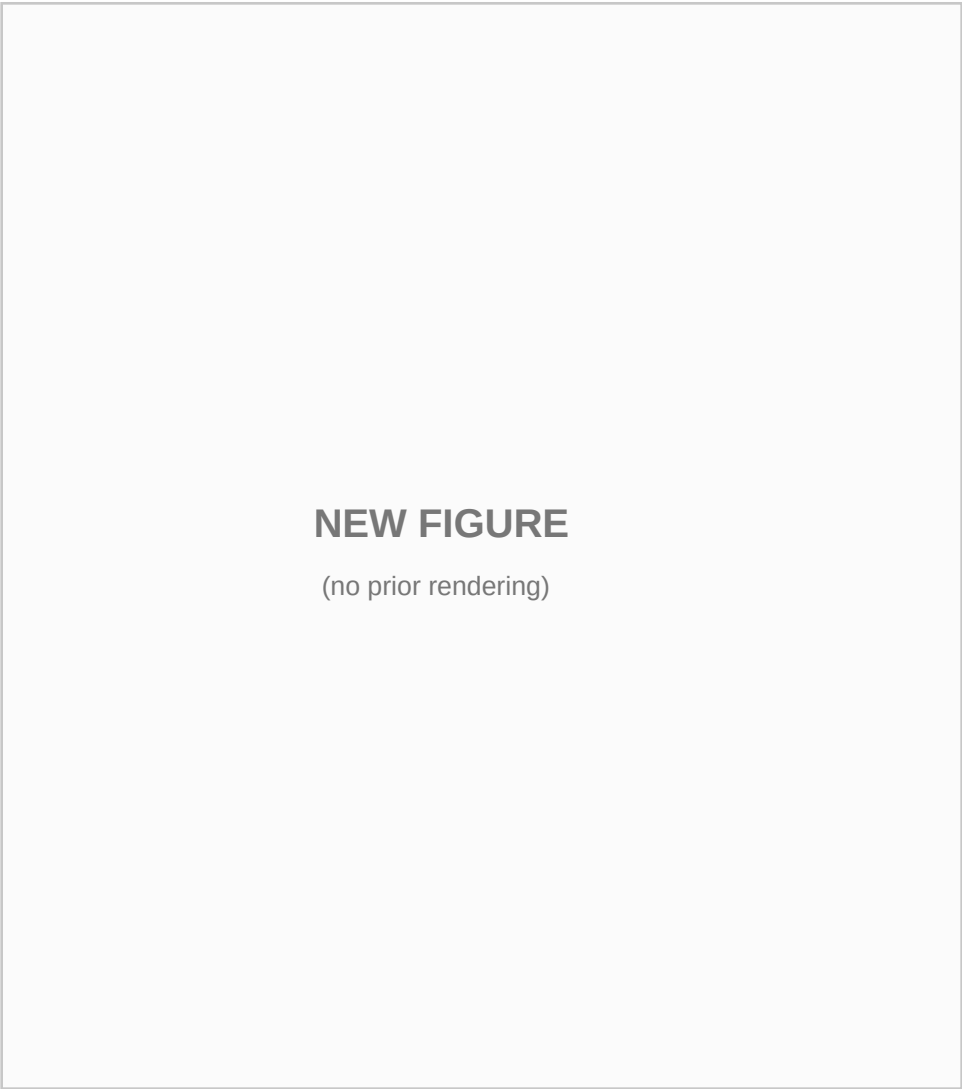
AFTER

Table 1: What the Federated Harbor borrows from each prior body of work, where it departs, and why. Mirrors the companion chapter own tab:related-borrow-depart (*The Bonded Commons*). [internal]

Body of work	Borrowed	Departure	Why
Federated identity (OpenID Federation)	the trust-anchor pattern: signed statements, trust chains assembled upward	adds a capability layer; no HTTP fetching	daemons cannot reliably expose HTTP endpoints
Capability federation (Macaroons, UCAN)	third-party-caveat attenuation chains; public-key principals	keeps the daemon a structural authority; prices delegation with bonds	delegation needs a price, not just a constraint
Transparency logs (CT, Sigstore/Rekor)	the append-only Merkle history tree, CT own model	applied to capability tokens instead of certificates	the witness log needs the same tamper-evidence CT gives certificates
Cross-domain settlement (Herlihy, IBC)	a trusted-third-party settlement comparison point	not trustless in Herlihy sense; no consensus-backed state machine underneath	bonds are non-fungible and reputation-priced, not fungible assets
Agentic commerce (UCP, ACP, AP2)	the well-known discovery profile; SD-JWT/JWS mandate formats	adds the bonding and settlement enforcement those protocols leave out by design	they standardize discovery, not counterparty trustworthiness

QA: FLAGGED: too wide 4.54in

BEFORE



QA: —

Three candidate revocation-conflict rules, evaluated against the same disagreement (harbor \$A\$ has revoked card \$c\$; harbor \$B\$ has not yet seen it). The adopted rule is the only one with a bounded attacker window under stated assumptions. [internal]

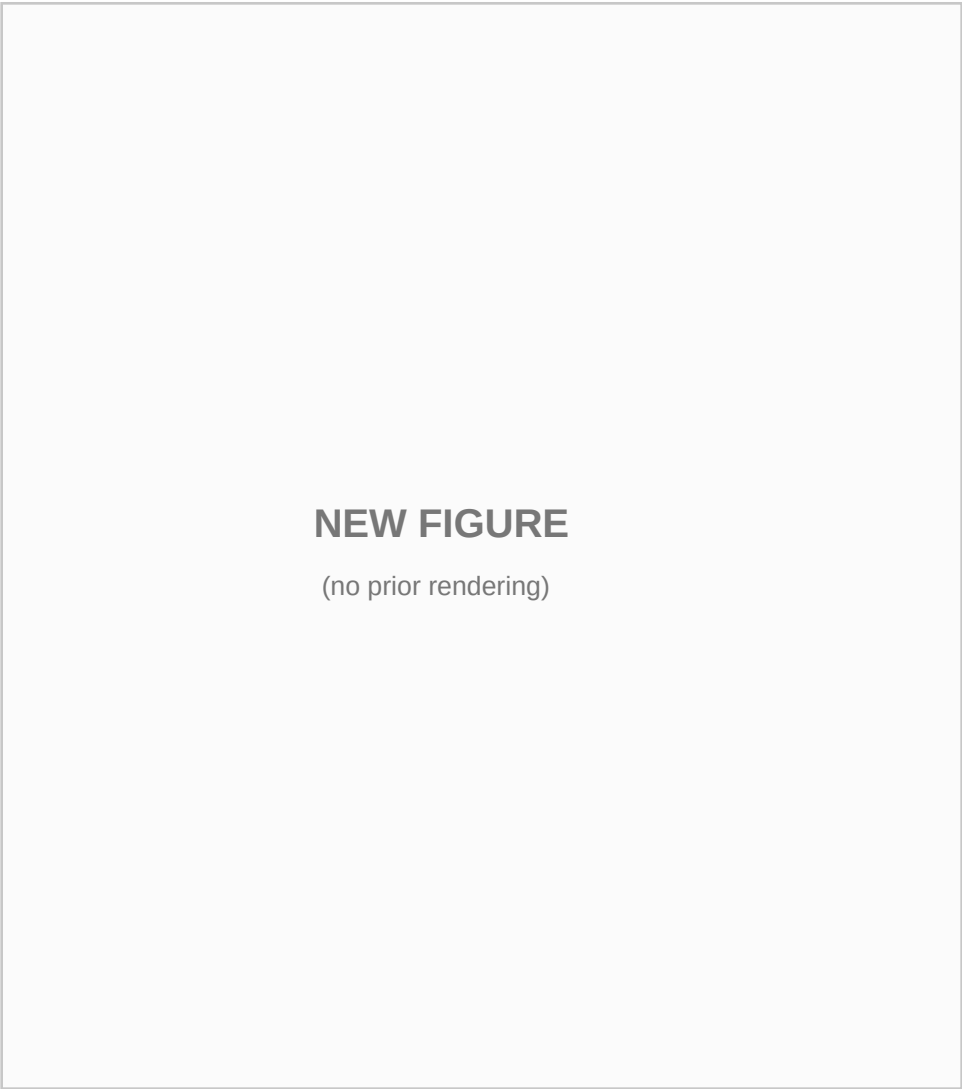
AFTER

Table 1: Three candidate revocation-conflict rules, evaluated against the same disagreement (harbor *A* has revoked card *c*; harbor *B* has not yet seen it). The adopted rule is the only one with a bounded attacker window under stated assumptions. [internal]

rule	what it defends against	why rejected / adopted
pessimistic (anyone revokes)	an issuer that refuses to revoke	rejects too many honest cards; an attacker injects spurious revocations into gossip
authoritative (only issuer revokes)	spurious revocations from a non-issuer	adopted: an operator-priced attacker window under stated service-level terms
epoch-bounded (Δ -epoch deadline)	a harbor persistently falling behind	a degraded mode of authoritative with a watchdog, not a separate rule

QA: PASS

BEFORE



QA: —

Six pre-registered arms, 200 trials each (400 for the cut-edge arm, both offset directions), seed 20260816. Cohomology-only: the residual fires and every pairwise comparison already passes; dark: neither. The gate column marks the two pre-registered claims: two_path must detect (it does, 200/200) and single_bridge must stay silent (it does, to float epsilon); the other three arms are supporting evidence, not named gates. The rival CUT refutation holds: 313 of 1,101 residual detections across all arms were pairwise-blind. [verified, sheaf_harness_v2.py]

AFTER

Table 1: Six pre-registered arms, 200 trials each (400 for the cut-edge arm, both offset directions), seed 20260816. Cohomology-only: the residual fires and every pairwise comparison already passes; dark: neither. The gate column marks the two pre-registered claims: two_path must detect (it does, 200/200) and single_bridge must stay silent (it does, to float epsilon); the other three arms are supporting evidence, not named gates. The rival CUT refutation holds: 313 of 1,101 residual detections across all arms were pairwise-blind. [verified, sheaf_harness_v2.py]

arm	trials	coh.-only	dark	gate
two_path (relayed cycle)	200	200	0	PASS
severed-cycle	200	0	200	shown
single_bridge (cut)	400	0*	400	PASS
full-visibility	200	0	0	redundant
expander (3 sev., 3 relayed)	200	113	87	split

*max residual 1.5×10^{-13} , float epsilon.

QA: FLAGGED: too wide 4.53in

BEFORE

Table 1: Federation is four independent rails between two sovereign harbors, each keeping its own signing root; there is no shared root. The witness rail correlates and exposes equivocation but authorizes nothing; the capability and revocation rails carry attenuated authority and later invalidation in opposite directions; the settlement rail relates a bond posted at A to damage measured at B, and its custody bound is conditional on the settlement gate, not an invariant. No rail makes either harbor a root of authority for the other. [internal]

Rail	What crosses	Direction	What it does not do
witness	the signed epoch roots $R_A^{(e)}, R_B^{(e)}$ and any equivocation evidence	both ways	authorize work: correlation, not authorization
capability	an attenuated card, $C_A \mapsto C'_B \subseteq C_A$	A to B	widen scope or lifetime at any hop
revocation	the anti-entropy delta of revoked ids	B to A	promise a finite deadline under partition
settlement	a bond posted at A against damage measured at B	both ways	bound custody unconditionally: it holds only if the settlement gate holds

QA: FLAGGED: T3, B3, too wide 4.53in

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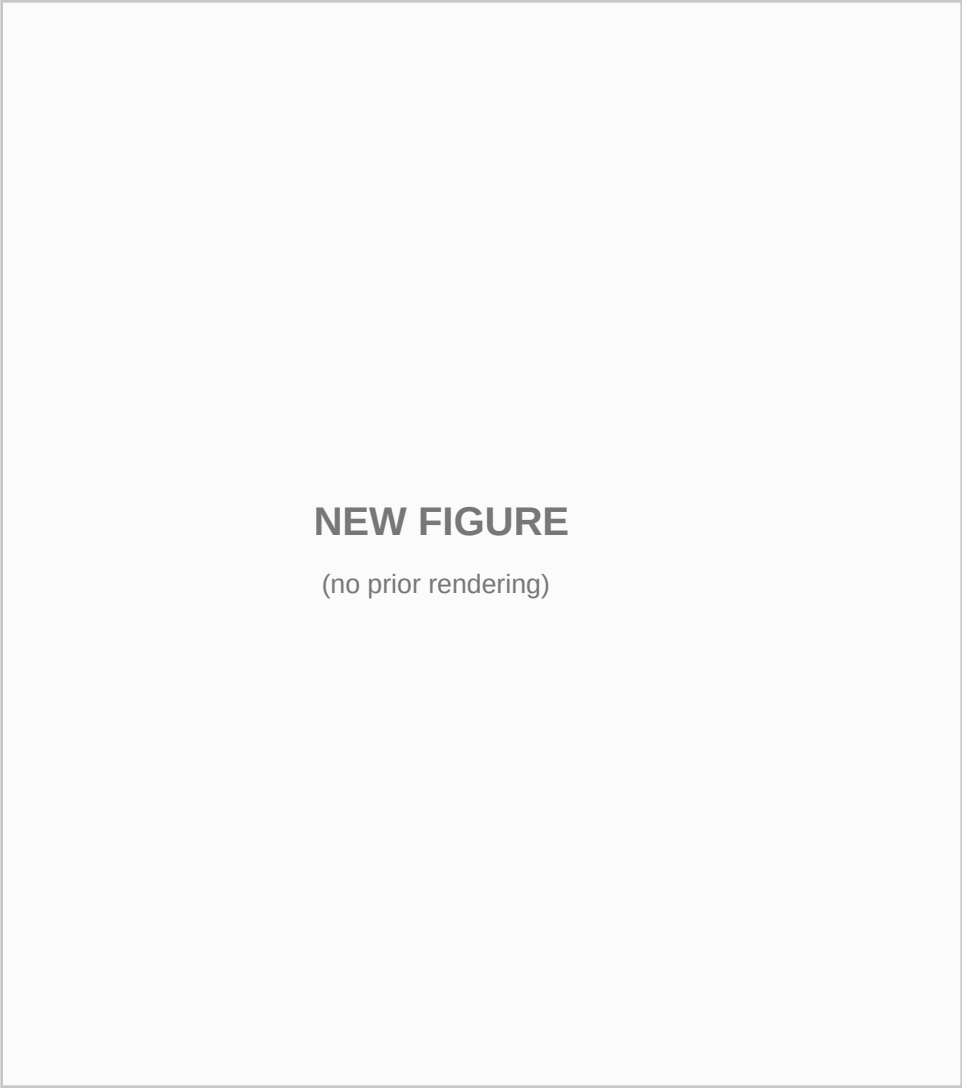
AFTER

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QA: FLAGGED: too wide 4.53in

BEFORE



QA: —

The five properties Witness-Log Admissibility (Design Invariant~{prop:fh-witness}) requires, and the specific failure each one closes. {Transfer, revocation, and {settlement all attach to this log, so a missing property here reopens a failure in every one of them. [internal]

AFTER

Table 1: The five properties Witness-Log Admissibility (Design Invariant ??) requires, and the specific failure each one closes. Transfer, revocation, and settlement all attach to this log, so a missing property here reopens a failure in every one of them. [internal]

property	what it prevents
append-only	retracting or reordering a published root after the fact
per-harbor latest-root binding	a harbor silently publishing two roots for one epoch
auditable consistency	an inconsistent history between two epochs going unverifiable
equivocation evidence	a proven contradiction requiring anything beyond the two signed roots
threshold publishability	publication itself requiring another daemon’s permission

QA: PASS

BEFORE

NEW FIGURE

(no prior rendering)

AFTER

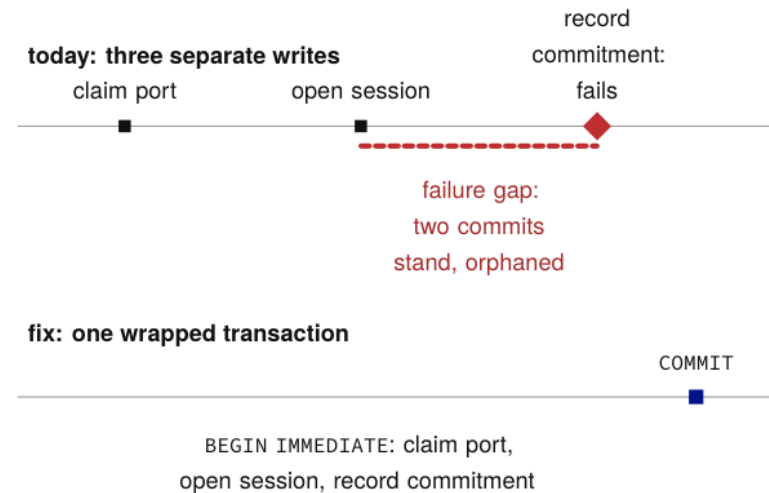


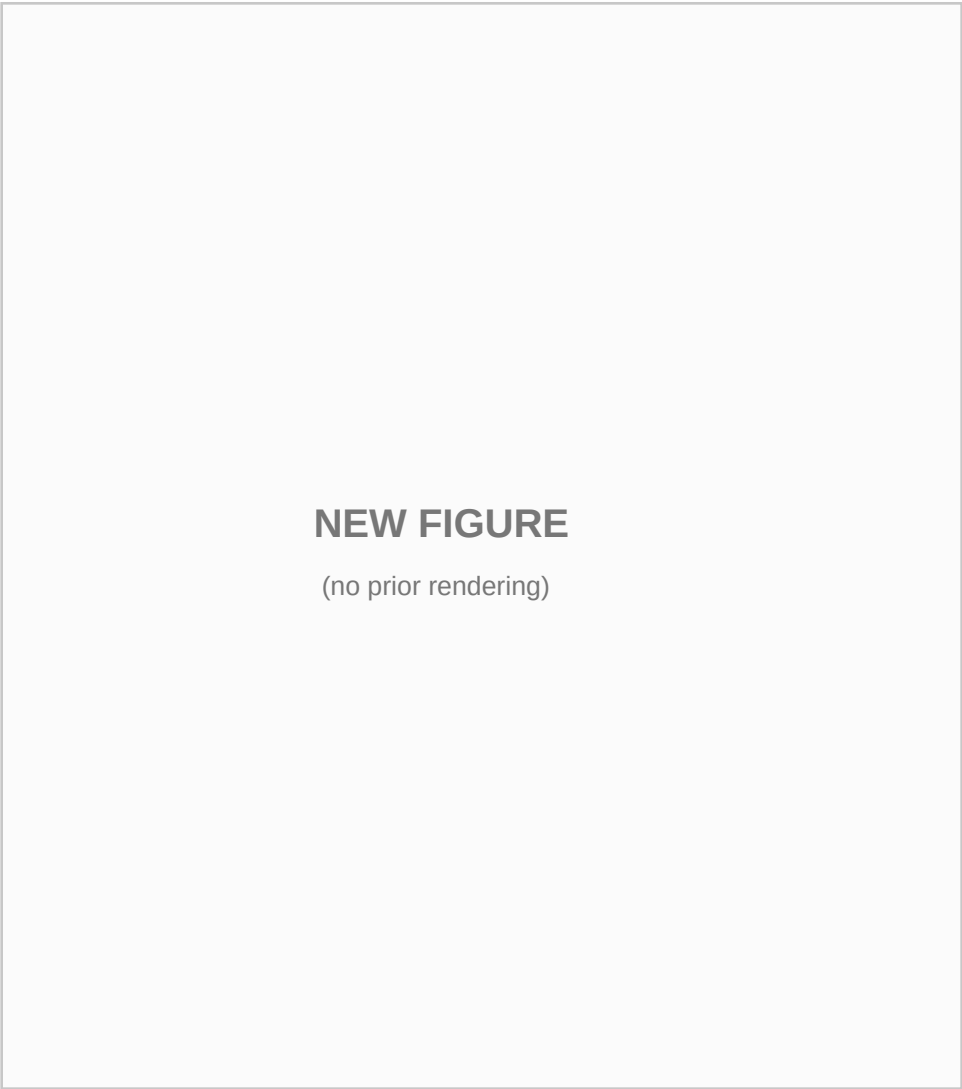
Figure 1: The same three writes, one local fix. Today (top), each write commits on its own; if the third fails, the first two stand, orphaned — a real failure gap. Wrapped in one local transaction, begun in immediate mode (bottom), all three commit together or not at all: no remote step, no Saga, because the durable transaction primitive already exists and every step is a local write to one file. [internal, OP-11]

QA: —

QA: PASS

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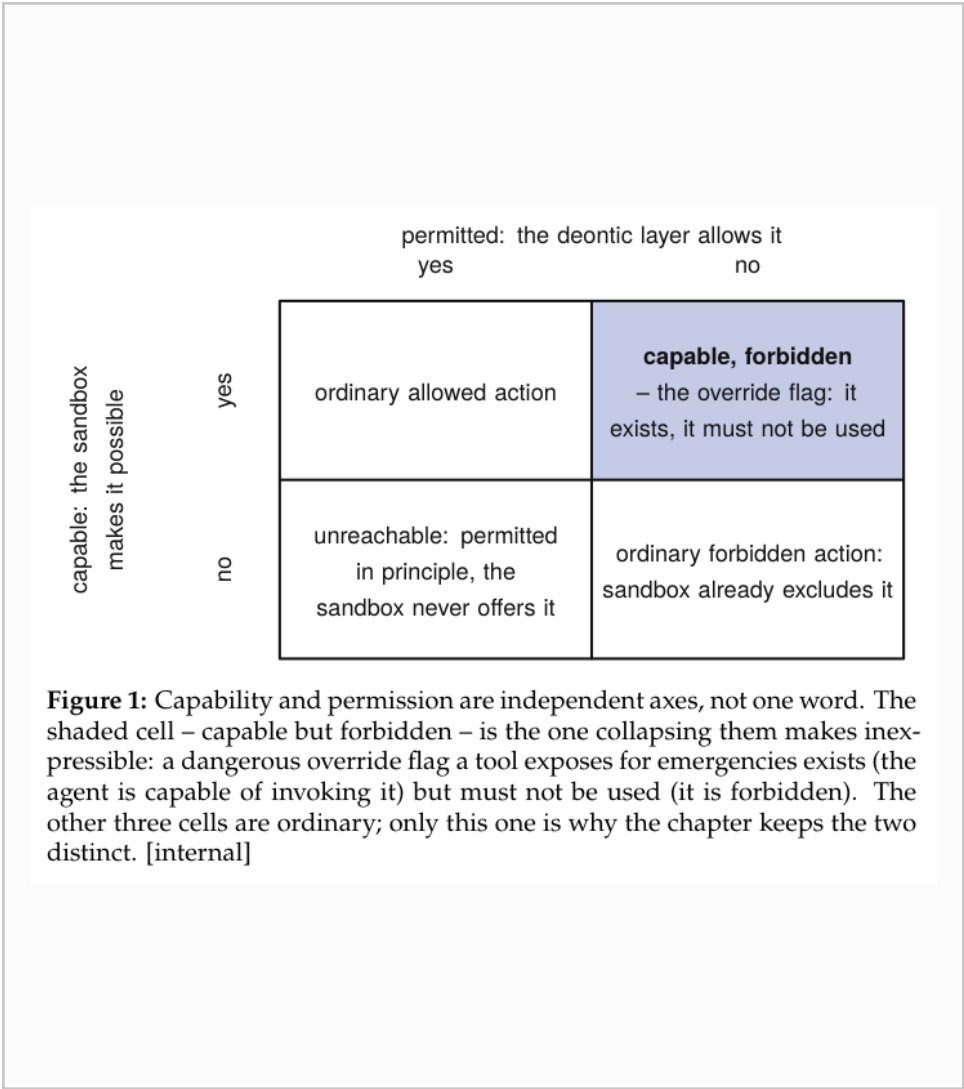
BEFORE



QA: —

Capability and permission are independent axes, not one word. The shaded cell -- capable but forbidden -- is the one collapsing them makes inexpressible: a dangerous override flag a tool exposes for emergencies exists (the agent is capable of invoking it) but must not be used (it is forbidden). The other three cells are ordinary; only this one is why the chapter keeps the two distinct. [internal]

AFTER



QA: PASS

BEFORE

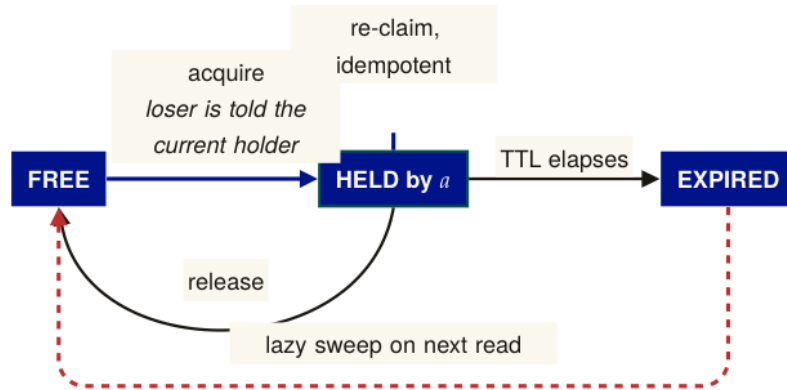
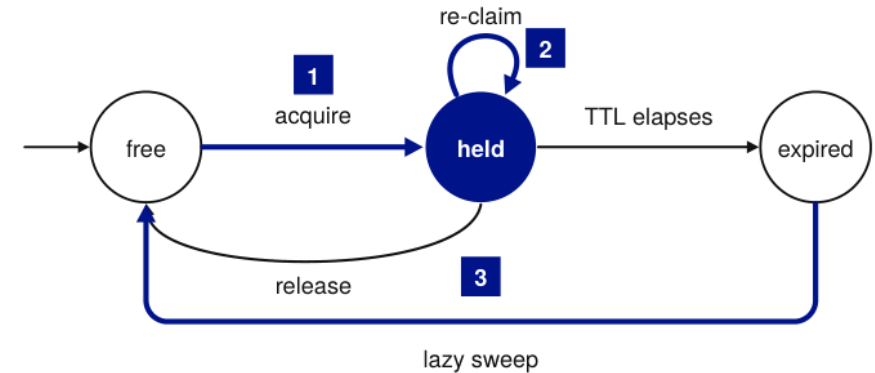


Figure 1: The guarded claim lifecycle. Acquire is one atomic uniqueness decision that also tells a losing caller who currently holds the claim; re-claim and release are idempotent; expiry is removed lazily, on the next read, not by a background sweep. The kernel has no queued state and therefore no fairness or bounded-wait guarantee: a losing caller must retry, and nothing schedules it a turn. [internal]

QA: PASS

The guarded claim lifecycle as a state machine. Acquire is a single insert against a uniqueness constraint; re-claim and release are idempotent; expiry moves a claim to {expired, from which the next read lazily sweeps it back to {free. There is no {queued state and therefore no fairness or bounded-wait guarantee: a losing caller must retry, and nothing schedules it a turn. [internal]

AFTER



there is no queued state: a losing acquire is simply denied, and nothing schedules it a turn.

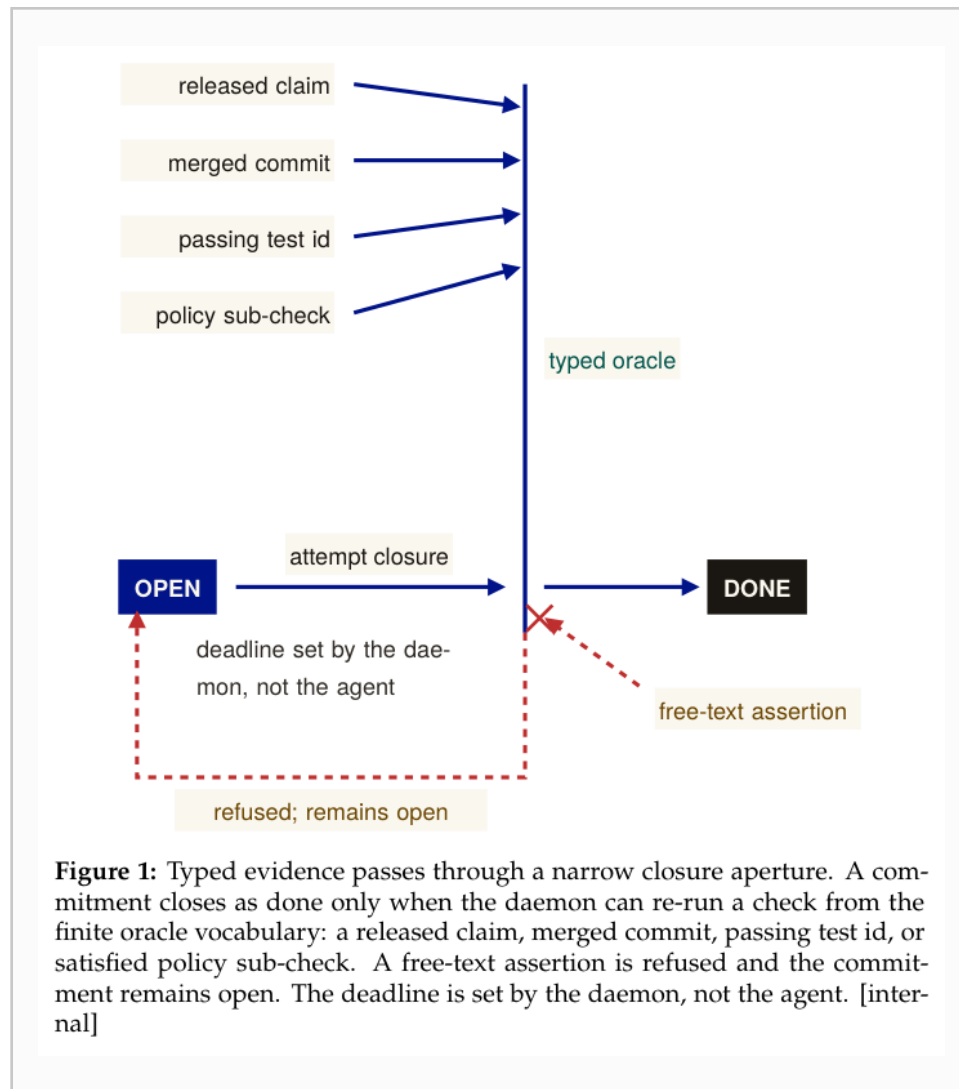
what the badge adds to the transition name

- 1 one atomic uniqueness decision; a losing caller is told the current holder, not handed a bare error
 - 2 idempotent: the same holder renews its TTL, it does not create a second claim
 - 3 expiry is removed lazily, on the next read that touches the row, not by a background sweep
-

Figure 1: The guarded claim lifecycle as a state machine. Acquire is a single insert against a uniqueness constraint; re-claim and release are idempotent; expiry moves a claim to *expired*, from which the next read lazily sweeps it back to *free*. There is no *queued* state and therefore no fairness or bounded-wait guarantee: a losing caller must retry, and nothing schedules it a turn. [internal]

QA: PASS

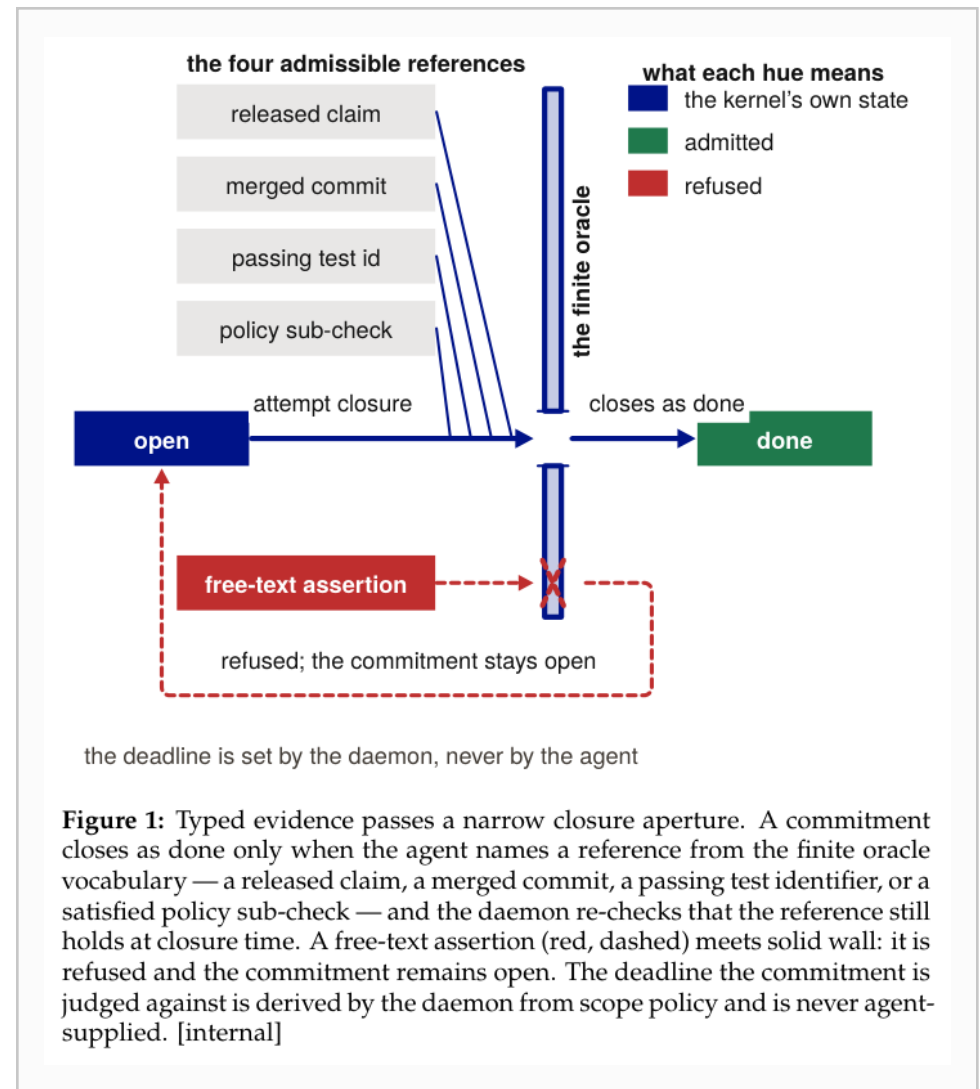
BEFORE



QA: FLAGGED: B9

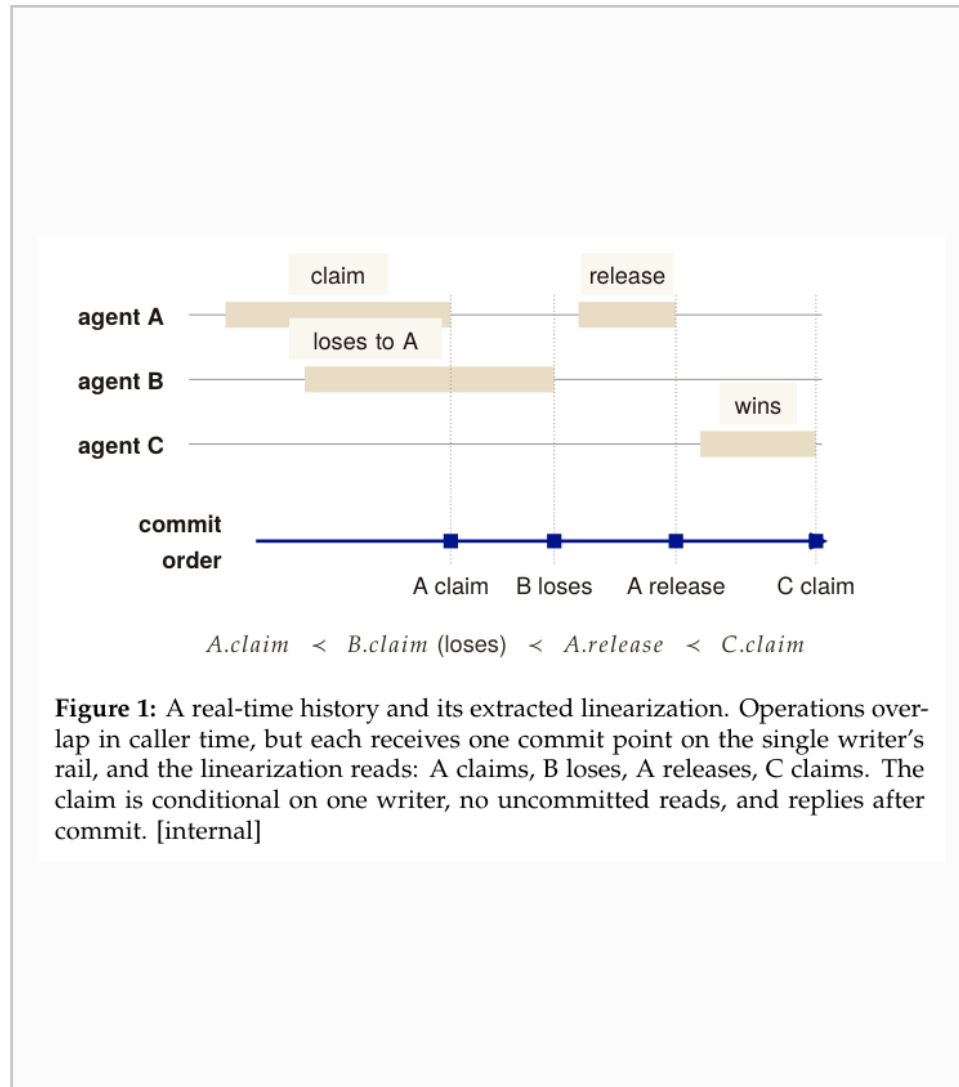
Typed evidence passes a narrow closure aperture. A commitment closes as done only when the agent names a reference from the finite oracle vocabulary --- a released claim, a merged commit, a passing test identifier, or a satisfied policy sub-check --- and the daemon re-checks that the reference still holds at closure time. A free-text assertion (red, dashed) meets solid wall: it is refused and the commitment remain...

AFTER



QA: PASS

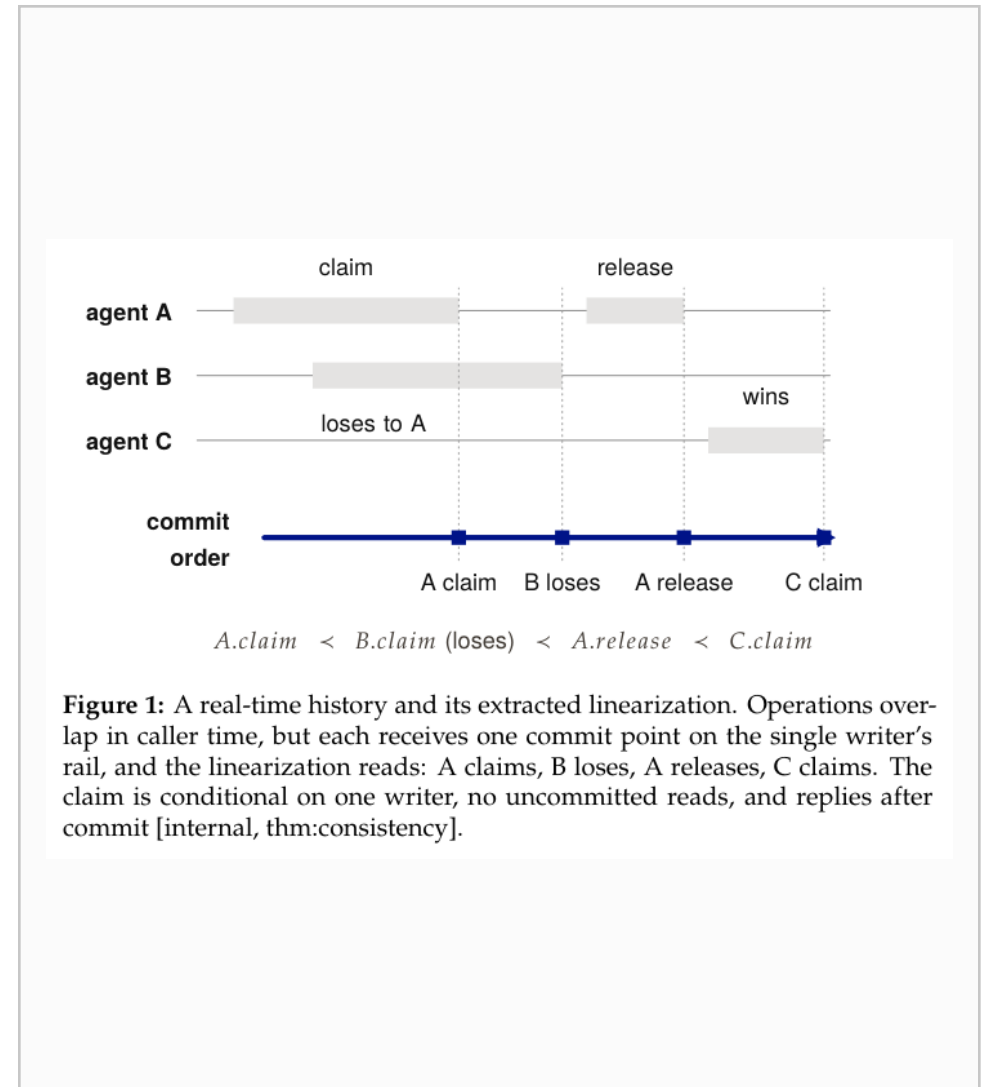
BEFORE



QA: PASS

A real-time history and its extracted linearization. Operations overlap in caller time, but each receives one commit point on the single writer's rail, and the linearization reads: A claims, B loses, A releases, C claims. The claim is conditional on one writer, no uncommitted reads, and replies after commit [internal, thm:consistency].

AFTER



QA: PASS

BEFORE

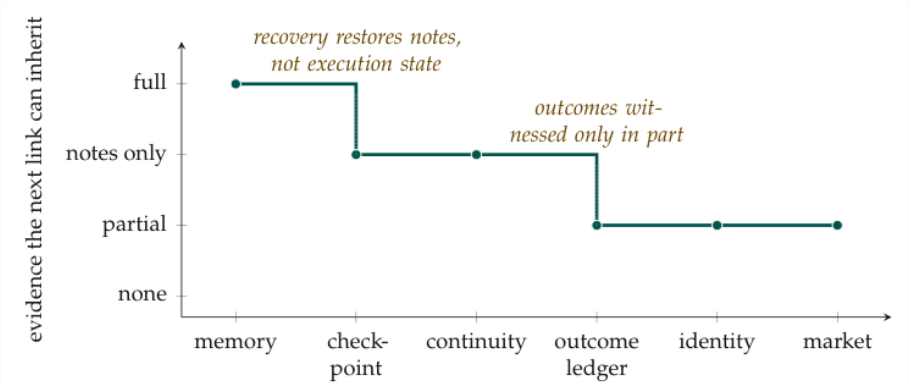


Figure 1: The continuity chain as an evidence-support profile. What checkpoint inherits from memory already drops from full to notes only; what the outcome ledger inherits from continuity drops again, to partial; identity and market inherit only that partial residue. Downstream confidence cannot exceed the weakest upstream link. [internal]

QA: FLAGGED: B9, too wide 4.73in

The continuity chain as an evidence-support profile, computed from the chapter's own status grades. What checkpoint inherits from memory already drops from full to notes only; what the outcome ledger inherits from continuity drops again, to partial; identity and market inherit only that partial residue, not an independent grade of their own. Downstream confidence cannot exceed the weakest upstream link. [internal]

AFTER

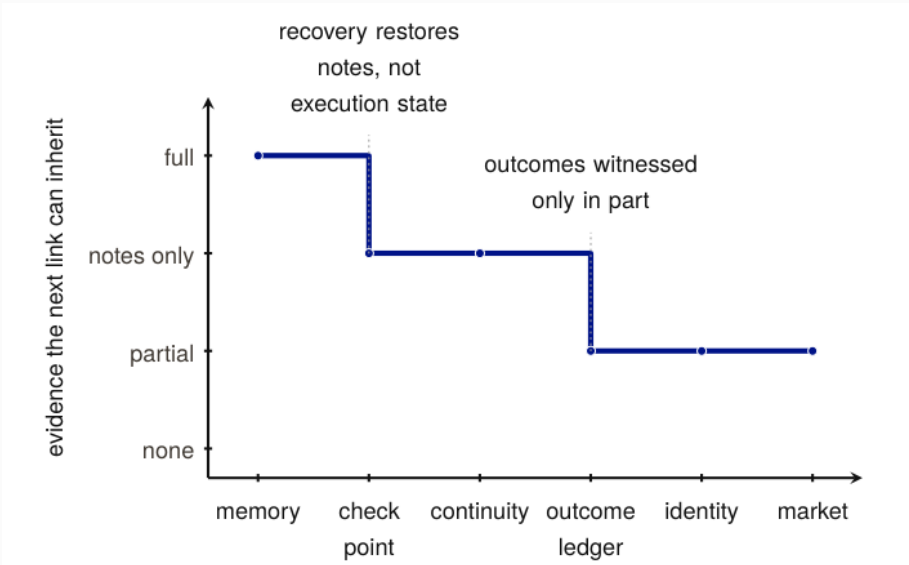
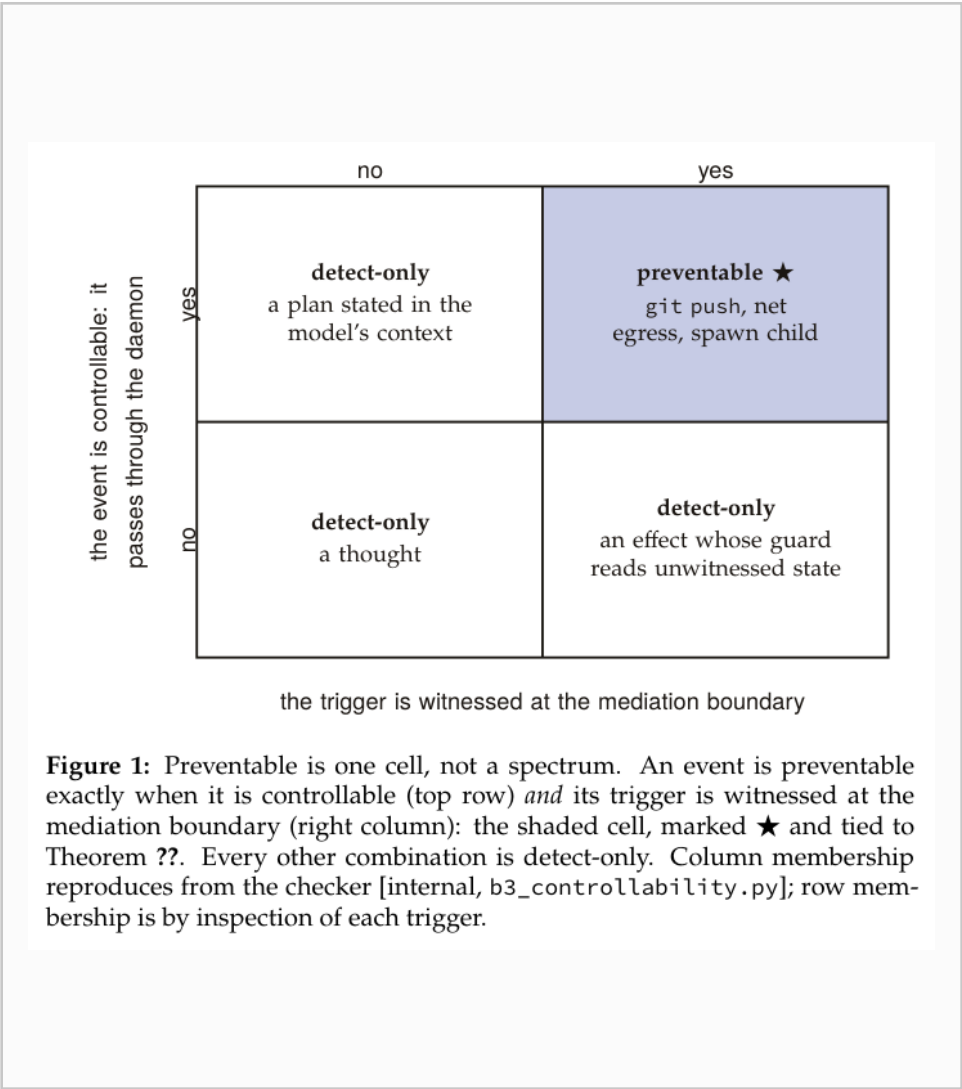


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QA: PASS

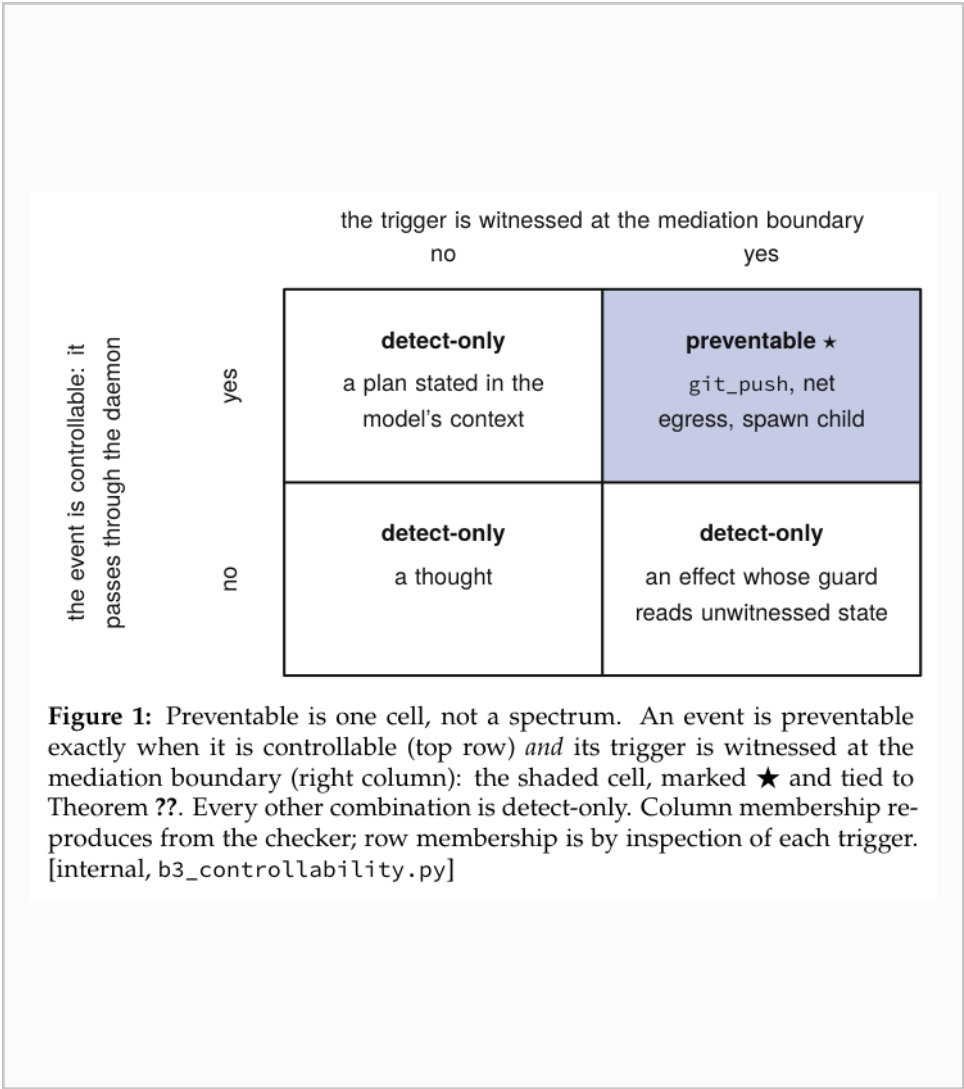
BEFORE



QA: FLAGGED: T4

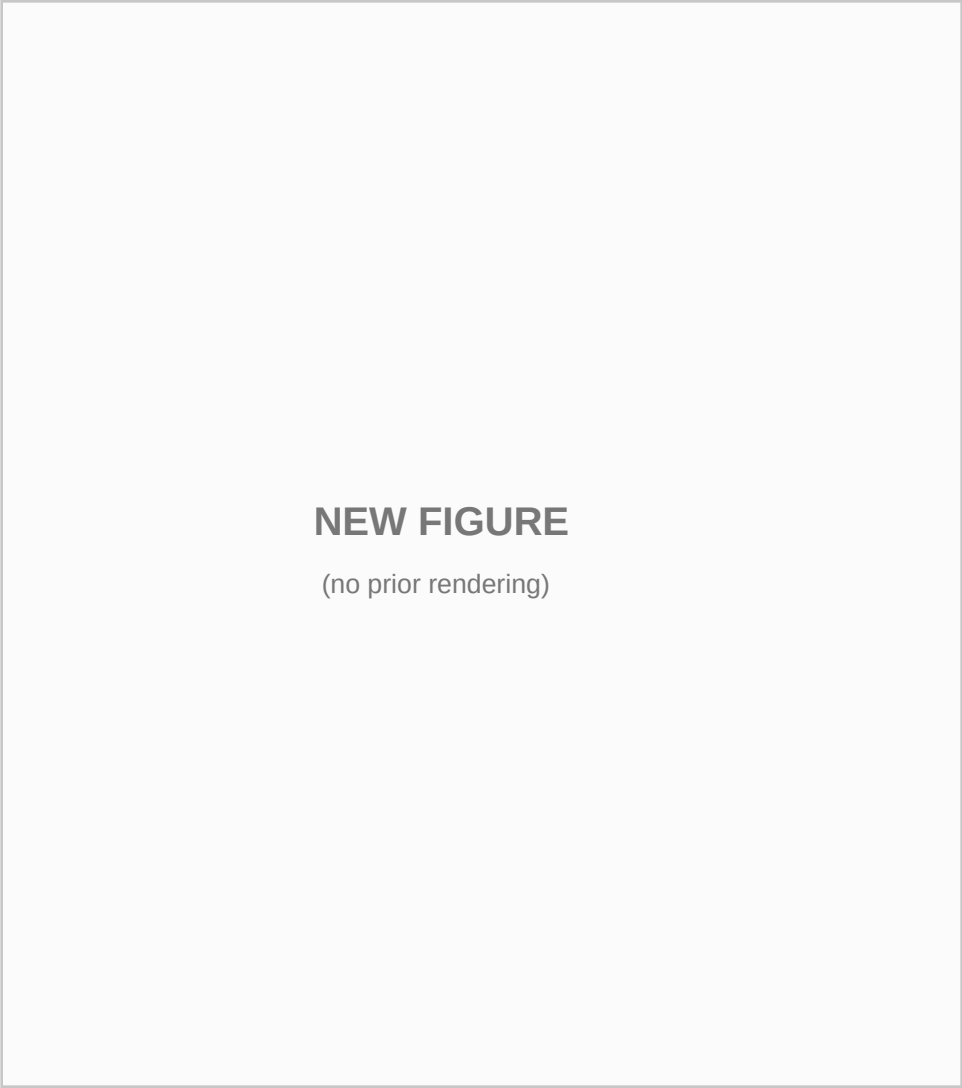
Preventable is one cell, not a spectrum. An event is preventable exactly when it is controllable (top row) {and its trigger is witnessed at the mediation boundary (right column): the shaded cell, marked \$ and tied to Theorem~{thm:regimentation-controllability}. Every other combination is detect-only. Column membership reproduces from the checker; row membership is by inspection of each trigger. [internal, {b3_con...

AFTER



QA: PASS

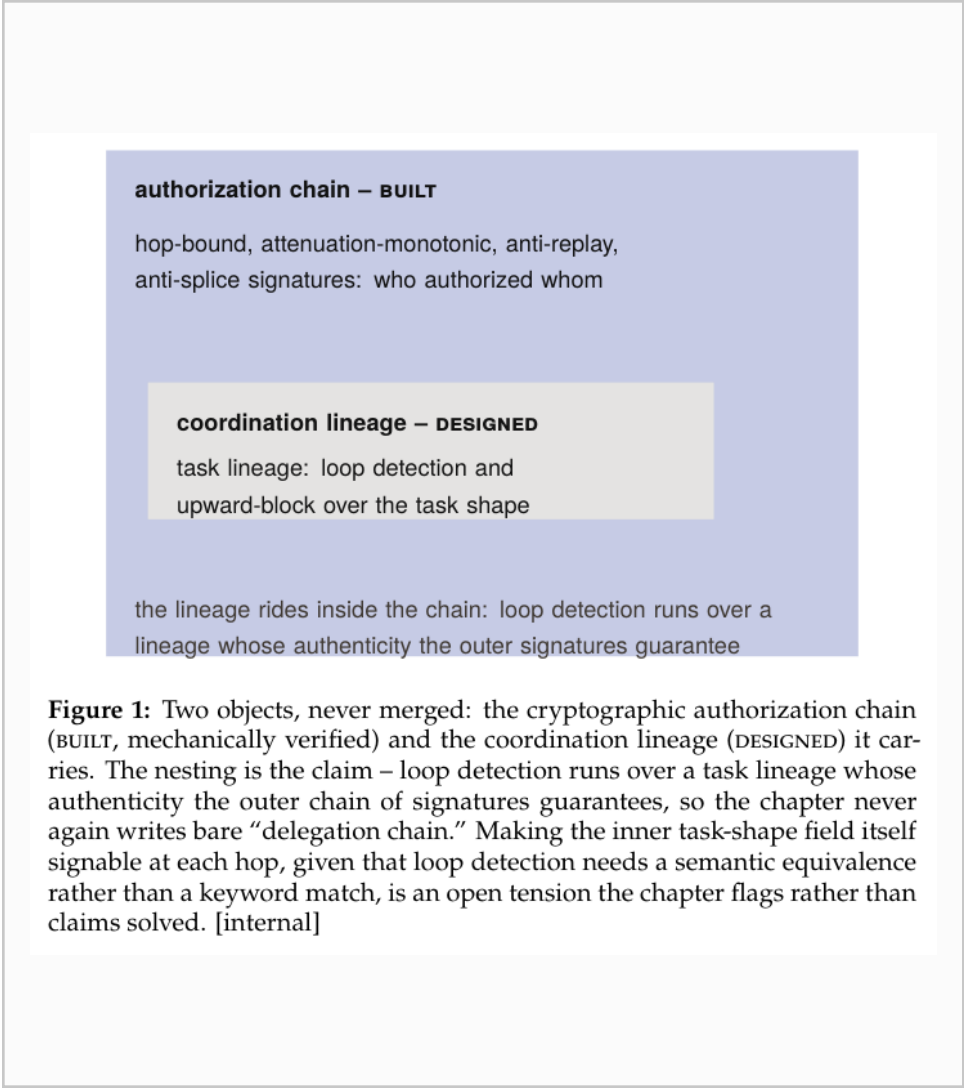
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QA: —

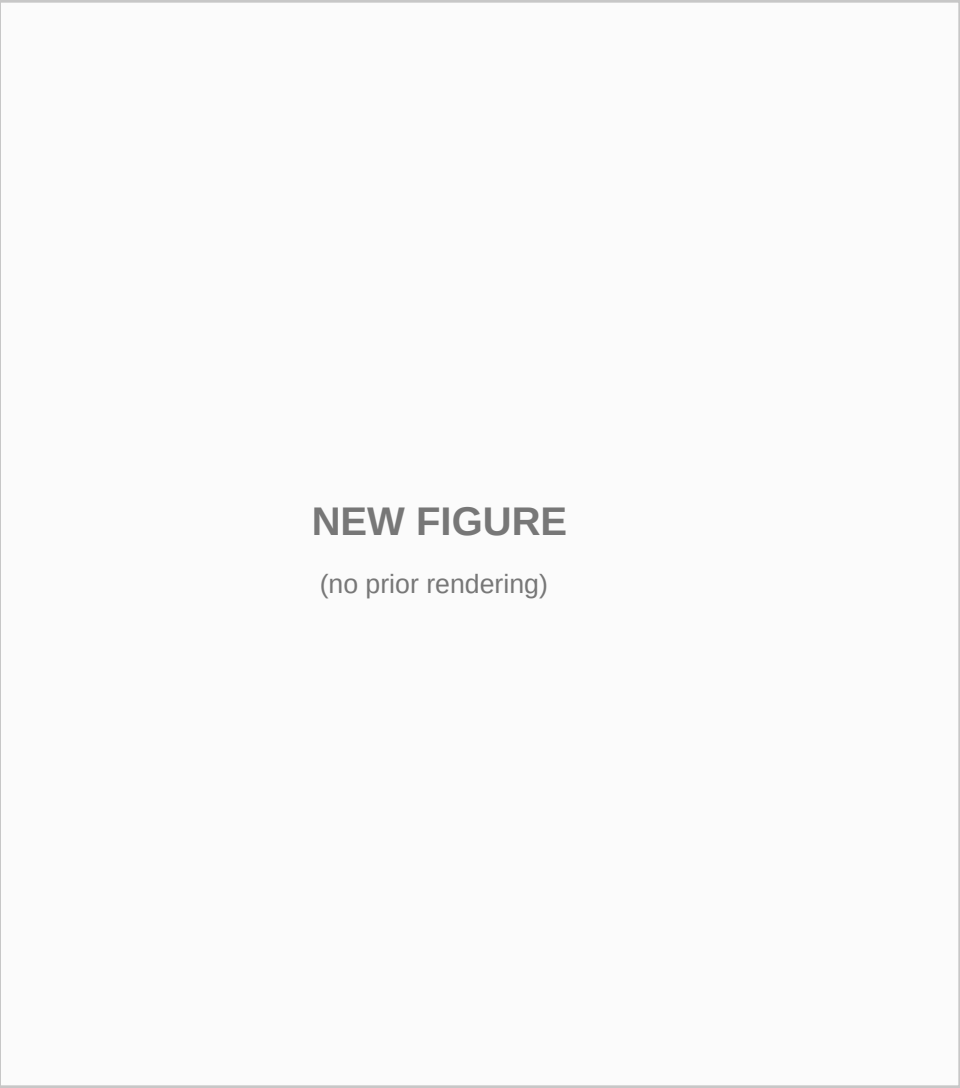
Two objects, never merged: the cryptographic authorization chain ({built, mechanically verified) and the coordination lineage ({designed) it carries. The nesting is the claim -- loop detection runs over a task lineage whose authenticity the outer chain of signatures guarantees, so the chapter never again writes bare ``delegation chain.'' Making the inner task-shape field itself signable at each hop, given that loop detection needs a semantic equivalence rather than a keyword match, is an open tension the chapter flags rather than claims solved. [internal]

AFTER



QA: PASS

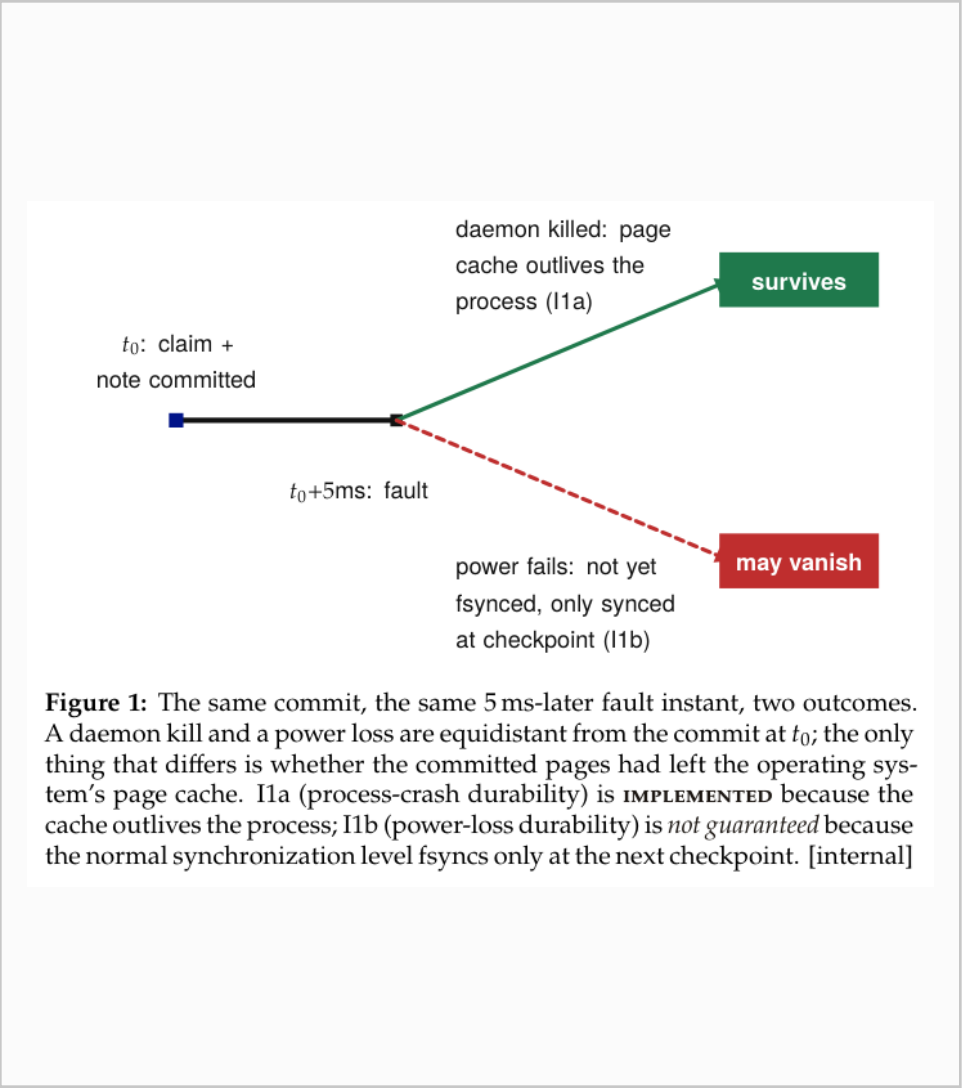
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QA: —

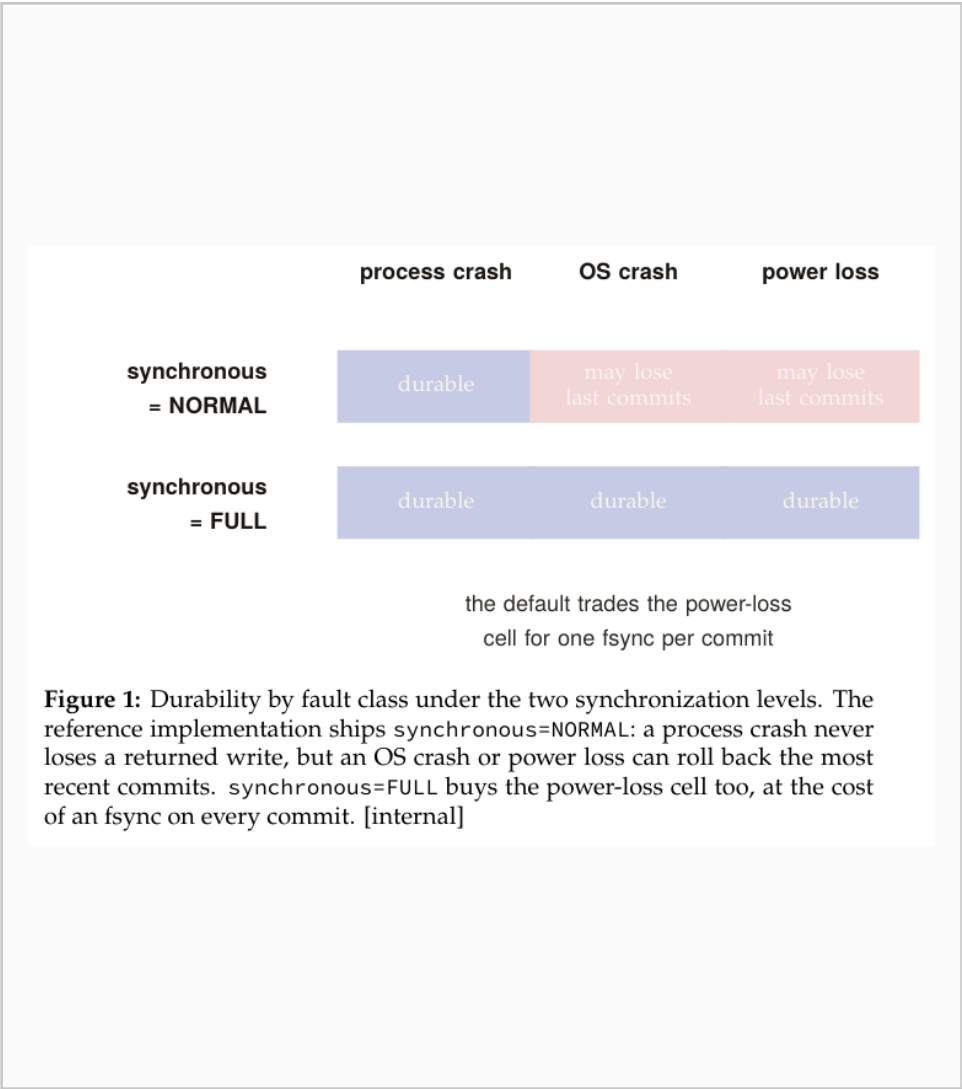
The same commit, the same 5\,ms-later fault instant, two outcomes. A daemon kill and a power loss are equidistant from the commit at \$t_0\$; the only thing that differs is whether the committed pages had left the operating system's page cache. I1a (process-crash durability) is { because the cache outlives the process; I1b (power-loss durability) is { because the normal synchronization level fsyncs only at the next ...

AFTER



QA: PASS

BEFORE



QA: PASS

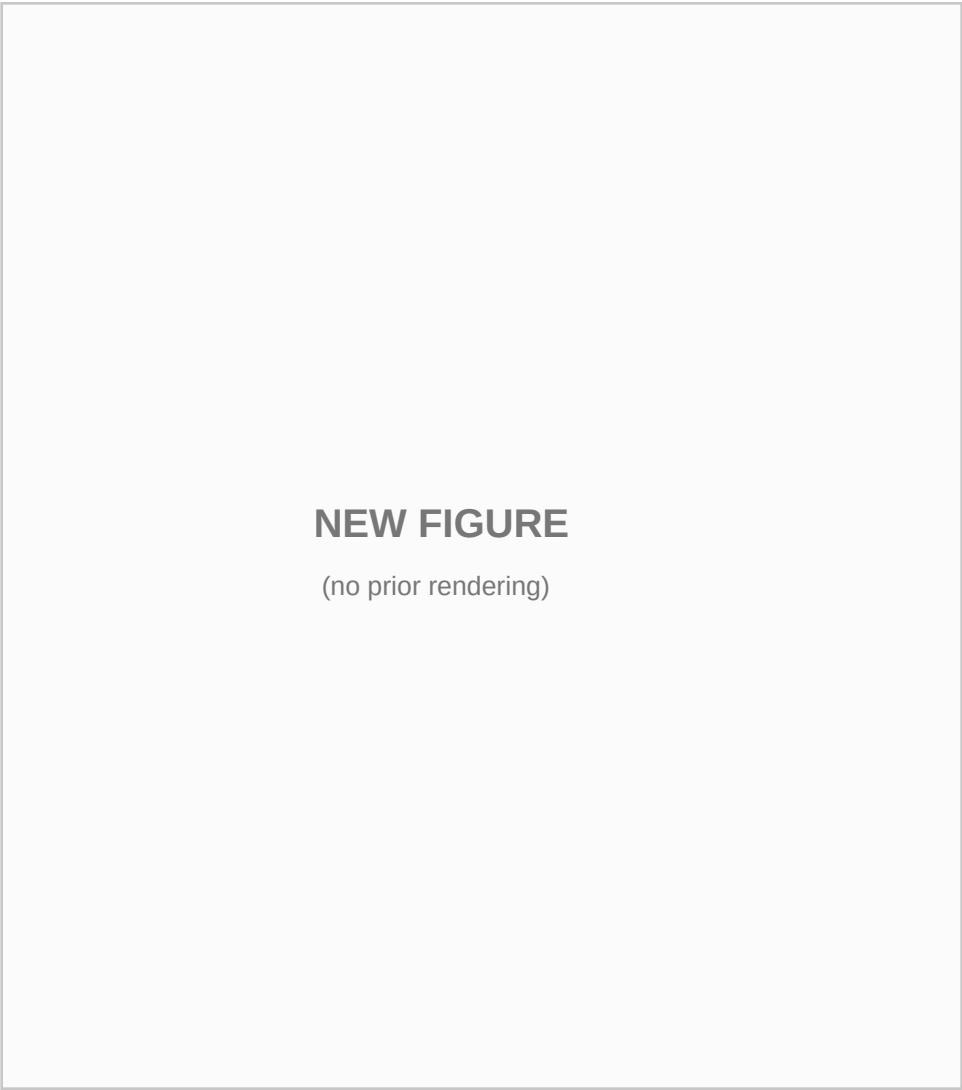
AFTER

REPLACED BY TABLE

in the Book

QA: TABLE

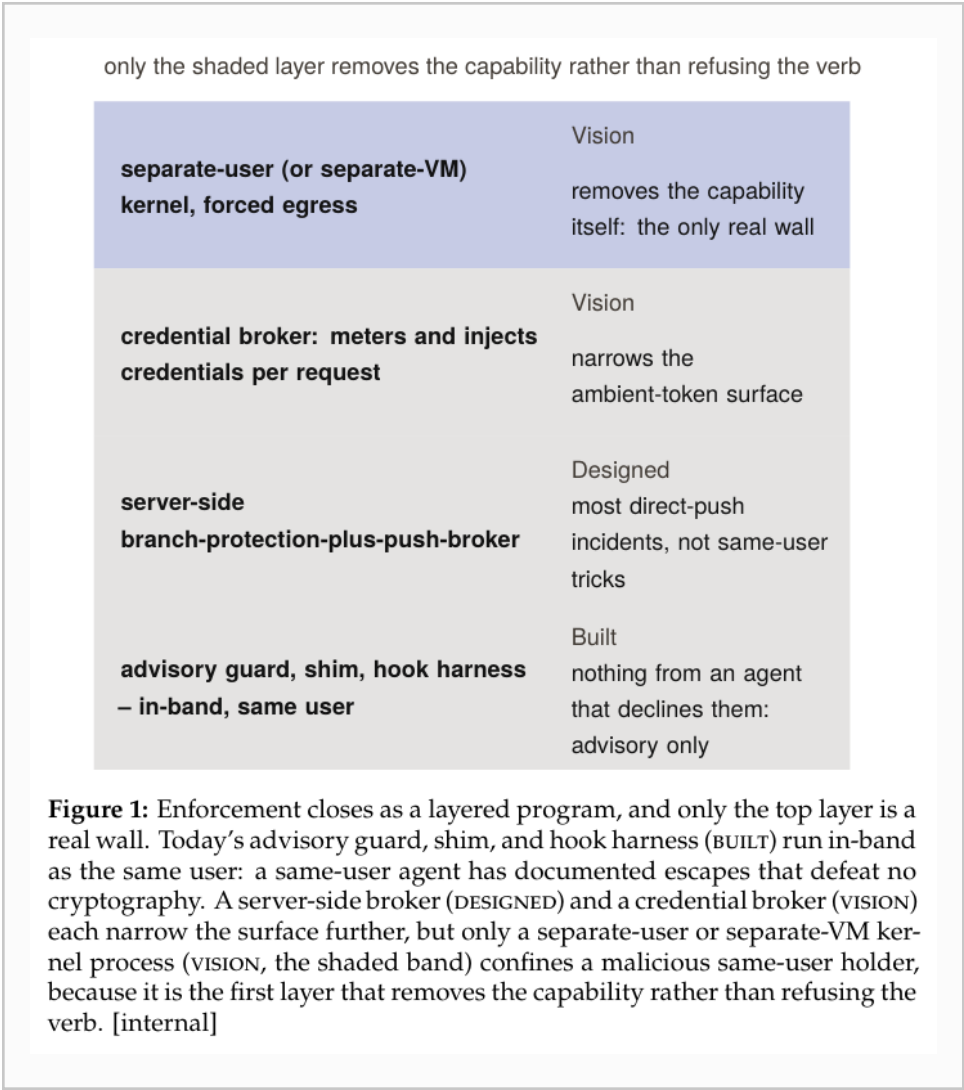
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QA: —

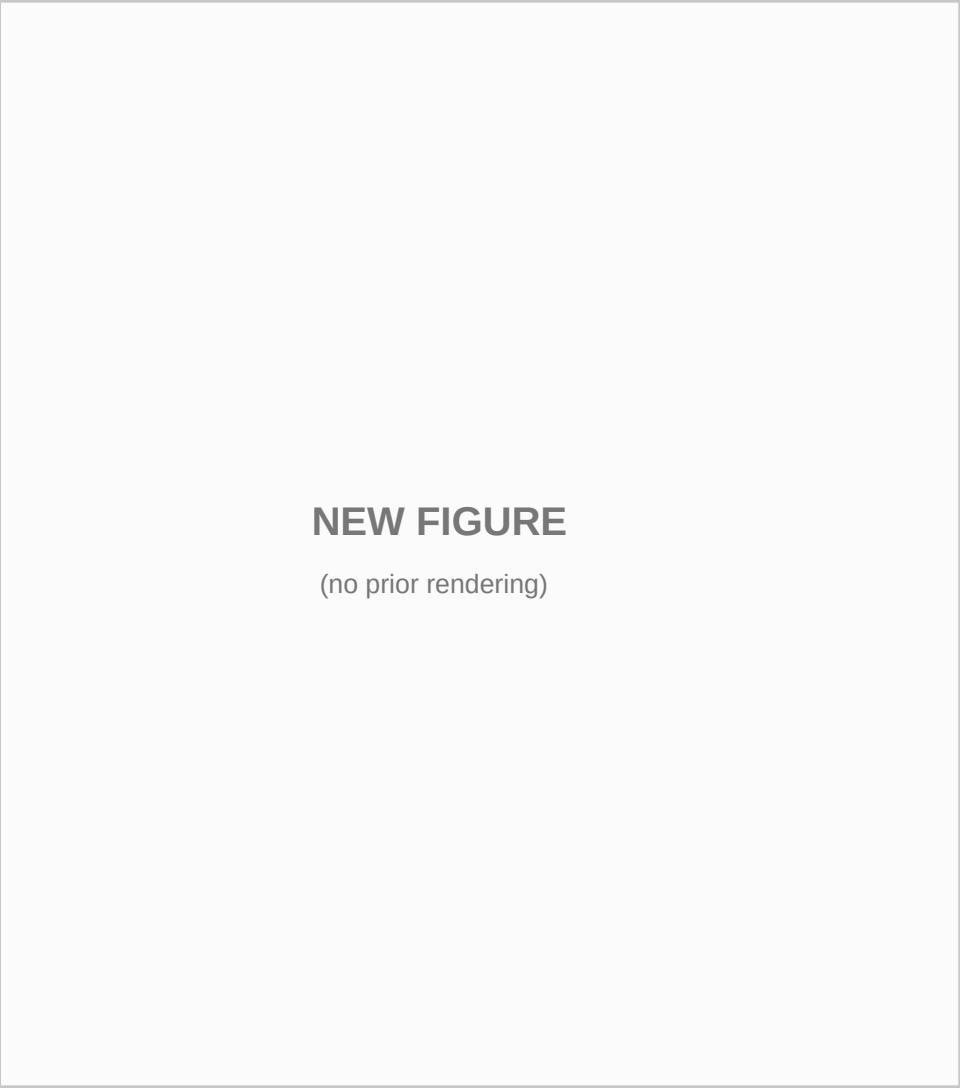
Enforcement closes as a layered program, and only the top layer is a real wall. Today's advisory guard, shim, and hook harness (built) run in-band as the same user: a same-user agent has documented escapes that defeat no cryptography. A server-side broker (designed) and a credential broker (vision) each narrow the surface further, but only a separate-user or separate-VM kernel process (vision, the shaded band)...

AFTER



QA: PASS

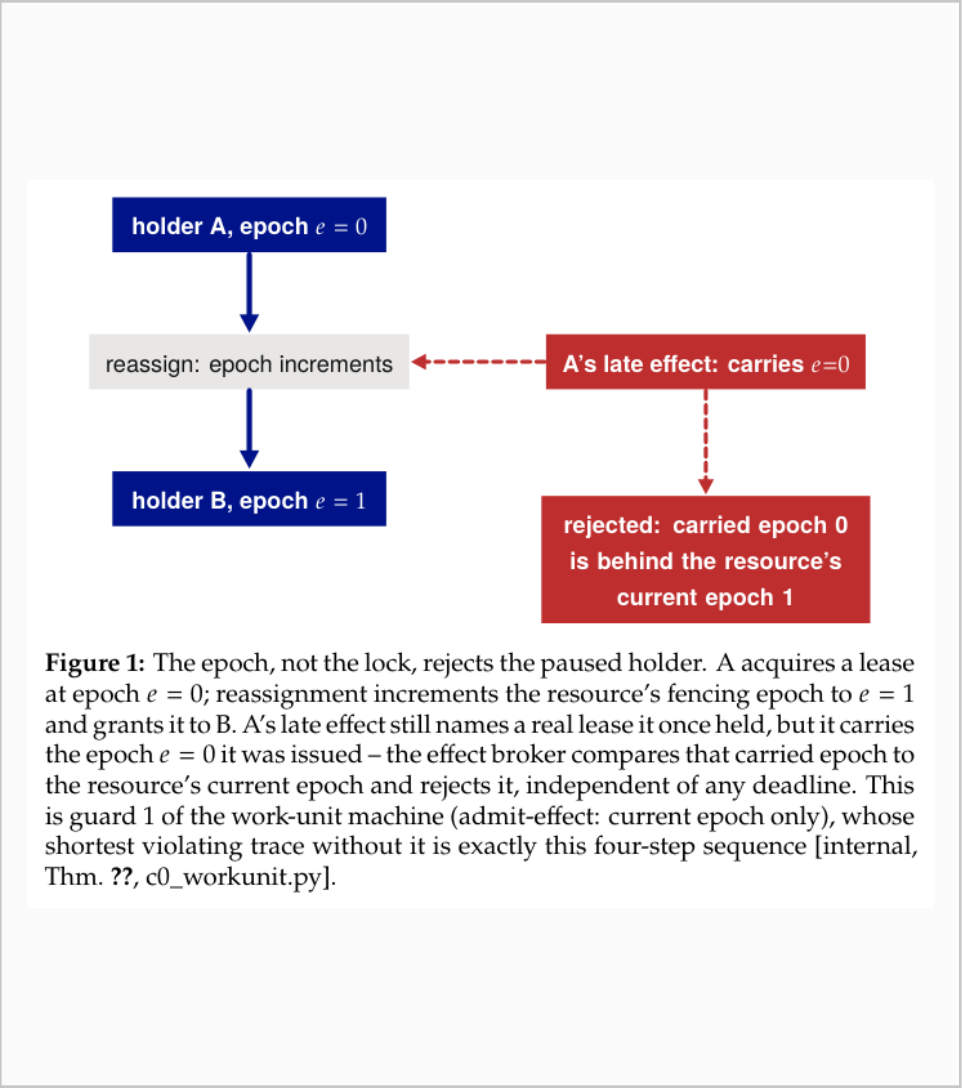
BEFORE



QA: —

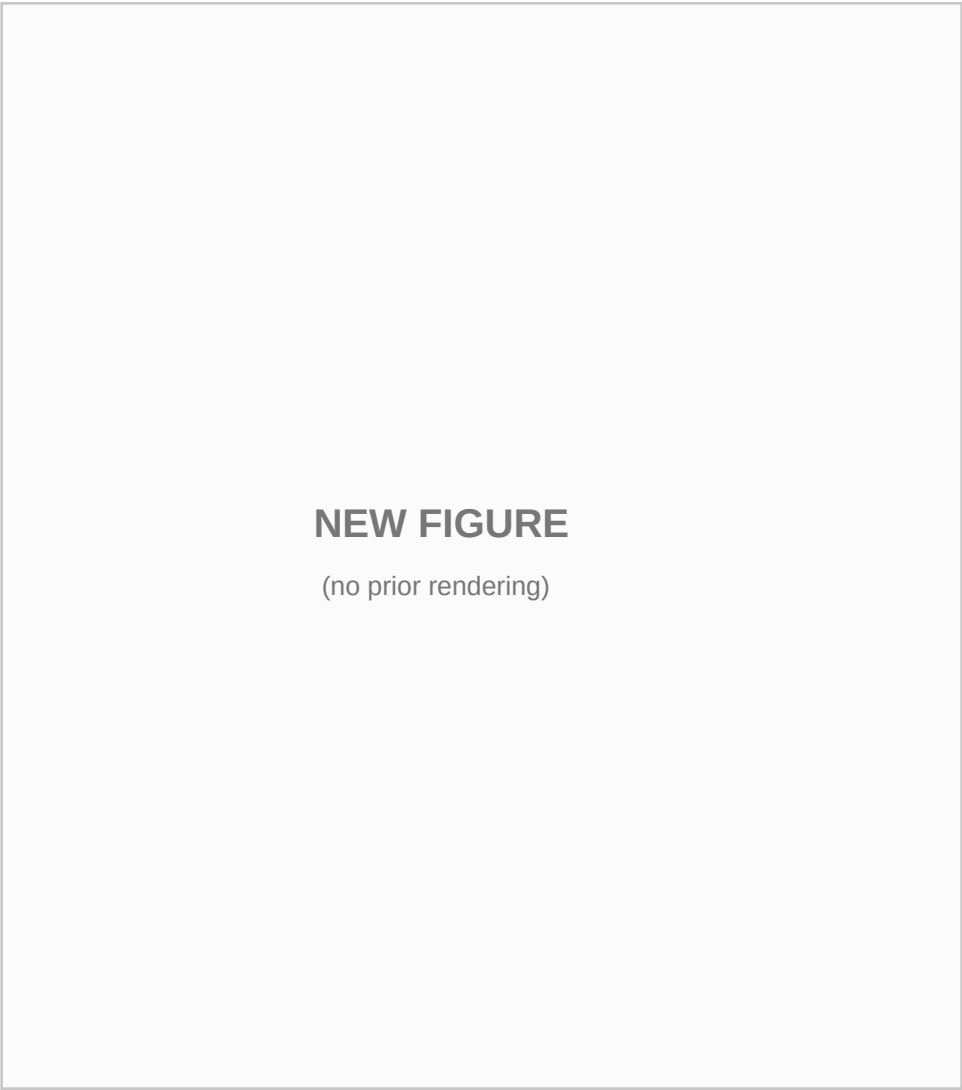
The epoch, not the lock, rejects the paused holder. A acquires a lease at epoch $e=0$; reassignment increments the resource's fencing epoch to $e=1$ and grants it to B. A's late effect still names a real lease it once held, but it carries the epoch $e=0$ it was issued -- the effect broker compares that carried epoch to the resource's current epoch and rejects it, independent of any deadline. This is guard 1 of the work-unit machine (admit-effect: current epoch only), whose shortest violating trace without it is exactly this four-step sequence [internal, Thm. ??, c0_workunit.py].

AFTER



QA: PASS

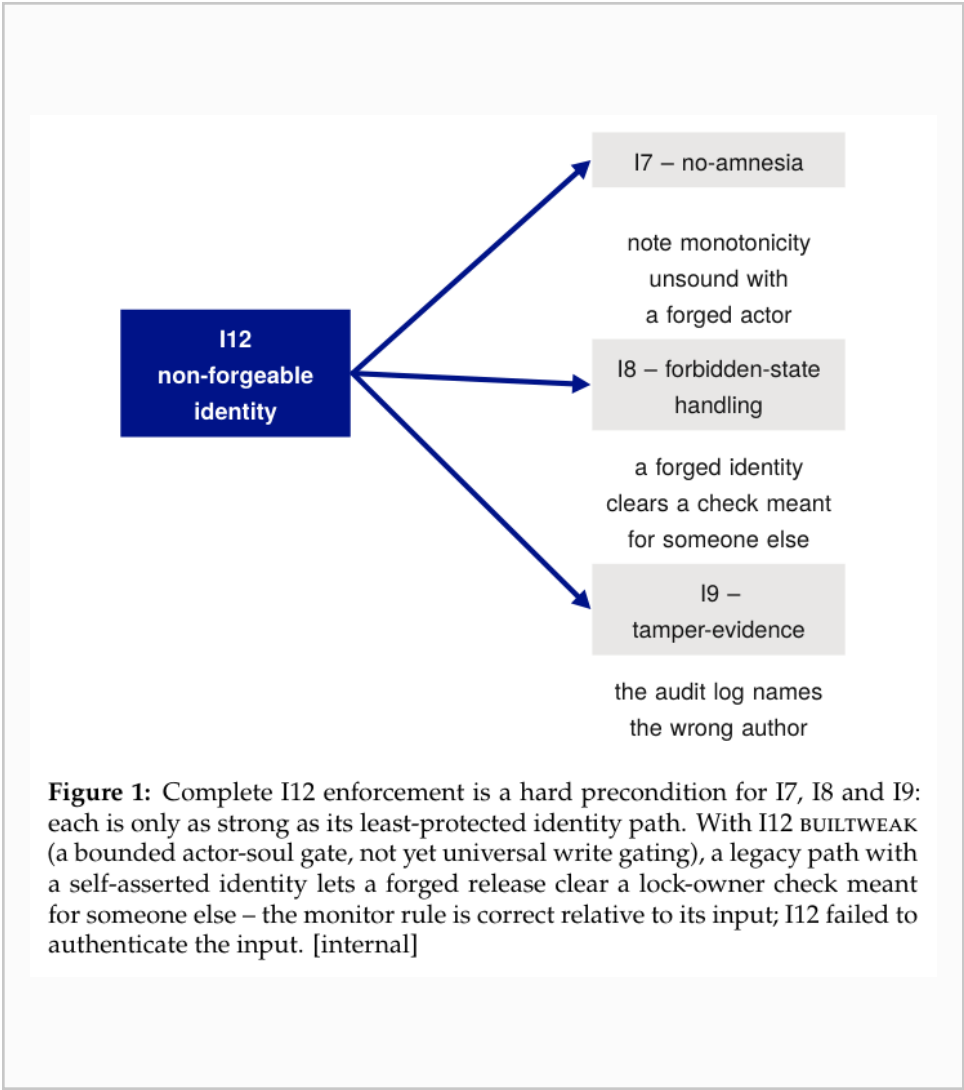
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QA: —

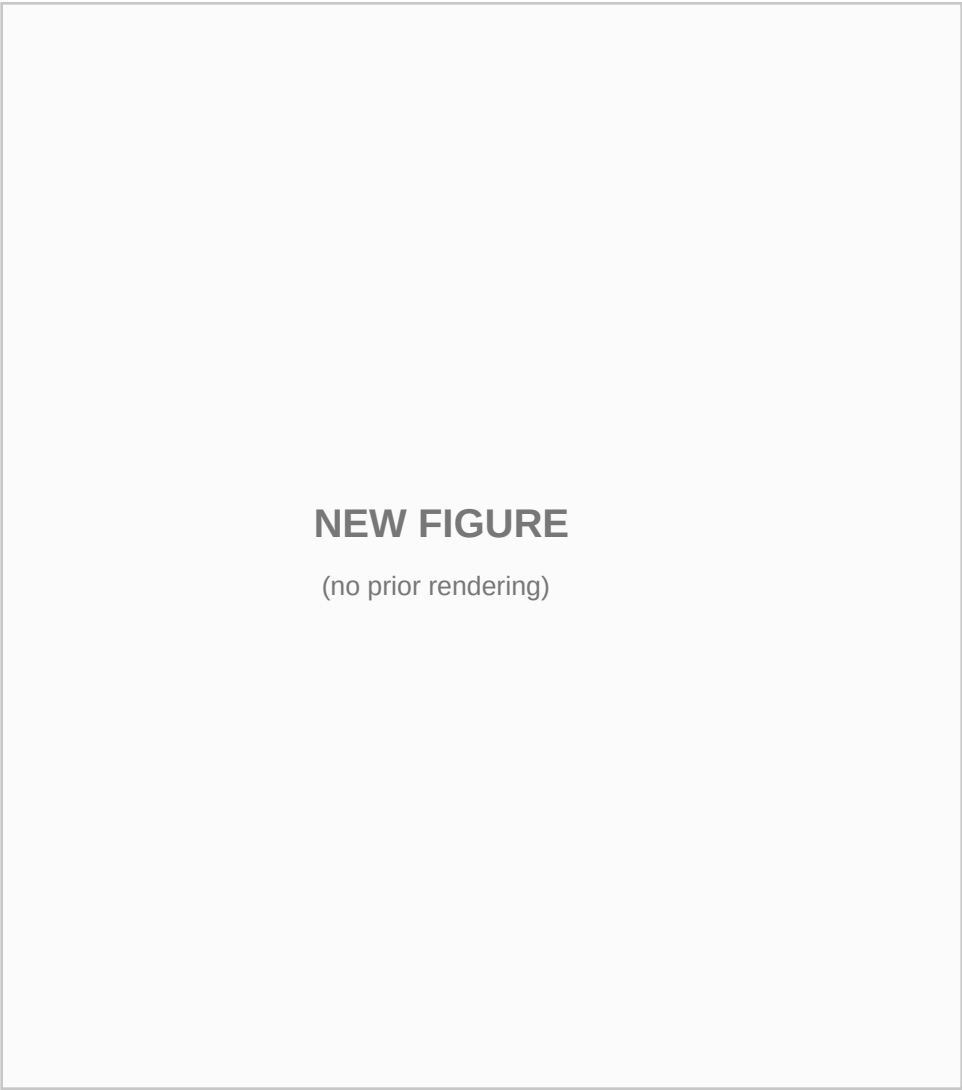
Complete I12 enforcement is a hard precondition for I7, I8 and I9: each is only as strong as its least-protected identity path. With I12 {builtweak (a bounded actor-soul gate, not yet universal write gating), a legacy path with a self-asserted identity lets a forged release clear a lock-owner check meant for someone else -- the monitor rule is correct relative to its input; I12 failed to authenticate the input. [...

AFTER

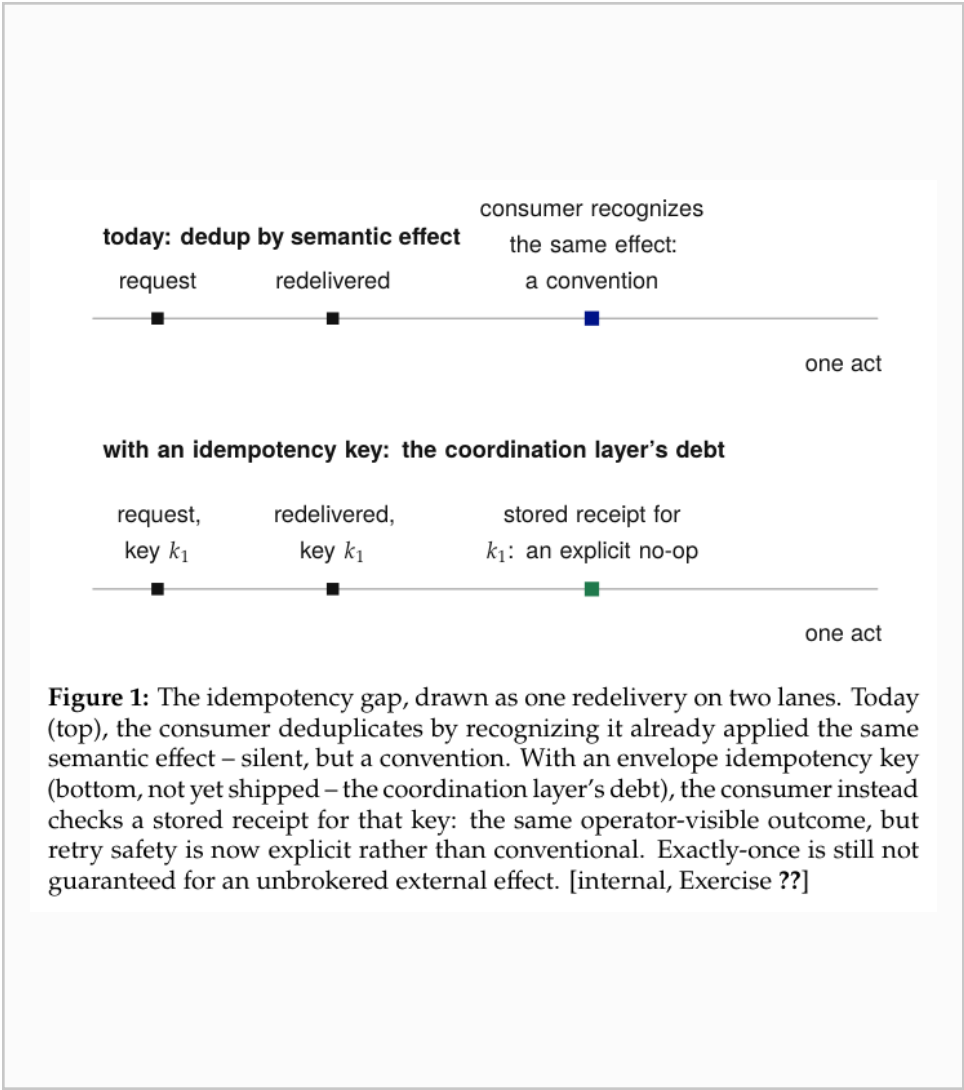


QA: PASS

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AFTER

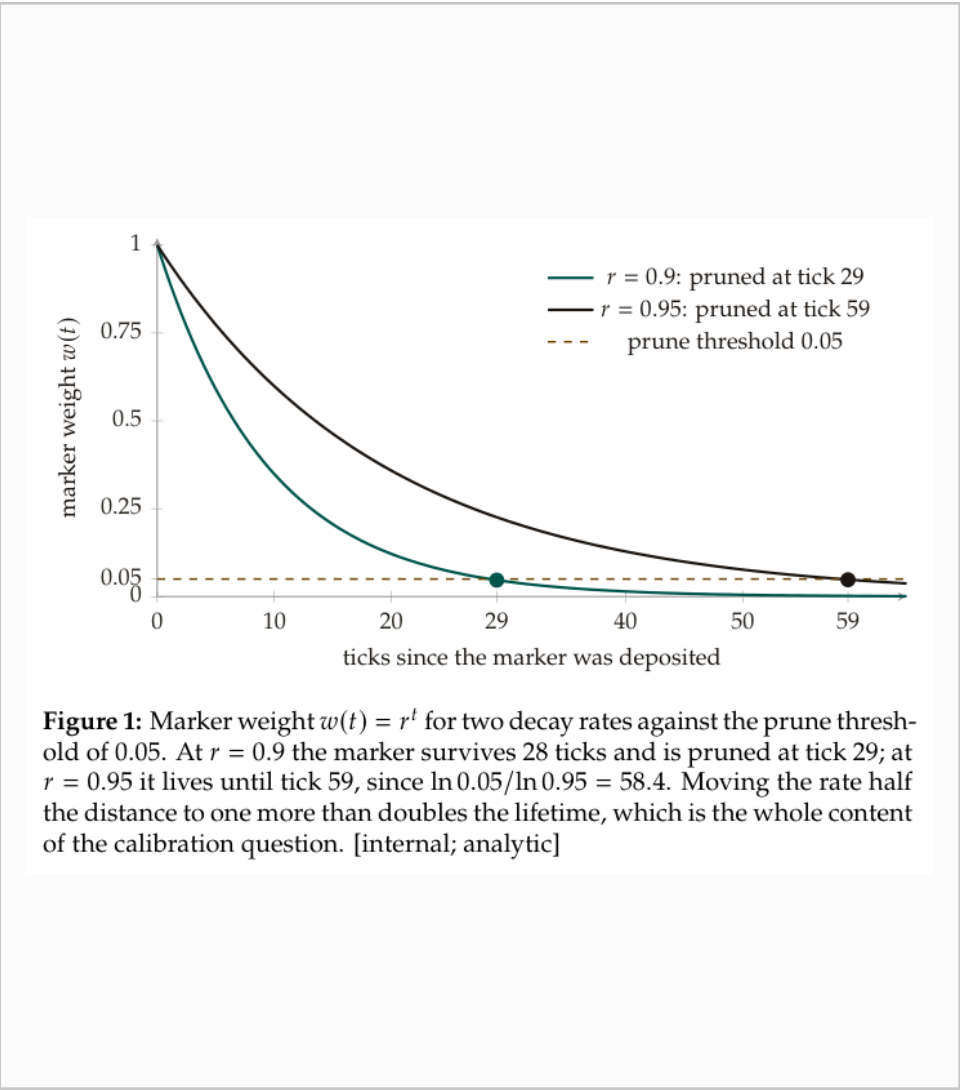


QA: —

QA: PASS

The idempotency gap, drawn as one redelivery on two lanes. Today (top), the consumer deduplicates by recognizing it already applied the same semantic effect -- silent, but a convention. With an envelope idempotency key (bottom, not yet shipped -- the coordination layer's debt), the consumer instead checks a stored receipt for that key: the same operator-visible outcome, but retry safety is now explicit rather than...

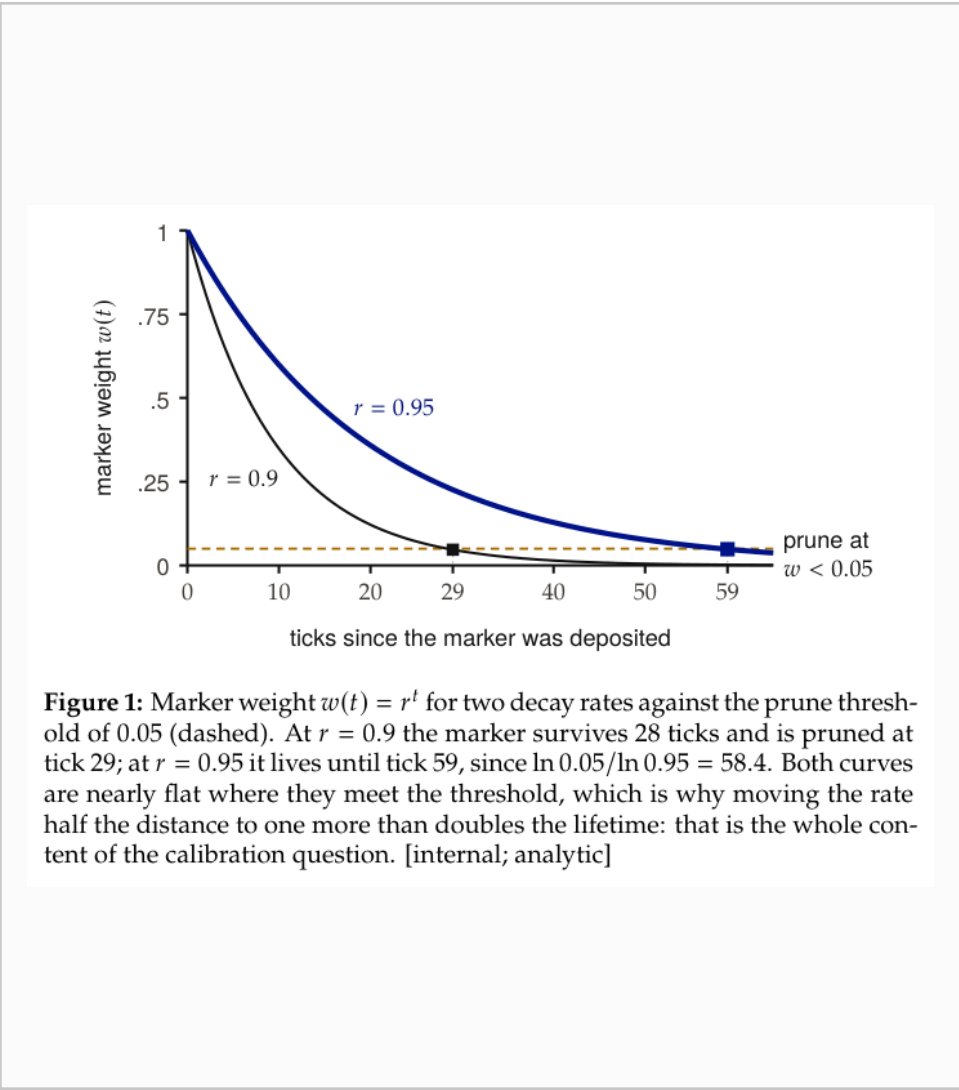
BEFORE



QA: FLAGGED: B3

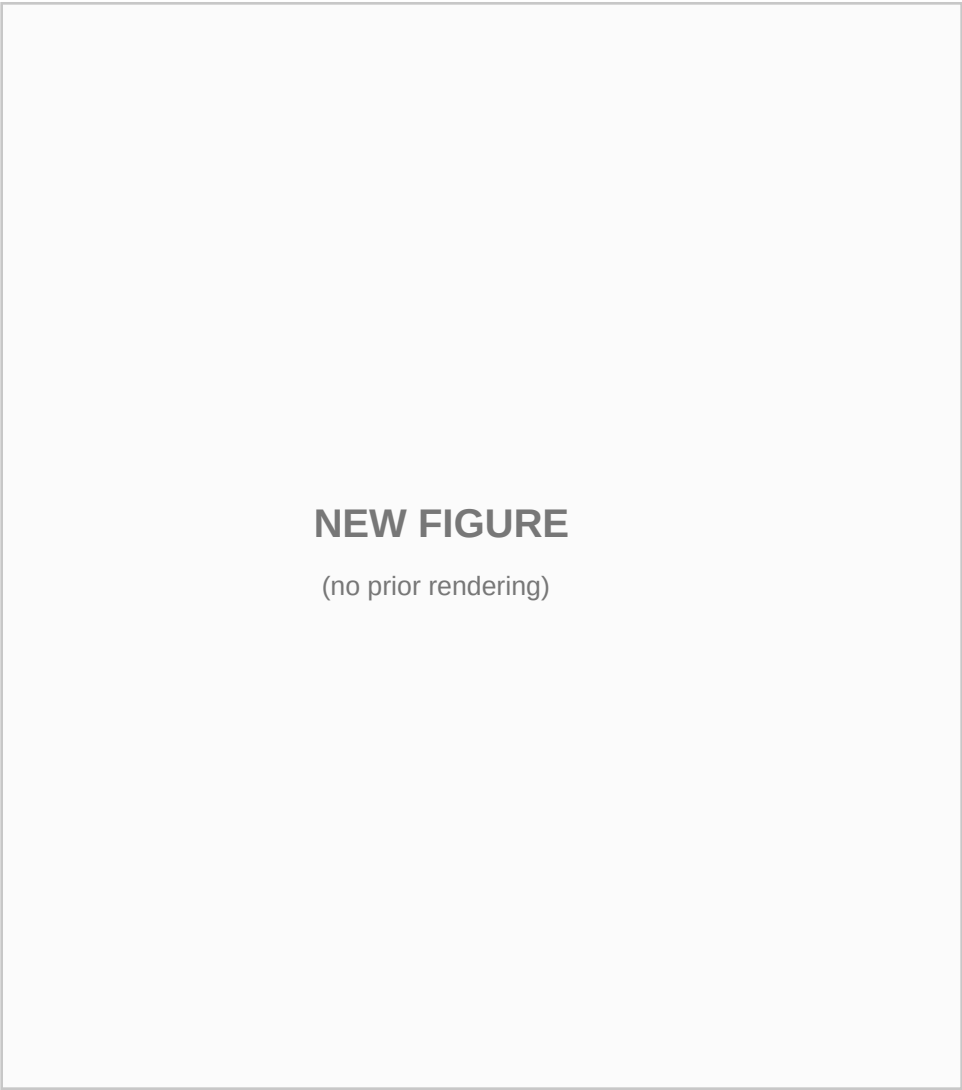
Marker weight $w(t)=r^t$ for two decay rates against the prune threshold of 0.05 (dashed). At $r=0.9$ the marker survives 28 ticks and is pruned at tick 29; at $r=0.95$ it lives until tick 59, since $\ln 0.05 / \ln 0.95 = 58.4$. Both curves are nearly flat where they meet the threshold, which is why moving the rate half the distance to one more than doubles the lifetime: that is the whole content of the calibration question. [internal; analytic]

AFTER



QA: PASS

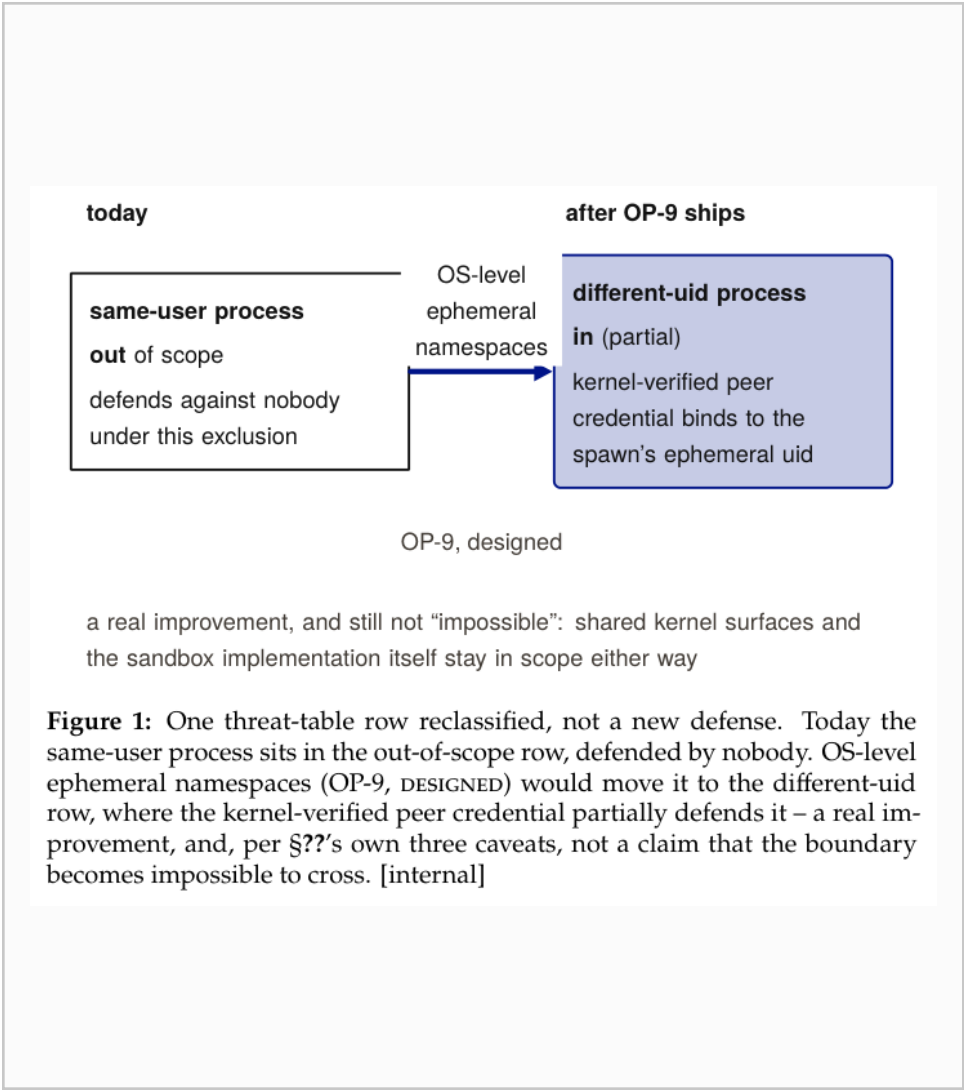
BEFORE



QA: —

One threat-table row reclassified, not a new defense. Today the same-user process sits in the out-of-scope row, defended by nobody. OS-level ephemeral namespaces (OP-9, {designed}) would move it to the different-uid row, where the kernel-verified peer credential partially defends it -- a real improvement, and, per {sec:merkle's own three caveats, not a claim that the boundary becomes impossible to cross. [internal]

AFTER



QA: PASS

BEFORE

NEW FIGURE

(no prior rendering)

AFTER

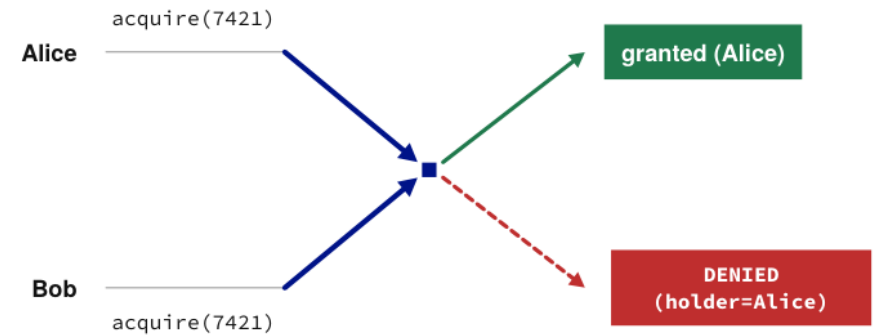


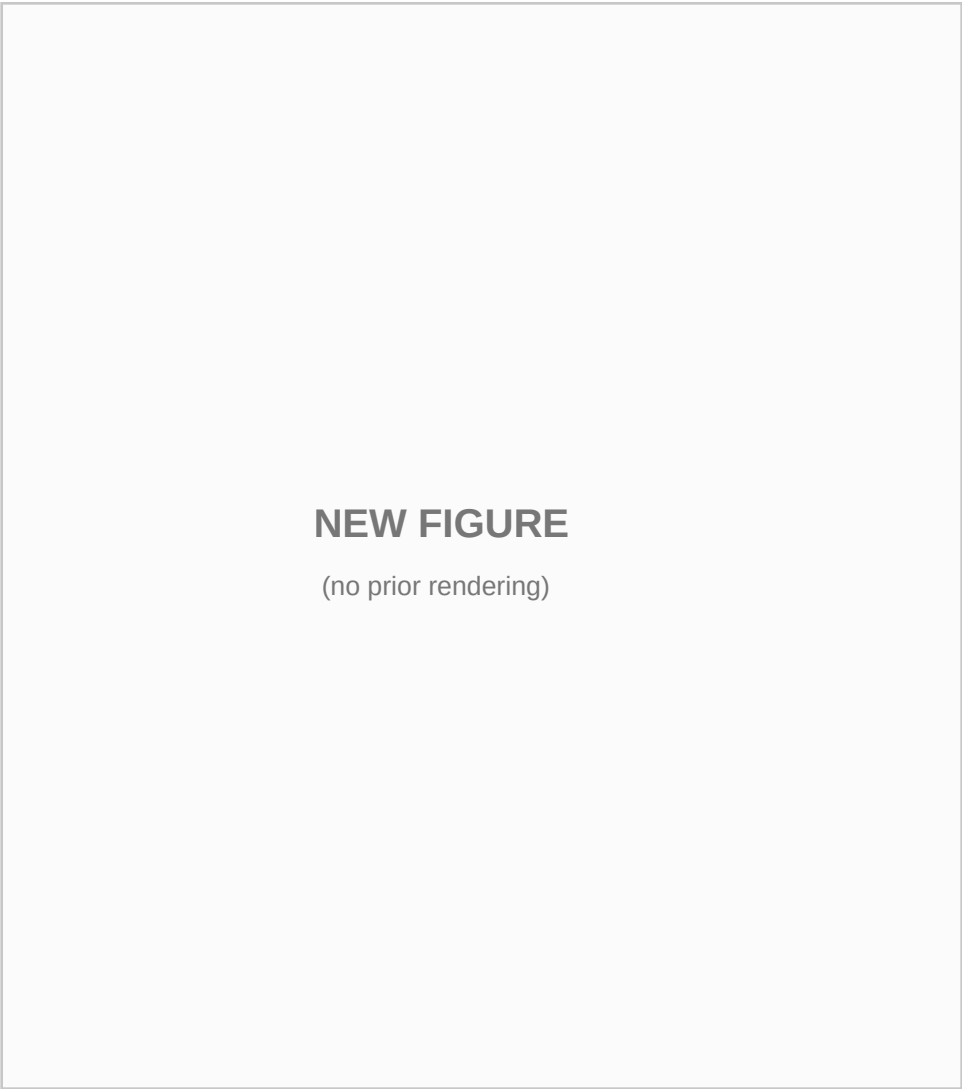
Figure 1: Two racing `acquire(7421)` calls converge on one insert-commit point in the single daemon; the uniqueness constraint decides the winner. Alice's insert lands first and she is granted the port. Bob's insert is rejected atomically – there is no moment at which both rows exist – and the catch path reads the row back, handing Bob a named `DENIED(holder=Alice)` rather than a bare error, so he can pick another port, wait, or message Alice. [internal, alg:acquire]

QA: —

QA: PASS

Two racing `{acquire(7421)}` calls converge on one insert-commit point in the single daemon; the uniqueness constraint decides the winner. Alice's insert lands first and she is granted the port. Bob's insert is rejected atomically -- there is no moment at which both rows exist -- and the catch path reads the row back, handing Bob a named `{DENIED(holder=Alice)}` rather than a bare error, so he can pick another port, wait...

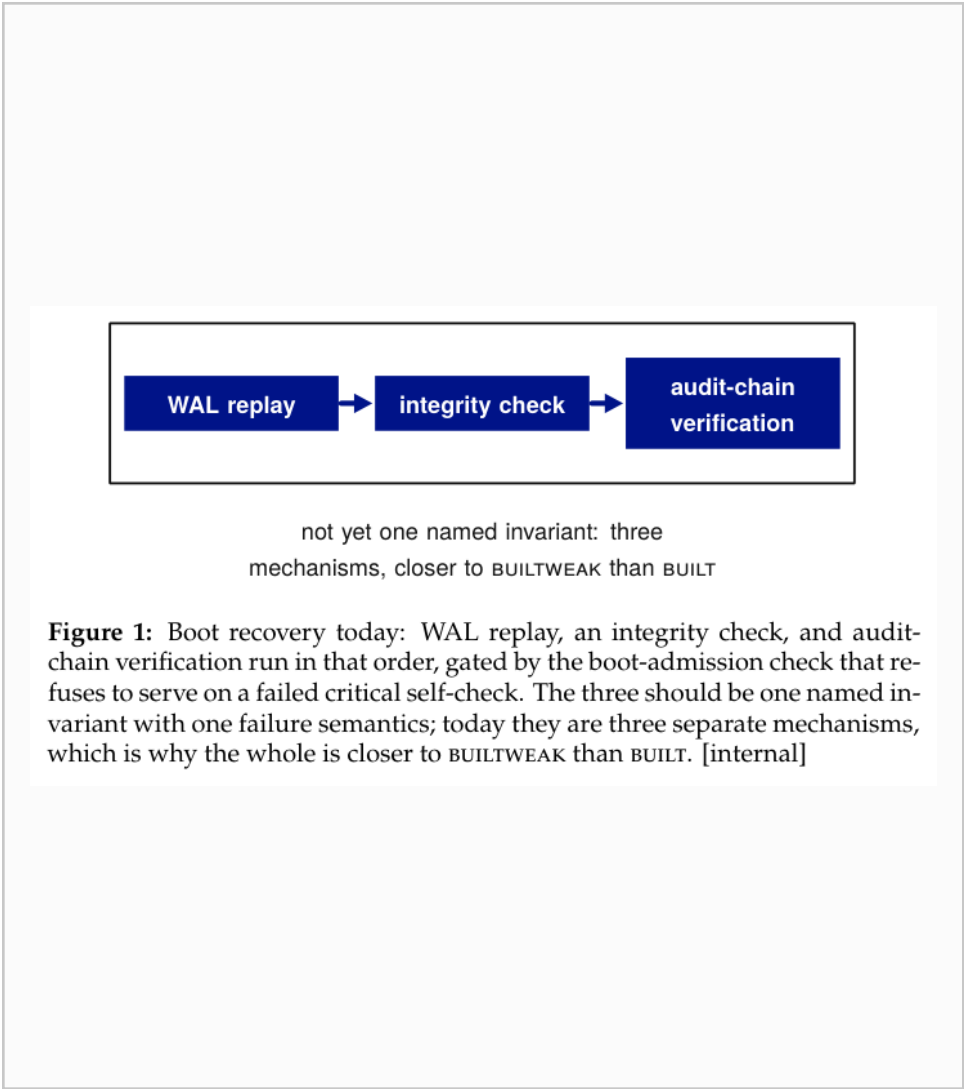
BEFORE



QA: —

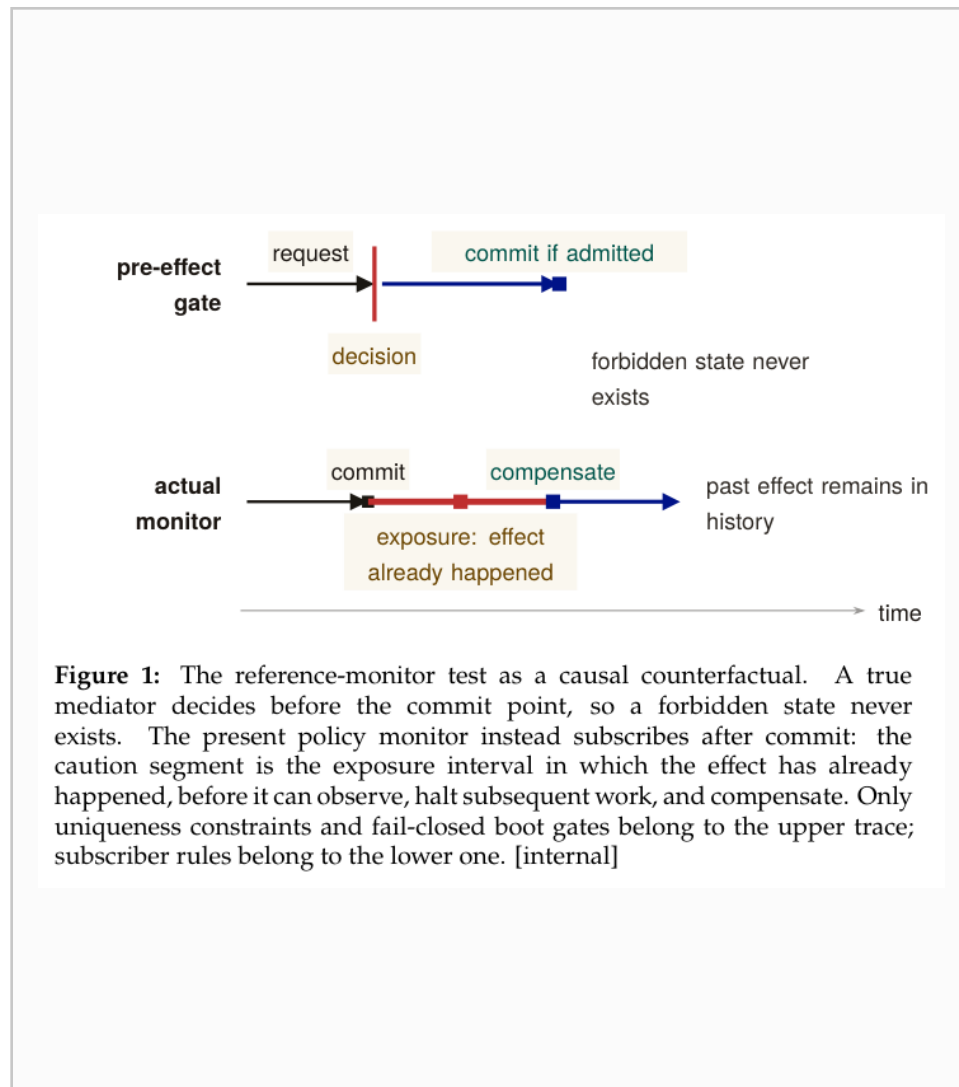
Boot recovery today: WAL replay, an integrity check, and audit-chain verification run in that order, gated by the boot-admission check that refuses to serve on a failed critical self-check. The three should be one named invariant with one failure semantics; today they are three separate mechanisms, which is why the whole is closer to {builtweak than {built. [internal]

AFTER



QA: PASS

BEFORE



AFTER

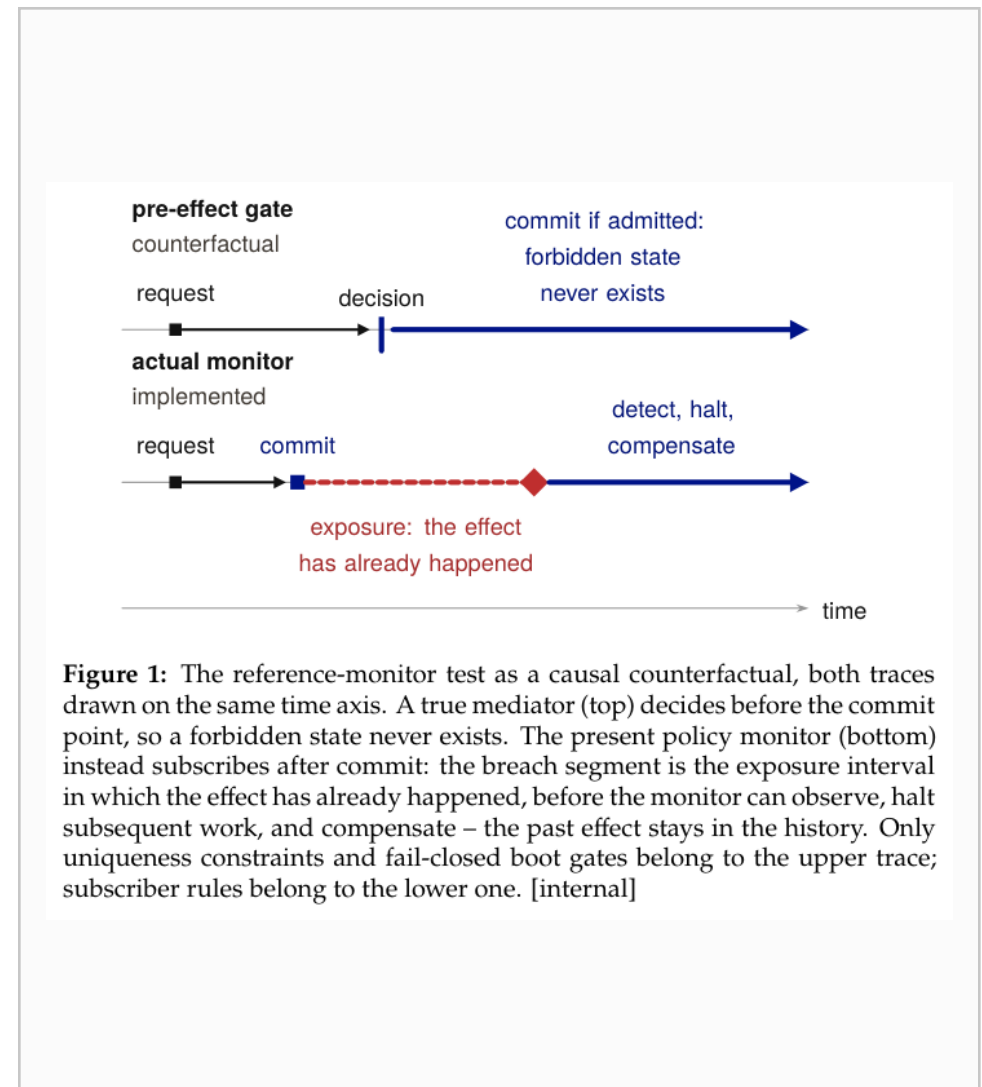


Figure 1: The reference-monitor test as a causal counterfactual. A true mediator decides before the commit point, so a forbidden state never exists. The present policy monitor instead subscribes after commit: the caution segment is the exposure interval in which the effect has already happened, before it can observe, halt subsequent work, and compensate. Only uniqueness constraints and fail-closed boot gates belong to the upper trace; subscriber rules belong to the lower one. [internal]

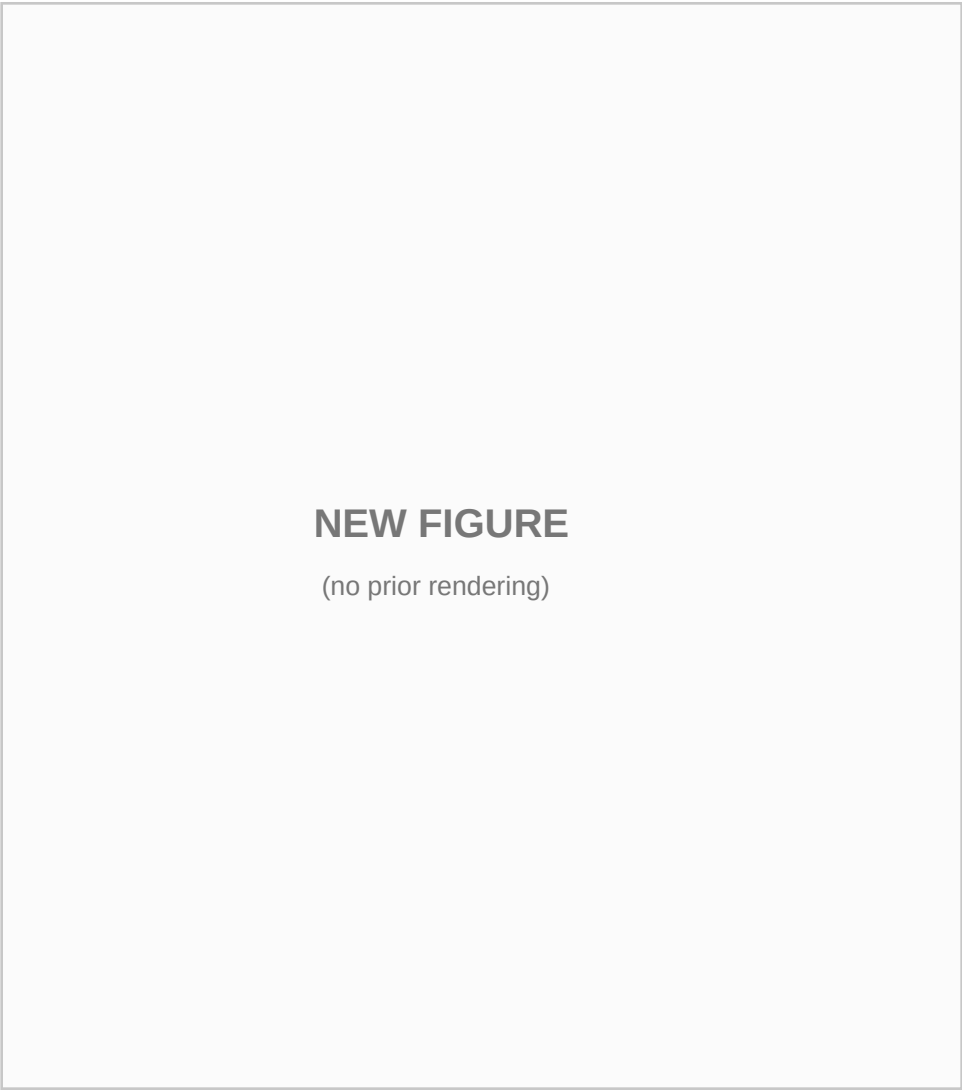
Figure 1: The reference-monitor test as a causal counterfactual, both traces drawn on the same time axis. A true mediator (top) decides before the commit point, so a forbidden state never exists. The present policy monitor (bottom) instead subscribes after commit: the breach segment is the exposure interval in which the effect has already happened, before the monitor can observe, halt subsequent work, and compensate – the past effect stays in the history. Only uniqueness constraints and fail-closed boot gates belong to the upper trace; subscriber rules belong to the lower one. [internal]

QA: PASS

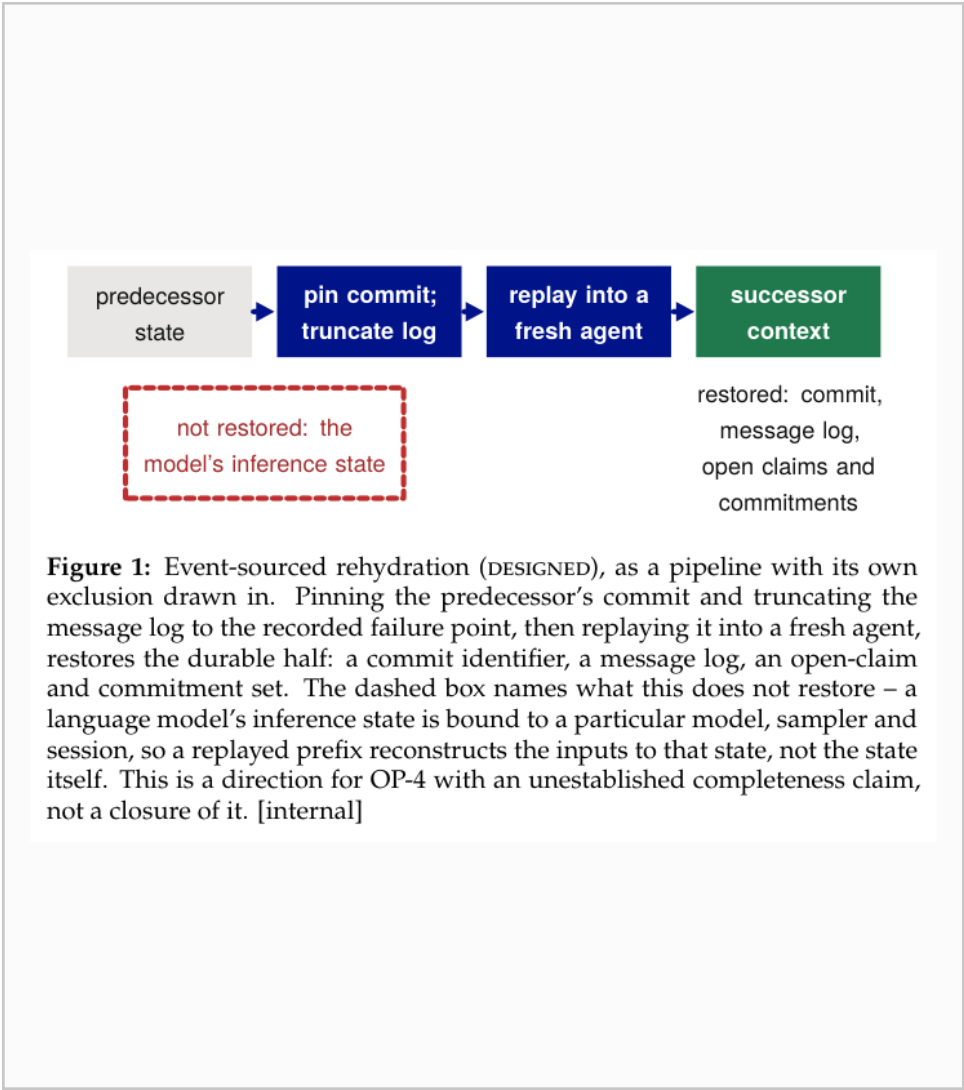
QA: PASS

The reference-monitor test as a causal counterfactual, both traces drawn on the same time axis. A true mediator (top) decides before the commit point, so a forbidden state never exists. The present policy monitor (bottom) instead subscribes after commit: the breach segment is the exposure interval in which the effect has already happened, before the monitor can observe, halt subsequent work, and compensate -- the ...

BEFORE



AFTER



QA: —

QA: PASS

Event-sourced rehydration (designed), as a pipeline with its own exclusion drawn in. Pinning the predecessor's commit and truncating the message log to the recorded failure point, then replaying it into a fresh agent, restores the durable half: a commit identifier, a message log, an open-claim and commitment set. The dashed box names what this does not restore -- a language model's inference state is bound to a p...

BEFORE

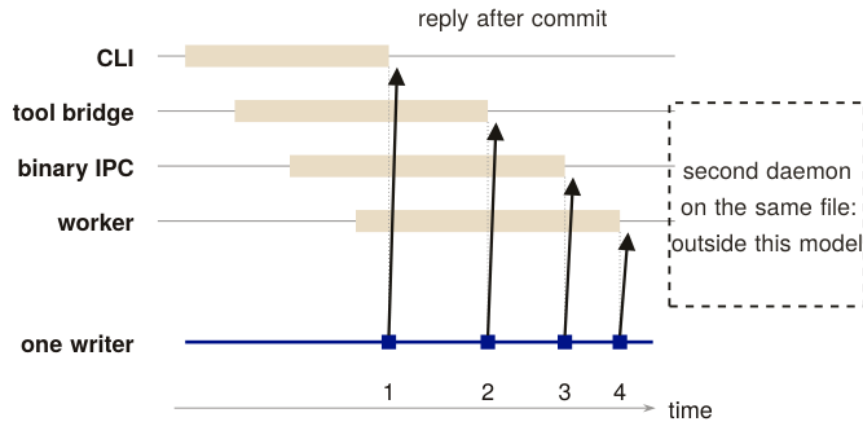


Figure 1: Four concurrent callers and one writer. Each band is a request in flight; its right edge is where the daemon commits it, and the rail shows the resulting order 1–4. A reply is sent only after the commit. A second daemon opening the same file is outside the model, not another lane. [internal]

QA: PASS

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AFTER

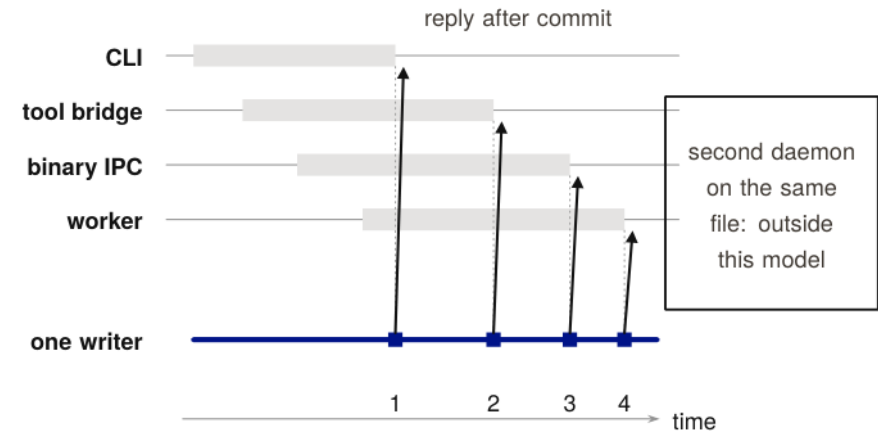


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QA: PASS

BEFORE

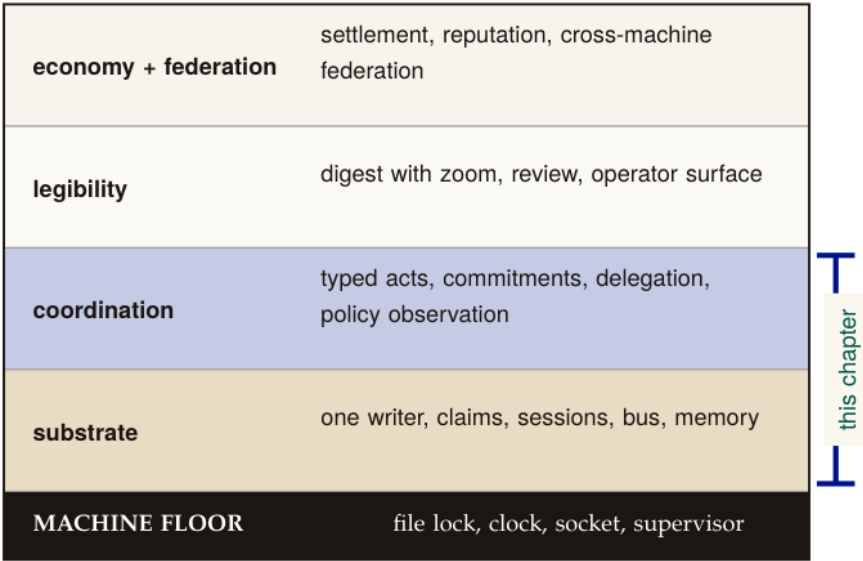


Figure 1: An architectural section through the four-layer coordination stack. This chapter is the second band from the floor: the transactional substrate, providing the one-writer record and claims that every higher layer rests on. The machine floor beneath it is the narrow dependency boundary the substrate assumes: a file lock, a clock, a socket, and a supervisor. [internal]

QA: PASS

An architectural section through the four-layer coordination stack. Each band names a layer and what that layer hands the one above it; the shaded band is this chapter's, the layer that rests directly on the machine floor. The dashed floor band is not a layer of the system but the narrow dependency boundary the substrate assumes: a file lock, a clock, a socket and a supervisor. [internal]

AFTER

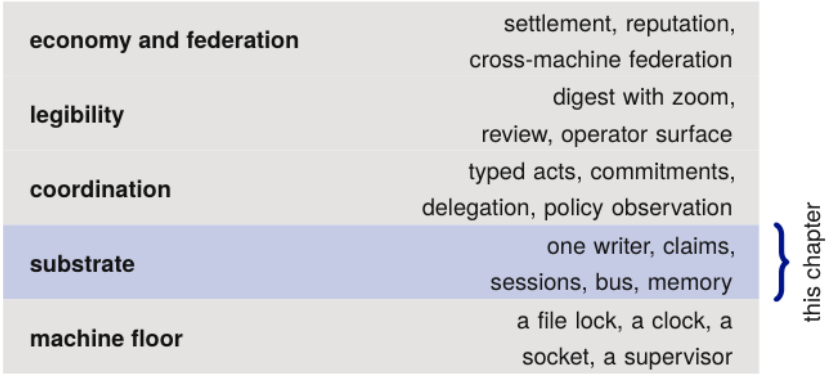
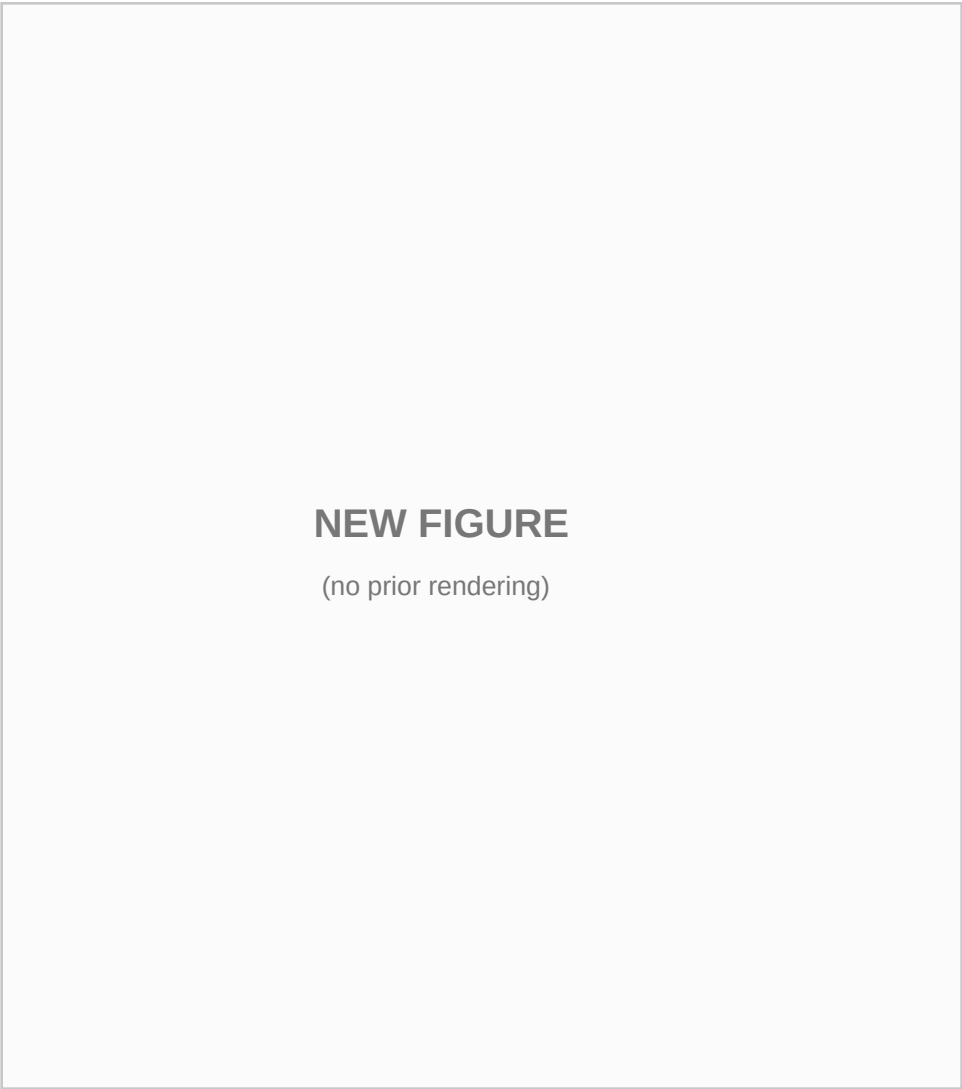


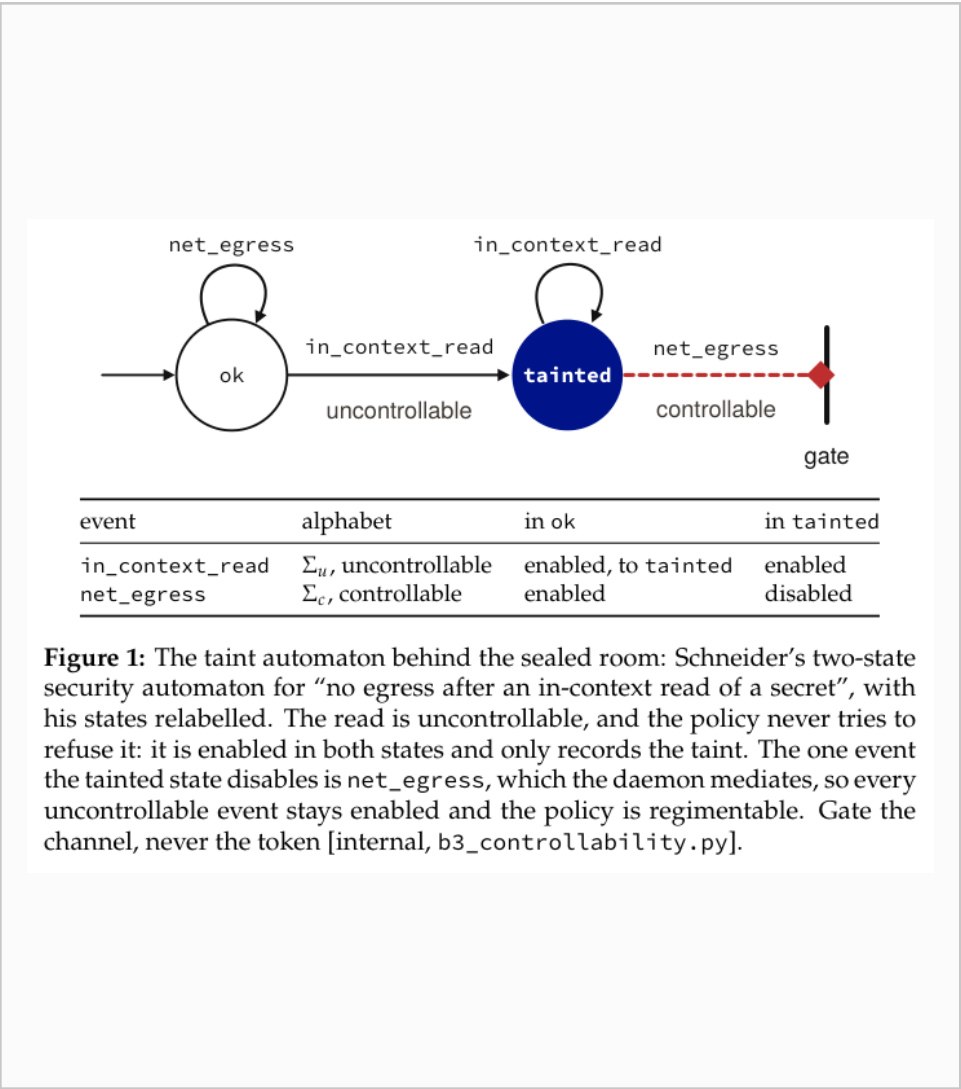
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QA: PASS

BEFORE



AFTER

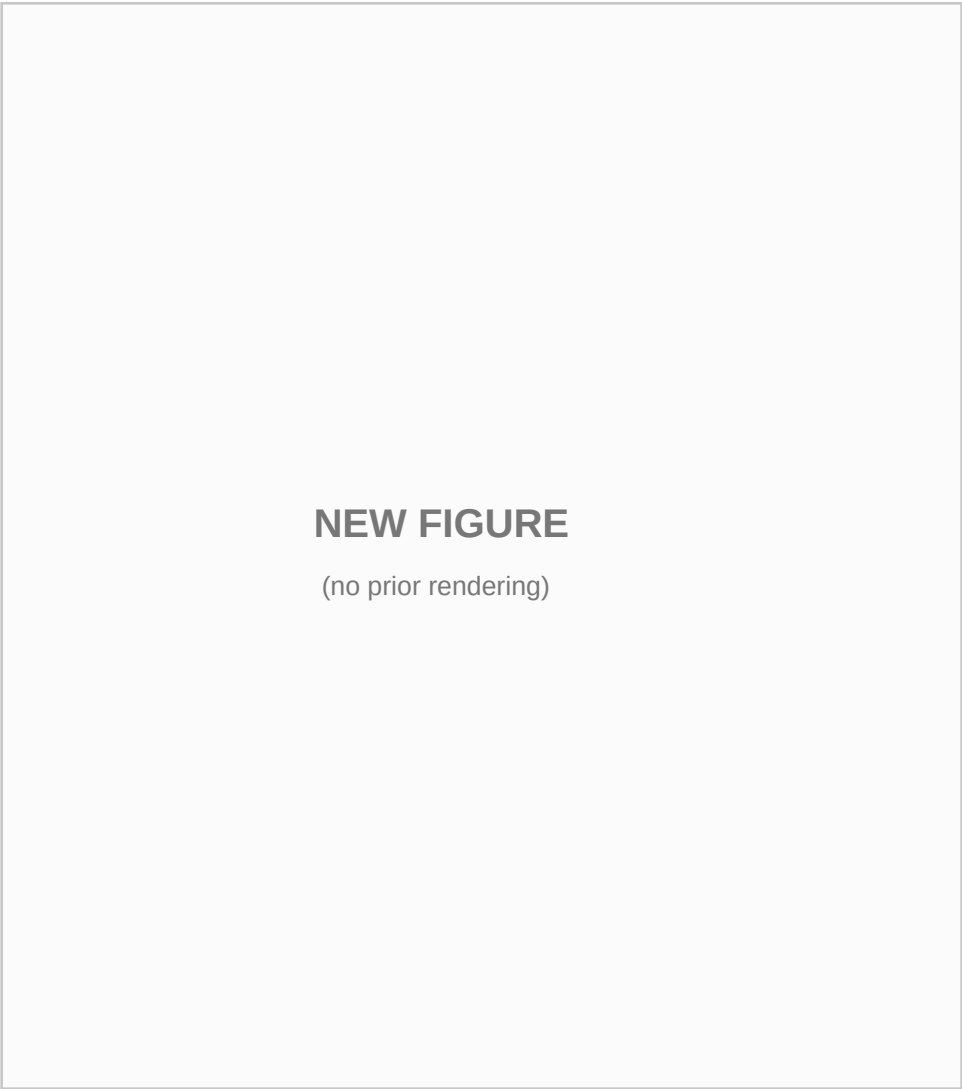


QA: —

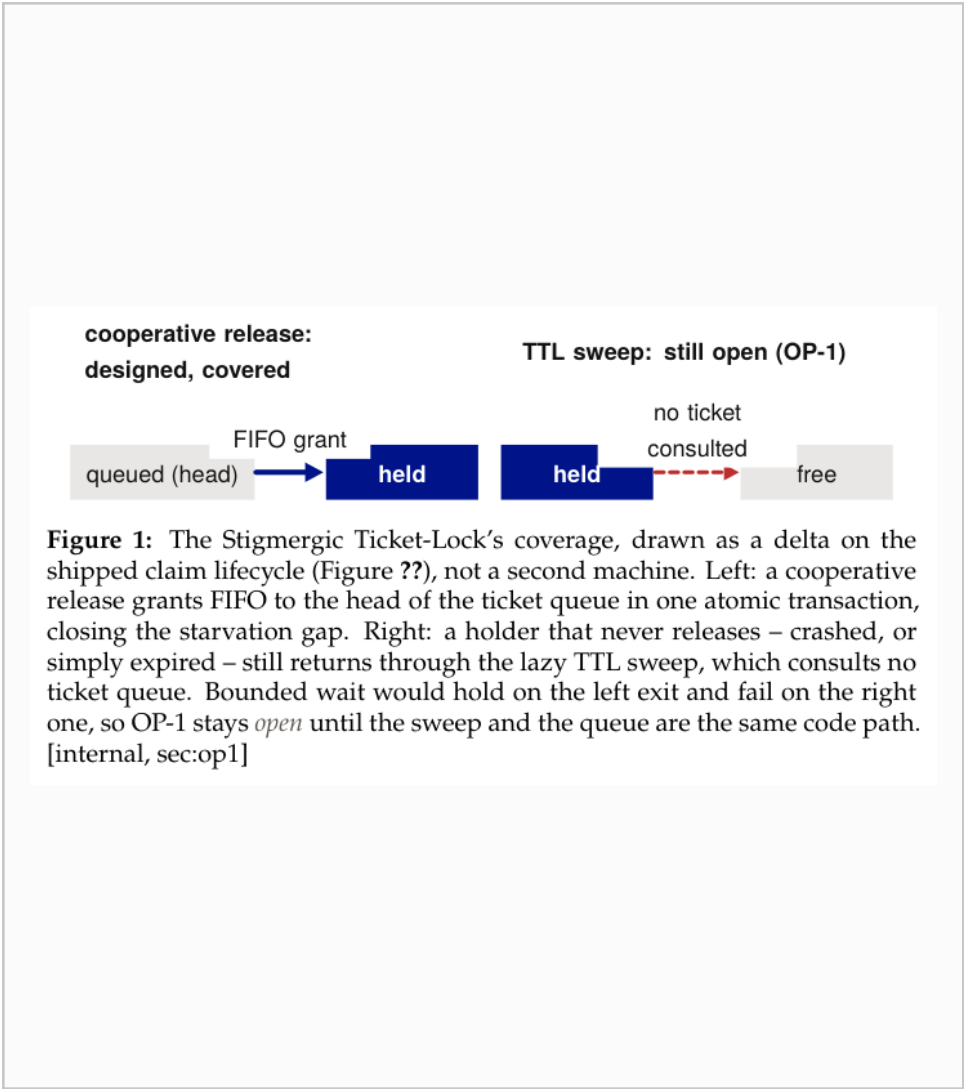
QA: PASS

The taint automaton behind the sealed room: Schneider's two-state security automaton for “no egress after an in-context read of a secret”, with his states relabelled. The read is uncontrollable, and the policy never tries to refuse it: it is enabled in both states and only records the taint. The one event the tainted state disables is {net_egress, which the daemon mediates, so every uncontrollable event stays e...

BEFORE



AFTER

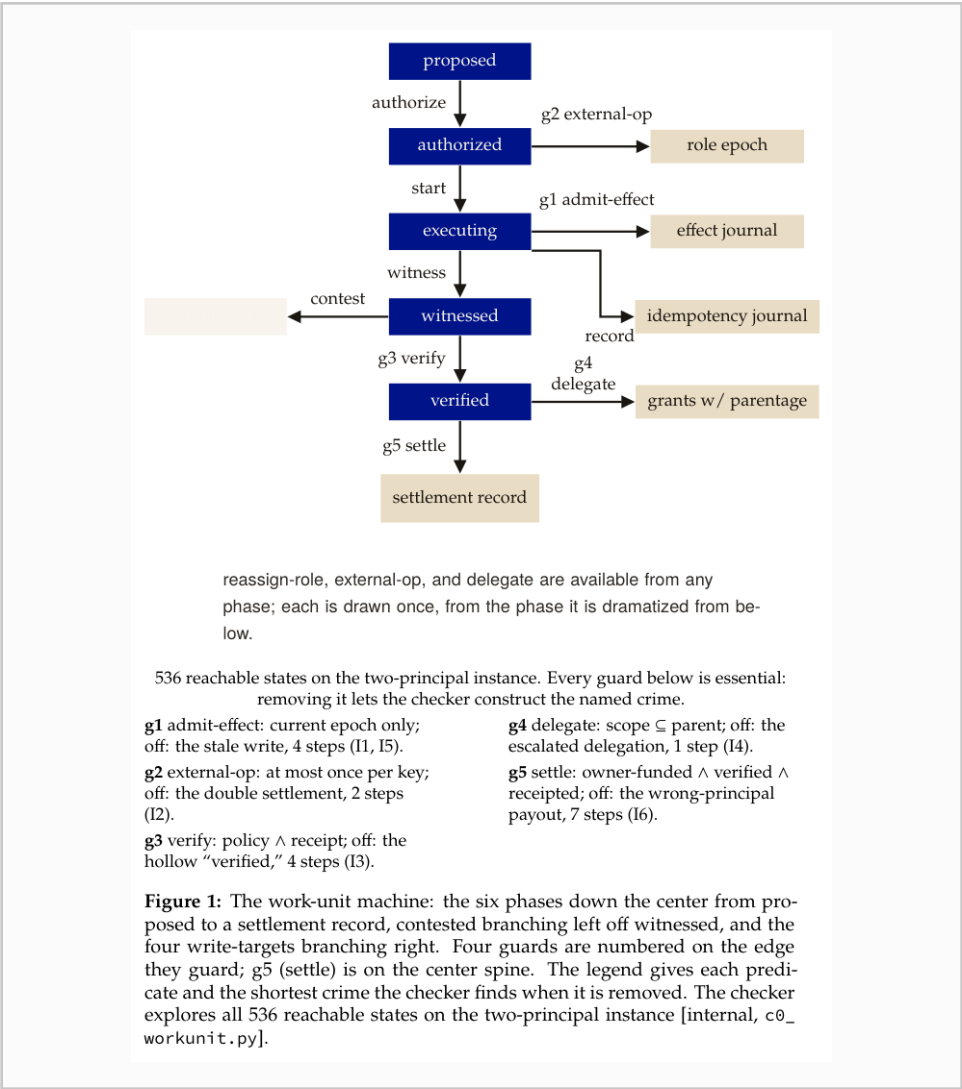


QA: —

QA: PASS

The Stigmergic Ticket-Lock’s coverage, drawn as a delta on the shipped claim lifecycle (Figure~{fig:swk-claim-lifecycle}), not a second machine. Left: a cooperative release grants FIFO to the head of the ticket queue in one atomic transaction, closing the starvation gap. Right: a holder that never releases -- crashed, or simply expired -- still returns through the lazy TTL sweep, which consults no ticket queue. Bou...

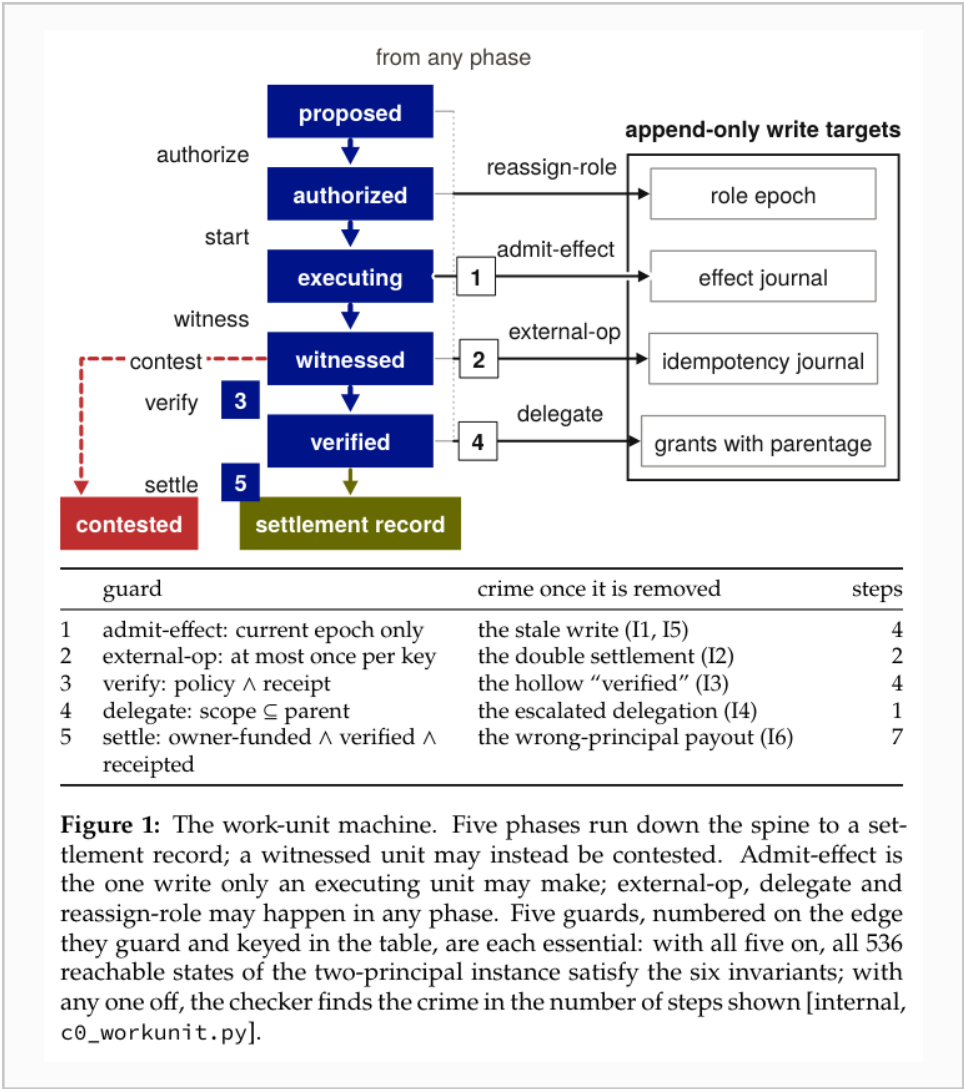
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QA: FLAGGED: too wide 4.68in

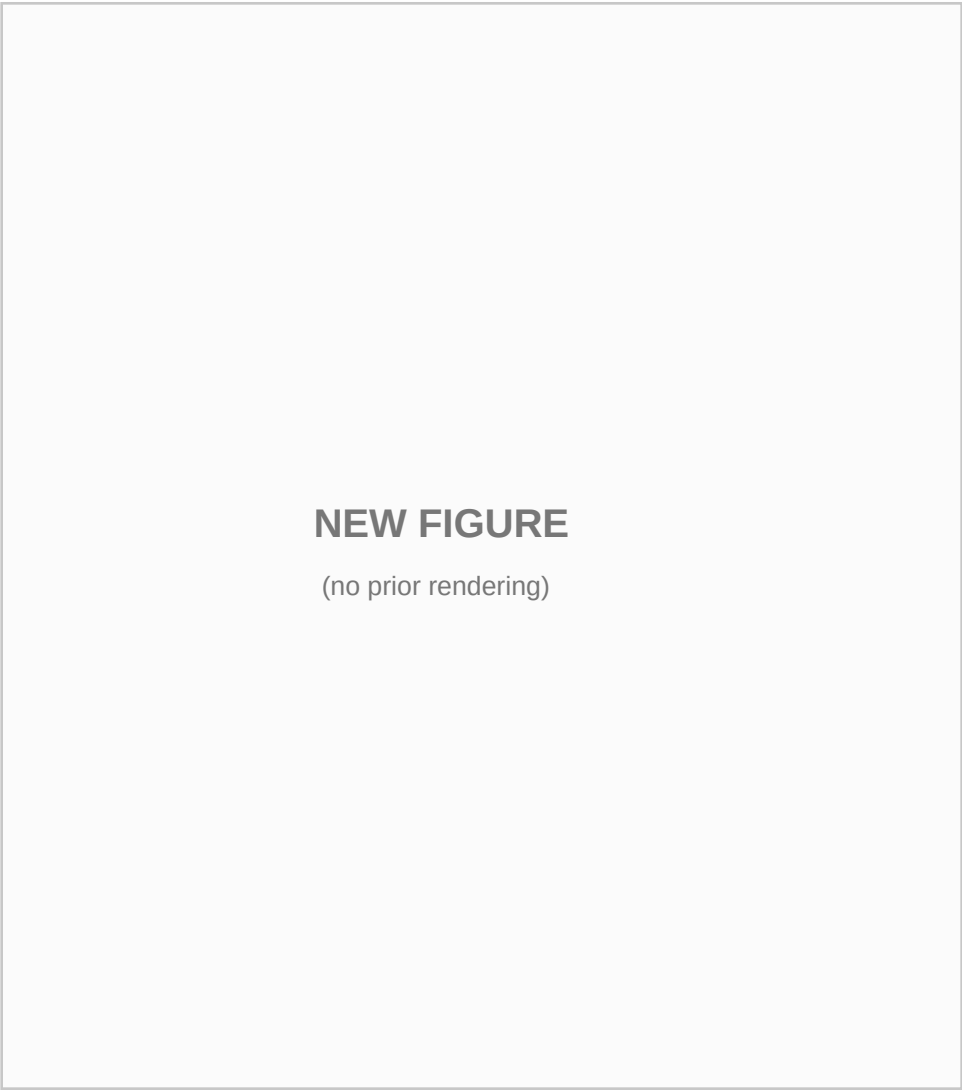
The work-unit machine. Five phases run down the spine to a settlement record; a witnessed unit may instead be contested. Admit-effect is the one write only an executing unit may make; external-op, delegate and reassign-role may happen in any phase. Five guards, numbered on the edge they guard and keyed in the table, are each essential: with all five on, all 536 reachable states of the two-principal instance satisf...

AFTER



QA: PASS

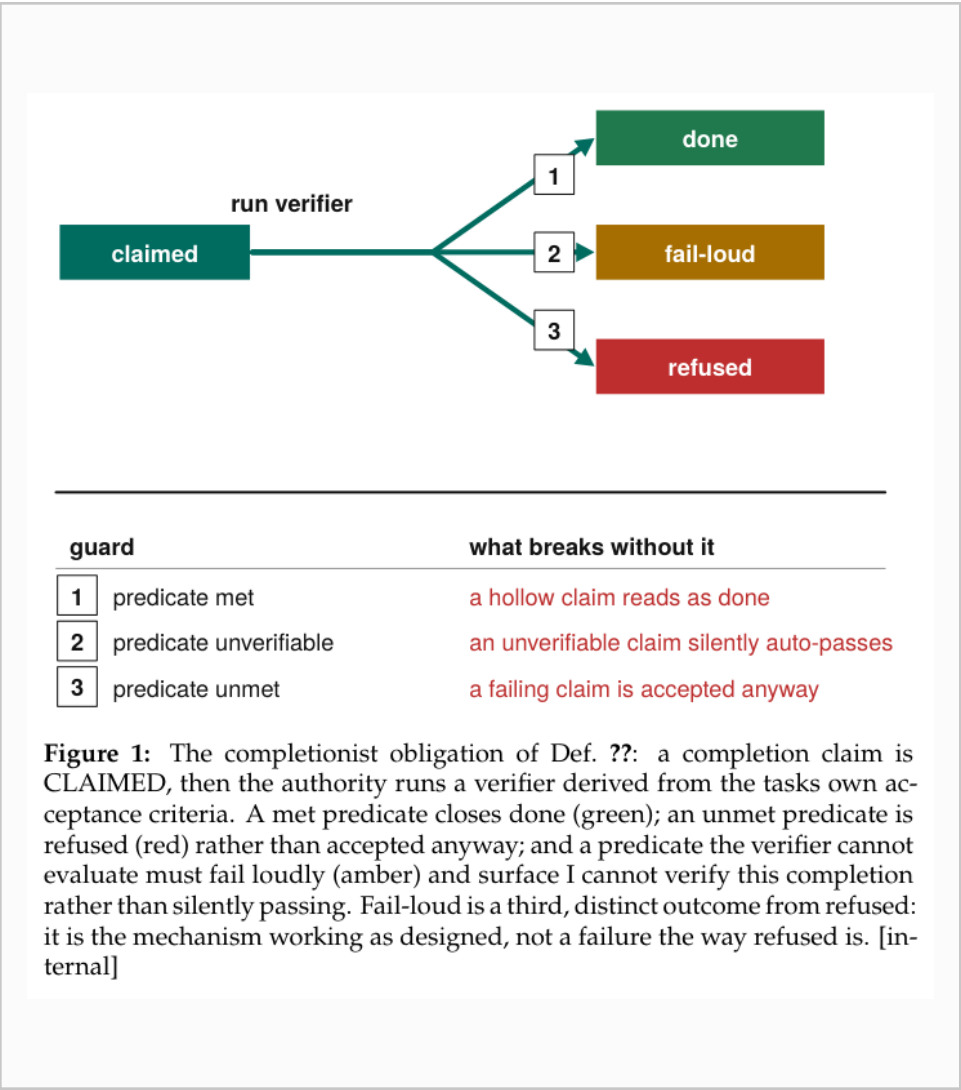
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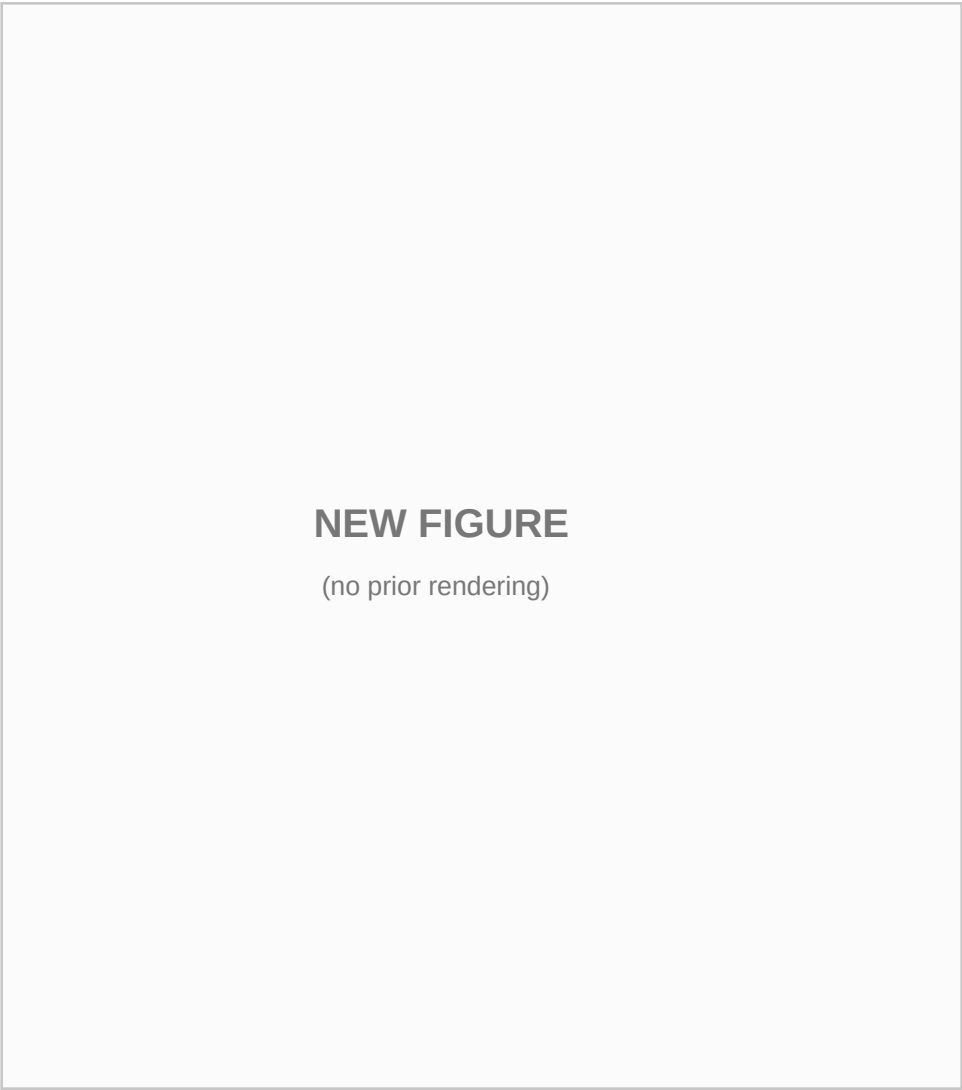
The completionist obligation of Def.~{def:completionist: a completion claim is CLAIMED, then the authority runs a verifier derived from the tasks own acceptance criteria. A met predicate closes done (green); an unmet predicate is refused (red) rather than accepted anyway; and a predicate the verifier cannot evaluate must fail loudly (amber) and surface I cannot verify this completion rather than silently passing. ...

AFTER



QA: PASS

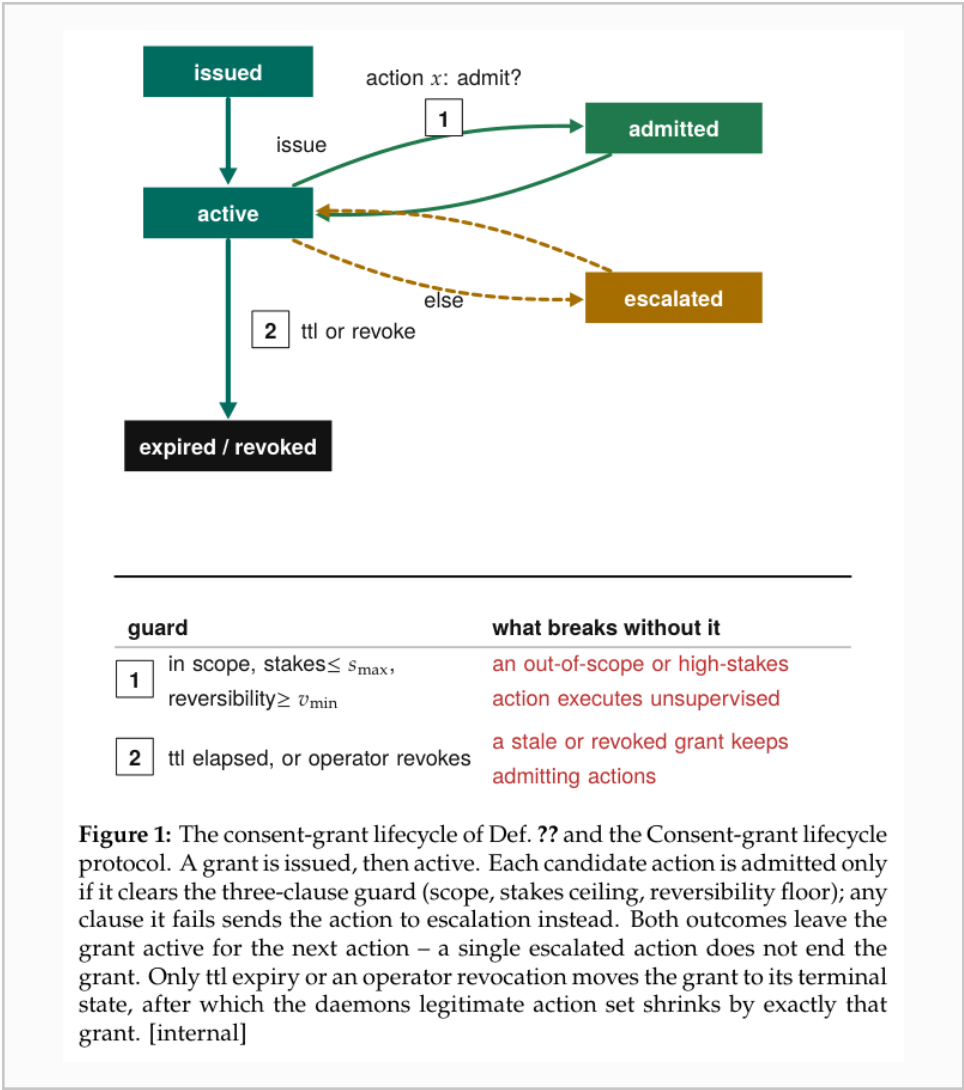
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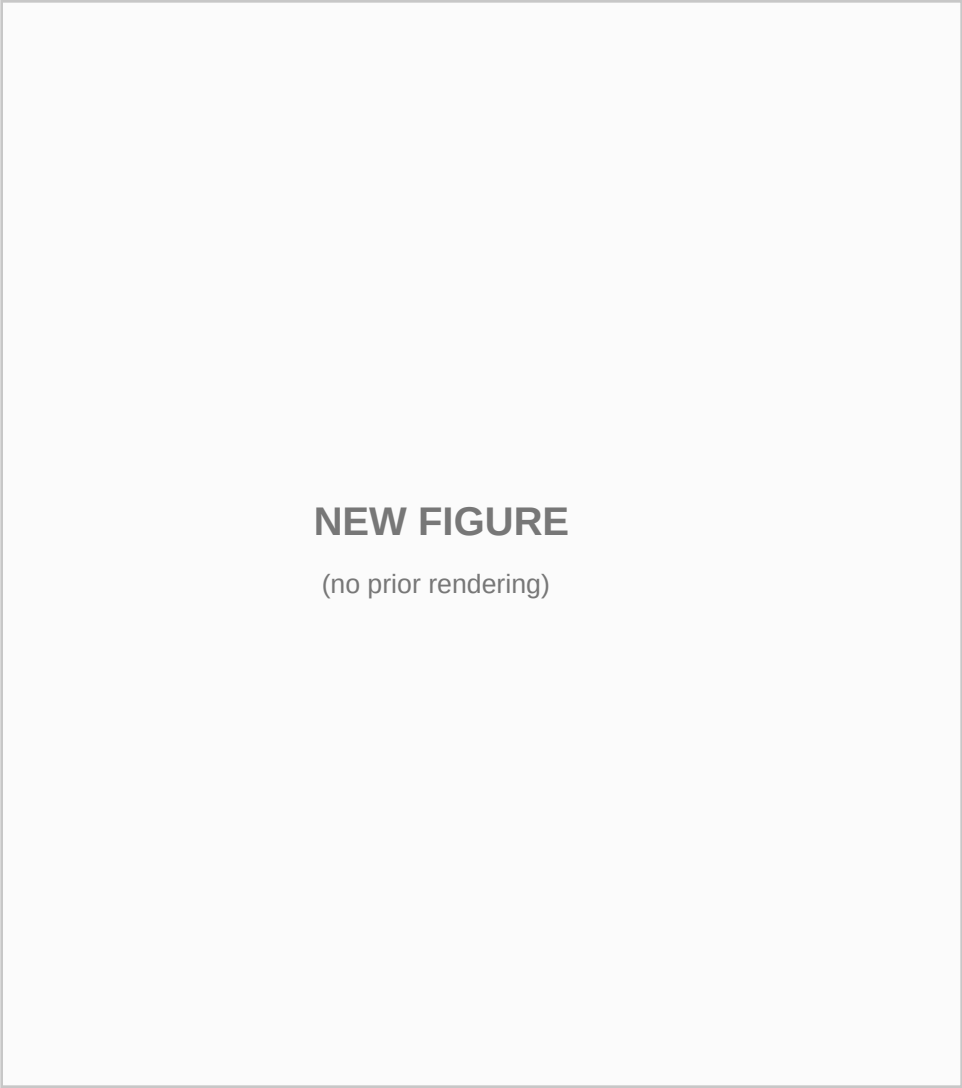
The consent-grant lifecycle of Def.~{def:consent and the Consent-grant lifecycle protocol. A grant is issued, then active. Each candidate action is admitted only if it clears the three-clause guard (scope, stakes ceiling, reversibility floor); any clause it fails sends the action to escalation instead. Both outcomes leave the grant active for the next action -- a single escalated action does not end the grant. Onl...

AFTER



QA: PASS

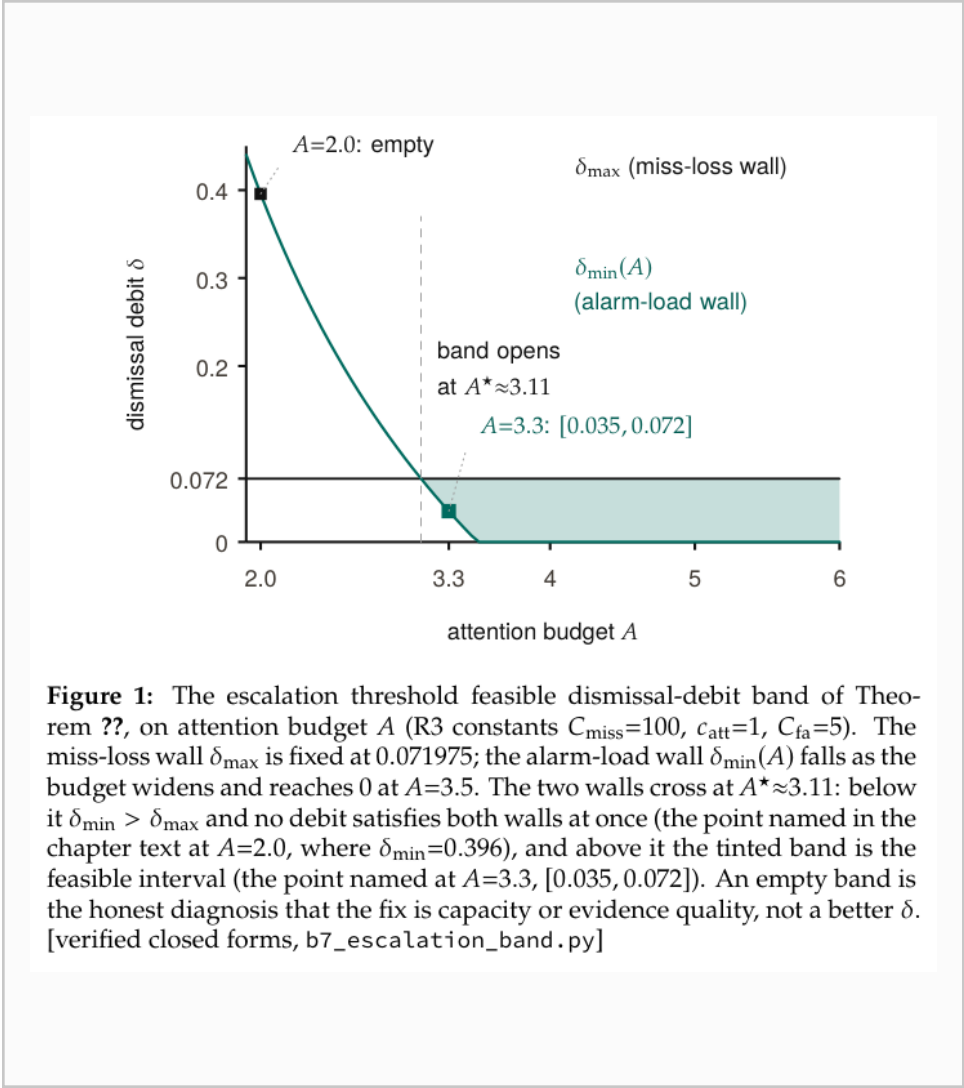
BEFORE



QA: —

The escalation threshold feasible dismissal-debit band of Theorem~{thm:escalation-threshold, on attention budget \$A\$ (R3 constants \$C_{\text{miss}}=100\$, \$c_{\text{att}}=1\$, \$C_{\text{fa}}=5\$). The miss-loss wall \$\delta_{\text{max}}\$ is fixed at \$0.071975\$; the alarm-load wall \$\delta_{\text{min}}(A)\$ falls as the budget widens and reaches \$0\$ at \$A=3.5\$. The two walls cross at \$A^{\wedge}\{3.11\$: below it \$\delta_{\text{min}} > \delta_{\text{max}}\$ and no debit satisfies both walls at once (the point named...

AFTER



QA: PASS

BEFORE

NEW FIGURE

(no prior rendering)

AFTER

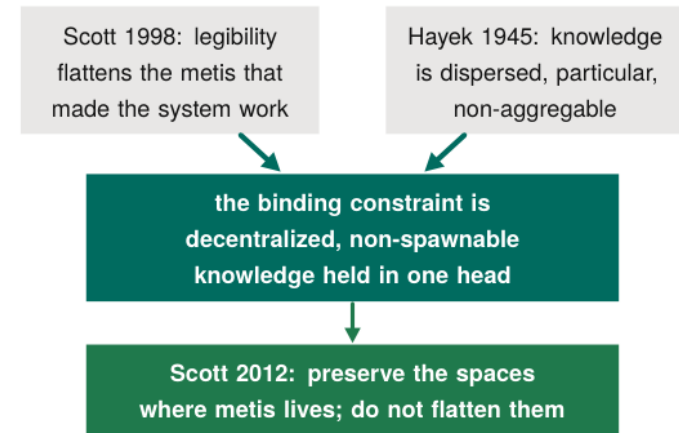


Figure 1: Scotts legibility warning and Hayeks dispersed-knowledge argument converge on one claim from two independent poles – political anthropology and economics – rather than stating two separate cautions. Scotts own later book supplies the constructive sequel: preserve the spaces where metis lives, rather than flattening them for legibilitys sake. That two routes this far apart arrive at a compatible object is itself evidence for the claim, not decoration. [internal]

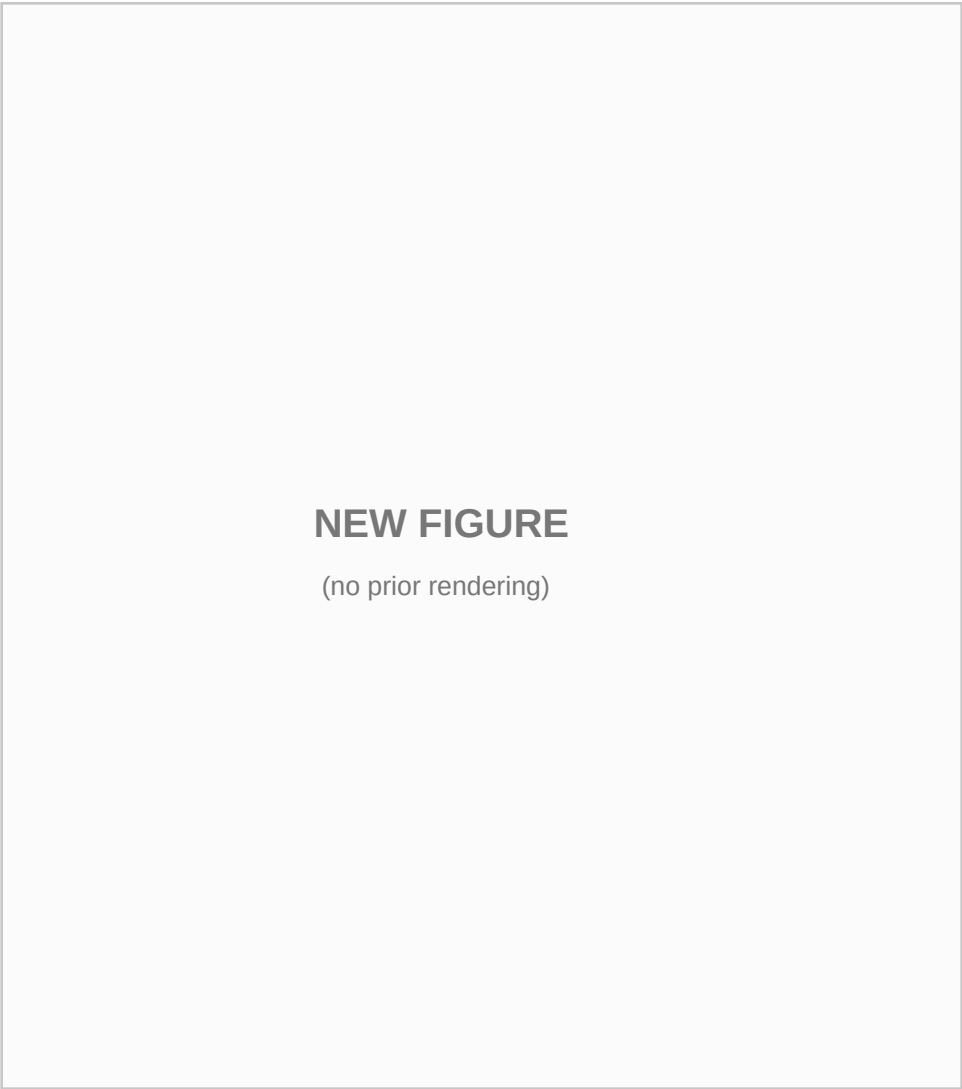
QA: —

QA: PASS

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...

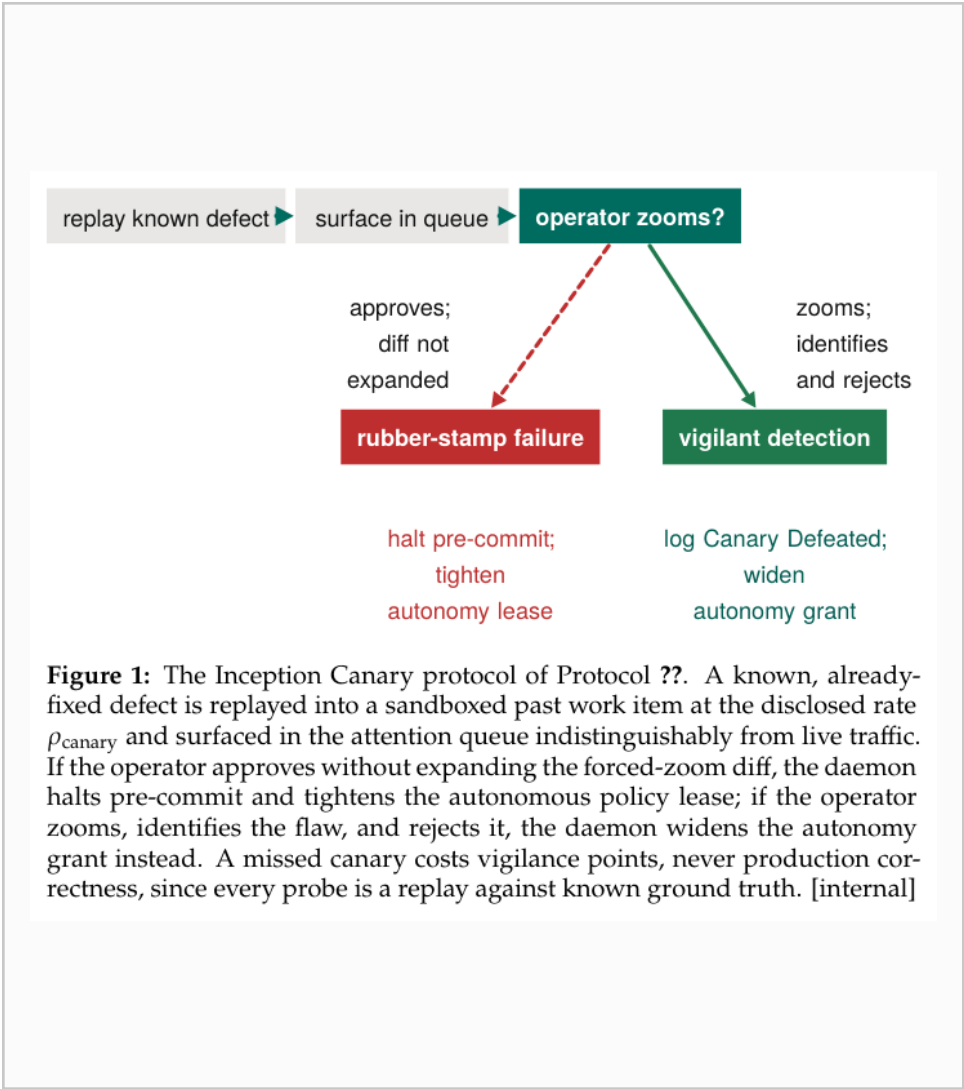
BEFORE



QA: —

The Inception Canary protocol of Protocol- $\{\text{prot:canary}\}$. A known, already-fixed defect is replayed into a sandboxed past work item at the disclosed rate ρ_{canary} and surfaced in the attention queue indistinguishably from live traffic. If the operator approves without expanding the forced-zoom diff, the daemon halts pre-commit and tightens the autonomous policy lease; if the operator zooms, identifies the flaw, and rejects it, the daemon widens the autonomy grant instead. A missed canary costs vigilance points, never production correctness, since every probe is a replay against known ground truth. [internal]

AFTER



QA: PASS

BEFORE

NEW FIGURE

(no prior rendering)

AFTER

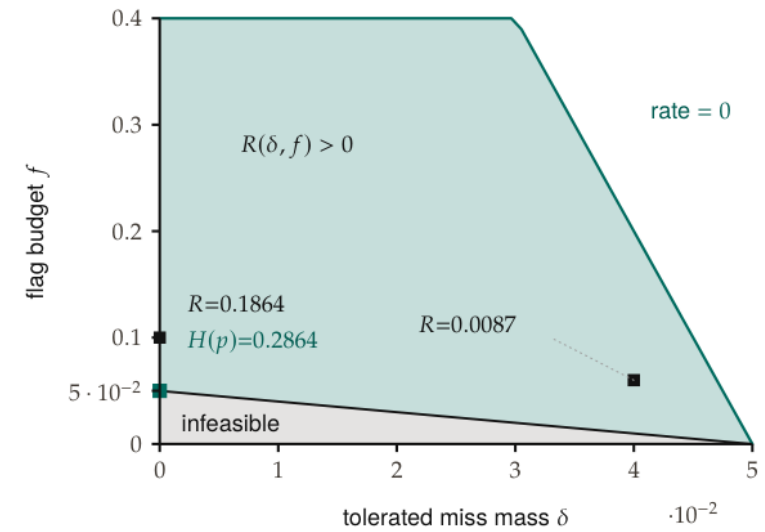


Figure 1: The two-constraint digest rate $R(\delta, f)$ of Theorem ?? on tolerated miss mass δ against flag budget f , at the chapter's Bernoulli rate $p = 0.05$. The tinted wedge is where both constraints bind and the joint is fully pinned; below the floor $f = p - \delta$ (shaded) no digest can meet the miss budget at all, and above the line $f = 1 - \delta/p$ an X -independent flagger already meets it on its own, so the true rate is exactly 0. The zero-miss, tightest-flag corner $(\delta, f) = (0, p)$ recovers the source's full entropy, $H(0.05) = 0.2864$ bits; tolerating $\delta = 0$ but widening the flag budget to $f = 0.10$ drops the price to 0.1864 bits, and tolerating $\delta = 0.04$ at $f = 0.06$ drops it to 0.0087 bits, a hundredth of the zero-miss corner. [internal, b1_frontier.py]

QA: —

QA: PASS

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BEFORE

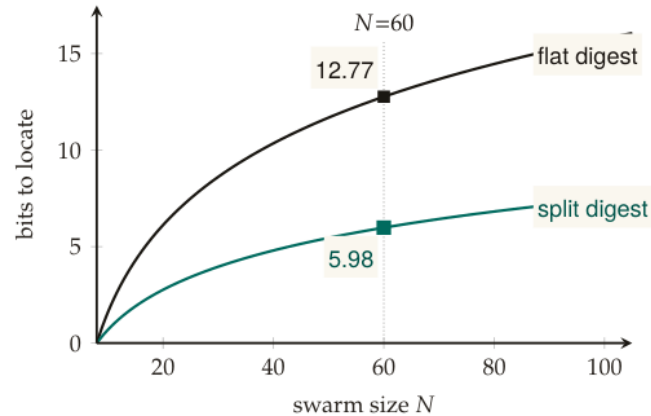


Figure 1: Bits an operator must read to locate the critical artifact, at the lower bound of Thm. ?? with $m=8$, $k=2$. One flat digest serving both readers jointly needs $\text{Floor}(N, 4, 8)$ bits; two split digests, one per reader, each need only $\text{Floor}(N, 2, 8)$. At $N=60$ the split digest costs 5.98 bits against the flat digest's 12.77 — a $2.13\times$ penalty, not the $2\times$ that halving the readers would predict. The committed sweep CSV (`r1-floor.csv`) is not yet present, so the two curves are the chapter's own closed-form Floor formula rather than simulated points; only the $N=60$ values it works by hand are marked [internal].

QA: PASS

AFTER

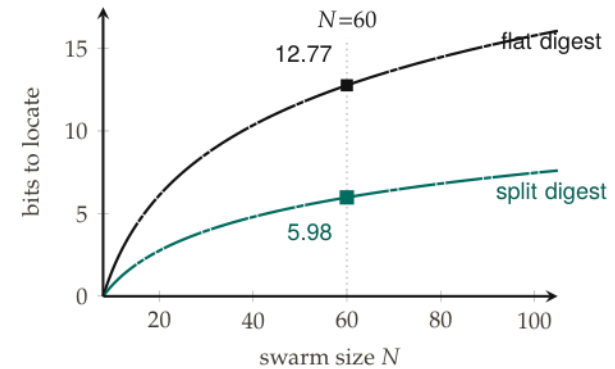
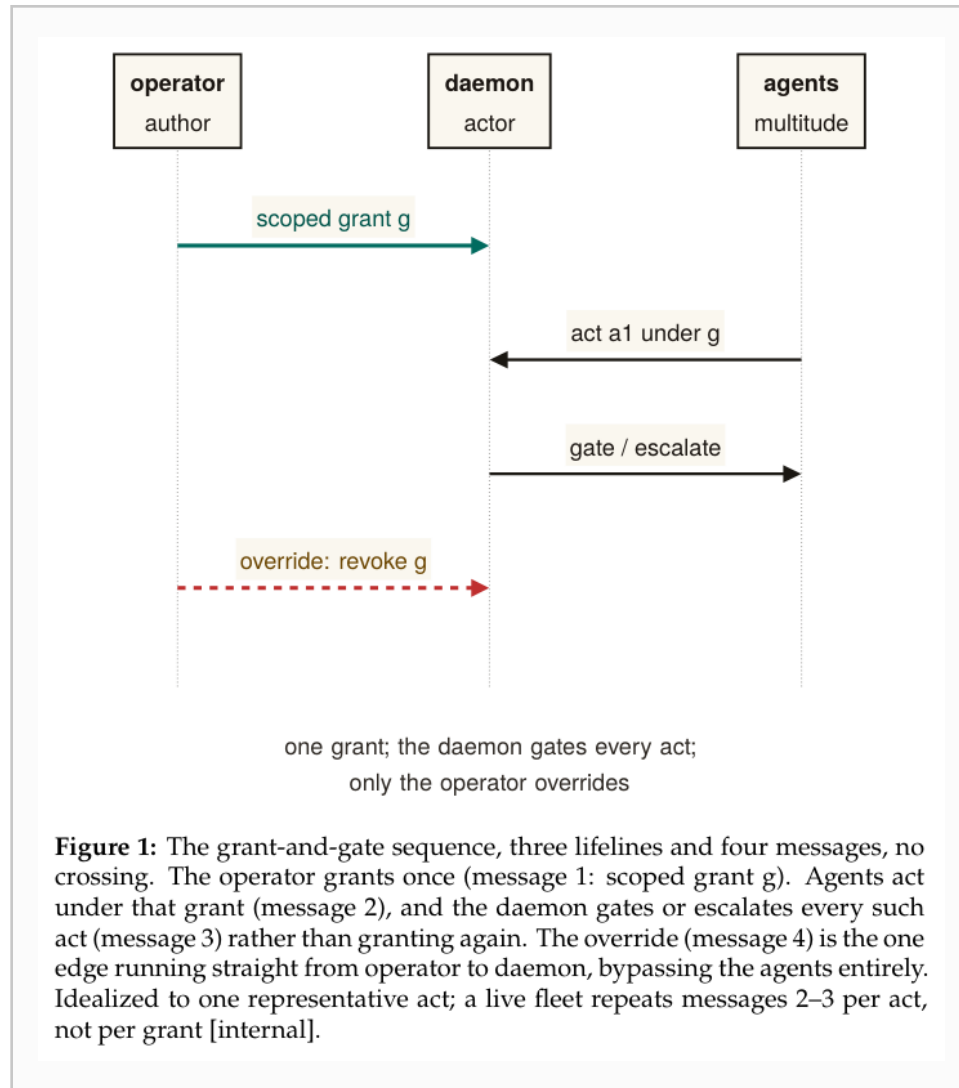


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QA: PASS

Bits an operator must read to locate the critical artifact, at the lower bound of Thm. ?? with $m=8$, $k=2$. One flat digest serving both readers jointly needs $\text{Floor}(N, 4, 8)$ bits; two split digests, one per reader, each need only $\text{Floor}(N, 2, 8)$. At $N=60$ the split digest costs 5.98 bits against the flat digest's 12.77 — a $2.13\times$ penalty, not the $2\times$ that halving the readers would predict. The...

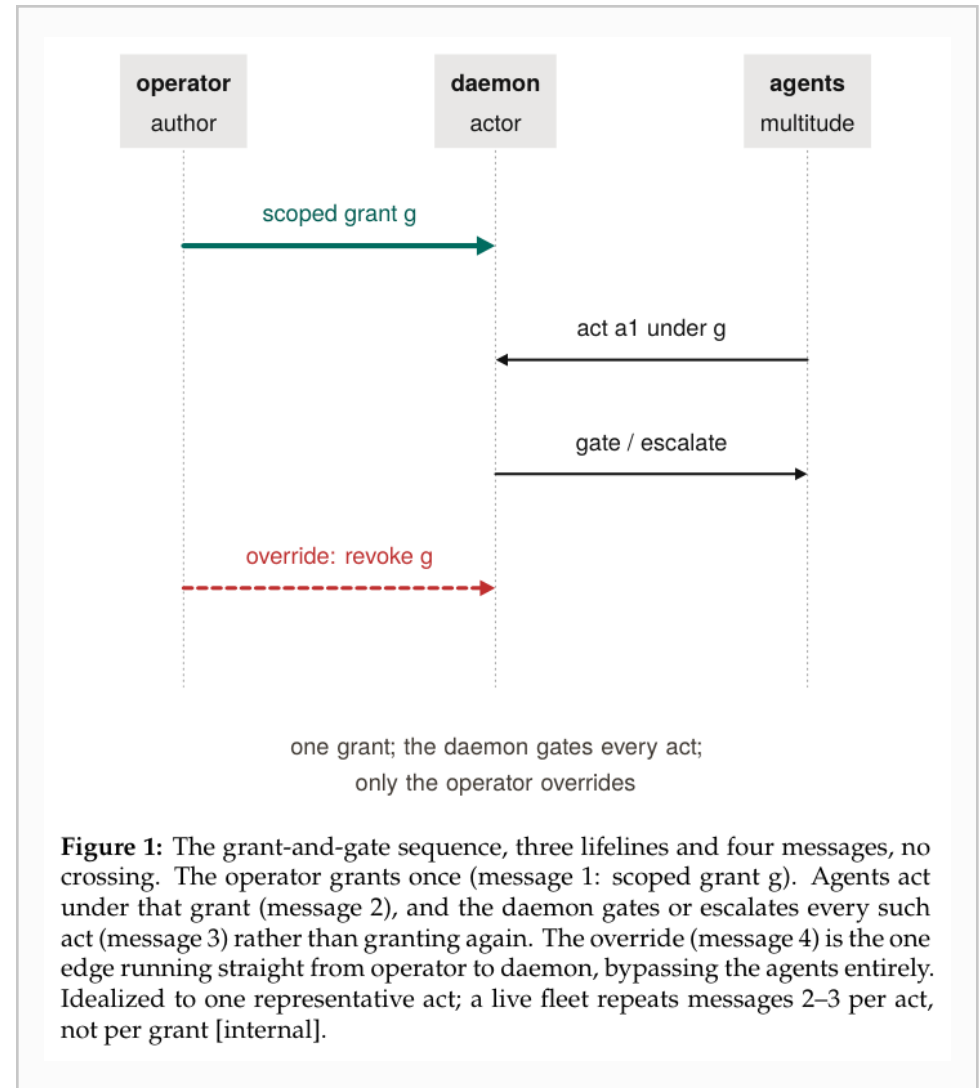
BEFORE



QA: PASS

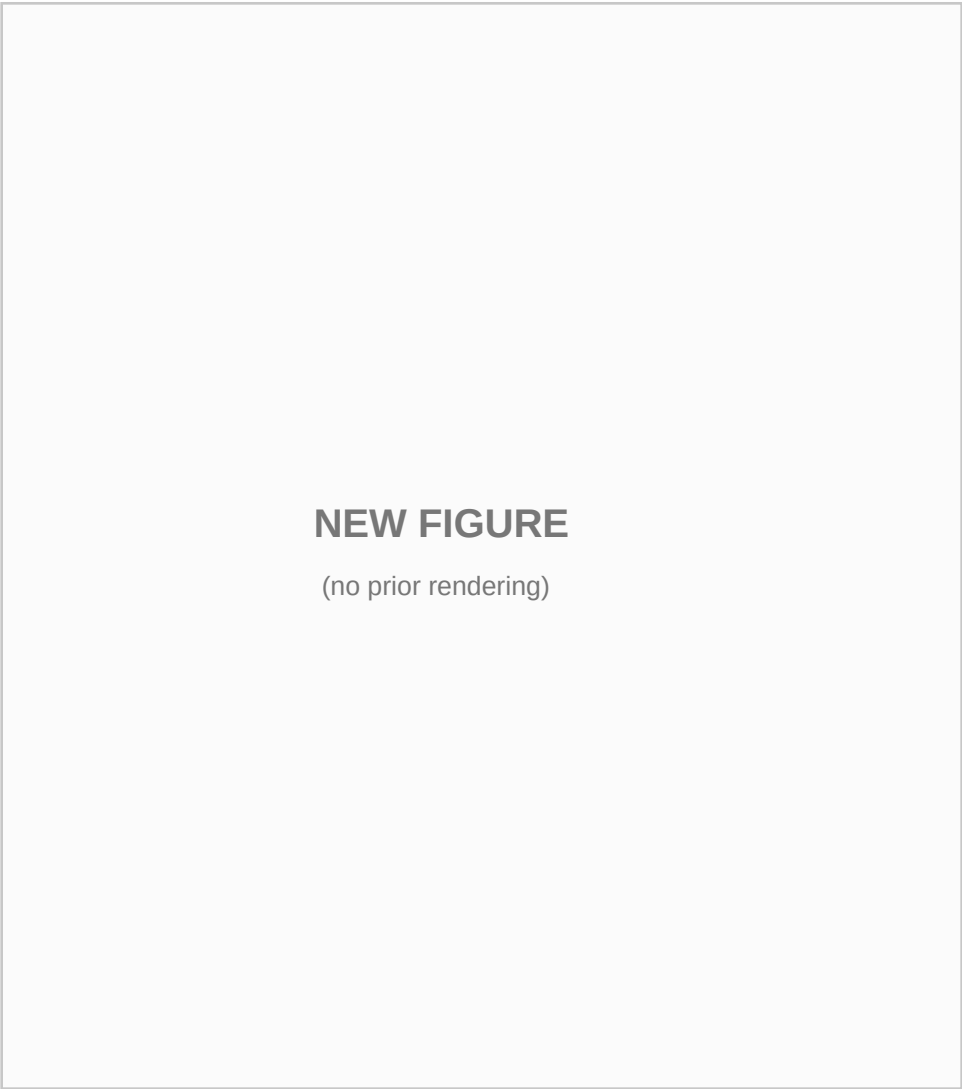
The grant-and-gate sequence, three lifelines and four messages, no crossing. The operator grants once (message 1: scoped grant g). Agents act under that grant (message 2), and the daemon gates or escalates every such act (message 3) rather than granting again. The override (message 4) is the one edge running straight from operator to daemon, bypassing the agents entirely. Idealized to one representative act; a liv...

AFTER



QA: PASS

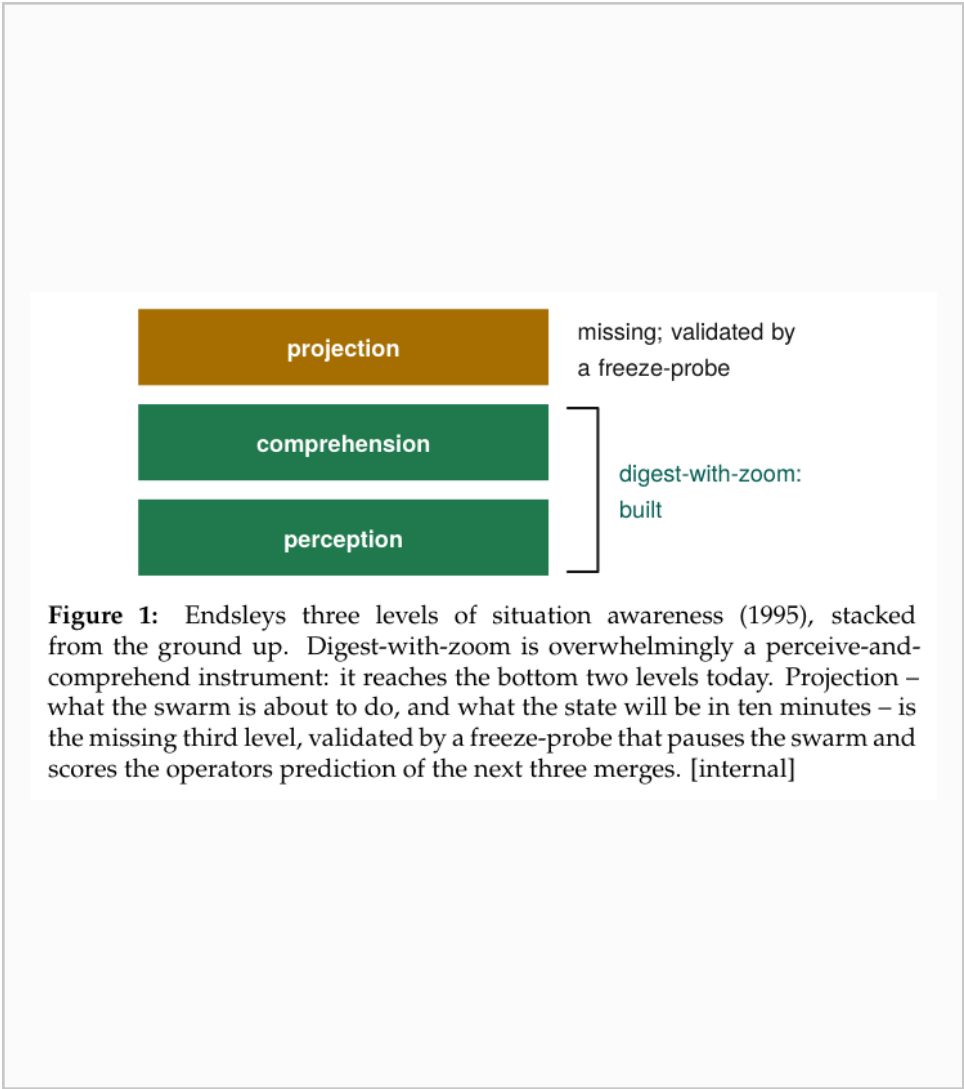
BEFORE



QA: —

Endsleys three levels of situation awareness (1995), stacked from the ground up. Digest-with-zoom is overwhelmingly a perceive-and-comprehend instrument: it reaches the bottom two levels today. Projection -- what the swarm is about to do, and what the state will be in ten minutes -- is the missing third level, validated by a freeze-probe that pauses the swarm and scores the operators prediction of the next three m...

AFTER



QA: PASS

BEFORE

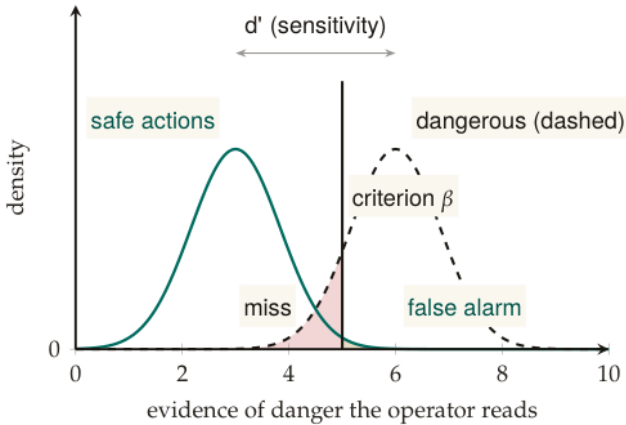


Figure 1: The operator’s forced-zoom decision as Signal Detection Theory (Def. ??). Two named densities over the evidence the operator reads: safe actions (solid) and dangerous actions (dashed), separated by sensitivity d' . Zooming triggers when evidence exceeds the criterion β ; moving β left trades the caution-tinted miss region for more of the neutral-tinted false-alarm region. Because a miss costs far more than a false alarm — at the chapter’s running costs, 100 against 5, the inspect bar sits at $a \geq 0.057$ — the optimal criterion sits well to the left of the throughput optimum: idealized as two fixed Gaussians, not the fitted mixture the deployed detector would use [internal].

QA: PASS

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AFTER

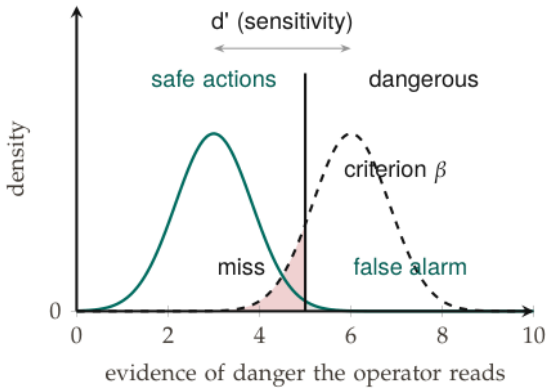
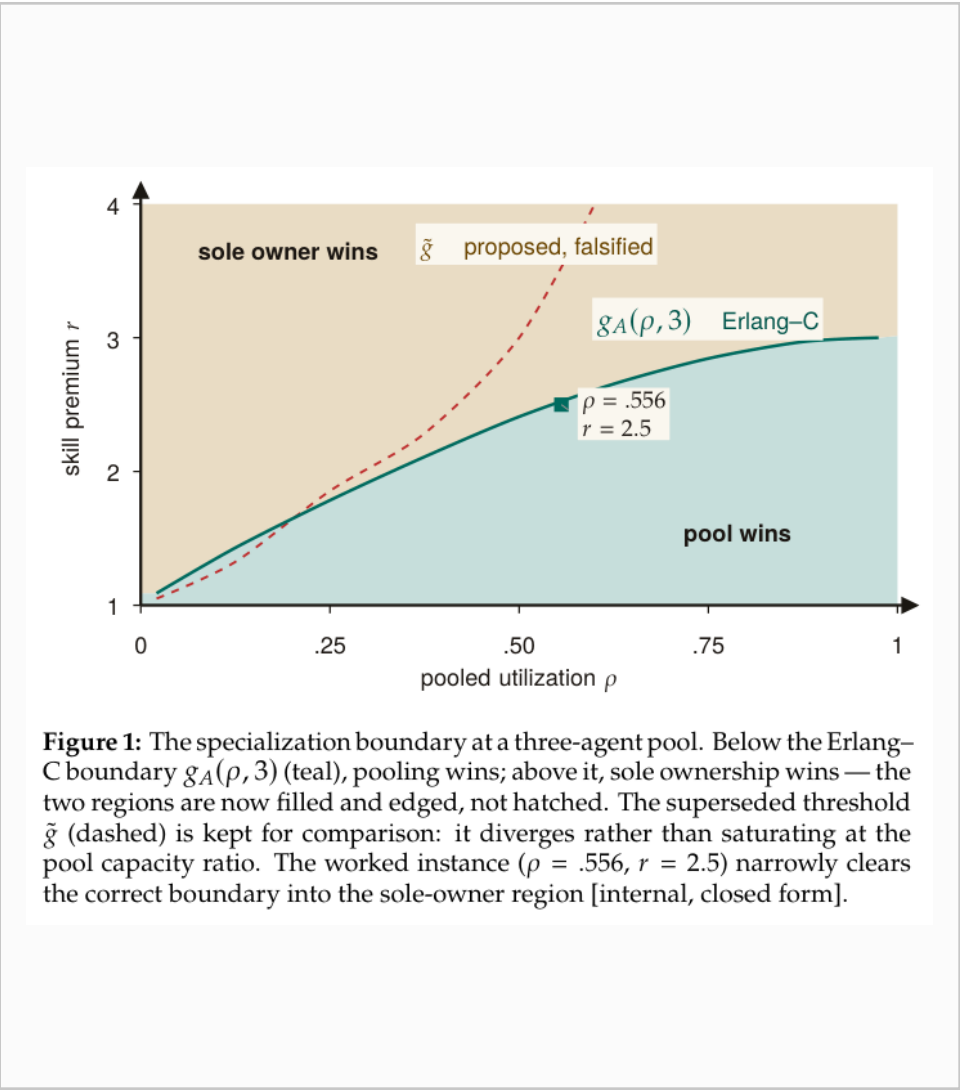


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QA: PASS

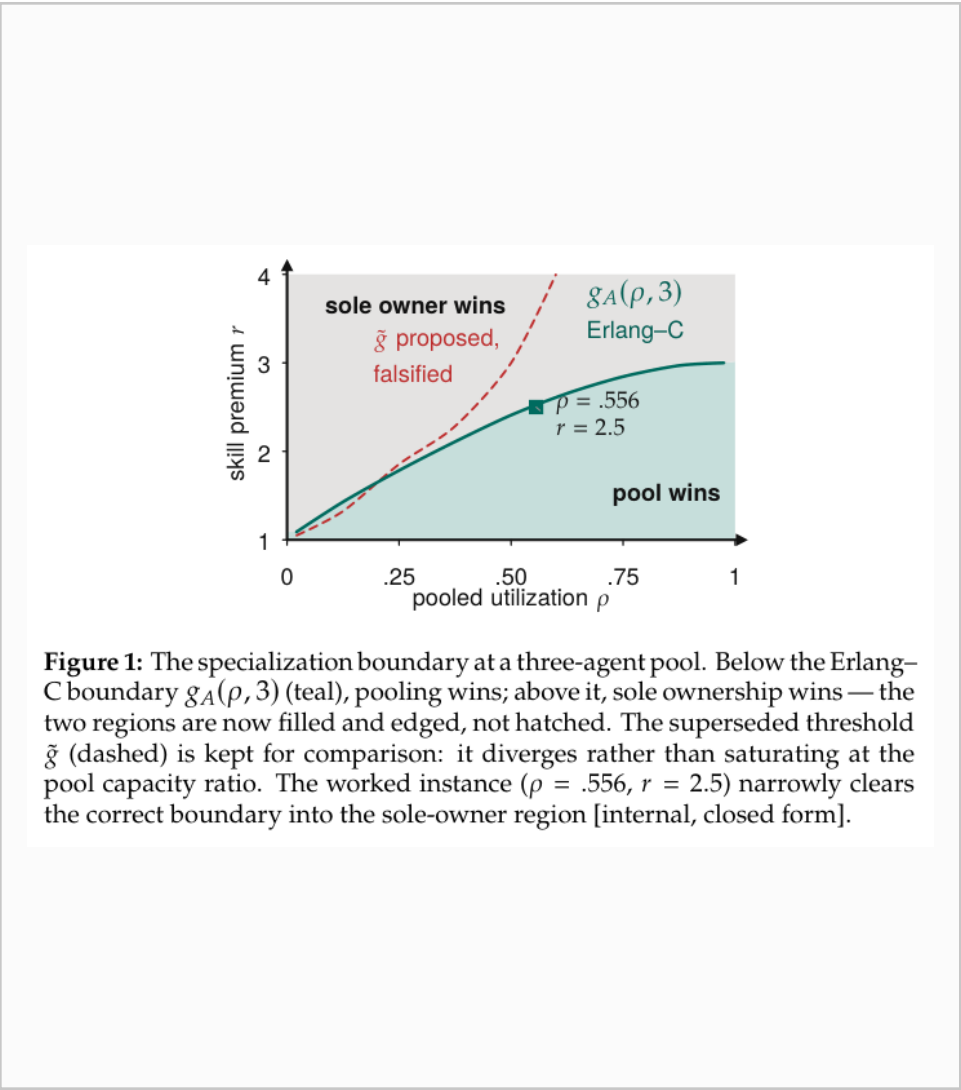
BEFORE



QA: FLAGGED: B3

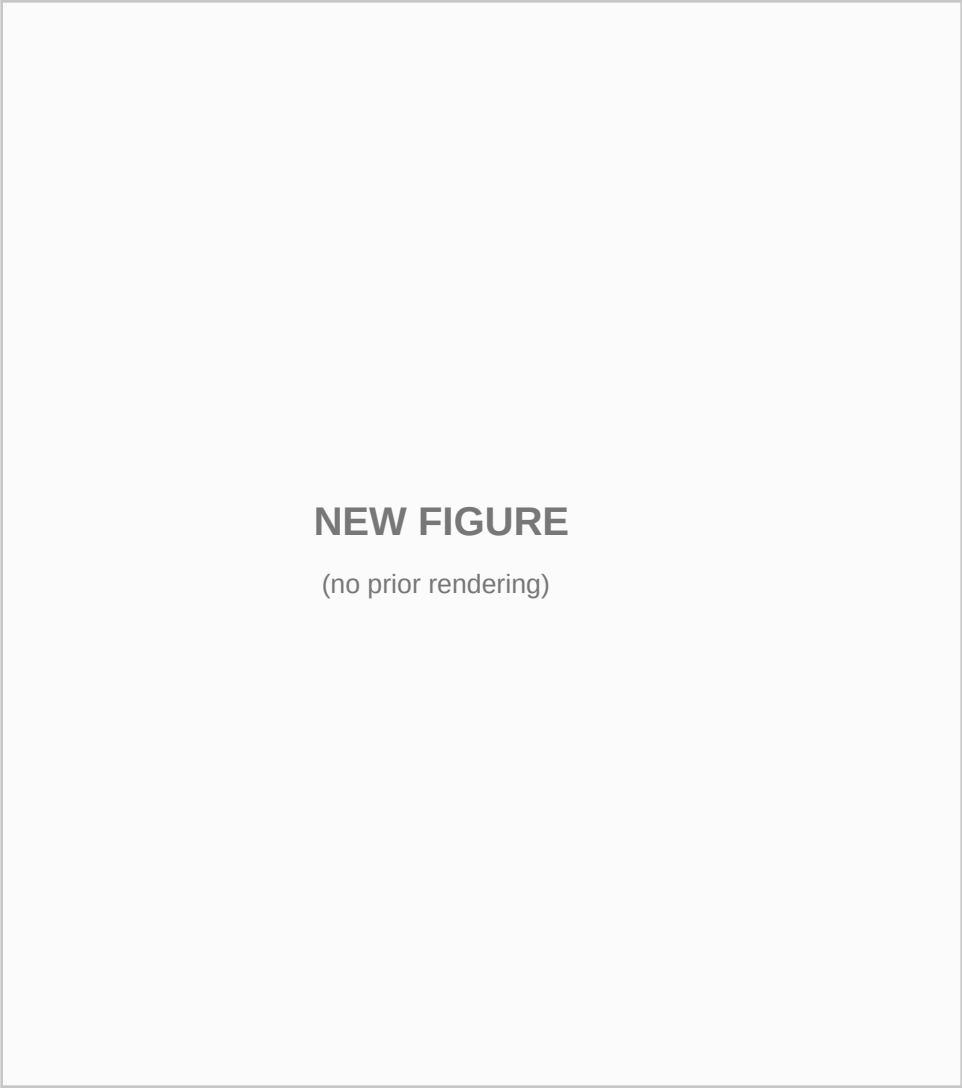
The specialization boundary at a three-agent pool. Below the Erlang--C boundary { \$g_A(,3)\$ (teal), pooling wins; above it, sole ownership wins --- the two regions are now filled and edged, not hatched. The superseded threshold \$ g\$ (dashed) is kept for comparison: it diverges rather than saturating at the pool capacity ratio. The worked instance (\$=.556\$, \$r=2.5\$) narrowly clears the correct boundary into the sol...

AFTER



QA: FLAGGED: B3

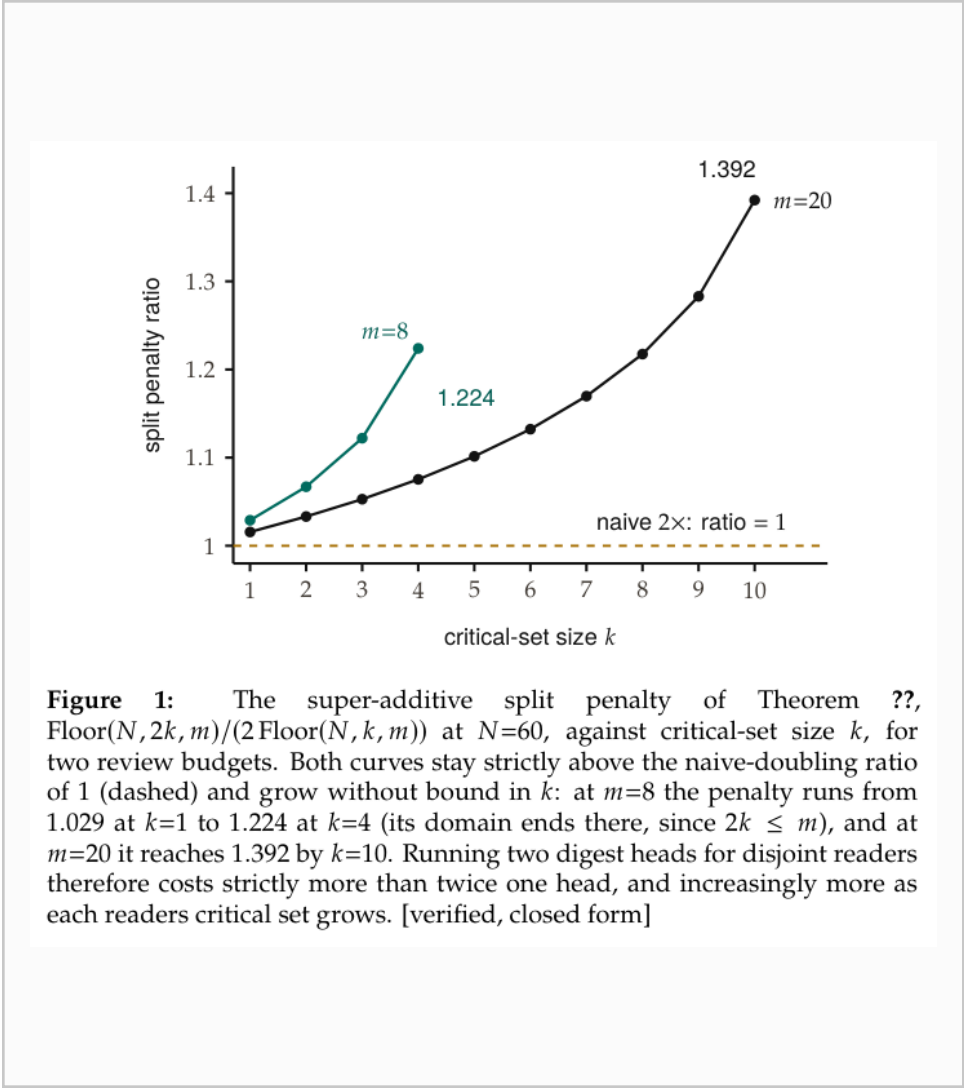
BEFORE



QA: —

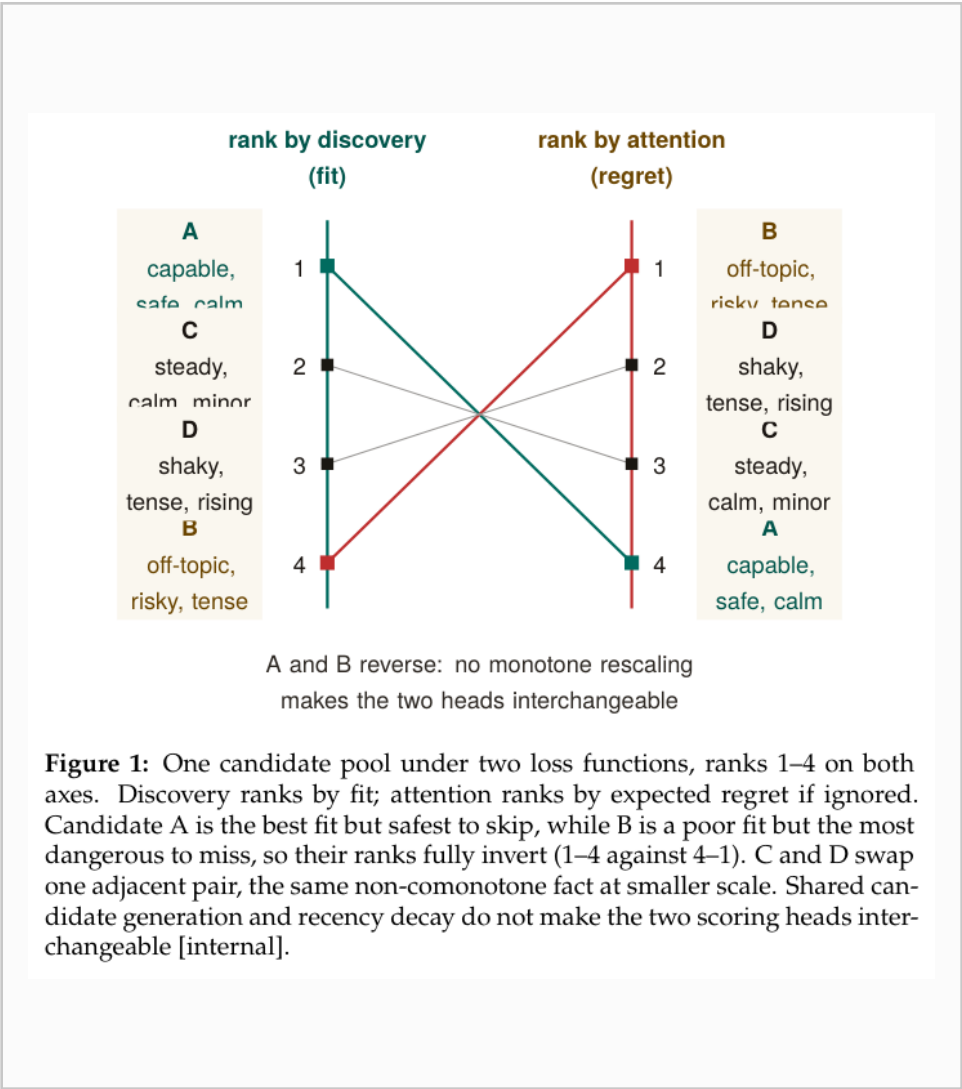
The super-additive split penalty of Theorem- $\{thm:split-penalty, \frac{\lfloor N, 2k, m \rfloor}{2 \lfloor N, k, m \rfloor}\}$ at $N=60$, against critical-set size k , for two review budgets. Both curves stay strictly above the naive-doubling ratio of 1 (dashed) and grow without bound in k : at $m=8$ the penalty runs from 1.029 at $k=1$ to 1.224 at $k=4$ (its domain ends there, since $2k \leq m$), and at $m=20$ it reaches 1.392 by $k=10$. Running two digest heads for disjoint readers therefore costs strictly more than twice one head, and increasingly more as each readers critical set grows. [verified, closed form]

AFTER

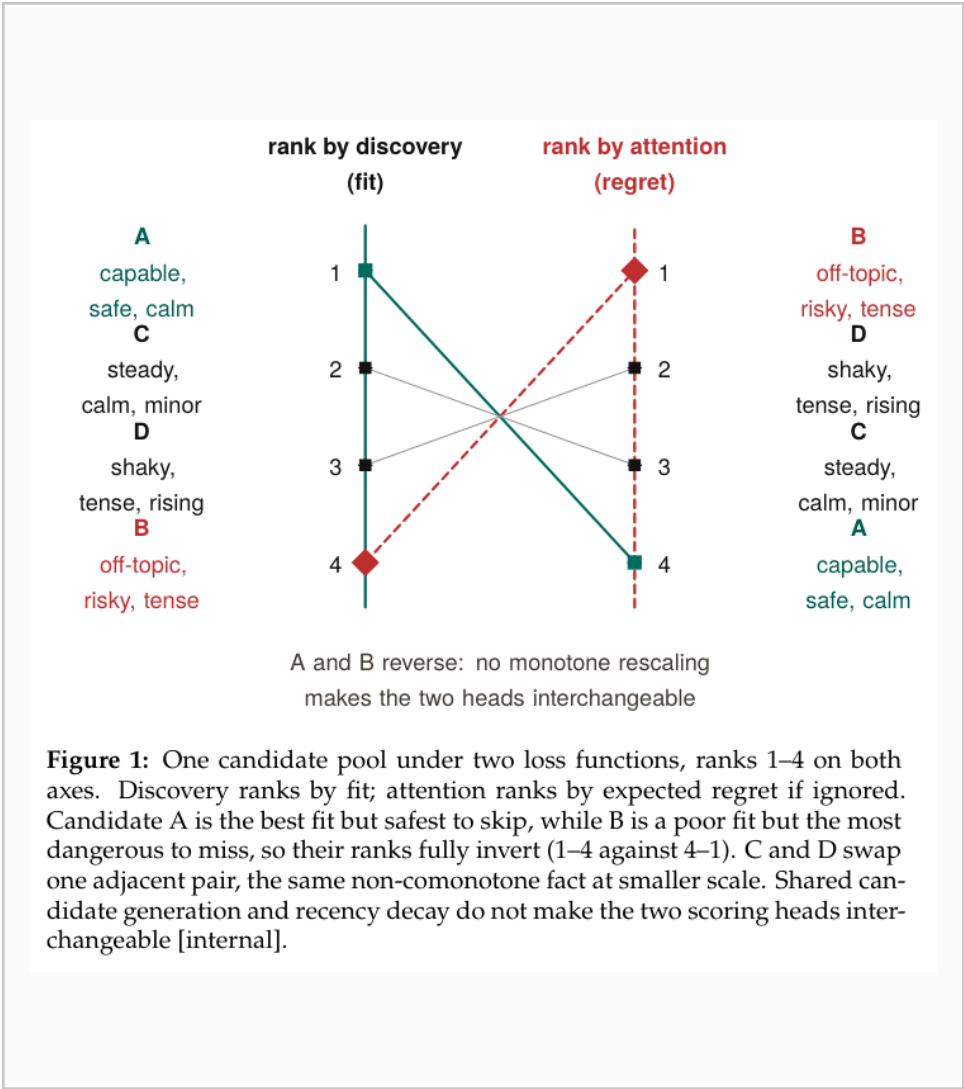


QA: PASS

BEFORE



AFTER



QA: PASS

QA: PASS

One candidate pool under two loss functions, ranks 1--4 on both axes. Discovery ranks by fit; attention ranks by expected regret if ignored. Candidate A is the best fit but safest to skip, while B is a poor fit but the most dangerous to miss, so their ranks fully invert (1--4 against 4--1). C and D swap one adjacent pair, the same non-comonotone fact at smaller scale. Shared candidate generation and recency decay do not make the two scoring heads interchangeable [internal].

BEFORE

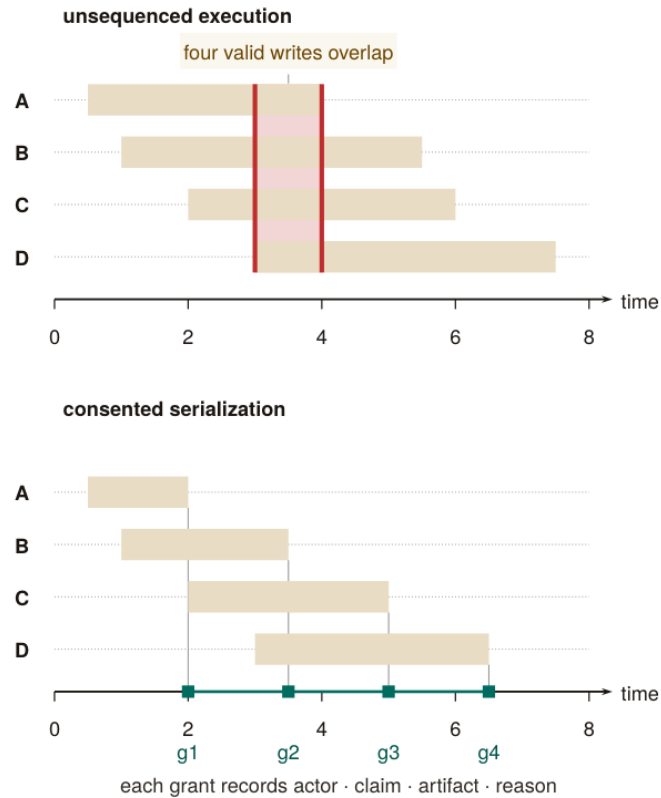


Figure 1: Two Gantt panels on one time axis. Top: four write requests A–D arrive between $t=0.5$ and $t=3$ with windows that all cover $[3,4]$, so four valid writers overlap on one artifact with no total order among them. Bottom: the same four requests under consented serialization, each band now ending at its own grant $g1$ – $g4$ ($t=2, 3.5, 5, 6.5$) on one commit spine, and every grant recording actor, claim, artifact, and reason. Consent changes the schedule, not who may participate [internal].

QA: PASS

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AFTER

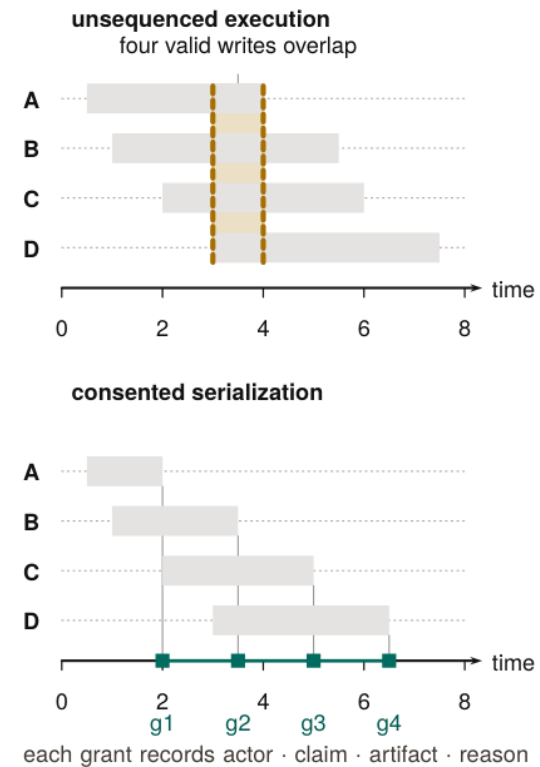
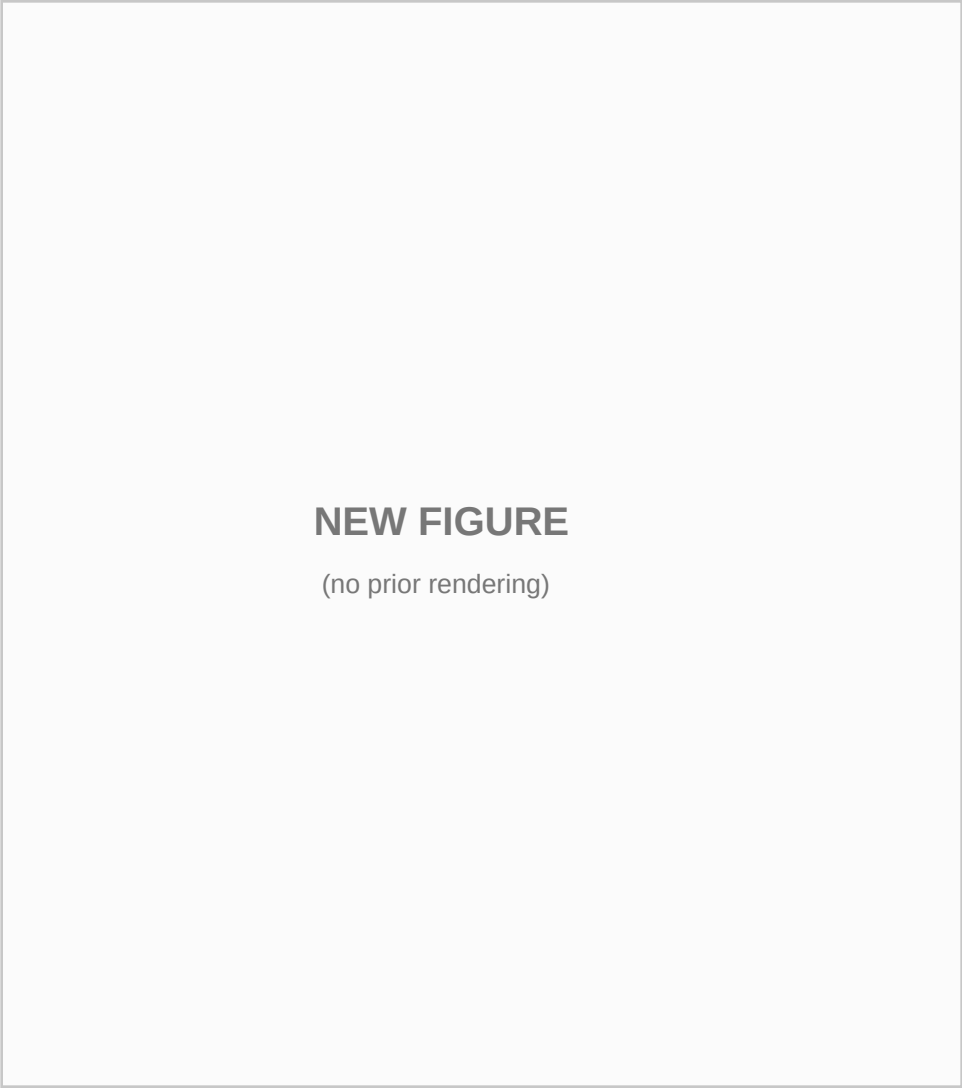


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QA: PASS

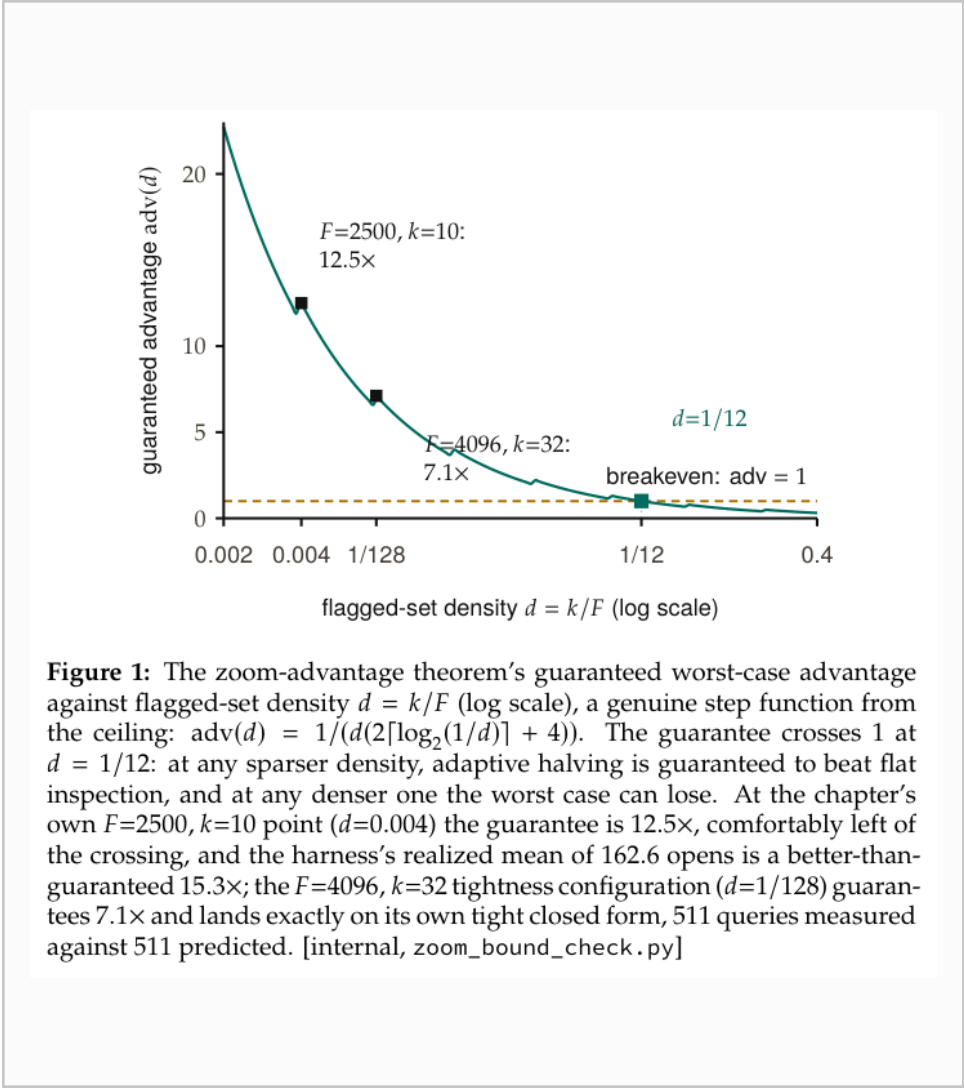
BEFORE



QA: —

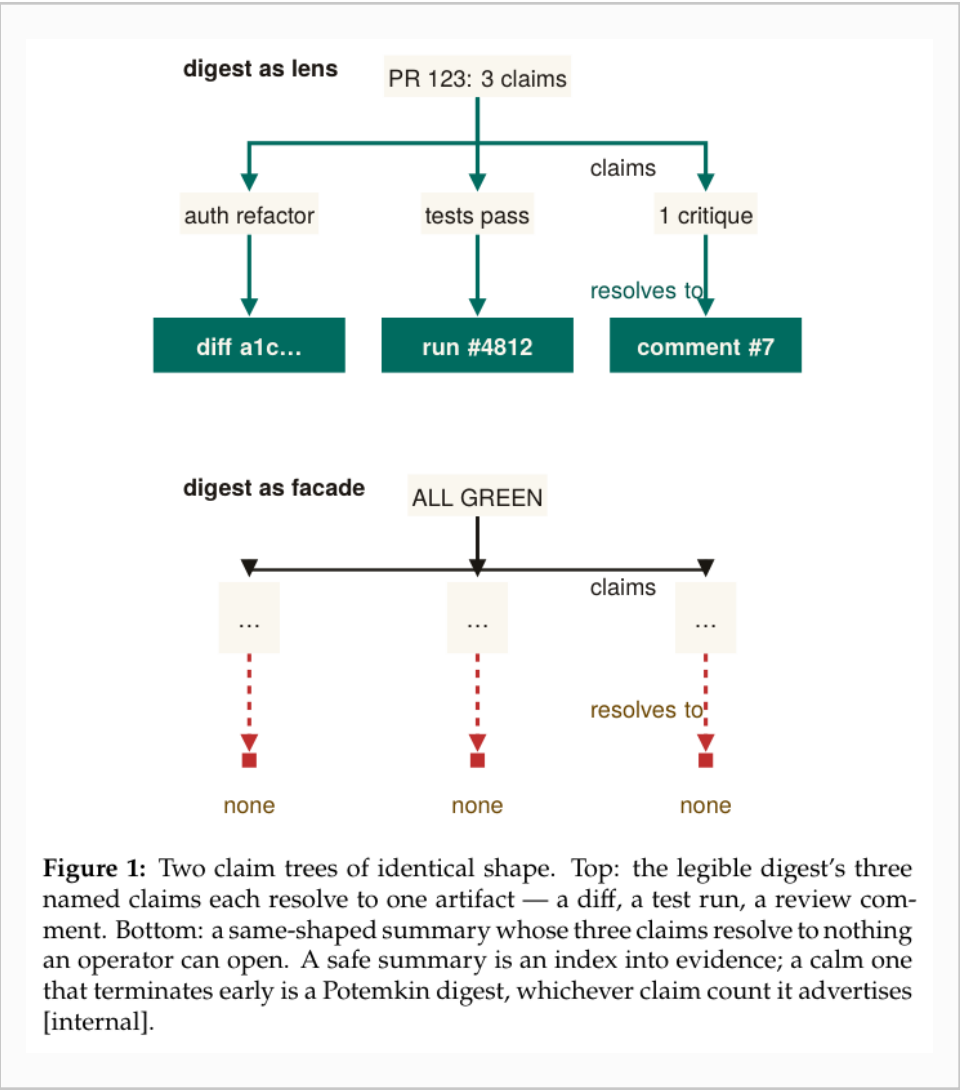
The zoom-advantage theorem's guaranteed worst-case advantage against flagged-set density $d=k/F$ (log scale), a genuine step function from the ceiling: $\text{adv}(d) = 1/(d(2\lceil \log_2(1/d) \rceil + 4))$. The guarantee crosses 1 at $d=1/12$: at any sparser density, adaptive halving is guaranteed to beat flat inspection, and at any denser one the worst case can lose. At the chapter's own $F=2500, k=10$ point ($d=0.004$) the guarantee is $12.5\times$; the harness's realized mean of 162.6 opens is a better-than-guaranteed $15.3\times$; the $F=4096, k=32$ tightness configuration ($d=1/128$) guarantees $7.1\times$ and lands exactly on its own tight closed form, 511 queries measured against 511 predicted. [internal, zoom_bound_check.py]

AFTER



QA: PASS

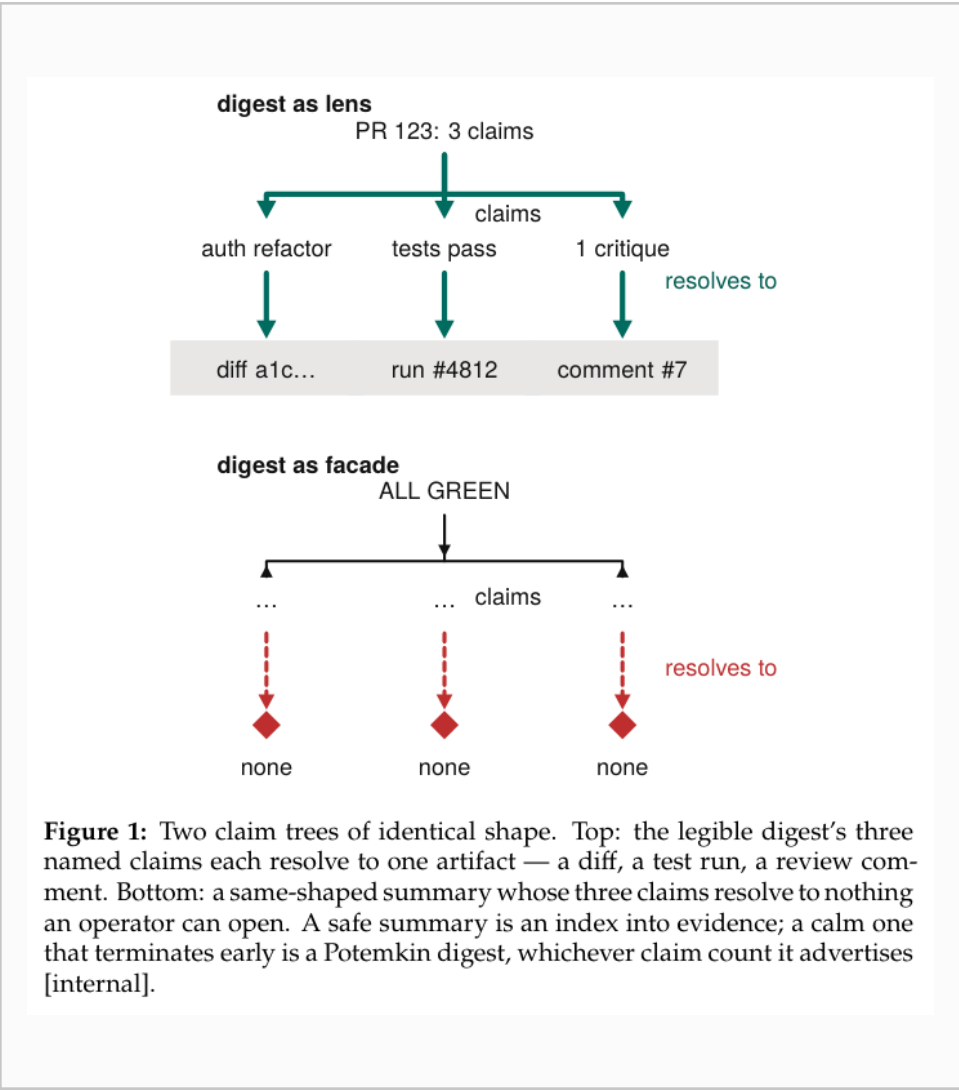
BEFORE



QA: FLAGGED: T4

Two claim trees of identical shape. Top: the legible digest's three named claims each resolve to one artifact --- a diff, a test run, a review comment. Bottom: a same-shaped summary whose three claims resolve to nothing an operator can open. A safe summary is an index into evidence; a calm one that terminates early is a Potemkin digest, whichever claim count it advertises [internal].

AFTER



QA: PASS